INDUSTRY STARTER PACKS · THE ARCHITECTURE WORKBOOK

Pharmaceuticals

Running the ADM in a pharmaceutical company — a step-by-step workbook for the repository you have been handed

The companion volume. Not something to read — something to work through. Every phase of the ADM as numbered steps: what you produce, which sheet of the repository it goes in, which columns, who signs it, how you know it is done, and the way it usually goes wrong. Written against a real populated metamodel and a real comparison tool, not against the standard in the abstract. And it carries the layer no other volume in this series needs: what each phase looks like when the systems themselves are regulated, and validation is a state you maintain rather than a thing you did.

Use it with the learner guide. South African lens. October 2026.

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CHAPTER 1

How to use this workbook

What you were handed, what is wrong with it, how to read a step, and the one thing that makes this volume different from its siblings: the systems themselves are regulated, so every phase changes.

The three files, and what each is actually for

You were handed three files. They are not three views of the same thing, and confusing them wastes the first week.

`Pharmaceuticals_TOGAF_Metamodel.xlsx` is a populated architecture repository for **Calderon Therapeutics**, an invented pharmaceutical company. 105 sheets. It is the thing this workbook works in: almost every step tells you which of its sheets to open and which columns to fill. Treat it as a baseline you inherited from somebody who left, because that is how it behaves. The skeleton is good and most of the flesh is generic, and telling the two apart is the skill this volume teaches [1].

`Pharmaceuticals_Architecture_Comparison_Tool_TEMPLATE.xlsx` is a baseline-versus-target diff tool, empty by design. You fill the baseline side from the repository and the target side from your own Phase B to D work, and it shows you the delta. It computes nothing the repository does not already hold, so do not load it before the repository is honest.

`Pharmaceuticals_ArchiMate4_Views.pptx` is a set of worked sample views. Use it to see what a finished view looks like before you draw your own. Item 16 takes the repository apart in ArchiMate 4 and tells you which rows have no element at all.

Three things none of them contains, which you should know before your first meeting: no validation state anywhere, no periodic-review date anywhere, and no named person anywhere. Every `Owner` column in the application sheet holds an organisation unit, and every `Performed by` cell in the role sheet reads `Person` [2][3]. In an industry where a release decision is signed by a named individual with a legal duty, that is not a cosmetic gap. It is the gap.

> This volume is a workbook. Open it at the step you are on, do the step, close it. Print the forms and write on them. Nothing here is written to be read front to back, and the one section that is — the next six pages — exists only so the rest makes sense.

The shape, in numbers you can check in ten minutes

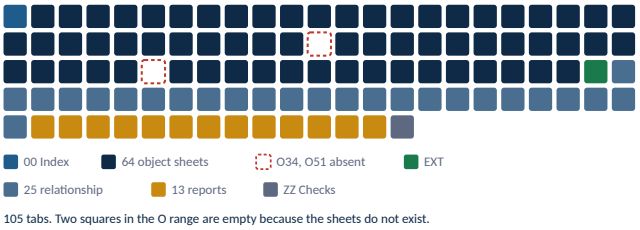


Figure 1.1 The **105 tabs**, in file order: the index, 64 object sheets with **`O34` and `O51` absent**, the industry extension, 25 relationship sheets, 13 reports and `ZZ Checks`. A loop from `O01` to `O66` fails twice.

105 tabs. They fall into six groups, and the group is readable from the prefix.

Group	Tabs	What it holds
`00 Index`	1	Contents page, with a per-sheet record count
`O01` – `O66`	64	Object sheets: one object type each
`EXT Industry Extension`	1	Six object types this sector needs beyond the TOGAF core
`R01` – `R25`	25	Relationship sheets, resolving between object sheets
`RPT01` – `RPT13`	13	Reports, computed over both
`ZZ Checks`	1	A fifteen-row self-assessment

Sixty-four object sheets numbered to 66, because **`O34` and `O51` do not exist**. There is no `O34 Something` hiding behind a filter; the tabs are absent. A loop from `O01` to `O66` fails twice, and a reader who assumes 66 object types will report two empty ones that are not there. `00`

`Index` lists 65 rows — the 64 `O` sheets plus `EXT`, which the index counts as the sixty-fifth object type [1].

The relationship sheets are the useful half and the half most people never open. `R06 Application interfaces with` is the interface register: 36 rows, and its `Managed` and `Logged` columns are the only honest statement in the repository about integration. `R09 Requirement satisfied by Bu` and `R13 Work Package delivers Build` both resolve their `Building block` column against **`O55 Building Block`**, the 37-row union sheet with a `Domain` column — not against the four domain sheets `O35`, `O44`, `O45` and `O46`. Say which sheet you mean every time you write "building block" [1].

One more trap. `O46 Technology Building Block` has **two columns headed `Ref`**. The second holds `T01`–`T07`, a key space nothing else in the file uses; call it `Ref (T)` when you must distinguish them. Nothing resolves against it, which is itself worth a question.

Row 4 is the header row

Every object sheet, relationship sheet and report has the same five-row opening, and getting it wrong is the most common way to waste an afternoon.

1	**Pharmaceuticals — TOGAF Content Metamodel, populated**
2	Calderon Therapeutics · Business domain · 12 records
3	*(blank)*
4	**Ref | Organisation unit | Type | Head | Headcount**
5	ORG-01 | Executive | Group | Chief Executive | 151
6	ORG-02 | Finance | Corporate | Chief Financial Officer | 265

Read row 1 as the header and every column is named after the sheet title.
Headers are row 4. Data starts at row 5. The first column is always Ref.

Figure 1.2 The five-row opening of every object sheet, shown for `O09 Organization Unit`. **Headers are row 4 and data starts at row 5** — `header=3` in pandas, `min_row=4` in openpyxl.

Row 1 is the sheet title. Row 2 is a subtitle reading `Calderon Therapeutics · <domain> domain · <n> records`. Row 3 is blank. **Row 4 is the header row. Row 5 is the first record.** The first column is `Ref` on every object sheet.

Any tool that defaults to the first row as the header gets nonsense: one column named after the sheet title and the rest unnamed. In pandas that is `header=3`; in openpyxl, `iter_rows(min_row=4)`. Write the constant down once at the top of your script and never type the number again by hand.

The reports will still bite you. `RPT04 Interface Catalogue` has 37 data rows and the thirty-seventh is **a sentence** — the string `19 of 36 interfaces are unmanaged and 31 carry no logging.` sitting alone in the `Ref` column. It is not alone in doing that: **nine of the thirteen reports end in a prose footer** — `RPT01`, `RPT04`, `RPT05`, `RPT06`, `RPT08`, `RPT09`, `RPT10`, `RPT11` and `RPT13`. Only `RPT02`, `RPT03` and `RPT12` end cleanly, and `RPT07 Gap Analysis` is empty, so it cannot [4]. The rule is therefore not "watch out for `RPT04`" but: **on every report, test whether the last populated row has one cell filled, and drop it if it does.** Then your interface count is 36 and not 37, and your orphan count is two and not three.

Twenty-three tab names are cut, and one ends in a space

Excel truncates a sheet name at 31 characters without warning, and whoever built this file wrote longer names. Twenty-three tabs are cut mid-word:

Tabs cut at 31 characters
`O26 Logical Technology Componen`, `O27 Physical Technology Compone`
`R02` – `R10`, nine tabs, each ending mid-phrase
`R13` – `R20`, eight tabs
`R23 Increment delivered by Proj`, `R24 Compliance Assessment tests`, `R25 Application supports Produc`
`RPT06 Requirements Traceability` — exactly 31, so you cannot tell whether it was cut

And one is worse than cut. **`'R04 Business Service supported '` ends in a space.** A lookup on `'R04 Business Service supported'` fails with a key error you will read three times before you see it.

The rule that survives all of this: **match on the prefix, never on the full phrase.** `startswith('R04')` works. `== 'R04 Business Service supported by Application'` does not, because that tab name has never existed in this file. Everything else is intact: the other 62 object

sheets, `R01`, `R11`, `R12`, `R21`, `R22`, and twelve of the thirteen reports.

The headline on the front page is wrong

`00 Index` row 2 reads: `Calderon Therapeutics · 65 object types · 25 relationship types · 850 records · 437 relationship instances`. `RPT01 Dashboard` repeats the same four figures and adds fourteen more; `ZZ Checks` repeats them again against a threshold column [1][4]. Count them yourself and three of the four do not hold.

Claim	Stated	Counted	Verdict
Object types populated	65	**At most 63**	Wrong. `O29 Gap` and `O59 Compliance Assessment` have a header row and nothing under it
Records	850	**814**	Wrong by 36
Relationship types	25	25 sheets, **23 with rows**	Half true. `R12` and `R24` are empty
Relationship instances	437	**437**	Correct. Summed across `R01` – `R25` it lands exactly

The record count is the interesting one, because the repository contradicts itself **inside a single sheet**. Row 2 of `00 Index` says 850. Rows 15 to 79 of the same sheet give a count per sheet — `13 records`, `6 records`, `14 records` and so on — and those sixty-five numbers **sum to 814**, which is what the sheets actually hold [1]. So the detail is right and the headline is wrong, and nobody ever added up the detail.

Where did 36 come from? `R06 Application interfaces with` has exactly 36 rows, and it is a relationship sheet, so its rows are not records. Treat that as a hypothesis, not a fact: nothing in the file states the arithmetic, and you cannot prove intent from a number that happens to match. What you can state in a steering meeting is the part you counted — 814 records, 437 relationship instances, two empty object sheets, two empty relationship sheets, one empty report.

> **Read `ZZ Checks` as a worked example of a dashboard that is honest about nine things and wrong about the rest.** The interface figures, the support-currency figures, the security figures and the work-package figures all hold up against the underlying sheets. `Object types populated: 65` and `Records: 850` do not. And `RPT01`'s `Gaps overdue: 0` is not wrong so much as **meaningless** — `O29 Gap` is empty, so nothing can be overdue. A zero against an empty source is the most dangerous number on any dashboard, and here it sits directly above a work-package line that is correctly reporting three overruns [4].

One structural note. The two sheets are not the same sheet. `ZZ Checks` carries fifteen checks with a `Threshold` column; `RPT01 Dashboard` carries eighteen measures with a `What it means` column. Fourteen lines are common. `ZZ Checks` alone carries `Records validated against a real organisation`; `RPT01` alone carries `Applications`, `Interfaces`, `Gaps overdue` and `Level 1 capabilities`. Say which sheet you took a figure from, because a reviewer who checks the other will not find it [4].

Step 1.1 — Count the repository before you quote it

Produce. A one-page count sheet: rows per object sheet, per relationship sheet and per report, and the four headline figures recomputed from those counts.

You need. The repository, any reader that lets you set the header row, and forty minutes. No workshop, no stakeholder. Do this before you meet anybody.

> **`No sheet — this stays on paper`**
>
> This count does not go into the repository, and you are not to create a sheet for it: the four new sheets this volume authorises are `O67` – `O70`. Fill **Form 1-A** and staple it to the Statement of Architecture Work when you issue it in item 3.

How.

1. List the tab names exactly as the file gives them, trailing space on `'R04 Business Service supported '` included, and record which 23 are cut at 31 characters.
2. For every sheet, read the header from row 4 and the data from row 5 down. Count rows where the first cell is populated.
3. On every `RPT` sheet, check whether the last populated row has only one cell filled. If it does, it is a prose footer, not a record. Drop it and note which sheets needed that.
4. Sum the object-sheet counts plus `EXT`. Sum the relationship-sheet counts. Count the reports that hold at least one record.
5. Compare your four totals with `00 Index` row 2, `RPT01` rows 5 to 8 and `ZZ Checks` rows 5 to 8. Write the differences in words, not as a percentage.
6. Name the empty sheets: `O29`, `O59`, `R12`, `R24`, `RPT07`.

Signed by. Nobody. It is your working paper and needs no approval. Keep it: every number you quote for the next ninety days traces back to it, and you will be asked where a figure came from.

Done when. You can state the record count, the relationship-instance count, the empty sheets and the truncated tab names from memory, and explain in one sentence how your figure differs from the repository's headline.

Shows up in. Nowhere in the repository, which is the point. It shows up in what you say: when somebody quotes 850 records as evidence the estate is documented, you say 814, and name the empty sheets.

Goes wrong when. You load the file with the default header row, get sixty-five tables whose first column is named after the sheet title, and conclude the repository is broken. It is not broken; you read it from the wrong row. The second failure is quieter: you accept the headline, build a Phase A vision on 850 records, and discover in Phase G that the compliance sheet meant to carry your evidence has never had a row in it.

Form 1-A — Repository count sheet

Feeds no sheet. It stays on paper, filed with the Statement of Architecture Work. One row per sheet group plus a total. A first pass needs about fourteen rows. Signed by nobody.

Sheet or group	Claimed	Counted	Difference	Empty sheets	Checked on
EXAMPLE — delete: `001 – 066 plus EXT`	850	814	36	`029 , 059`	

Step 1.2 — Test the dashboard's claims one at a time

Produce. An annotated copy of `ZZ Checks` and `RPT01 Dashboard` with each line marked *holds*, *wrong*, or *meaningless against an empty source*.

You need. Step 1.1's count sheet and the underlying sheets named below. Nothing else.

> **`ZZ Checks` and `RPT01 Dashboard`**
> Headers on row 4. `ZZ Checks` columns are `Check`, `Result`, `Threshold`; `RPT01` columns are `Measure`, `Value`, `What it means`. Both are reports, not records, so your annotations go on paper or in a separate file, never over the report.

How.

1. Take the interface lines first, because they hold. `Interfaces: 36` against `R06`'s 36 rows. `Unmanaged: 19` against `R06.Managed`, whose values are the words `Managed` and `Unmanaged`, not `Yes` and `No`. `No logging: 31` against `R06.Logged`, where five of 36 read `Yes`. `Direct database reads: 6` against `R06.Type` [5].
2. Take the ones that hold against small sheets: two platform components past support in `O27 Physical Technology Compone`, two absent controls in `O41 Security Control`, four entities with no lineage in `O20 Data Entity`, three work packages past their end date in `O30 Work Package`.
3. Mark the two that are wrong: `Object types populated` and `Records`.
4. Mark `Gaps overdue: 0` meaningless, and write why: the source sheet is empty.
5. Mark `Vacant architecture roles: 0` as true but uninformative — every row of `O10 Actor` reads `Filled`, and there are eighteen rows, none of which is a pharmaceutical job [3].
6. Keep the annotated page. It is the most useful artefact you own in week one.

Look it up. Volume 1, chapter 19, section *The checks that take an afternoon*.

Signed by. Nobody yet. Item 2 puts this in front of `GB-01 Architecture Board`, defined in `O57 Governance Body` — but it is not a governance artefact until a governance body meets.

Done when. Every line on both sheets carries one of the three marks, and you can give the underlying sheet and column for each without looking.

Shows up in. `RPT01 Dashboard` itself, once your corrections land in the object sheets: the record count moves as you add rows, to a number you can defend.

Goes wrong when. You dismiss the whole dashboard because two lines are wrong. Nine of fifteen checks hold. A reviewer who has heard you call the dashboard useless, and then finds the interface figures are exactly right, stops believing the rest of your findings — and the interface figures are the ones you most need them to believe.

The six objects the industry runs on are declared and empty

`EXT Industry Extension` is three columns and six rows. It names the object types a pharmaceutical company needs beyond the TOGAF core, with a sentence each on why the core is not enough [6]:

Ref	Industry extension type	What the sheet says it holds
`X01`	Compound	A molecule or biologic under investigation, with its programme
`X02`	Study / Trial	A clinical investigation with its protocol, sites and phase
`X03`	Batch / Lot	A manufactured quantity with genealogy and disposition
`X04`	Regulatory Submission	A dossier to an authority with its status and commitments
`X05`	GxP Computerised System	A validated system with its validation state and owner
`X06`	Adverse Event	A safety event with causality, seriousness and reportability

The sheet holds **no instances of any of them** — a declaration of six object types and a population of zero. `00 Index` nonetheless counts `EXT` as the sixty-fifth populated object type and credits it with six records: the six type declarations themselves.

Five of the six have a partial home elsewhere: `O20 Data Entity` carries entities called `E01 Compound / Molecule`, `E02 Study / Trial`, `E09 Batch / Lot`, `E06 Regulatory Submission` and `E08 Adverse Event`. They are data entities, not the industry objects, and their

`Master system` column is wrong eight times out of fourteen — item 5 takes that apart. But `X05 GxP Computerised System` has **no home at all**: no entity, no column for validation state, no sheet listing which systems are in GxP scope [2][7]. That is why item 2 opens on `X05` and creates `O67 Validated System`. You cannot write a vision with a credible date, classify an application, qualify infrastructure, plan a migration or hold a release gate without it.

Step 1.3 — Prove the six industry objects are empty, and say so in writing

Produce. Two sentences for your findings log, each with a sheet reference, stating that the six industry object types are declared and unpopulated and that `X05` has no equivalent anywhere.

You need. `EXT`, `O20 Data Entity`, `O23 Application Component`, and the willingness to write a negative finding before you have a solution.

> **`EXT Industry Extension`**
>
> Three columns: `Ref`, `Industry extension type`, `What it holds and why the core metamodel is not enough`. Six rows, `X01` to `X06`. You add no rows here — it declares types. The instances go in `O67` and, for the other five, in the sheets items 4 to 7 name.

How.

1. Read all six rows of `EXT` and copy the `Ref` and type name into your log.
2. Search every object sheet for a row whose `Ref` starts `X`. There are none outside `EXT`.
3. In `O20 Data Entity`, list the five entities that carry industry-object names and note their `Ref`. Note that there is no entity for a validated or computerised system.
4. In `O23 Application Component`, read the ten column headers. Confirm there is no validation-state column, no GxP flag and no review date. The columns are `Ref`, `Application`, `Type`, `Description`, `Disposition`, `Criticality`, `Health`, `In service`, `Users`, `Owner`.
5. Write the two sentences. Name the sheets. Do not soften them and do not attach a remediation estimate yet.

Look it up. Volume 1, chapter 9, section *The object most architects leave out: the validated system itself*; and chapter 11, section *The capability no other industry has: the register of validated systems*.

Signed by. You, as the author of the findings log. It becomes a governance artefact in item 2, Step 2.1, once the question of whether an architecture decision is a quality record has been answered.

Done when. The two sentences exist, each names a sheet, and neither contains the word "opportunity".

Shows up in. `RPT13 Orphan Analysis`, indirectly — it finds two applications no business service maps to, and cannot find the larger problem, that no sheet maps anything to a validation state.

Goes wrong when. You read `EXT` as six populated object types because the index says six records, and report the industry extension as in place. It is in place as a schema and absent as data, and the difference is the whole of items 2 to 7.

Step 1.4 — Agree the four new sheets and their numbers before anybody types

Produce. A one-line decision fixing `O67` to `O70` to four named sheets, and a note that no fifth sheet is to be created.

You need. Step 1.3's findings and fifteen minutes with whoever controls the file. If nobody controls the file, that is your first governance finding.

> **`The four sheets this volume tells you to create`**
>
Sheet	Defined in	What it carries
`O67 Validated System`	item 2	The register: one row per computerised system, with its GxP regimes and validation state
`O68 Periodic Review Schedule`	item 11	When each system's validated state is next re-proven, and by whom
`O69 Retention Clock`	item 5	Per record class: the period, the event the clock starts at, the instrument
`O70 Regulated Role`	item 2	The statutory roles, with a named person and what each signs for

How.

1. Fix the numbering now. `O67` to `O70`, in that order, those four names. The numbers are free because the object range stops at `O66`.

2. Do not fill the holes at `O34` and `O51`. Anything you put there reads as part of the original TOGAF content metamodel set, and a later reader will trust it more than it deserves.
3. Write the four sheet names into `00 Index` below the `EXT` row, with `0 records` against each, so the contents page tells the truth about what exists and what is planned.
4. Agree who may add a sheet. One person. Record the name.
5. Note in your log that items 2, 5 and 11 define these sheets in full and every other item only references them. If you find a fifth new sheet proposed anywhere, it is a defect.

Signed by. `GB-01 Architecture Board`, per its mandate in `O57 Governance Body`, which reads *Holds G1 and G2; approves dispensations with an expiry.* The board is the only body in the repository whose mandate covers the repository's own structure, through `GB-05 Architecture Function`, whose mandate reads *maintains the repository* [8].

Done when. `00 Index` lists 69 object-sheet rows: the 64 that exist, `EXT`, and the four you have reserved — and the four reserved rows say `0 records` rather than being quietly omitted.

Shows up in. `00 Index`, and then in every count you do from here on. Your record total will not change, because reserving a sheet adds no records.

Goes wrong when. Two people create a validated-system register in the same fortnight, one called `O67 Validated System` and one called `O67 GxP Systems` with eight columns. Agree the numbering and the columns on the same day, or you will merge two registers in Phase D with a vendor in the room.

How to read a step

Every numbered step has the same eight parts in the same order. Learn them once and you can read a step in thirty seconds and work it in an hour.

Label	What it tells you
Produce.	The artefact. One sentence. If you cannot name the artefact, the step is not ready
You need.	Inputs: prior steps, people, and the volume 1 chapter that teaches the theory
Where it lands.	A bordered sheet block naming the sheet and the columns, with what goes in each non-obvious one
How.	Six numbered actions or fewer, each something you do rather than consider
Signed by.	The body from `057 Governance Body`, or the named role. Where the real signatory is a regulated role the repository lacks, the step says so
Done when.	A testable condition, not a feeling
Shows up in.	The `RPT` sheet where the work becomes visible, and what the number should look like
Goes wrong when.	The real failure mode, named concretely, usually from the step before

Two of those labels do the work that makes this volume different from its siblings.

Signed by is where a regulated estate shows through. In most industries the signatory is a governance body and the only question is which one. Here it is sometimes a named individual with a statutory duty — the Responsible Pharmacist, the person who releases a batch, the officer responsible for pharmacovigilance — and **none of those roles exists in this repository** [3]. So every step needing a regulated signature says who should sign, says the role is absent, and says who signs in the meantime. Item 2 creates `070 Regulated Role` to end that.

Shows up in keeps you honest. A step whose output never appears in a report has not been done; it has been discussed. Item 14 catalogues all thirteen reports.

Some steps carry a **Look it up.** line naming the volume 1 chapter and the exact section heading behind the step. Volume 1 is 246 pages across 23 items and this volume does not repeat it: the reasoning is there, the procedure is here.

The two things you write in

Two blocks are meant to be written on with a pen.

A **form** is a bordered card whose columns are the real columns of a real sheet, in sheet order, spelled as the sheet spells them. Each carries a form number, the sheet it feeds, how many rows a first pass needs, and who signs it. Where a sheet has more than six columns the form splits into A, identify, and B, assess, and says which columns you type straight into Excel. Where it feeds a sheet the repository does not have, it says so on its face — here, `067` to `070`.

An **ask** is questions with ruled space under each: what to put to a person, in their words, before you open the sheet. Almost every defect in this repository is a column filled by pattern rather than by someone who knew the answer.

THREE QUESTIONS TO ANSWER BEFORE YOU WORK ANY STEP IN THIS VOLUME

Which systems in this company hold a GxP record, and who told you so?

Who is permitted to change this repository, and where is that written down?

When this workbook and volume 1 disagree with the repository, which one does your client believe?

How this workbook maps onto the ADM

Eleven of the twenty items are a phase. The other nine are orientation, reference and the fill-in-able material you will use again on your second engagement.



Figure 1.3 The one-page map. Each ADM phase, the item of this workbook that covers it, and the sheets that item works in. A `+` marks a sheet you create: `O67` and `O70` in item 2, `O69` in item 5, `O68` in item 11.

The map above is the one page to photocopy. Read it as: this phase, this item, these sheets. A `+` marks a sheet you create. Four notes on the ordering, which is not quite the ADM's.

Phase C is two items, because data and application fail differently here: data on classification and mastering, application on the absence of validation state. **Phase D is "Technology Architecture and the validated estate"**, because the distinction between qualification and validation lands there, and it is the one architects most often collapse. **Phase G is the longest chapter in the volume**, because it contains the release decision and that is signed by a person with a legal duty. And **Requirements Management is an item of its own**, because the user requirement specification lives there — the first document in the validation V model, and the same artefact as a TOGAF requirement with the evidence burden attached.

Item	Covers	Principal sheets
1	Orientation: the files, the shape, the counting	`00 Index`, `ZZ Checks`, `RPT01`, `EXT`
2	Preliminary: standing, roles, bodies, principles, tailoring	`O57`, `O58`, `O05`, `O32`, `O09`–`O11`, `+O67`, `+O70`
3	Phase A: vision, scope, stakeholders, the date problem	`O01`–`O04`, `O06`, `O31`, `O53`, `O54`
4	Phase B: replacing a generic corporate	`O09`–`O11`, `O13`–`O17`, `O35`, `R22`
5	Phase C data: classification, mastering, retention	`O20`, `O36`, `O37`, `O42`, `O43`, `+O69`
6	Phase C application: the portfolio's contradictions	`O23`, `O24`, `O45`, `R04`–`R06`, `RPT02`
7	Phase D: qualification against validation	`O25`–`O28`, `O46`, `O48`–`O50`, `RPT10`
8	Phase E: gaps, packages, two live alternatives	`O29`, `O30`, `O52`, `O54`, `O55`, `O66`
9	Phase F: migration as a validation activity	`O39`, `O40`, `O53`, `O56`, `R15`
10	Phase G: compliance assessment and the release signature	`O33`, `O58`, `O59`, `O60`, `R24`
11	Phase H: periodic review and change control	`O61`, `O62`, `O63`, `+O68`
12	Requirements Management and the user requirement specification	`O06`–`O08`, `R08`, `R09`, `RPT06`

Item	Covers	Principal sheets
13	Sheet-by-sheet reference, all 105 tabs	every sheet
14	The thirteen reports, and which ones embarrass you	`RPT01` – `RPT13`
15	Workshop packs, checklists, ninety-day calendar	`009` – `011` , `013` , `020` , `023`
16	The repository in ArchiMate 4, and what has no element	`007` , `011` , `055`
17	Where to look in volume 1	—
18	Every form, printable, in one place	all four new sheets
19	The principles and standards starter set	`005` , `032`
20	The maturity assessment, scoring absences	`ZZ Checks` , every sheet

Why a regulated estate changes every phase

Here is the thing that makes this volume different from the four siblings in this series, said once, plainly, so that the per-phase warnings later make sense.

In every other industry, the architecture governs the systems. In this one, **the systems are themselves regulated.** A computerised system that holds a GxP record is validated, which means its lifecycle, its change control, its access model, its audit trail and its decommissioning are all regulated. "Just patch it" does not exist. Neither does an unattended dependency bump, nor a tenant the vendor upgrades on its own cadence, nor a configuration change made by an administrator without a change record [9].

Three consequences you will feel in every phase.

A change is a validation event before it is a backlog item. An impact assessment, a documented risk evaluation, re-testing proportionate to the risk, approval by quality assurance, and an update to the validation documentation set. That is not overhead around the change; it is the change. A plan that treats it as a percentage uplift on effort is wrong by a multiple on the systems that matter.

Validation is a state you maintain, not a thing you did. A system is validated today and validated-with-an-overdue-review in eighteen months, having had nothing done to it. Periodic review keeps the state true, and the repository has no column for a review date — so it cannot tell you about a single system whose validated state has quietly lapsed.

The signatory is often a person, not a committee. Batch release, submission, expedited safety report: each has a named individual behind it, in some cases with a professional registration and a personal duty. An architecture that routes those decisions to a board has misread the industry.

And one consequence for you, which is the subject of item 2. **The architecture function has no standing in the quality system.** Five governance bodies sit in `057` and not one is a quality body — no change control board, no validation committee, no data integrity forum [8]. So the question you answer in week one is whether an architecture decision is a quality record. If it is, it is controlled, versioned, approved and retained, and it binds. If it is not, nobody has to follow it, and everything in this volume is advice.

Look it up. Volume 1, chapter 13, sections *The system itself is regulated* and *Which GxP, and for which record*. Those pages are the teaching half of this volume. Read them before item 2 and again before item 7.

Two invented companies, kept apart

Calderon Therapeutics is the invented company in the repository you were given. Its data is real within this workbook: when a step shows a filled-in sheet, the values come from Calderon. It appears nowhere in volume 1.

Veldtkruin Group is volume 1's invented group — a generics manufacturer and a contract developer, with systems named Kelpfield, Sentinel, Rooibos and the rest. It appears here only when a step points back at volume 1.

Both are inventions. Never mix their data, and never quote a Calderon figure as an industry benchmark: the figures are internally consistent and describe no real organisation [1].

Before you turn the page

Four things to have done before you work item 2. You have counted the repository and know your four numbers differ from its headline. You have read volume 1 chapter 13. You know which 23 tab names are cut and that one ends in a space. And you have accepted that the most important sheet in a pharmaceutical architecture repository is the one this repository does not have.

Item 2 creates it.

THE CHAPTER IN TEN LINES

- Three files: a 105-sheet repository, an empty comparison tool, a set of sample views. Only the first one is populated.
- Headers are on row 4 and data starts on row 5. The first column is Ref on every object sheet.
- There are 64 object sheets, not 66. O34 and O51 do not exist, and a loop over the range fails twice.
- Twenty-three tab names are cut at 31 characters, and R04 ends in a space. Match on the prefix, never the full phrase.
- The repository claims 850 records and holds 814. The per-sheet counts on 00 Index sum to 814, so the sheet contradicts its own subtitle.
- It claims 65 populated object types and has at most 63: O29 Gap and O59 Compliance Assessment are empty.
- The 437 relationship instances figure is correct. Count the rest yourself before you quote any of it.
- Nine of the thirteen reports end in a prose footer. Drop the last row of any report whose final row has one cell filled.
- Gaps overdue reads zero because the gap sheet is empty. A zero against an empty source is the most dangerous figure on a dashboard.
- EXT declares six industry object types and holds no instances of any of them. X05, the GxP computerised system, has no home anywhere in the file.
- No sheet carries a validation state, a periodic-review date or a named person. In this industry those three absences are the finding.
- Four new sheets are authorised: O67 and O70 in item 2, O69 in item 5, O68 in item 11. There is no fifth.
- A step has eight parts. Signed by and Shows up in are the two that keep a regulated estate honest.

SOURCES FOR THIS CHAPTER

- **1.** Pharmaceuticals_TOGAF_Metamodel.xlsx, sheet 00 Index — row 2 subtitle and the per-sheet contents list at rows 14 to 79; the sixty-five per-sheet record counts sum to 814 against the 850 stated in row 2
- **2.** Pharmaceuticals_TOGAF_Metamodel.xlsx, sheet O23 Application Component, header row 4 and rows 5 to 16 — ten columns, no validation state, no GxP flag, no review date, and an Owner column holding organisation units
- **3.** Pharmaceuticals_TOGAF_Metamodel.xlsx, sheets O10 Actor rows 5 to 22 and O11 Role rows 5 to 22 — the same eighteen generic corporate titles in both, every Performed by cell reading Person, every Status reading Filled
- **4.** Pharmaceuticals_TOGAF_Metamodel.xlsx, sheets ZZ Checks rows 4 to 21 and RPT01 Dashboard rows 4 to 24 — fifteen checks against eighteen measures, fourteen lines common to both; and the trailing single-cell prose row present on RPT01, RPT04, RPT05, RPT06, RPT08, RPT09, RPT10, RPT11 and RPT13
- **5.** Pharmaceuticals_TOGAF_Metamodel.xlsx, sheet R06 Application interfaces with, rows 5 to 40 — 36 rows; Managed takes the values Managed (17) and Unmanaged (19), Logged takes Yes (5) and No (31), Type includes six Direct DB read rows
- **6.** Pharmaceuticals_TOGAF_Metamodel.xlsx, sheet EXT Industry Extension, rows 5 to 10 — six industry object types declared, X01 to X06, with no instances held anywhere in the workbook
- **7.** Pharmaceuticals_TOGAF_Metamodel.xlsx, sheet O20 Data Entity, rows 5 to 18 — fourteen entities including E01 Compound / Molecule, E02 Study / Trial, E06 Regulatory Submission, E08 Adverse Event and E09 Batch / Lot, and no entity for a computerised or validated system
- **8.** Pharmaceuticals_TOGAF_Metamodel.xlsx, sheet O57 Governance Body, rows 5 to 9 — five bodies, GB-01 to GB-05, none of them a quality body; GB-05 Architecture Function carries the mandate to maintain the repository
- **9.** Pharmaceuticals — a starter pack for enterprise architects (volume 1), chapter 13, GxP, validation and data integrity, sections The system itself is regulated and Which GxP, and for which record

CHAPTER 2

Standing the practice up in a regulated estate

The architecture function has no standing in the quality system and the repository cannot name one validated system. Ten steps: decide whether your decisions are quality records, build the two registers everything else depends on, and sign the list of phases you are not allowed to skip.

Open on the sheet with nothing in it

Open `EXT Industry Extension` and read row 9. `X05 GxP Computerised System` — *a validated system with its validation state and owner*. Now search the other 104 tabs for one instance of it. There are none [1].

That is where Preliminary starts in a pharmaceutical company. Not with a stakeholder map, not with a tooling decision. With the fact that **the repository cannot name a single validated system, say what state it is in, or say who owns it** — while most of the twelve applications in `O23 Application Component` hold GxP records [2].

The rest of Preliminary is ordinary TOGAF. This item spends its length on the four things that are not: whether your decisions are quality records, who signs in this industry, the two sheets you create today, and which parts of the ADM a regulated company may not skip. Calderon Therapeutics is invented and describes no real company [1].

The question you answer in week one

The architecture function in this repository has no standing in the quality system. Read `O57 Governance Body`: five bodies — Architecture Board, Investment Committee, Data Governance Council, Executive Committee, Architecture Function. **None is a quality body.** No change control board, no computerised system validation committee, no data integrity forum, no mandate containing the word quality [3].

So when you write an architecture decision — *the master for a batch record is the manufacturing execution system, and no other system may originate one* — what kind of document is that?

A quality record is controlled: a document number, a version, an approver, a retention period, a change history, a place in the document system inspectors read. It binds, and a project that ignores it is in breach of a controlled document. It also costs you: every revision goes through document control, and you cannot quietly correct a diagram.

Not a quality record means advice: no approver with a duty, nobody obliged to follow it. That is the status quo in most companies, and it is why the previous architect's target state is on a shared drive and the estate does not resemble it.

> **The GxP overlay for the Preliminary Phase.** Decide this in week one, record it with a date, and do it before you produce anything. An architecture decision that is not a quality record cannot constrain a validated system: a validated system changes only through change control, and change control consumes only controlled documents. Leave the question open and you spend Phases B to D producing artefacts with no mechanism to take effect, then find out in Phase G, when a work package you never saw delivers against a design you never approved.
>
> Most companies land on the middle answer: standards and principles become controlled documents, decision records do not. Take it explicitly, and write down which outputs are controlled.

Look it up. Volume 1, chapter 13, section *What the architect is actually expected to produce*; and chapter 19, section *Week one: find out what is validated*, whose order is this item's order.

BEFORE YOU ANSWER, ASK THESE FOUR QUESTIONS OF THE HEAD OF QUALITY

Is an architecture standard a controlled document today? Under what document number?

Who approves a controlled document, and how long does an approval take?

When a change control record cites a design, which document does it cite?

If the architecture function disappeared tomorrow, which of its outputs would an inspector still be shown?

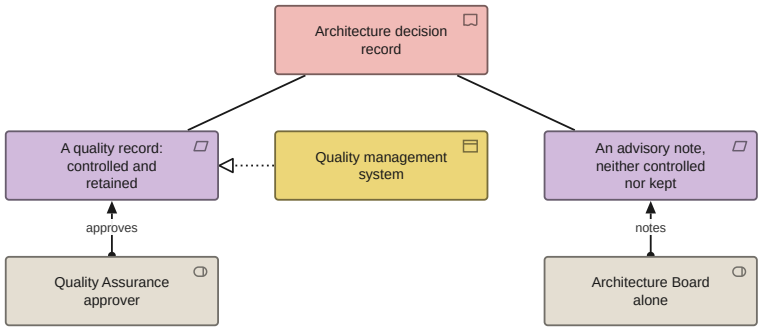


Figure 2.1 The week-one decision, drawn. The left branch binds and costs you document control; the right branch is free and nobody has to follow it. The element is `Role` — there is no layer-prefixed variant in ArchiMate 4, and no Constraint element, so each branch is expressed as a `Requirement`.

Step 2.1 — Decide whether an architecture decision is a quality record, and record the decision

Produce. One dated governance decision stating which classes of architecture output are controlled quality records.

You need. Thirty minutes with the Head of Quality and the answers above. If there is no Head of Quality there is no quality system, which is a larger finding.

> **`O60 Governance Decision`**
>
> Four columns on row 4: `Ref`, `Decision`, `Body`, `Status`. Seven rows, `DEC-01` to `DEC-07`, four of them `Outstanding` [4]. **The sheet has no date column**, which is why nothing in it can be late. Add `Decided on` and `Review due`, both `YYYY-MM-DD`. Then write your row.
>
> - `Decision` — *Architecture standards and principles are controlled quality records; decision records are not*, or the opposite, in one sentence with no conjunction.
> - `Body` — see **Signed by** below, because the right body does not exist yet.
> - `Status` — `Approved` or `Outstanding`. Still `Outstanding` at the end of week one is itself the finding you report.

How.

1. Ask Quality which document class an architecture standard would fit, and the approval lead time for that class, in days.
2. Decide which outputs are controlled. Principles, standards and the reference architecture are the usual set; views and decision records usually are not.
3. Write it as one sentence a person who was not in the room can act on.
4. Add the two date columns to `O60`, then add `DEC-08`.
5. Add `DEC-09`, recording that `O67` and `O70` are to be created. Step 1.4 reserved the numbers; this is the authority to populate them.
6. Tell the Head of Quality, the Chief Information Officer, `ACT-08` and whoever runs change control.

Signed by. A quality body, and there is not one. So `GB-04 Executive Committee`, whose mandate reads *Escalation point when bodies cannot agree* — the only one in `O57` broad enough [3]. Record that you used an escalation body because the correct one does not exist, and that Step 2.6 constitutes it. Do not let `GB-01 Architecture Board` decide this about itself.

Done when. `O60` has a `DEC-08` row with a `Decided on` date, and you can say in one sentence which of your outputs bind.

Shows up in. None of the thirteen reports: there is no governance report in the repository. `RPT09 Regulatory Coverage` is the nearest, and it reports obligations, not decisions [5].

Goes wrong when. The decision is taken informally — "of course architecture is governed" — and nobody writes a document number. Six months later a project asks which document your standard is, there is no answer, and the standard drops out of the change record. The opposite failure is as bad: every output becomes a quality record, the approval lead time turns out to be six weeks, and you stop producing anything.

The two sheets you create today

`O67 Validated System` and `O70 Regulated Role`, defined here once and in full. Every other item references them and none redefines them, so a different column set anywhere is a defect.

One note on prefixes, because getting it wrong costs a rework. `O67` uses `VAL-01` onward, not `VS-`: `O14 Value Stream` holds `VS01` to `VS05`, and a mixed key space breaks every lookup you write. `O70` uses `RGR-01`. Neither prefix appears anywhere else in the 105 sheets [6].

Step 2.2 — Define `O67 Validated System`

Produce. A new sheet, `O67 Validated System`, twelve columns, row-4 header convention, no rows yet.

You need. Step 1.4's numbering, Step 2.1's decision, and `O23 Application Component`. Not a workshop — thirty minutes in Excel.

`O67 Validated System` — the sheet the repository does not have

Create it after `O66 Architecture Increment` with the five-row convention: title, subtitle, blank, headers on row 4, record on row 5. Twelve columns, in order:

Column	What goes in it
`Ref`	`VAL-01` onward. Never `VS-`: that key space belongs to `O14 Value Stream`
`Application`	Copied character for character from `O23.Application`, so the two sheets join
`GxP`	**Which regimes**, not yes or no: `GMP`, `GCP`, `GLP`, `GDP`, `GVP`, `GDocP`, comma-separated, or `None`
`Validation state`	One of `Not in scope`, `In scope, not validated`, `Validated`, `Validated, review overdue`, `Under revalidation`, `Decommissioned, records retained`
`Last periodic review`	`YYYY-MM-DD` or `Never`. `Never` is legitimate and common
`Next review due`	`YYYY-MM-DD`. A date, never a frequency — that belongs in `O68`
`System owner`	A named person who can be rung up, never an organisation unit
`Process owner`	The named person in the business whose records these are
`Validation lead`	The named validation engineer. No such role exists in `O11 Role` yet
`URS reference`	The user requirement specification's document number
`Audit trail reviewed`	`Yes` with a date, `Partial`, `No`, or `No trail`
`Decommissioning plan`	`Yes` with a document reference, `No`, or `Not applicable`

How.

1. Type the twelve headers into row 4 exactly as above. Spelling matters: item 13 and every form in item 18 reference these names.
2. Set `GxP` as text, not a checkbox. A boolean flag cannot answer *which GxP, and for which record* — the question that decides the evidence burden on everything under it.
3. Restrict `Validation state` to the six values. A free-text column fills with `In progress`, which is not a state, and the register stops being countable.
4. Add the sheet to `00 Index` with `0 records`.
5. Do not add a `Criticality` column. `O23` has one, and two registers with the same column diverge inside a quarter.
6. Leave it empty. Step 2.3 fills it with Quality in the room.

Look it up. Volume 1, chapter 13, sections *Which GxP, and for which record* and *The validation master plan and the deliverable set*. Volume 1, chapter 11, section *The capability no other industry has: the register of validated systems*.

Signed by. `GB-05 Architecture Function`, whose `O57` mandate reads *Publishes criteria, assesses compliance, maintains the repository* [3]. The structure needs no quality signature; the content does.

Done when. The sheet has twelve headers on row 4, `Validation state` restricted to six values, and a row in `00 Index`.

Shows up in. No report yet. Item 14 adds a validated-estate view over `O67`.

Goes wrong when. Somebody adds the register as extra columns on `023`. It looks tidier and it is wrong: a validated system is not always an application. An instrument with embedded firmware and a spreadsheet performing a GMP calculation are both in scope and neither is in `023`.

Form 2-A — Validated system register, part one: identify

Feeds `067 Validated System`, a sheet the repository does not have — create it first, per Step 2.2. Six of the twelve columns. A first pass needs twelve to eighteen rows, because instruments and spreadsheets are systems too. Signed by the Head of Quality.

Ref	Application	GxP	Validation state	System owner	URS reference
VAL-01	EXAMPLE — delete this row. MES (GMP)	GMP, GDocP	Validated, review overdue		

Form 2-B — Validated system register, part two: assess

Feeds `067 Validated System`. The other five columns, keyed by `Ref`. `Process owner` is the one column on no form: type it straight into Excel, because it changes more often than anything else on the sheet. Signed by the Head of Quality and the system owner jointly.

Ref	Last periodic review	Next review due	Validation lead	Audit trail reviewed	Decommissioning plan
VAL-01	EXAMPLE — delete. Never			No	No

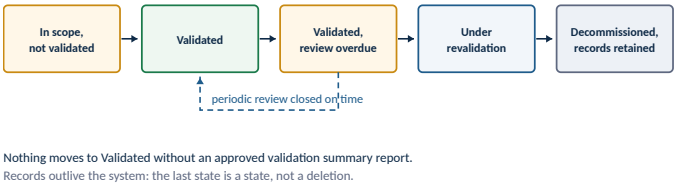


Figure 2.2 `067.Validation state`, the five states a system in GxP scope can be in, plus `Not in scope` for one that holds no GxP record. The dashed return is periodic review, which is the only thing that moves a system back from **review overdue** — item 11 builds `068` around it.

Step 2.3 — Populate `067` from `023`, and let Quality decide the scope

Produce. A populated `067 Validated System`, one row per computerised system, plus a note of the in-scope systems absent from `023`.

You need. Forms 2-A and 2-B printed, the Head of Quality, the validation lead if there is one, two hours. The validation master plan names the systems the company *believes* are in scope, which is a first draft before you speak to anyone.

> **`O67 Validated System, rows 5 onward`**
>
> Fill `Ref`, `Application`, `GxP`, `Validation state`, `System owner` and `URS reference` in the room; the review dates and `Audit trail reviewed` afterwards, from documents. `Application` must match `O23.Application` exactly where a counterpart exists; rows with none take the system's own name.

How.

1. Put the twelve rows of `O23` on the wall: `A01 ELN / LIMS`, `A02 CTMS`, `A03 EDC`, `A04 Safety Database`, `A05 RIM`, `A06 eTMF`, `A07 MES (GMP)`, `A08 QMS`, `A09 ERP`, `A10 Serialisation`, `A11 Statistical Computing`, `A12 Medical Information` [2].
2. For each, ask Quality one question: **which records in this system are GxP records?** Not *is it validated*. The record decides the scope, not the software.
3. Write the regimes into `GxP`: expect `GMP` on `A07`, `A08`, `A10`; `GCP` on `A02`, `A03`, `A06`; `GVP` on `A04`; a mixture on `A01`. Three are contestable and not yours to settle: `A05`, `A09`, `A12`.
4. Argue `A12` out loud, because it teaches the rule. The moment somebody types a patient's symptom into a medical enquiry note, the system has created a potential adverse event report. Nothing about the software changed — the record changed.
5. Then ask the question that produces the extra rows: **what else holds a GxP record?** Instrument software, a validated spreadsheet, a temperature monitor, a vendor-hosted portal. None is in `O23`; each gets a `VAL-` row.
6. Count the rows reading `Never` in `Last periodic review` and `No` or `No trail` in `Audit trail reviewed`. Those two numbers are your Phase A evidence.

Look it up. Volume 1, chapter 13, section *Which GxP, and for which record*; and chapter 12, section *What is actually installed*, for what these twelve products are.

Signed by. The **Head of Quality**, not the Architecture Board. Validation scope is a quality decision, and signing it yourself is the fastest way to lose the engagement. `O09 Organization Unit` has no Quality unit and `O11 Role` no quality role, so until Step 2.5 the signature line reads *Head of Quality* (*role not in repository*), countersigned by `ACT-07 Chief Compliance Officer`, the closest filled role in `O10 Actor` [7].

Done when. Every row has a `GxP` value naming regimes or `None`, a `Validation state` from the six, and at least one row has no counterpart in `O23`. If every row maps one-to-one onto `O23`, you asked question 5 and did not wait for the answer.

Shows up in. `RPT02 Application Portfolio`, by contrast: still twelve applications, still a `Health` column, still no validation state. The difference between the two sheets is the argument you take to Phase A.

Goes wrong when. You count the GxP systems yourself from the descriptions and report a number. There is no GxP column in the repository, so no count can be read out of the file. Any figure is a judgement, and if it is yours rather than Quality's it will be reversed in front of a client.

Nobody in this repository does a pharmaceutical job

`O10 Actor` and `O11 Role` hold the **same eighteen** titles — Chief Executive, four domain architects, Head of Delivery, Data Steward, Internal Auditor and the rest. Every `O11.Performed by` cell reads `Person`; every `O10.Status` reads `Filled`. And `R21 Role performed by Actor` joins `ROL-01` to `ACT-01` straight down the index, which is why the lists are identical: one list entered twice and joined by position [7].

There is no Responsible Pharmacist, no qualified person, no QA release officer, no clinical research associate, no pharmacovigilance officer, no validation engineer, no regulatory affairs manager, no study monitor, no quality assurance auditor. `ZZ Checks` reports `Vacant architecture roles: 0`, true and uninformative: the eighteen filled roles sign nothing in this industry [8].

`O09 Organization Unit` is the same story with a worse twist. Twelve units: Executive, Finance, Risk & Compliance, Human Resources, Technology, Data & Analytics, Operations, Commercial, **Customer Service**, Supply & Logistics, Internal Audit, Legal. No Quality. No Regulatory Affairs. No Pharmacovigilance. No Manufacturing. No Clinical Development. Legal has 240 people and Customer Service 211 against Technology's 116, which no pharmaceutical company has looked like [9].

The twist is in `R22 Org Unit owns Capability`. The capability map is right: `O13` decomposes the correct ten level-1 capabilities. `R22` then assigns them to the units **in index order** — `ORG-01` → `C01`, `ORG-02` → `C02`, on down. So Human Resources owns Regulatory Affairs, Technology owns Pharmacovigilance, Data & Analytics owns Manufacturing & Supply, Operations owns Quality Assurance & Control, Customer Service owns Medical Affairs [10].

That is a positional join, the same defect as the rotated `Master system` column in `O20 Data Entity`. A repository can be fully populated, internally consistent and completely untrue, and no automated check catches it. The capability map already names the five functions the

organisation sheet omits; nobody read the two sheets side by side.

Step 2.4 — Define `O70 Regulated Role`

Produce. A new sheet, `O70 Regulated Role`, seven columns, one row per role carrying a statutory or quality-system duty.

You need. Step 1.4's numbering, volume 1 chapter 16 open for the instruments, and volume 1 chapter 5 for what these people do.

> **`O70 Regulated Role` — the second sheet the repository does not have**
>
> After `O67`, same five-row convention. Seven columns:
>
> | Column | What goes in it |
> |---|---|
> | `Ref` | `RGR-01` onward. Unused anywhere in the 105 sheets |
> | `Role` | The statutory job title, not a corporate grade. `Responsible Pharmacist`, not `Head of Quality` |
> | `Statutory basis` | The instrument creating the duty, named and dated. Where none exists and the duty is the quality system's, `Quality system only` |
> | `Named person` | A person's name. A regulated role with no name is the finding; write the row with the cell blank |
> | `Deputy` | The named deputy. One-deep is the normal state, and it is a risk |
> | `Registration number held` | Where the duty requires a professional registration; `Not required` where it does not |
> | `Signs for` | What this role signs: batch release, a submission, an expedited report, a validation summary report, a review outcome, a disposal |

How.

1. Create the sheet, seven headers on row 4, and add it to `00 Index`.
2. Fill `Statutory basis` **by copying from volume 1 chapter 16**, which carries the instrument, its number and its date. Never from memory or from a workshop.
3. Leave `Named person` blank where you have no name, and count the blanks. That count is the one finding here an executive reads without being asked twice.
4. Fill `Signs for` from the decision, not the job description. A role that signs nothing belongs in `O11 Role`.
5. Keep `Registration number held` off anything you print: type it into Excel and restrict the sheet, because a professional registration number is a personal identifier.
6. For each `RGR-` row, note which `O67.VAL-` rows it signs for. A validated system with no regulated signatory is your Phase G problem, a year early.

> **Never fill `Statutory basis` from memory.** The instruments, their numbers and their dates are in volume 1, chapter 16, sections *The spine: Act 101 of 1965 and SAHPRA, Licensing, and the Responsible Pharmacist* and *Pharmacy law*. Volume 1 is dated, which is why it is the source. Where a duty is contractual or comes from the company's own quality system, `Quality system only` is the correct value, not a gap.

Look it up. Volume 1, chapter 3, section *Qualified Person versus Responsible Pharmacist* — they are not the same role and conflating them in `O70` will embarrass you. Volume 1, chapter 13, section *The Responsible Pharmacist*; and chapter 5, section *The interview order*.

Signed by. `GB-05 Architecture Function` for the sheet; the **Head of Quality and the Responsible Pharmacist jointly** for the content, because `Signs for` states their own authority. Neither role exists in `O10 Actor`, so the signature block names them and records that they are absent.

Done when. The sheet has seven headers, every row a `Signs for` value, and the count of blank `Named person` cells is in your findings log.

Shows up in. Nowhere, deliberately: a report over `O70` would publish names and registration numbers. Restrict the sheet and quote only counts.

Goes wrong when. `O70` becomes a second copy of `O11 Role` with longer titles. The test is `Signs for`: eighteen generic corporate titles sign nothing a regulator would ask about.

Form 2-C — Regulated role register

Feeds `O70 Regulated Role`, a sheet the repository does not have — create it first, per Step 2.4. Six of the seven columns. `Registration number held` is deliberately **not** here: type it into Excel and restrict the sheet, because a printed form leaves the room. Eight to twelve rows is a first pass. Signed by the Head of Quality and the Responsible Pharmacist.

Ref	Role	Statutory basis	Named person	Deputy	Signs for
RGR-01	EXAMPLE — delete. Responsible Pharmacist	From volume 1 ch 16			Batch release

Step 2.5 — Put the pharmaceutical organisation into `O09`, `O10` and `O11`

Produce. Five new organisation units, the regulated and operational roles added to `O10` and `O11`, every `Performed by` cell examined, and `R22` re-resolved.

You need. `O70` from Step 2.4, volume 1 chapter 5, the organisation chart the payroll system produces rather than the induction pack's, and two hours.

> **`O09`, `O10`, `O11` and `R22 Org Unit owns Capability`**
>
> `O09`: `Ref`, `Organisation unit`, `Type`, `Head`, `Headcount` — twelve rows, add from `ORG-13`. `O10`: `Ref`, `Actor`, `Type`, `Status`; `O11`: `Ref`, `Role`, `Type`, `Performed by` — both run to row 22, add from `ACT-19` and `ROL-19`. `R22`: `Org unit`, `Unit name`, `Capability`, `Capability name` — ten rows, index-joined, wrong in five [9][10].
>
> - `O09.Type` — values in use are `Group`, `Corporate`, `Line`, `Assurance`. Quality is `Assurance`; Regulatory Affairs and Pharmacovigilance are `Line` in most companies. Ask, do not assume.
> - `O11.Type` — add `Quality`, `Regulatory`, `Clinical`, `Safety`, `Manufacturing`, `Validation`. The seven existing values name no pharmaceutical function.
> - `O11.Performed by` — the permitted answers are a person, an organisation unit or a collaboration. Every cell reads `Person`, which is why the column carries no information [7].
> - `O10.Status` — keep `Filled` and `Vacant`, add `Shared with another site`, the real answer for a Responsible Pharmacist at a two-site company.

How.

1. Add `ORG-13 Quality`, `ORG-14 Regulatory Affairs`, `ORG-15 Pharmacovigilance`, `ORG-16 Manufacturing`, `ORG-17 Clinical Development`. Five rows, ten minutes.
2. Re-resolve `R22` against `O13`'s level-1 rows: `C04` → `ORG-14`, `C05` → `ORG-15`, `C06` → `ORG-16`, `C07` → `ORG-13`, `C03` → `ORG-17`.
3. For every `RGR-` row in `O70`, add the matching `O11` role with `Type` set to the function and the matching `O10` actor with a real `Status`. Then add the roles that sign nothing but do the work: study monitor, data manager, quality assurance auditor, batch record reviewer, validation engineer.
4. Correct `Performed by` on all eighteen existing rows. `ROL-16 Data Steward` is usually a collaboration; `ROL-08 Chief Enterprise Architect` is a person.
5. Extend `R21 Role performed by Actor` and break the index-order habit: the join is `ROL-19` to whichever actor performs it, not to `ACT-19`.
6. Leave `ORG-09 Customer Service` and the eighteen generic roles in place, marked, for Phase B. Do not fix the headcounts: a corrected number you cannot source is worse than an obviously wrong one.

Look it up. Volume 1, chapter 5, sections *Quality and manufacturing*, *Regulatory affairs* and *A role-to-system map*.

Signed by. `GB-04 Executive Committee` for the units, the Head of Quality for the regulated roles, the Chief Operating Officer (`ACT-03`) for the operational ones. Record that no body in `O57` has a mandate over role definition — `GB-03 Data Governance Council` designates data owners and nothing else — because that absence is why `O11` looks as it does [3].

Done when. `O09` has seventeen rows; `O11` holds a role of `Type: Quality` and one of `Type: Validation`; no `Performed by` cell is unexamined; and neither `R21` nor `R22` still joins every row to its own index number.

Shows up in. `ZZ Checks` row 17 and `RPT01 Dashboard` row 19, both still reporting zero vacant architecture roles while your count of vacant *regulated* roles is not zero. Quote yours and say where it came from. `RPT03 Capability Heatmap` will not change, which is the point: it was always right about the capabilities and silent about who owns them.

Goes wrong when. You rename `ORG-03 Risk & Compliance` to `Quality` because it is nearly right. It is not: quality assurance owns batch release and the validated state, a corporate risk function neither. Add a unit; do not repurpose one. The second failure is adding forty roles at once so nobody recognises the list — add the ones that sign something first.

Governance, gates, principles and standards

Four sheets, four steps. The gates are good and you keep them. The other three need pharmaceutical content the repository has none of.

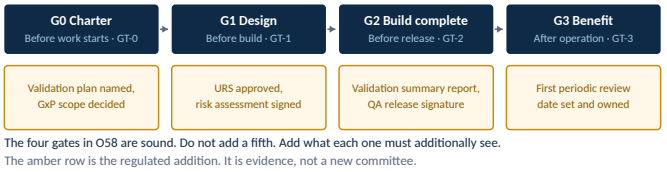


Figure 2.3 `O58 Governance Gate`, **extended.** The four gates are sound and you keep them. The amber row is what each gate must additionally see once the system under it is validated. No fifth gate: the release decision is `G2`.

Step 2.6 — Constitute the three governance bodies `O57` does not have

Produce. Three rows in `O57 Governance Body`: a change control board, a computerised system validation committee, a data integrity forum.

You need. Step 2.1's decision and the existing committee structure. Most companies already have a change control board that is not in the repository, which is a different problem from not having one.

`O57 Governance Body`

Five columns: `Ref`, `Governance body`, `Cadence`, `Membership`, `Mandate`. Five rows, `GB-01` to `GB-05`. Add three:

Ref	Governance body	Cadence	Mandate, in the sheet's own style
`GB-06`	Change Control Board	Weekly	Approves change to a validated system; no change proceeds without a record
`GB-07`	Computerised System Validation Committee	Monthly	Approves validation scope, plans and summary reports; owns `O67`
`GB-08`	Data Integrity Forum	Quarterly	Reviews audit-trail review coverage and non-production data

How.

1. Find out whether these bodies exist first. If a change control board meets every Tuesday, your job is to record it in `O57`, not constitute it.
2. Write `Cadence` from the meeting invitation, not the terms of reference.
3. Set `Membership` to roles from `O11` and `O70` by reference. A list of names goes stale in a quarter.
4. Give `GB-07` ownership of `O67` in its `Mandate`. A register with no owning body decays within two review cycles.
5. Give none of them a gate. `O58`'s four are held by `GB-01` and `GB-02`; a fifth gate-holder is how a governance regime becomes a queue.
6. Record in `O60` which you created and which you documented.

Signed by. `GB-04 Executive Committee`: a new standing body is an executive act, and `GB-01 Architecture Board`'s mandate — *Holds G1 and G2; approves dispensations with an expiry* — does not reach it [3].

Done when. `O57` has eight rows, each new cadence comes from a real invitation, and `GB-07`'s mandate names `O67`.

Shows up in. `O60 Governance Decision` and `O61 Governance Change Request`, twelve rows against five bodies today. After this step the body column has somewhere credible to point.

Goes wrong when. You create three bodies that never meet. An empty governance row is worse than a missing one, because the next reader believes it.

Step 2.7 — Bind the four gates to validation evidence

Produce. `O58 Governance Gate` with `What it tests` extended to name the validation evidence each gate must see, and `O59 Compliance Assessment` ready to receive it.

You need. `O58`, `O59`, `O31 Deliverable`, and the validation master plan if it exists.

> **`O58 Governance Gate` and `O59 Compliance Assessment`**
>
> `O58`: `Ref`, `Gate`, `When`, `What it tests`. Four rows, `GT-0` to `GT-3`, and they are sound [11]. `O59`: `Ref`, `Building block`, `Block name`, `Acceptance criterion`, `Gate`, `Verdict`, `Evidence` — **empty, and `R24 Compliance Assessment tests` empty with it**, which is why `O63 Control Objective` can record `CO-01` as `Evidenced` with nothing behind it [5].

How.

1. Do not add a fifth gate. Four is right, and the release gate is `G2`.
2. Extend `GT-0`: the GxP scope decision is recorded in `O67` and a validation plan is named.
3. Extend `GT-1`: the user requirement specification is approved and the risk assessment signed. Item 12 covers why the specification and the TOGAF requirement are the same artefact.
4. Extend `GT-2`: the validation summary report exists and carries the quality release signature. Name the signatory from `O70`, not `O57`.
5. Extend `GT-3`: the first periodic review date is set and owned. Item 11 builds `O68` for it.
6. Write one `O59` row per acceptance criterion for the first work package you govern, `Verdict` blank. An empty `O59` is a sheet; an `O59` with blank verdicts is a queue somebody must clear.

Look it up. Volume 1, chapter 13, section *The V-model, and how it actually runs*.

Signed by. `GB-01 Architecture Board` for `G1` and `G2` per its own mandate, with `GB-07` from Step 2.6 co-signing the validation evidence at `G2`. The release signature is not the board's: it belongs to the named role in `O70`, and item 10 is about nothing else.

Done when. All four `What it tests` cells name a validation artefact and `O59` has at least one row.

Shows up in. `O33 Architecture Contract`: six of eight read *Not yet assessable* and two *Non-compliant* [12]. The first value is honest only while `O59` is empty.

Goes wrong when. The gates stay generic and the validation evidence travels a separate route through Quality. Two governance paths over one change is how a system gets released twice and validated once.

Step 2.8 — Put the regulator into `O05 Principle`

Produce. Three principles in `O05 Principle`, in the sheet's own columns, that the existing twenty-four do not say.

You need. `O05`, and item 19, which carries the full starter set. This step adds the minimum Preliminary cannot proceed without.

> **`O05 Principle`**
>
> Six columns: `Ref`, `Principle`, `Category`, `Statement`, `Owner`, `Status`. Twenty-four rows, `PRN-01` to `PRN-24`, every one owned by `Architecture Board`, every `Category` one of `Business`, `Application`, `Data`, `Technology`, `Security`. Nineteen `Approved`, five `Draft`. The generic set: **not one mentions validated state, data integrity, patient safety, retention or the regulator** [13]. Add a sixth `Category` value, `Regulatory`.

Ref	Principle	Category	Statement
`PRN-25`	Validated State is Maintained	Regulatory	A validated system stays validated only while its reviews and change records are current
`PRN-26`	Integrity is a Property of the Record	Regulatory	The record carries its own attribution, timestamp and history; a report cannot restore them
`PRN-27`	The Patient's Data is the Most Protected Class	Regulatory	Subject data outranks every other classification in the estate

How.

1. Add the `Regulatory` category first. Without it the three land under `Business` and read as aspiration.
2. Write the rows with `Status: Draft` and `Owner: Architecture Board`, matching the sheet, then take them to the board.
3. Read `PRN-03 Compliance is Designed, Not Added` and `PRN-14 Classification Travels with Data` first. Both are good and both are nearly the point; your three say the part they leave out.
4. Do not touch the five `Draft` rows — `PRN-05`, `PRN-10`, `PRN-15`, `PRN-20`, `PRN-24`. Clearing them is not this week's fight.
5. Leave the other three pharmaceutical principles — retention runs from an event, an interface into a validated system is inside its validation scope, the dictionary version is part of the datum — to item 19, which writes them in these same columns.
6. Note which of your three contradicts something in the existing twenty-four. If none does, you have written restatements.

Signed by. `GB-01 Architecture Board`, which owns every row in the sheet; the `Status` column is the record of its approval.

Done when. `O05` has twenty-seven rows, three of `Category: Regulatory`, and each statement is a sentence you could lose an argument with.

Shows up in. `GT-1` in `O58`, which tests that a design *meets principles, standards and reference architecture*. A principle no gate tests is decoration.

Goes wrong when. The three are written as obligations — *we must comply with Annex 11*. A principle is a design rule you apply without being asked; the obligation already lives in `O63 Control Objective` and `O01 Driver`.

Step 2.9 — Add the GxP standards `O32` does not have

Produce. Five rows in `O32 Standard`: computerised system validation, electronic records and signatures, data integrity, audit-trail review, periodic review.

You need. `O32`, plus volume 1 chapter 13 and appendix 22 for which standards exist.

> **`O32 Standard`**
>
> Five columns: `Ref`, `Standard`, `Category`, `Status`, `Conformance`. Sixteen rows, categories `Integration`, `Data`, `Security`, `Platform`, `Delivery`, `Governance`. **Not one is a GxP standard.** Two rows contradict themselves: `STD-02 Event schema standard` is `Deprecated` and `Conformant`, `STD-07 Logging and monitoring standard` is `Deprecated` and in `Breach` — a deprecated standard can be neither, because it is not in force [14]. Add a `Validation` category and five rows from `STD-17`.

How.

1. Add the category `Validation`, then `STD-17 Computerised system validation standard`, `STD-18 Electronic records and signatures standard`, `STD-19 Data integrity standard`, `STD-20 Audit trail review standard`, `STD-21 Periodic review standard`.

2. Set `Status` to `Emerging` and `Conformance` to `Not assessed` for all five. Both values are already in use and both are honest on day one.
3. Resolve `STD-02` and `STD-07` while you are here: reinstate them as `Current` or clear the conformance cell. A deprecated standard with a verdict teaches every reader that the column means nothing.
4. Check `STD-04 Data retention standard`, `Current` and `Conformant`, against `STD-15 Records management standard`, `Deprecated` and `Conformant`. Retention here runs from an event and periods differ by record class, so one conformant retention standard is improbable. Item 5 builds `O69`.
5. Do not write version numbers into `Standard` from memory. The versions move; volume 1 carries them with dates.
6. Extend `R18 Standard applies to Applica` for the five new rows, pointing at the `O67` systems rather than all twelve of `O23`.

Look it up. Volume 1, chapter 13, section *21 CFR Part 11 and EU GMP Annex 11*; and appendix 22, section *How to check a standard's version*.

Signed by. `GB-01 Architecture Board` publishes standards per `GB-05`'s mandate. `GB-07` from Step 2.6 owns the content of the five, because a validation standard Quality has not agreed is an IT preference.

Done when. `O32` has twenty-one rows, five `Category: Validation`, and neither `STD-02` nor `STD-07` carries a conformance verdict against a deprecated status.

Shows up in. `RPT12 Standards Conformance`, straight off `O32`: sixteen rows today, twenty-one after, five more `Not assessed`. The report looks worse. That is what an honest baseline does.

Goes wrong when. The five are written as documents nobody owns. A standard with no owning body, no conformance route and no gate that tests it is a filename.

Step 2.10 — Tailor the ADM, and record the tailoring

Produce. One approved table stating which parts of the ADM this company runs in full, compresses or skips, plus a governance decision recording it.

You need. Everything above, the quality manual, and `O31 Deliverable` — which phases twenty TOGAF deliverables correctly and is the best-built sheet for this purpose [15].

> **`O31 Deliverable` and `O60 Governance Decision`**
>
> `O31` columns: `Ref`, `Deliverable`, `Phase`, `Status`. Twenty rows phased `A` to `G` plus `RM`, thirteen `Approved` and seven `Draft`. What is absent: no validation plan, no user requirement specification, no validation summary report. Add `DLV-21 Validation Plan` at `G`, `DLV-22 User Requirement Specification` at `RM` and `DLV-23 Validation Summary Report` at `G`, and record the tailoring as `DEC-10` in `O60`.

What a pharmaceutical company may compress, and what it may not:

ADM element	Decision	Why
Preliminary capability assessment, Phase A business scenarios	**Compress both**	The company already assesses itself for inspection, and `O14`'s five value streams are genuinely pharmaceutical. Validate, do not re-derive. Item 20 runs the assessment properly later
Phase B: GxP marking of each process	**Cannot skip**	The marking decides the evidence burden on every system under the process, and `O16` is twenty placeholder rows
Phase C data: classification	**Cannot skip**	A clinical-practice and data-protection obligation before an IT one. `O20` classifies subject data `Internal`
Phase D: qualification apart from validation	**Cannot skip**	Infrastructure is qualified, the system on it validated. Different owners, evidence and documents
Phase E: alternatives analysis	**Cannot skip**	`O54` marks `ALT-B Recommended` and `ALT-E Partly adopted`. Two live answers is not a decision
Phase F: migration verification	**Cannot skip**	A migration into a validated system verifies value **and meaning**
Phase G: compliance assessment	**Cannot skip**	`O59` is empty and `O63` records evidence anyway. This is the gap the regime exists to close
Phase H: periodic review	**Cannot skip**	It keeps the validated state true, and no column holds its due date
Requirements Management	**Cannot skip**	The user requirement specification is the first document in the V model
Architecture Contract (`O33`)	**Cannot skip**	The only artefact binding a supplier's validation obligation. Six of eight read *Not yet assessable*

How.

1. Work the table line by line against this company and change the verdicts that do not apply. Three or four of the ten usually move.
2. Add an eleventh line for anything the quality system already does that the ADM would duplicate.
3. Add `DLV-21` to `DLV-23` with `Status: Draft`.
4. Record `DEC-10 Adopt the tailored ADM for a regulated estate` in `O60`, with a `Decided on` date and a `Review due` date twelve months out.
5. Publish the table as an annex to `DLV-02 Statement of Architecture Work`, which `O31` already carries as `Approved` at phase `A`.
6. Give one person the job of refusing to skip a *cannot skip* line under schedule pressure, and write their name beside it.

Look it up. Volume 1, chapter 19, section *The ninety-day shape.*

Signed by. `GB-01 Architecture Board` for the tailoring, `GB-07` for the three validation deliverables, and `GB-02 Investment Committee` for anything that changes the cost of a phase — its mandate covers `G0` funding [3].

Done when. `O60` holds `DEC-10` with a date, `O31` holds twenty-three rows, and the table is an annex to an approved deliverable rather than a slide.

Shows up in. `RPT08 Roadmap`, indirectly: the phases you did not compress get realistic dates. Three of the twelve in `O30 Work Package` are already past their end date [16].

Goes wrong when. The tailoring is written as a list of things to skip and not as a list of things that cannot be. The second list survives contact with a deadline only if it is signed.

What Preliminary leaves you with

Two new sheets, `O67 Validated System` and `O70 Regulated Role`, populated enough to argue from. Three governance bodies, created or documented. Five organisation units, and the knowledge that the relationship sheet joined the old ones to the capability map by position. Three principles and five standards that mention the regime the company operates under. A tailoring decision with a date on it.

And one sentence for the steering meeting: *we have a register of validated systems, and these are the ones whose validated state we cannot evidence.* That is Phase A's opening line.

THE CHAPTER IN TEN LINES

- EXT declares X05 GxP Computerised System and the repository holds no instance of it anywhere.
- Decide in week one whether an architecture decision is a quality record, and record the decision with a date.
- O57 Governance Body has five bodies and not one of them is a quality body. No change control board, no validation committee, no data integrity forum.
- O60 Governance Decision has no date column. Add Decided on and Review due before you write a row in it.
- Create O67 Validated System with twelve columns, keyed VAL-01. Not VS- : that key space belongs to O14 Value Stream.
- The GxP column names regimes, not yes or no. Which GxP and for which record decides the evidence burden on everything under it.
- Quality decides validation scope, not you. There is no GxP column in the repository, so any count you quote is a judgement.
- Create O70 Regulated Role with seven columns, keyed RGR-01. Copy the statutory basis from volume 1 chapter 16, never from memory.
- O10 and O11 hold the same eighteen generic titles, joined by index in R21, and every Performed by cell reads Person.
- R22 joins organisation units to capabilities in index order, so Human Resources owns Regulatory Affairs and Technology owns Pharmacovigilance.
- O05 has twenty-four generic principles and none mentions validated state, data integrity, retention or the regulator.
- O32 has sixteen IT standards and not one GxP standard. Two of them are deprecated and carry a conformance verdict anyway.
- Tailor the ADM in writing, and sign the list of phases you cannot skip rather than the list you can.

SOURCES FOR THIS CHAPTER

- **1.** Pharmaceuticals_TOGAF_Metamodel.xlsx, sheet EXT Industry Extension, rows 5 to 10 — X01 Compound, X02 Study / Trial, X03 Batch / Lot, X04 Regulatory Submission, X05 GxP Computerised System, X06 Adverse Event, declared as object types with no instances held in any of the other 104 sheets; and sheet 00 Index rows 1 to 12 on the illustrative nature of the data
- **2.** Pharmaceuticals_TOGAF_Metamodel.xlsx, sheet O23 Application Component, header row 4 and rows 5 to 16 — twelve applications A01 to A12; the ten columns carry no GxP flag, no validation state and no review date, and the Owner column holds organisation units
- **3.** Pharmaceuticals_TOGAF_Metamodel.xlsx, sheet O57 Governance Body, rows 5 to 9 — GB-01 Architecture Board, GB-02 Investment Committee, GB-03 Data Governance Council, GB-04 Executive Committee, GB-05 Architecture Function, with their cadences and mandates as quoted
- **4.** Pharmaceuticals_TOGAF_Metamodel.xlsx, sheet O60 Governance Decision, rows 5 to 11 — seven decisions, DEC-01 to DEC-07, four of them Outstanding; four columns only, with no date column of any kind
- **5.** Pharmaceuticals_TOGAF_Metamodel.xlsx, sheets O59 Compliance Assessment and R24 Compliance Assessment tests — both carry a header row on row 4 and no records; sheet O63 Control Objective rows 5 to 11, where CO-01 records Evidence: Evidenced; and sheet RPT09 Regulatory Coverage, which reproduces O63 and ends in a single-cell prose row
- **6.** Pharmaceuticals_TOGAF_Metamodel.xlsx — reference prefixes in use across all 105 sheets, read from the Ref column of each object sheet; VAL- and RGR- appear nowhere, while VS01 to VS05 are in use on sheet O14 Value Stream rows 5 to 9
- **7.** Pharmaceuticals_TOGAF_Metamodel.xlsx, sheets O10 Actor rows 5 to 22, O11 Role rows 5 to 22, and R21 Role performed by Actor rows 5 to 22 — the same eighteen titles in both object sheets, joined ROL-01 to ACT-01 straight down the index, every Performed by cell reading Person and every Status reading Filled
- **8.** Pharmaceuticals_TOGAF_Metamodel.xlsx, sheet ZZ Checks rows 4 to 21 and RPT01 Dashboard rows 4 to 24 — Vacant architecture roles reported as 0 against a threshold of 0
- **9.** Pharmaceuticals_TOGAF_Metamodel.xlsx, sheet O09 Organization Unit, rows 5 to 16 — twelve units with Head and Headcount; Legal 240, Customer Service 211, Technology 116, Data & Analytics 23, and no Quality, Regulatory Affairs, Pharmacovigilance, Manufacturing or Clinical Development unit
- **10.** Pharmaceuticals_TOGAF_Metamodel.xlsx, sheets R22 Org Unit owns Capability rows 5 to 14 and O13 Business Capability rows 5 to 104 — the ten level-1 capabilities C01 to C10 assigned to ORG-01 to ORG-10 in index order, placing Regulatory Affairs under Human Resources, Pharmacovigilance under Technology, Manufacturing & Supply under Data & Analytics, Quality Assurance & Control under Operations and Medical Affairs under Customer Service
- **11.** Pharmaceuticals_TOGAF_Metamodel.xlsx, sheet O58 Governance Gate, rows 5 to 8 — GT-0 G0 Charter, GT-1 G1 Design, GT-2 G2 Build complete, GT-3 G3 Benefit, with their When and What it tests columns as quoted
- **12.** Pharmaceuticals_TOGAF_Metamodel.xlsx, sheet O33 Architecture Contract, rows 5 to 12 — eight contracts against WP-01 to WP-08, six Not yet assessable and two Non-compliant
- **13.** Pharmaceuticals_TOGAF_Metamodel.xlsx, sheet O05 Principle, rows 5 to 28 — twenty-four principles PRN-01 to PRN-24, every Owner reading Architecture Board, nineteen Approved and five Draft, categories limited to Business, Application, Data, Technology and Security
- **14.** Pharmaceuticals_TOGAF_Metamodel.xlsx, sheet O32 Standard, rows 5 to 20 — sixteen standards STD-01 to STD-16, no GxP or validation standard among them; STD-02 Deprecated and Conformant, STD-07 Deprecated and in Breach, STD-04 Current and Conformant, STD-15 Deprecated and Conformant
- **15.** Pharmaceuticals_TOGAF_Metamodel.xlsx, sheet O31 Deliverable, rows 5 to 24 — twenty TOGAF deliverables correctly phased A to G plus RM, thirteen Approved and seven Draft, with no validation plan, user requirement specification or validation summary report among them
- **16.** Pharmaceuticals_TOGAF_Metamodel.xlsx, sheet O30 Work Package, rows 5 to 16 — twelve packages; WP-02, WP-04 and WP-10 recorded as Past end date

CHAPTER 3

Architecture Vision

The phase where you find out whether the date you were given is arithmetically possible.

What this phase has to do here

Phase A turns a request into a scope you can be held to [16]. Here it does one more thing, and that thing decides whether the rest of this workbook is usable: it fixes **which systems hold regulated records**, because that single fact sets the cost and the duration of every change you later propose.

`O31 Deliverable` carries three Phase A rows — `DLV-01 Architecture Vision`, `DLV-02 Statement of Architecture Work` and `DLV-03 Stakeholder Map` — and all three already read `Approved` [5]. Nobody told you by whom or on what date, because `O31` has four columns and none of them is an approver or a date. Treat the three rows as empty. You are writing them.

Two companies appear in this volume and both are inventions. **Calderon Therapeutics** is the one in the repository you were handed. **Veldtkruin Group** is volume 1's worked example and appears here only in cross-references. Never move a number between them.

> **THE VISION CANNOT PROMISE A DATE THAT A VALIDATION CYCLE DOES NOT ALLOW**
>
> This is the Phase A form of the GxP overlay, and it is the one that gets architects fired.
>
> A go-live date for a non-regulated system says when software will work. A go-live date for a GxP computerised system says **when a validation summary report will be signed**, and the signature is not yours. Between the two sits a documented sequence that does not compress on request: user requirements, functional and configuration specification, risk assessment at system and function level, supplier assessment, installation, operational and performance qualification executed with evidence, deviation records against the protocols, a migration plan and report where records move, operating procedures, training records for every user and administrator, and the summary report carrying the release-for-use statement [12].
>
> Three parts of it do not compress. **The evidence** — a test executed twice because the first run raised a deviation takes twice as long, and you cannot know in advance which ones will. **The approval queue** — the quality function that approves protocols is small and is also preparing for an inspection. **The retrospective work** on systems already running, which is nobody's project and everybody's dependency.
>
> So Phase A here produces two dates: when the architecture is ready, and the earliest date it could lawfully be in use. Publish only the first and you have published a fiction with your name on it. Put both in `DLV-01` and the arithmetic behind the second in `DLV-02`, where it can be argued with.
>
> **Look it up.** Volume 1, chapter 13, *Computerised system validation, as it is actually practised*, and *The validation master plan and the deliverable set*.

What the repository already claims, and what it cannot support

Read these before your first meeting. Each is a sentence an executive may already have seen, and each is wrong in a way you will be asked about.

Where it is said	The claim	What the sheets support
`RPT01 Dashboard`, `ZZ Checks`	850 records, 65 object types populated	814 records over the 64 `O` sheets plus `EXT`; two sheets are empty, so at most 63 types carry data [10]
`O04 Measure` row 5	`MSR-01 Obligations with evidence`, baseline 54%	`O63 Control Objective` shows two of seven `Evidenced` — 29% [3][4]
`O63` row 5	`CO-01 GMP / GCP / GLP / GVP` is `Evidenced`	`O59 Compliance Assessment`, which would hold that evidence, has no rows [4]

One claim is correct and worth saying out loud: `O31`'s twenty deliverables are phased sensibly, `O58 Governance Gate`'s four gates are sound, and `O06 Requirement` opens on `REQ-001 Evidence GMP / GCP / GLP / GVP`, the right first requirement for this industry [5][9].

ANSWER THESE BEFORE YOU OPEN THE WORKBOOK AGAIN

Which of Calderon's twelve applications hold GxP records, and who told you so?

Who signs a validation summary report in this company, and how many are in the queue?

What date were you given, and was validation duration in the estimate behind it?

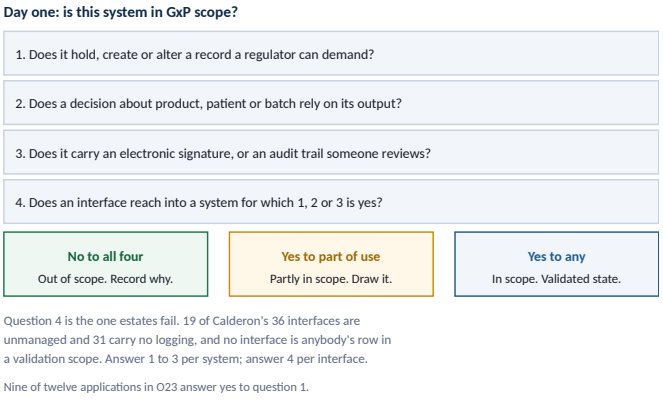


Figure 3.1 The classification you do on **day one**, before anything else in Phase A. Question 4 is the one that is routinely answered wrongly, because an interface into a validated system is inside its validation scope and is almost never anybody's row.

Step 3.1 — Split the application portfolio into GxP and not, on day one

Produce. A twelve-row split of `O23 Application Component` into GxP and not-GxP, with the record class that makes each one regulated — the first population of `O67 Validated System`.

You need. `O23 Application Component` rows 5–16; the quality assurance manager and the validation lead; the site procedure set.

> **`O67 Validated System` — a sheet the repository does not have; item 2 defines it**
>
> Columns: `Ref`, `Application`, `GxP`, `Validation state`, `Last periodic review`, `Next review due`, `System owner`, `Process owner`, `Validation lead`, `URS reference`, `Audit trail reviewed`, `Decommissioning plan`. In Phase A fill the first three and `System owner`; the rest belong to Phase C and to item 2's register build.
>
> `O23`'s ten columns run `Ref`, `Application`, `Type`, `Description`, `Disposition`, `Criticality`, `Health`, `In service`, `Users`, `Owner` — **no validation-state column, no GxP flag, no periodic-review date**. Do not add columns to `O23`. Create `O67` and join on `Ref`.

How.

1. List the twelve applications from `O23`. Nine are GxP on their face: `A01 ELN / LIMS`, `A02 CTMS`, `A03 EDC`, `A04 Safety Database`, `A05 RIM`, `A06 eTMF`, `A07 MES (GMP)`, `A08 QMS`, `A10 Serialisation`.
2. For each, write the record class that makes it regulated, not the regime letters. "Subject data and source records" is useful; "GCP" is not.
3. Take `A09 ERP`, `A11 Statistical Computing` and `A12 Medical Information` to the validation lead one at a time. Each is regulated for part of what it does, and the part is the answer.
4. Record `Partly in scope` where true and name the boundary in one sentence. A declared boundary is a valid architectural act; an undeclared one is a finding.
5. Ask for the validation summary report and last periodic review for three systems of your choosing. What comes back is your real baseline.
6. Put the sheet in front of quality assurance and ask them to disagree in writing.

Signed by. The quality assurance manager with the validation lead. `O57 Governance Body` has five bodies and **none is a quality body** — no change control board, no validation committee [9]. Until item 2's governance step lands, no body can sign this, so two

named people sign and `GB-01 Architecture Board` minutes it.

Done when. All twelve rows carry `Yes`, `No` or `Partly in scope`, and each `Partly in scope` row carries a one-sentence boundary.

Shows up in. Nowhere yet, and that is the finding. `RPT02 Application Portfolio` reproduces `O23`'s columns and cannot show validation state. Note it in `DLV-02` as a reporting gap.

Goes wrong when. You accept the quality function's register without testing it. The usual state is two registers that disagree, and the disagreement is found by an inspector rather than by you.

Look it up. Volume 1, chapter 19, *Week one: find out what is validated*; chapter 13, *What the architect is actually expected to produce*.

Form A-1 — GxP classification of the application portfolio

Feeds `O67 Validated System`, which you are creating. Twelve rows for Calderon; a real first pass needs twenty to sixty, because the grain is one system instance in one environment for one defined use, not one application. The remaining `O67` columns are typed straight into Excel. Signed by the quality assurance manager and the validation lead.

Ref	Application	GxP	Validation state	Last periodic review	System owner
VAL-01	EXAMPLE — delete this row	Yes	Validated	2024-03	

Step 3.2 — Rewrite the driver list so it names the jurisdiction you work in

Produce. An amended `O01 Driver` carrying the South African obligations the sheet lacks, with an honest `Origin` and `Owner` on every row.

You need. `O01 Driver` rows 5–17; `O63 Control Objective` rows 5–11; the regulatory affairs manager.

> **`O01 Driver` — `Ref`, `Driver`, `Type`, `Description`, `Origin`, `Owner`, `Pressure`**
>
> Thirteen rows. `DRV-01` to `DRV-07` are `Regulatory`, all `External`, all owned by `Chief Compliance Officer`, all `Pressure: High`. `DRV-08` to `DRV-13` are `Business`, all `Internal`, all owned by `Group Executive`, all `Medium` [1].
>
> Two defects to fix in passing. **`Group Executive` is not an actor** — it is in neither `O10 Actor` nor `O11 Role`, so six of thirteen drivers are owned by a name the repository does not carry [7]. And `DRV-13 Sustainability reporting` reads `Origin: Internal` while its description is an external reporting obligation.

How.

1. Read the seven regulatory drivers. `DRV-02 21 CFR Part 11 and Annex 11` and `DRV-03 Falsified Medicines / DSCSA` are United States and European instruments. They may apply through a partner or an export market, but they are not the licence this company operates under.
2. Add South African rows: medicine registration, variation and renewal under the Medicines and Related Substances Act [11]; manufacturing and wholesaling licensing with the Responsible Pharmacist named [12]; single exit price declaration and its annual adjustment; scheduling; pharmacy licensing under the Pharmacy Act [15]; and processing of health and special personal information under POPIA [14].
3. Write each `Description` as the duty, not the instrument. "Declare one exit price per presentation and adjust only when permitted" tells a designer something; "comply with s 22G" does not.
4. Set `Owner` to a person who exists in `O10 Actor`, or record that no such actor exists. Do not invent one.
5. Keep the seven inherited rows and mark their `Origin` as `External — partner or export market`. Deleting them loses the reason they were written.
6. Re-run `R08 Requirement derives from Dr…` and confirm each new driver reaches a row of `O06 Requirement`. Thirty-two requirements derive from thirteen drivers today; your new rows will have none.

Signed by. The regulatory affairs manager — the one interview that converts a driver list from generic to specific, and the second person you should see.

Done when. `O01` carries at least six South African rows, each with an owner who exists in `O10`, and no driver in `R08` has zero requirements.

Shows up in. `RPT09 Regulatory Coverage`, which today carries `O63`'s seven EU and US framed obligations and no South African one. After this step it carries both sets, and the South African rows all read as unevidenced, which is accurate.

Goes wrong when. You treat the inherited seven as the obligation set because they look authoritative. They belong to a company headquartered somewhere else.

Look it up. Volume 1, chapter 16, *The spine: Act 101 of 1965 and SAHPRA*, and *POPIA and health data*.

Form A-2 — Driver register, South African rows

Feeds `O01 Driver`. Use the next free `DRV-` number; the sheet ends at `DRV-13`. Six to ten rows is a first pass. `Description` is typed straight into Excel. Signed by the regulatory affairs manager, countersigned by the chief compliance officer.

Ref	Driver	Type	Origin	Owner	Pressure
DRV-14	EXAMPLE — delete this row	Regulatory	External		High

Step 3.3 — Test the goals and objectives against this industry's business model

Produce. An amended `O02 Goal` and `O03 Objective`, with the two rows that belong to another industry replaced.

You need. `O02 Goal` rows 5–10; `O03 Objective` rows 5–18; `R11 Goal realised by Objective`.

> **`O02 Goal — Ref, Goal, Description, Owner, Status; O03 Objective — Ref, Objective, Goal, Horizon, Status`**
>
> Six goals, fourteen objectives, two to three objectives per goal through `R11` [2].
>
> `GOL-01 Protect the licence to operate` is the right first goal here, and its description — "Meet every obligation with evidence, not assertion" — is the right description. Keep it.
>
> `GOL-03 Grow revenue from existing customers`, with `OBJ-05 Single view of customer across channels` and `OBJ-06 Cross-sell rate improved from baseline`, is not. A prescription medicine is not cross-sold, and in South Africa it may not be promoted to the public at all.

How.

1. Keep `GOL-01`, `GOL-04 Make the estate changeable`, `GOL-05 Use data as an asset` and `GOL-06 Operate resiliently`. They survive translation into this industry.
2. Replace `GOL-03`. The pharmaceutical equivalents are portfolio progression, registration coverage and supply reliability. Pick the one the chief commercial officer reports on.
3. Replace `OBJ-05` and `OBJ-06` with objectives that have a pharmaceutical denominator: registrations maintained against their renewal dates; batches released first time without deviation; expedited safety reports filed inside the clock.
4. Check `O02.Owner` against `O10 Actor`. `GOL-03`'s owner `Chief Commercial Officer` is not one of `O10`'s eighteen rows [7].
5. Re-point `R11` so every goal carries two or three objectives and every objective names one goal.
6. Leave `Status` alone. Every goal reads `Active`, which is plausible; changing it without authority is worse than leaving it wrong.

Signed by. The chief executive, who owns `GOL-01`. In practice the chief commercial officer signs the row you replaced.

Done when. No goal or objective uses a word from retail or banking, and every `O02.Owner` resolves to a row of `O10 Actor` or is marked as not existing.

Shows up in. `RPT06 Requirements Traceability`, indirectly: `REQ-022` to `REQ-027` derive from `DRV-08 Cost pressure` and need re-pointing once the goals move.

Goes wrong when. You rewrite all six. Nobody then recognises the set, the executive disowns it and Phase A restarts.

Look it up. Volume 1, chapter 4, *Four customers at once*; chapter 6, *The portfolio, not the product, is the unit*.

Step 3.4 — Reconcile the two numbers the repository gives for the same measure

Produce. One agreed definition and one agreed baseline for `MSR-01 Obligations with evidence`, written where a reader of either sheet will find it.

You need. `O04 Measure` row 5; `O63` rows 5–11; `R10 Objective measured by Measu…`; the chief compliance officer.

> **`O04 Measure — Ref, Measure, Unit, Domain, Baseline, Cadence`**
>
> Sixteen measures. `MSR-01 Obligations with evidence` is a percentage with a baseline of **54** and a monthly cadence [3].
>
> `O63 Control Objective` holds seven obligations with an `Evidence` column reading `Evidenced`, `Partial`, `In breach`, `In breach`, `In breach`, `Evidenced`, `Partial`. Two of seven evidenced is **29%**. Counting `Partial` as a half gives 43%. **Neither arithmetic produces 54** [4].

How.

1. Decide what counts. Obligations fully evidenced over obligations recorded is the defensible rule, because it is the rule an inspector applies.
2. Write the rule into the `Measure` cell: `Obligations with evidence (fully evidenced ÷ recorded)`. A measure whose denominator is not in its name gets recomputed differently by the next person.
3. Set `Baseline` to 29 and date it. `O04` has no date column, so put the date in the `Measure` text or create a column, and say which in `DLV-02`.
4. Then fix the other half: `CO-01` reads `Evidenced` while `O59 Compliance Assessment`, the only sheet that would hold that evidence, is empty [4]. Either the evidence is outside the repository — name where — or `CO-01` is `In breach` and the baseline is 14%.
5. Check the orphans. `R10` maps fourteen objectives to fourteen measures; `MSR-15 Architecture compliance rate` and `MSR-16 Initiatives through a design gate` measure nothing [3].
6. Agree the number with the chief compliance officer before it reaches a slide. A corrected number that surprises its owner in a board pack costs you the phase.

Signed by. The chief compliance officer, who owns all seven regulatory drivers in `O01`.

Done when. One number holds across `O04` and `O63`, the counting rule is written down, and the two unmeasured measures are mapped or struck.

Shows up in. `RPT01 Dashboard` and `RPT09 Regulatory Coverage`, which then agree. A dashboard that disagrees with its own source sheet is the first thing a sceptical executive finds.

Goes wrong when. You change 54 to 29 quietly. Someone then asks why compliance fell twenty-five points in a month.

Look it up. Volume 1, chapter 20, *Regulator to evidence.*

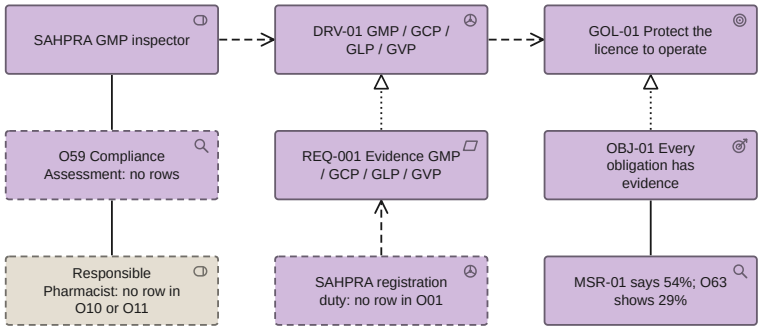


Figure 3.2 The Phase A motivation chain as the repository holds it. The three **dashed** elements are the ones with no row anywhere: the sheet that would hold the compliance evidence, the role that signs for release, and the obligation this company actually operates under.

Step 3.5 — Draw the stakeholder map, including the people the repository has no row for

Produce. `DLV-03 Stakeholder Map`: one page, every stakeholder with their concern, the artefact that answers it and the evidence they will ask for — four of them not in the organisation chart.

You need. `O09 Organization Unit` rows 5–16; `O10 Actor` rows 5–22; `O11 Role` rows 5–22; `R21 Role performed by Actor`.

> **`O70 Regulated Role` — a sheet the repository does not have; item 2 defines it**
>
> Columns: `Ref`, `Role`, `Statutory basis`, `Named person`, `Deputy`, `Registration number held`, `Signs for`. In Phase A fill it for the three roles that can stop your architecture: the Responsible Pharmacist, the QA release officer and the pharmacovigilance officer.
>
> `O10 Actor` and `O11 Role` hold the **same eighteen** generic corporate titles, `R21` maps them one-to-one, and every `O11.Performed by` cell reads `Person` [7]. No Responsible Pharmacist, no qualified person for pharmacovigilance, no QA release officer, no validation engineer, no regulatory affairs manager. Phase B rebuilds these sheets; Phase A needs those three roles named.

How.

1. Take the eighteen titles from `O10` as given. They are the architecture and executive audience and they are mostly real.
2. Add the three regulated roles to `O70` with the person's name, not the title. The Responsible Pharmacist resides in the Republic, is registered with the Pharmacy Council, and there is not more than one per licence — so this is a name and a registration number, not a function [12][15].
3. Add the four external stakeholders that appear in no sheet: SAHPRA, which runs four inspections against four record sets and allows thirty calendar days to respond to a report [13]; the South African Pharmacy Council; the Information Regulator; and the research ethics committee for anything clinical.
4. Give each stakeholder one concern and one artefact. "Can you show me the validated-system register" is a concern; "compliance" is not.
5. Mark who has a veto. The Responsible Pharmacist has one over release. Hearing it in Phase A is cheap; hearing it in Phase G is not.
6. Keep the map to one page. A stakeholder map that needs two pages has become a contact list.

Signed by. The chief enterprise architect (`ACT-08`), with the Responsible Pharmacist countersigning the row that describes their own veto.

Done when. Every stakeholder has a concern, an artefact and a named person, and the four external bodies are on the page.

Shows up in. `O31 DLV-03` moves from `Approved` to a document that exists. Record the previous state in your decision log.

Goes wrong when. You map the organisation chart. `O09`'s twelve units include no Quality, no Regulatory Affairs, no Pharmacovigilance, no Manufacturing and no Clinical Development [8]. That chart will not lead you to the people who matter.

Look it up. Volume 1, chapter 5, *The regulator is a stakeholder with a data demand*, and *The interview order*.

Form A-3 — Regulated role register, Phase A subset

Feeds `O70 Regulated Role`, which you are creating. Three rows in Phase A, six to twelve by the end of Phase B. `Registration number held` is typed into Excel, not written on a printed form. Signed by the quality assurance manager; the Responsible Pharmacist row is countersigned by its holder.

Ref	Role	Statutory basis	Named person	Deputy	Signs for
RR-01	EXAMPLE — delete this row	Medicines Act licence			Batch release

Step 3.6 — Replace the constraints and assumptions with the ones this estate imposes

Produce. An amended `O07 Constraint` and `O08 Assumption`, and a corrected `O52 Implementation Factor` row, so the plan is not built around a trading peak.

You need. `O07 Constraint` rows 5–12; `O08 Assumption` rows 5–10; `O52 Implementation Factor` rows 5–12.

> **`O07 Constraint` — `Ref`, `Constraint`, `Description`, `Type`**
>
> Eight rows. `CON-04 Regulatory change in flight` and `CON-05 Data residency` are real here. `CON-06 Change freeze windows`, described as "Peak periods are closed to change", is not: the closed windows here are the product quality review cycle, the stability pull schedule and an inspection.
>
> **Note for item 16.** ArchiMate 4 has no Constraint element. Drawn, these eight rows become `Requirement` elements. Volume 1 chapter 17 has the rule.

How.

1. Keep `CON-01`, `CON-03`, `CON-04`, `CON-05`, `CON-07` and `CON-08`. They hold here.
2. Rewrite `CON-06`. The real freeze set: the annual product quality review window; any period in which an inspection is notified or in progress; a stability pull or a batch in process on the affected system; and a database lock on an active trial.
3. Add the two constraints the sheet lacks: a change to a validated system is a change-control record before it is a backlog item; and a variation may be required before a business change is lawful, its class deciding whether you proceed on notification or wait for approval.
4. Correct `O52` row 7. `IF-03 Peak trading freeze` is a `Constraint` on the plan and is retail vocabulary. Replace it rather than adding a second row — a reader who inherits it plans around a peak that does not exist [6].
5. Move `ASM-05 Regulatory interpretation remains stable` from `Open` to `At risk`. Interpretation of the computerised-systems rules is in active revision.
6. Add the missing assumption: that the validation function has capacity for the programme's protocol load. If you cannot evidence it, mark it `At risk` and carry it into `DLV-02`.

Signed by. `GB-01 Architecture Board` for the constraint set; the quality assurance manager for the freeze windows, because they own the cycle that creates them.

Done when. No `O07` constraint references a trading peak, every freeze window names the regulated activity behind it, and `O52` has no retail row.

Shows up in. `RPT08 Roadmap`, later. A roadmap built on `IF-03` as written schedules work into an inspection window.

Goes wrong when. You add the GxP constraints and leave `IF-03` in place. The plan then has two freeze regimes and the delivery manager follows the shorter one.

Look it up. Volume 1, chapter 7, *The Annual Product Quality Review*, and *The cadence, which is the spine of this chapter*.

Step 3.7 — Do the date arithmetic before you promise a date

Produce. A one-page date test: for each system the vision touches, the earliest date it could lawfully be in use and the three largest contributors to it.

You need. Step 3.1's classification; the validation master plan; the validation lead; the change and deviation backlog.

> **No sheet holds this, and that is the finding**
>
> `O30 Work Package` carries `Cost`, `Start` and `End`; `O66 Architecture Increment` carries `Lands`. Neither carries a validation duration, a qualification window or a periodic-review date. Record the arithmetic as an appendix to `DLV-02`, and from item 11 maintain it in `O68 Periodic Review Schedule`.

How.

1. For each GxP system in scope, ask the validation lead for three numbers: how long the last validation of that system took, the current protocol approval queue, and the open change records against it.
2. Add the retrospective load. Systems already running with no current validation summary report are work that happens before, not after, the change you are planning.
3. Convert the backlog into a date, not a risk. "Six protocols ahead of us at roughly three weeks each" is a date. "Validation capacity is a risk" is not.
4. Lay the result against the date you were given. Where they differ, write the difference in weeks and name what would close it: more validation resource, a smaller first increment, or a later date.
5. Pick the smallest first increment still worth doing. Here that is usually a non-GxP system, or a reporting layer honestly declared out of scope.
6. Put both dates in the vision, labelled.

Signed by. The validation lead, whose numbers these are and who is the scarcest person in the building. Where there is no validation lead, that absence is your first finding and the chief compliance officer signs.

Done when. Every GxP system in scope has an earliest lawful in-use date with three named contributors, and the vision quotes the later date.

Shows up in. `RPT08 Roadmap`, once Phase E runs. Today `O30` has three work packages past their end date — `WP-02`, `WP-04`, `WP-10` — which is what happens when dates skip this step.

Goes wrong when. The validation lead gives you an average. An average hides the one system with a twelve-month qualification dependency, and that is the system the vision is about.

Look it up. Volume 1, chapter 13, *Periodic review*; chapter 3, *Validation, qualification, verification*.

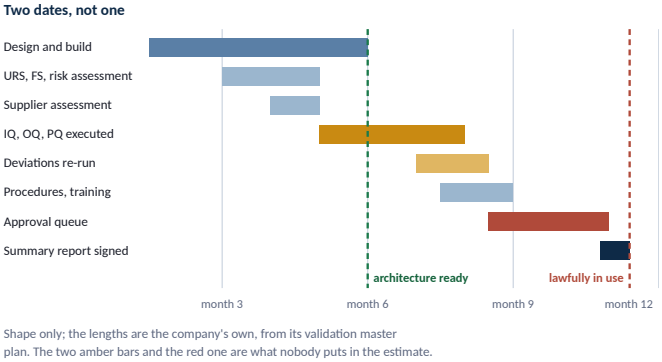


Figure 3.3 Why a regulated vision carries two dates. The **architecture-ready** date is when the design is done; the **lawfully-in-use** date is when a validation summary report has been signed. Step 3.7 produces the second one.

Step 3.8 — Write the one-page Architecture Vision

Produce. `DLV-01 Architecture Vision`: one page, nine blocks, no appendix.

You need. Steps 3.1 to 3.7; `O02 Goal`; `O13 Business Capability` level 1; `O54 Architecture Alternative`.

> **O31 Deliverable — Ref, Deliverable, Phase, Status**
>
> Set DLV-01 to the status your governance supports. It reads Approved with no approver and no date [5]. If GB-01 has approved your document, write Approved; otherwise Draft, and let the gate change it.

How.

1. Write the nine blocks, each at most four sentences: why now; the obligation set; what is in and out of scope; the value streams from O14 the work touches; the level-1 capabilities from O13 and the three you will move; the two dates from step 3.7; the chosen alternative from O54; what you will not do; and what you need from whom.
2. Name the regulated boundary in the scope block. "Nine of twelve applications hold GxP records" belongs on the page, in those words.
3. Use O54's own options. ALT-B Foundations first at $46m is Recommended and ALT-E Compliance led at $31m is Partly adopted. Two live answers is not a recommendation; pick one and close the other.
4. Put the measure from step 3.4 on the page with its corrected baseline, and nothing else numeric except the two dates.
5. Keep it to one page. If the one-page version is unconvincing, the architecture is unconvincing, and a second page does not fix that.
6. Read it aloud to the regulatory affairs manager before the board sees it. They will find the sentence that is not true.

Signed by. GB-02 Investment Committee, which holds G0 funding, with GB-01 Architecture Board recommending [9]. The chief executive owns GOL-01 and should be in the room.

Done when. One page, nine blocks, two dates, one alternative, and the regulatory affairs manager has read it.

Shows up in. O60 Governance Decision, where DEC-01 Adopt the recommended architecture alternative reads Outstanding today. O60 has no date column, so record the date in the minute.

Goes wrong when. The vision promises a compliance outcome rather than a mechanism. "Full Part 11 compliance by Q4" is unfalsifiable; "audit trail review scheduled, performed and recorded on nine systems, evidenced monthly" can be checked.

Look it up. Volume 1, chapter 2, *How to tell which archetype you are in, in the first week.*

Step 3.9 — Write the Statement of Architecture Work with the regulated-estate caveats in it

Produce. DLV-02 Statement of Architecture Work, with seven caveats a non-regulated statement would not need.

You need. Steps 3.1 to 3.8; O57; O58; O33 Architecture Contract.

> **O31 Deliverable row 6, and O33 Architecture Contract — Ref, Architecture contract, Work package, Compliance status**
>
> O33 has eight rows against O30's twelve work packages, so four packages have no contract. Name that in the statement rather than finding it at GT-2 G2 Build complete.

How.

1. Write the ordinary content: scope, approach, deliverables, resources, timetable, acceptance criteria, and the change-control route for the statement itself.
2. Caveat one, the validated estate. Nine of twelve applications hold regulated records; a change to any is a change-control record and a validation event; the dates here assume step 3.7's queue.
3. Caveat two, the four sheets the repository lacks — O67 Validated System, O68 Periodic Review Schedule, O69 Retention Clock, O70 Regulated Role. Say who creates each, in which phase.
4. Caveat three, the quality system's standing. State plainly whether an architecture decision is a quality record here, because if it is not, nobody has to follow it. Item 2 takes the decision; the statement records it.
5. Caveats four to six: O29 Gap and O59 Compliance Assessment are empty, so no gap analysis or compliance evidence is inherited; O20 Data Entity and O43 Master Data Rule disagree on the master system for every entity they share, which Phase C must resolve; and the signatory for release is a named person with a legal duty who is not in the repository.
6. Caveat seven, what you will not produce: not a validation methodology, not level-3 work instructions. Both belong to quality.

Signed by. GB-02 Investment Committee approves, GB-01 Architecture Board recommends, the quality assurance manager signs the caveats. Where the real signatory is the Responsible Pharmacist, record their agreement in the minute; they do not sign architecture documents.

Done when. All seven caveats are in the body, each with a named owner, and each missing sheet has a creating phase.

Shows up in. GT-0 G0 Charter, which tests that "Scope, acceptance criteria and contract exist" [9]. This is the document it tests.

Goes wrong when. The caveats go into an appendix, which is where a caveat goes to be ignored. Put them in the body, numbered, above the timetable.

Look it up. Volume 1, chapter 19, *What to say when you do not know*; chapter 16, *Who inspects, and what they ask for*.

Step 3.10 — Take Phase A through the gate and record what you could not establish

Produce. A minuted `GT-0` decision, an updated `O60 Governance Decision`, and a written list of what you could not establish.

You need. `DLV-01`, `DLV-02`, `DLV-03`; `O58`; `O60`; `O62 Delivery Risk`.

> **`O60 Governance Decision — Ref, Decision, Body, Status`**
>
> Seven rows, four `Outstanding`: `DEC-01` adopt the alternative, `DEC-02` select the integration product, `DEC-03` fund compliance mechanisms, `DEC-04` designate data owners [9]. **The sheet has no date column**, so nothing can be overdue — which is how `RPT01` reports zero gaps overdue with a straight face.

How.

1. Put all three deliverables to `GB-01`, then to `GB-02` for G0. Two bodies, two minutes.
2. Close `DEC-01`: one alternative, recorded, with the rejected ones named. Two live answers in `O54` is how a programme funds both.
3. Add a date to `O60` — a column if you may, the decision text if not — and a review date for each `Outstanding` row.
4. Move `RSK-01 Integration product not selected` out of `O62 Delivery Risk`. Its likelihood is `Certain`, which makes it an issue; `O52` already holds it as `IF-01 Vendor selection not made`. Three `O62` rows read `Certain`.
5. Write the "could not establish" list: no validation state on any system; no periodic-review date anywhere; no quality body in `O57`; no South African obligation in `O63`; no evidence behind `CO-01`.
6. Publish it with the vision. It is the most credible page you produce in Phase A.

Signed by. `GB-02 Investment Committee` for G0; `GB-01 Architecture Board` for the decision log.

Done when. `GT-0` is minuted, `DEC-01` closed, every `Outstanding` decision carries a review date, and the list is attached to `DLV-01`.

Shows up in. `RPT01 Dashboard`. Its "Gaps overdue: 0" line stays at zero until `O29 Gap` carries rows in Phase E. Say so in the minute, so nobody reads zero as good news.

Goes wrong when. You pass the gate without the "could not establish" list, and Phase B then discovers the architecture function has no standing in the quality system. That belongs in Phase A's minute.

Look it up. Volume 1, chapter 13, *The system itself is regulated*.

What Phase A hands to Phase B

Artefact	Where it lives	Who owns it
GxP / not-GxP split of twelve applications	`O67 Validated System`, new	Quality assurance manager
Driver list with the South African obligations	`O01 Driver`, amended	Regulatory affairs manager
One `MSR-01` baseline with its counting rule	`O04`, `O63`	Chief compliance officer
Three regulated roles, named	`O70 Regulated Role`, new	Quality assurance manager
Two dates: architecture ready, lawfully in use	Appendix to `DLV-02`	Validation lead
Seven regulated-estate caveats	`DLV-02`	Chief enterprise architect
What could not be established	Attached to `DLV-01`	Chief enterprise architect

Phase B rebuilds the business layer around rows three and four. Its reason is on its first page: nobody in the repository does a pharmaceutical job [7][8].

THE CHAPTER IN TEN LINES

- Phase A in a regulated estate has one extra job: fix which systems hold regulated records, on day one.
- O31 already marks all three Phase A deliverables Approved, with no approver column and no date column. Treat them as empty.
- Nine of Calderon's twelve applications are GxP systems, and O23 has no validation-state column, no GxP flag and no periodic-review date.
- O01 Driver is thirteen rows of EU and US obligations plus generic business pressure. There is no South African obligation in it.
- Six of thirteen drivers are owned by Group Executive, which is not an actor anywhere in the repository.
- MSR-01 records a baseline of 54% for obligations with evidence. O63 shows two of seven evidenced, which is 29%. Agree one number and write the counting rule into the measure.
- GOL-03 and its two objectives are about cross-selling to customers, which is not how a prescription medicine reaches a patient.
- The stakeholder map must carry four bodies that are in no sheet: SAHPRA, the Pharmacy Council, the Information Regulator and the research ethics committee.
- CON-06 says peak trading periods are closed to change. The real closed windows are the product quality review cycle, a notified inspection, a stability pull and a database lock.
- A regulated vision quotes two dates: architecture ready, and earliest lawful use. The second comes from the validation queue, not from the delivery plan.
- The Statement of Architecture Work carries seven caveats a non-regulated one would not need, and they go in the body, not an appendix.
- O60 Governance Decision has four outstanding decisions and no date column, which is why the dashboard can report zero overdue.
- The most credible page you produce in Phase A is the list of what you could not establish.

SOURCES FOR THIS CHAPTER

- **1.** Pharmaceuticals_TOGAF_Metamodel.xlsx, sheet O01 Driver, rows 5-17 (headers row 4); thirteen drivers, seven Regulatory and six Business, owners Chief Compliance Officer and Group Executive
- **2.** Pharmaceuticals_TOGAF_Metamodel.xlsx, sheets O02 Goal rows 5-10, O03 Objective rows 5-18 and R11 Goal realised by Objective rows 5-18
- **3.** Pharmaceuticals_TOGAF_Metamodel.xlsx, sheets O04 Measure rows 5-20 and R10 Objective measured by Measu... rows 5-18; MSR-15 and MSR-16 appear in no R10 row
- **4.** Pharmaceuticals_TOGAF_Metamodel.xlsx, sheets O63 Control Objective rows 5-11 (Evidence column: two Evidenced, two Partial, three In breach) and O59 Compliance Assessment, header row only
- **5.** Pharmaceuticals_TOGAF_Metamodel.xlsx, sheet O31 Deliverable, rows 5-24; twenty deliverables, thirteen Approved and seven Draft, phased A to RM with no Preliminary and no H rows
- **6.** Pharmaceuticals_TOGAF_Metamodel.xlsx, sheets O07 Constraint rows 5-12, O08 Assumption rows 5-10 and O52 Implementation Factor rows 5-12
- **7.** Pharmaceuticals_TOGAF_Metamodel.xlsx, sheets O10 Actor rows 5-22, O11 Role rows 5-22 and R21 Role performed by Actor rows 5-22; the same eighteen titles in both object sheets, Performed by reads Person on every row
- **8.** Pharmaceuticals_TOGAF_Metamodel.xlsx, sheet O09 Organization Unit, rows 5-16; twelve units, Legal at 240 and Customer Service at 211 against Technology at 116
- **9.** Pharmaceuticals_TOGAF_Metamodel.xlsx, sheets O57 Governance Body rows 5-9, O58 Governance Gate rows 5-8 and O60 Governance Decision rows 5-11
- **10.** Pharmaceuticals_TOGAF_Metamodel.xlsx, sheets ZZ Checks and RPT01 Dashboard; both claim 850 records and 65 object types populated, against 814 records counted across the 64 O sheets plus EXT
- **11.** Medicines and Related Substances Act 101 of 1965, consolidated to GG 40869 of 26 May 2017 (SAHPRA's published copy); cited as volume 1 cited it https://www.sahpra.org.za/wp-content/uploads/2019/09/Medicines-and-Related-Substances-Act_101-of-1965_Act_GG-40869_2017-05-26.pdf
- **12.** SAHPRA, SAHPGL-INSP-02 v10, Guidelines for Good Manufacturing Practice of Medicines, 13 April 2026; assigns batch-release duties and the duty to keep manufacturing and testing processes in a validated state to the Responsible Pharmacist. Cited as volume 1 cited it; re-check the version before quoting it to a client https://www.spotlightnsp.co.za/wp-content/uploads/2026/05/final-gmp-guideline-for-publication_20260424.pdf
- **13.** SAHPRA, SAHPGL-INSP-05 v2, Guideline on How to Respond to a GMP, GWP, GCP or GVP Inspection Report, 18 September 2022; one deadline of 30 calendar days from receipt, extensions only by prior written grant, no critical/major/minor classification scheme https://www.sahpra.org.za/wp-content/uploads/2022/09/SAHPGL-INSP-05_v2-Guideline-on-How-to-Respond-to-a-GMP-GWP-GCP-or-GVP-Inspection-Report.pdf
- **14.** Protection of Personal Information Act 4 of 2013, s 26, prohibition on processing of special personal information; the instrument volume 1 chapter 16 works through for health and trial data https://popia.co.za/section-26-prohibition-on-processing-of-special-personal-information/
- **15.** Pharmacy Act 53 of 1974, establishing the South African Pharmacy Council, with which the Responsible Pharmacist is separately registered; identified by title, as volume 1 identified it https://pharmcouncil.co.za/Media/Default/Pharmacy%20Act/Pharmacy%20Act%2053%20of%201974.pdf
- **16.** The Open Group, TOGAF Standard, Architecture Development Method phase descriptions, public library landing page.org was robots-disallowed for the research behind volume 1, so the phase content in this volume is taken from volume 1 and from the repository. Identified, not retrieved: The Open Group's public pages are robots-disallowed to automated retrieval, so no edition or wording is claimed from it.

CHAPTER 4

Business Architecture

The capability map knows what industry this is. The organisation chart does not.

The one sentence this item is built on

Nobody in the repository does a pharmaceutical job.

Not one of the eighteen titles in `O10 Actor`, not one of the twelve units in `O09 Organization Unit`, not one of the twenty rows in `O16 Business Process`. The capability map knows the industry: `O13` 's ten level-1 capabilities run Discovery Research through Supply Chain & Distribution, and they are right [3]. The organisation meant to perform them is a generic corporate with a Customer Service department of 211 people against a Technology department of 116 [1].

Phase B here is a rebuild on a sound skeleton [15]. Keep `O13` 's structure and `O14` 's five value streams; replace almost everything else.

Calderon Therapeutics is the invented company in your repository; **Veldtkruin Group** is volume 1's invented group, and appears here only in cross-references. Keep their data apart.

> **THE PROCESS MODEL MUST SAY WHICH STEPS ARE GXP AND WHICH ARE NOT**
>
> Every step you draw becomes an evidence burden on the systems beneath it in Phase C and Phase D. A step that creates, alters or decides on a regulated record puts its system inside a validation scope: specification, qualification, audit trail, access control, periodic review, retention. A step that does not, does not. That difference is six weeks against one week per change, for the rest of the estate's life.
>
> `O16 Business Process` has twenty rows, five columns — `Ref` , `Business process` , `Value stream` , `Description` , `Automation` — and **no GxP column and no regime column** [5]. Nothing downstream can ask which steps are regulated, because the question has no field. Add both in step 4.9, and mark the regime that compels the record — GLP, GCP, GMP, GVP or GDP — not the word "GxP", because the regime decides the inspection, the inspector and the record set.
>
> Mark the unregulated steps explicitly too. An unmarked step reads as regulated to a careful quality function and unregulated to a hurried delivery team, and both readings are expensive.
>
> **Look it up.** Volume 1, chapter 13, *Which GxP, and for which record*; chapter 20, *GxP regime to the record it compels*.

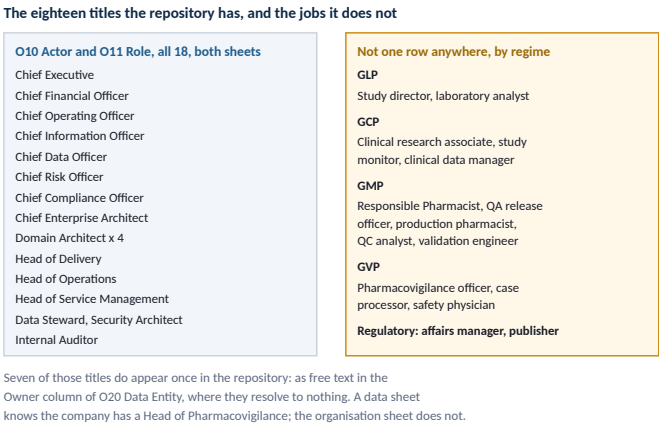


Figure 4.1 The Phase B finding in one picture. The left column is both `O10 Actor` and `O11 Role` , which hold the **same eighteen** titles. The right column is what a pharmaceutical company runs on, and none of it has a row.

The layer disagrees with itself before anybody models anything

`O16` 's twenty rows are named `<value stream> — stage N` and described as `Stage N of <value stream>` , four per stream [5]: a naming convention applied twenty times, not a process model. And `O14 Value Stream` 's `Stages` column reads 4, 5, 6, 4, 5 — **twenty-four stages** [4]. Four have no process row at all: `VS02` stage 5, `VS03` stages 5 and 6, `VS05` stage 5.

ANSWER THESE BEFORE YOUR FIRST WORKSHOP

Who in this company releases a batch, by name, and in which system?

Which organisation unit owns quality, and what is its headcount?

Step 4.1 — Run the five checks that tell you how bad the business layer is

Produce. A one-page finding sheet of five counted results, dated, to open your first workshop.

You need. `O09` rows 5–16; `O10` and `O11` rows 5–22; `R21`; `R22`; `O20` rows 5–18.

> **`Read-only. You are counting, not writing`**
>
> 1. **Units against regimes.** `O09` 's twelve units against the five GxP regimes: no Quality, no Regulatory Affairs, no Pharmacovigilance, no Manufacturing, no Clinical Development [1].
> 2. **Heads against actors.** Six of twelve `O09.Head` values appear in no `O10 Actor` row: Chief People Officer, Chief Commercial Officer, Head of Service, Head of Supply, Head of Internal Audit, General Counsel [1][2].
> 3. **Actor against role.** `O10` and `O11` hold the same eighteen titles in the same order, `R21` maps them one-to-one, and every `O11.Performed by` cell reads `Person` [2].
> 4. **Capability ownership.** `R22 Org Unit owns Capability` maps `ORG-01` → `C01` on to `ORG-10` → `C10` : a positional join, not an assignment [8].
> 5. **Where the real job titles are.** `O20 Data Entity` 's `Owner` column names Head of Pharmacovigilance, Head of Regulatory Affairs, Head of Clinical Development, Head of Preclinical Development, Head of Discovery Research, Head of Supply Chain and Head of Commercial. **Seven of its eight distinct owners exist nowhere else in the repository** [9].

How.

1. Do check 5 first. It changes the room, because it shows the repository already knows the jobs exist, in a data sheet, as text, resolving to nothing.
2. Count, do not characterise. "Six of twelve heads are not actors" survives a hostile reading; "the organisation model is poor" does not.
3. Put the headcounts on the page unchanged: Legal 240, Finance 265, Commercial 234, Customer Service 211, Operations 206, Technology 116, Data & Analytics 23 [1].
4. Propose no target organisation. Phase B describes; the business decides.
5. Date the sheet and name the file after the repository version. You will be asked in six months whether it has improved.

Signed by. Nobody. It is a finding sheet, not a deliverable. Circulate it to the chief enterprise architect (`ACT-08`) and the chief operating officer first.

Done when. Five numbers, each traceable to a sheet and a row range, on one page.

Shows up in. `RPT13 Orphan Analysis` , four rows today and more once roles and capabilities are tested against each other.

Goes wrong when. You present it as criticism of whoever built the repository. They built a sound metamodel and filled it with a generator. Say that, in those words.

Look it up. Volume 1, chapter 5, *Read this before you book the workshop*.

Step 4.2 — Rebuild `O09 Organization Unit` around the five regimes

Produce. An amended `O09` carrying the functions a pharmaceutical company is organised into, each with a head who exists in `O10` .

You need. Step 4.1; `O09` rows 5–16; `O17 Business Function` rows 5–14; the chief operating officer; human resources.

> **`O09 Organization Unit — Ref, Organisation unit, Type, Head, Headcount`**
>
> Twelve rows. Add from `ORG-13` upward; do not renumber. The units this company must have and does not: Quality Assurance, Quality Control, Regulatory Affairs, Pharmacovigilance, Manufacturing, Clinical Development, Preclinical Development, Medical Affairs.
>
> `O17 Business Function` already names ten of these — `BFN-01` to `BFN-10`, with names identical to `O13`'s ten level-1 capabilities [6]. The function list is right and the organisation list is wrong, so the fix is a transcription, not a design exercise.

How.

1. Copy the ten names from `O17` into `O09` as candidates, then ask the chief operating officer which are departments and which are activities inside one.
2. Split quality in two. Quality Assurance owns the quality system — deviations, change control, corrective actions, supplier qualification, audits; Quality Control runs the laboratory. Different heads, different systems, different inspection exposure.
3. Decide centralised against site-autonomous quality and record it in `Type`. It decides whether there is one quality system or one per site, and Phase C cannot answer that for you.
4. Get real headcounts, including the ones that embarrass the existing rows. A Quality function smaller than Legal is a finding.
5. Mark `Customer Service` for review. The enquiry-handling function here is Medical Information, a regulated adverse-event intake channel.
6. Set every `Head` to a title that exists in `O10 Actor`, or create the actor in step 4.3. No head may resolve to nothing.

Signed by. The chief operating officer, with human resources confirming headcounts. No `O57` body has a mandate over organisation design, so this is a business signature minuted at `GB-01 Architecture Board`.

Done when. `O09` carries Quality, Regulatory Affairs and Pharmacovigilance units, every `Head` resolves to an `O10` row, and `Type` answers the centralised-or-site question.

Shows up in. `R22`, which you repair in step 4.6, and `RPT03 Capability Heatmap` once ownership is real.

Goes wrong when. You add the units and leave the headcounts. A two-person Pharmacovigilance unit against a 211-person Customer Service unit is not wrong because of the chart.

Look it up. Volume 1, chapter 5, *Quality and manufacturing*; chapter 7, *Centralised quality against site-autonomous quality*.

Form B-1 — Organisation unit register

Feeds `O09 Organization Unit`, all five columns. Eight to twelve new rows is a first pass. `Type` is Group, Corporate, Line or Assurance here; add `Regulated` if the business agrees. Signed by the chief operating officer.

Ref	Organisation unit	Type	Head	Headcount
ORG-13	EXAMPLE — delete this row	Regulated	Head of Quality Assurance	

Step 4.3 — Separate actor from role, and stop `Performed by: Person`

Produce. An amended `O10 Actor` and `O11 Role` holding different things, and an `R21` that is no longer one-to-one.

You need. Step 4.2; `O10`, `O11` and `R21` rows 5–22.

> **`O10 Actor — Ref, Actor, Type, Status; O11 Role — Ref, Role, Type, Performed by`**
>
> Eighteen rows each, the same eighteen titles, `R21` mapping `ROL-01` → `ACT-01` through `ROL-18` → `ACT-18`, `Performed by` reading `Person` on all eighteen [2].
>
> A role is a part someone plays; an actor is who plays it. When both sheets hold the same list, neither carries information. The symptom is `Performed by: Person` — the column exists to say whether a role is performed by an individual, an organisation unit, a committee or an external party, and `Person` fits every row and distinguishes none.
>
> `Status` reads `Filled` on every `O10` row, which is why `ZZ Checks` reports zero vacant roles. The check is correct and tells you nothing: the vacant roles are the ones with no row.

How.

1. Keep `O10` as the actor list and add the people this company has: validation lead, QA release officer, quality control analyst, regulatory affairs manager, submission publisher, pharmacovigilance officer, case processor, safety physician, clinical research associate, clinical data manager, production pharmacist, supply planner.
2. Rebuild `O11` as roles, which are fewer than actors and outlive them. "Batch release" is a role; "production pharmacist, fill and finish site" is an actor.
3. Set `Performed by` honestly: `Person`, `Organisation unit`, `Committee` or `External party`. An audit-trail review role is performed by someone independent of the system's operation.
4. Re-point `R21` so a role may have several actors and an actor several roles. One-to-one is the shape of a generated sheet.
5. Mark `Status` as `Vacant` where it is. The validation lead and the qualified person for pharmacovigilance are routinely one deep or unfilled, and that is a delivery constraint.
6. Keep the four domain architects. `ACT-09` to `ACT-12` are real and you need them in Phase C.

Signed by. Human resources for the actor list, the chief operating officer for the role list. A role with a statutory basis belongs in `O70` and is signed by quality — step 4.4.

Done when. `O10` and `O11` hold different lists, `R21` is no longer one-to-one, and no `Performed by` cell reads `Person` unless one individual performs it alone.

Shows up in. `ZZ Checks`, "Vacant architecture roles", which should stop reading zero.

Goes wrong when. You add forty actors and no roles. The repository then holds a staff directory where a role model should be, and every later assignment points at someone who will leave.

Look it up. Volume 1, chapter 5, *A role-to-system map*, and *The interview order*.

Form B-2 — Role register

Feeds `O11 Role`, all four columns. One row per role actually performed in the company; twenty-five to forty for a mid-size manufacturer, so continue on a second copy. Transcribe into `O11` the same day, while you remember which answers were hedged. Signed by the chief operating officer.

Ref	Role	Type	Performed by
ROL-19	EXAMPLE — delete this row	Quality	Person

Step 4.4 — Record the regulated roles in `O70 Regulated Role`

Produce. `O70` filled for every role carrying a statutory duty, with a named person and deputy against each.

You need. Step 4.3; `O70 Regulated Role`, defined in item 2; the quality assurance manager; the Responsible Pharmacist.

> **`O70 Regulated Role` — a sheet the repository does not have; item 2 defines it**
>
> Columns: `Ref`, `Role`, `Statutory basis`, `Named person`, `Deputy`, `Registration number held`, `Signs for`.
>
> A regulated role is not a role with a compliance flavour: its holder is personally answerable to a body outside the company. Three to start with:
>
> - **The Responsible Pharmacist.** South Africa's batch-release role, accountable both to SAHPRA and to the Pharmacy Council, resident in the Republic, registered with the Council, not more than one per licence — and carrying an explicit duty that manufacturing and testing processes remain in a validated state [12][13]. That duty gives them a legitimate interest in your `O67 Validated System` register, your review dates and your open change records.
> - **The QA release officer**, where the company separates record review from certification.
> - **The pharmacovigilance officer**, with the responsible person and the back-up expected to reside in South Africa [14].

How.

1. Write the person's name, not the title. A row reading "the Responsible Pharmacist" is a diagram; a row with a name and a registration number is a record.
2. Fill `Deputy` on every row. A single-point regulated role with no named deputy is a licence risk, and in a mid-size company it is the normal state.
3. Fill `Signs for` with the decision, not the document: "certification and release of finished product", "expedited report submission", "database lock".
4. Cross-reference each row to its `O11` role from step 4.3, so the architecture model and the statutory model do not drift.
5. Put the Responsible Pharmacist's veto in writing, in their words, attached to the Phase B output. It is the cheapest risk reduction in this phase.
6. Do not put these rows in `O11` instead. `O11` has no statutory-basis and no registration column, and conflating the two loses what makes a regulated role different.

Signed by. The quality assurance manager, with each named holder confirming their own row. Neither `O57 Governance Body` nor any `O58` gate can approve this, because **there is no quality body in the repository**: no change control board, no validation committee, no data integrity forum [11].

Done when. Every regulated role has a named person, a deputy, a statutory basis and a `Signs for` entry, cross-referenced to an `O11` row.

Shows up in. Nothing yet. `RPT05 Data Ownership` keeps naming `O20`'s non-existent heads until Phase C repairs it.

Goes wrong when. You record the role and not the registration. The architecture then cannot answer the question an inspector opens with: who signed, under what authority, on what date.

Look it up. Volume 1, chapter 13, *The Responsible Pharmacist*; chapter 3, *Qualified Person versus Responsible Pharmacist*.

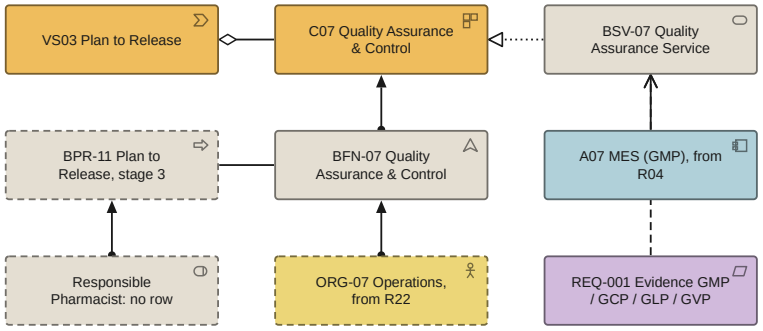


Figure 4.2 The Phase B object chain for one capability, as the repository resolves it. The three **dashed** elements are the broken links: a process that is a name only, an owner assigned by position, and the role that actually signs for release and has no row at all.

Step 4.5 — Keep the capability tree, replace its content, re-score level 1

Produce. A level-2 decomposition of three level-1 capabilities that describes work somebody does, and corrected scores on all ten level-1 rows with the rule written down.

You need. `O13` rows 5–104; `R01` rows 5–94; `RPT03 Capability Heatmap` rows 5–14; the heads of the three functions you pick.

> **`O13 Business Capability — Ref`, `Capability`, `Level`, `Parent`, `Description`, `Criticality`, `Current`, `Target`, `Disposition`**
>
> One hundred rows: ten level 1, thirty level 2, sixty level 3. `R01` resolves all ninety parent-child links without a break. **Keep that structure.** It is the best-built thing in the repository [3].
>
> Now test the content. Each level-2 name is a level-1 name plus one word from a pool of ten: Administration, Optimisation, Onboarding, Reporting, Assurance, Support, Planning, Execution, Monitoring, Design. Each level-3 name is a level-2 name plus "— Define" or "— Operate". `C05.02 Pharmacovigilance Reporting` is real; `C05.01 Pharmacovigilance Onboarding` fell out of the generator.
>
> Then the scores. `RPT03` reproduces the ten level-1 rows and computes `Gap` as target minus current. **Every gap is 1 and every disposition is `Sustain`** [10], so the heatmap ranks nothing. Three values to argue about: `C01 Discovery Research` is `Criticality: Low` in a company whose first value stream is Target to Candidate; `C08 Commercial & Market Access` is the most mature capability in the estate at `Current: 4` while `C03 Clinical Development` and `C05 Pharmacovigilance` sit at 2; and the scores do not roll up — `C01` is `Low` while its descendant `C01.01.02` is `Critical` [3][4].

How.

1. Pick three level-1 capabilities to redo: `C07 Quality Assurance & Control`, `C05 Pharmacovigilance`, `C04 Regulatory Affairs`. They are the three an inspector tests and the three described worst.
2. Decompose each with the function head, to level 2 only. `C07`'s real level 2 is batch release, deviation and investigation, corrective and preventive action, change control, supplier qualification, audit and self-inspection, training and document control, product quality review, complaint and recall.
3. Go to level 3 only where a system decision depends on it, and keep the `Ref` scheme so `R01` still resolves. Mark the branches you have not redone as generated.
4. Write the scoring rule before you re-score: one to five, each number defined by observable evidence. Score `Criticality` against consequence of failure, whose top here is patient harm, then licence loss, then supply interruption, then cost.
5. Set `Current` from evidence — the deviation backlog for `C07`, expedited reports filed inside the clock for `C05`, the variation queue for `C04` — and decide whether a parent's score derives from its children or is independent. Both are defensible; silence is not.
6. Make the gaps differ, and set `Disposition` from gap and criticality together, in the sheet's vocabulary of `Sustain` and `Build`.

Signed by. The head of each function you decompose, for their own branch; `GB-01 Architecture Board` for the structure and the scores. The chief enterprise architect signs the scoring rule, which outlives the scores.

Done when. Three branches have level-2 children that name work, `R01` still resolves every link, the ten gaps take three or more values, and the rule is written down.

Shows up in. `RPT03 Capability Heatmap`, which becomes usable, and `RPT01 Dashboard`'s "Level 1 capabilities: 10" line, which stays correct.

Goes wrong when. You redo all ten branches, or score them yourself over a weekend. The first is three months of workshops for seven branches nobody will use this year; the second produces scores that are defensible and unowned, and the first executive to dispute one discards the set.

Look it up. Volume 1, chapter 5, *Quality and manufacturing*; chapter 11, *Seven domains, and one that no other industry has*.

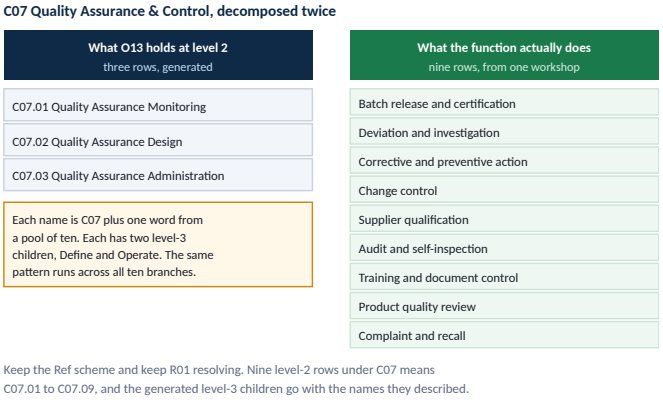


Figure 4.3 Step 4.5 in one page. `O13`'s structure is sound and its level-2 **content** is a suffix generator. The right-hand column is one workshop's output with the function head.

Step 4.6 — Reassign capability ownership, because `R22` is a positional join

Produce. A repaired `R22 Org Unit owns Capability` in which each level-1 capability is owned by the unit that performs it.

You need. Step 4.2's amended `O09`; `R22` rows 5–14; `O17` rows 5–14.

> **`R22 Org Unit owns Capability — Org unit, Unit name, Capability, Capability name`**
>
> Ten rows: `ORG-01` → `C01`, `ORG-02` → `C02`, on to `ORG-10` → `C10` [8]. Read what that produces:
>
> | Unit | Owns | Why it is wrong |
> |---|---|---|
> | `ORG-02` Finance | `C02` Preclinical Development | Finance does not run toxicology |
> | `ORG-04` Human Resources | `C04` Regulatory Affairs | The registration portfolio is not an HR asset |
> | `ORG-05` Technology | `C05` Pharmacovigilance | Making IT own safety reporting is how a clock gets missed |
> | `ORG-06` Data & Analytics | `C06` Manufacturing & Supply | A 23-person unit owns the plant |
> | `ORG-09` Customer Service | `C09` Medical Affairs | The nearest miss, and still wrong |
>
> Two of the ten are plausible: `ORG-08 Commercial` → `C08` and `ORG-10 Supply & Logistics` → `C10`. That is what a positional join looks like when the lists line up at the end.

How.

1. Point each capability at the unit that performs it, using `O17`'s function names as the bridge: `BFN-07 Quality Assurance & Control` performs `C07`, so the Quality Assurance unit owns it.
2. Name a person behind each unit through `O09.Head`. A capability owned by a department is owned by nobody when a decision is needed.
3. Let one unit own several capabilities, and give each capability exactly one owner. Shared ownership is the commonest cause of an unresolved Phase C decision.
4. Record the two capabilities nobody wants. There are always two, usually master data and the validated-system register.

5. Leave `ORG-11 Internal Audit` and `ORG-12 Legal` owning nothing. Assurance functions do not own operating capabilities, and the sheet is right to exclude them.
6. Check the result for rotation. If it still reads as a sequence, you have re-sorted rather than reassigned.

Signed by. The chief operating officer, with each owning unit head accepting their own rows. `GB-03 Data Governance Council` designates data owners and has no mandate here [11].

Done when. Every row names a unit that performs the capability, no unit owns a capability outside its function, and each row has a person behind it through `O09.Head`.

Shows up in. `RPT03 Capability Heatmap`, indirectly, and every later conversation where someone must decide.

Goes wrong when. You reassign in the repository and not in the business. A sheet saying Quality Assurance owns `C07` while the quality manager has not been told is worse than the positional join, because it looks settled.

Look it up. Volume 1, chapter 5, *Read this before you book the workshop*.

Step 4.7 — Reconcile the value streams with their own stage counts

Produce. Five value streams whose `Stages` count matches the process rows beneath them, and a `Capabilities` column that agrees with `R02`.

You need. `O14` rows 5–9; `R02 Value stream uses Capabilit…` rows 5–19; `O16` rows 5–24.

> **`O14 Value Stream – Ref`, `Value stream`, `Scope`, `Stages`, `Capabilities`**
>
> Five rows, and **they are genuinely pharmaceutical**: Target to Candidate, Candidate to Approval, Plan to Release, Signal to Action, Launch to Patient, each with an accurate `Scope` sentence [4]. Keep all five, their names and their scopes.
>
> Three defects underneath them:
>
> - `Stages` claims 4, 5, 6, 4, 5 — twenty-four. `O16` holds four rows per stream, twenty in all [4][5].
> - `O14.Capabilities` and `R02` agree on `VS01` to `VS03` and disagree on the other two. `O14` says `VS04 Signal to Action` uses `C01, C02, C03`; `R02` says `C04, C05, C06`. `R02` is closer to the truth, and it is a sliding window: each stream takes the next three capability numbers [4].
> - In `R02`, `C08 Commercial & Market Access`, `C09 Medical Affairs` and `C10 Supply Chain & Distribution` are used by **no value stream at all** — and `VS05` is called Launch to Patient.

How.

1. Fix the stage counts from the real process model, not the other way round. Count the stages you agree in step 4.8 and write that number into `Stages`.
2. Delete `O14.Capabilities` or make it a derived view of `R02`. One of the two must be the source.
3. Repair `R02` by reading each `Scope` sentence. Launch to Patient runs authorisation through reimbursement, supply and uptake, so it uses `C04`, `C08`, `C10` and `C09` — not `C05`, `C06`, `C07`.
4. Give `C08`, `C09` and `C10` a value stream. A capability used by no stream either belongs to one nobody drew or is not a capability.
5. Leave the five stream names alone. They are the one thing in this layer you would otherwise have had to invent.
6. Add nothing. Five is the right number here; a sixth usually turns out to be a stage of one of these.

Signed by. `GB-01 Architecture Board`, with the owning unit head for each stream from step 4.6.

Done when. `Stages` matches the process count for all five streams, one sheet holds the stream-to-capability mapping, and `C08`, `C09` and `C10` each appear in `R02`.

Shows up in. `RPT03`, and from Phase E `RPT08 Roadmap` — a work package scoped to a stream whose stage count is wrong is scoped wrongly by the same proportion.

Goes wrong when. You trust `O14.Capabilities` because it sits on the object sheet. The relationship sheet is closer to right here — the opposite of the usual rule — and you find that only by reading the `Scope` sentences.

Look it up. Volume 1, chapter 7, *The cadence, which is the spine of this chapter*.

Step 4.8 — Build the level-1 process model under one value stream

Produce. A level-1 process model for one value stream: five to twelve named steps, no swimlanes, replacing that stream's four `stage N` rows.

You need. Step 4.7; `O16` rows for that stream; the process owner.

> **`O16 Business Process — Ref`, `Business process`, `Value stream`, `Description`, `Automation`**
>
> Twenty rows, four per stream, each named `<value stream> — stage N` and described as `Stage N of <value stream>` [5]. The `Automation` column is filled — `Manual`, `Automated`, `Partly automated` — against processes that do not exist, so it records nothing.
>
> Replace the four rows for one stream, keep the `Ref` numbers in use, and add from `BPR-21` upward for stages beyond four. Name each row for the work: "Assemble and validate the submission sequence", not "stage 3".

How.

1. Start with `VS03 Plan to Release`, which `O14` says has six stages and `O16` gives four. The missing two are usually testing and certification, which are the two that matter.
2. Decide the level before you draw. Level 1 is one page, five to twelve steps, no lanes, and its purpose is to agree where the process starts and ends. Most arguments that look like level-2 arguments are unresolved level-1 boundaries.
3. Name the boundaries. Where does Plan to Release stop and Launch to Patient start? If two people answer differently, that is the finding.
4. Write each `Description` as what happens and what it produces: one sentence, active voice, artefact named.
5. Set `Automation` from observation. A step performed inside a validated system through a manual transcription is manual.
6. Repeat for one more stream only. Two streams modelled properly teaches the method; five modelled badly teaches nothing.

Signed by. The process owner for that value stream — the unit head from step 4.6 — and the quality assurance manager where any step touches a regulated record.

Done when. One value stream has five to twelve named steps, each with a one-sentence description and an observed `Automation` value, and its start and end boundaries are written down.

Shows up in. `O16`'s row count, which rises above twenty, and `R02`, which you can now test against the steps.

Goes wrong when. You model at level 2 because the lanes feel more real. You then argue all workshop about which system does a step whose existence nobody has agreed.

Look it up. Volume 1, chapter 8, *The four levels, and who reads each*.

Step 4.9 — Mark every step GxP or not, and name the regime

Produce. Two new columns on `O16` — `GxP` and `Regime` — filled for every process row you have written, and a one-page count of steps by regime.

You need. Step 4.8; the quality assurance manager; the validation lead; `O67 Validated System`.

> **`O16 Business Process` — two columns the sheet does not have**
>
> Add `GxP` with values `Yes`, `No` or `Partly`, and `Regime` with the regime that compels the record: GLP for preclinical laboratory work, GCP for anything involving a trial subject, GMP for manufacture and release, GVP for safety, GDP for distribution and wholesaling.
>
> **Why the regime and not just the flag.** The regime decides which inspection sees the step, which record set is demanded and which retention clock applies. A GMP step and a GCP step both read "yes" under a flag and share nothing else: different inspectors, different record sets, different retention, running to decades in the trial case.

How.

1. Mark each step against four questions. Does it create, alter or approve a record a regulator can demand? Does a decision about product, patient or batch rely on its output? Does it carry an electronic signature, or an audit trail someone reviews? Does it hand off into a system for which any of those is true?
2. Name the regime from the record, not the department. A sales representative's conversation that mentions an adverse event is a GVP step performed in a commercial function, and day zero runs from it.
3. Mark the `Partly` steps and say which part. "Approve the artwork" is regulated as to the approved text, not the layout review before it.
4. Mark unregulated steps explicitly as `No`. A blank cell reads as regulated to quality and unregulated to delivery.
5. Carry the marking into Phase C. Every system under a `Yes` step inherits a validation scope, and `O67` records that inheritance.
6. Count the result on one page: steps by regime, and the systems each regime reaches. That page makes your Phase D estimate credible.

Signed by. The quality assurance manager with the validation lead. This is a quality determination, not an architecture one: you propose, they decide. Record the date; the determination changes when the process does.

Done when. Every process row carries `Yes`, `No` or `Partly` and a regime, every `Partly` names its boundary, and the count by regime is on one page.

Shows up in. Phase C and Phase D, in every estimate; and `RPT09 Regulatory Coverage`, which can carry a process reference rather than only an obligation.

Goes wrong when. You mark everything `Yes` to be safe. That converts the estate into one validation scope, triples the evidence burden, and the marking is ignored within a year.

Look it up. Volume 1, chapter 13, *Which GxP, and for which record*; chapter 16, *The retention clocks*.

Form B-3 — Process register with the GxP marking

Feeds `O16 Business Process`. `GxP` and `Regime` are **two columns `O16` does not have** and that you are adding; `Description` is typed straight into Excel. Eight to twelve rows per value stream. Signed by the process owner, with `GxP` and `Regime` signed by the quality assurance manager.

Ref	Business process	Value stream	Automation	GxP	Regime
BPR-21	EXAMPLE — delete this row	VS03	Partly automated	Yes	GMP

Step 4.10 — Draw one process at level 2, with lanes that resolve to roles

Produce. One level-2 BPMN model, with every lane resolving to an `O11 Role` or an `O23 Application Component` row.

You need. Steps 4.3, 4.8 and 4.9; `O23` rows 5–16; `R04`; the integration designer.

> **`No sheet holds a process model, and none should`**
>
> Keep the model in your modelling tool and reference it from `O16` by name. Do not try to express swimlanes in the repository.
>
> One rule keeps the two from drifting: **every lane must correspond to an `O11 Role` or an `O23` application that exists.** If a lane has no counterpart, either the lane is wrong or the role model is incomplete — and here it will be the role model, which is why step 4.3 comes first. ArchiMate cannot draw a hand-off, having no swimlanes; use BPMN at level 2 only, and never model the same flow in both.

How.

1. Pick the process with the most hand-offs, not the most steps. Here that is batch release or adverse event intake; both cross four functions.
2. Draw the lanes first, from `O11` and `O23`. A lane called "the business" is one you have not understood.
3. Model the decision, not the approval. "QA approves" tells you nothing; what is decided, on what evidence, against what criterion, with what outcomes, tells you everything.
4. Mark the GxP boundary. Shade the regulated lanes, so a reader sees where the evidence burden starts.
5. Ask for hand-off failures explicitly: not "what goes wrong" but "when was this step last redone, and why". People remember occasions precisely and causes badly.
6. Bring the validation lead to the first workshop, not the last. They are usually the only person who can say whether the interface you have drawn is in anybody's validation scope, and it frequently is not.

Signed by. The process owner, with the quality assurance manager for the GxP banding. The documented process is a controlled document under the quality system, so a model that differs from it is a finding in its own right. Raise it; do not quietly correct it.

Done when. Every lane resolves to an `O11` or `O23` row, every gateway has a named criterion, and the regulated part of the flow is visibly distinct.

Shows up in. `RPT04 Interface Catalogue`, once your hand-offs are matched against `R06`'s thirty-six interfaces. Nineteen are unmanaged and thirty-one carry no logging, and the level-2 model is how you learn which sit inside a validated flow.

Goes wrong when. The model is drawn with whoever was available rather than the people who do the work. You get a confident, well-evidenced answer to a question you did not ask.

Look it up. Volume 1, chapter 8, *Running a process-modelling workshop in a GxP environment*, and *The notation exception*; chapter 17, *The BPMN exception, stated once*.

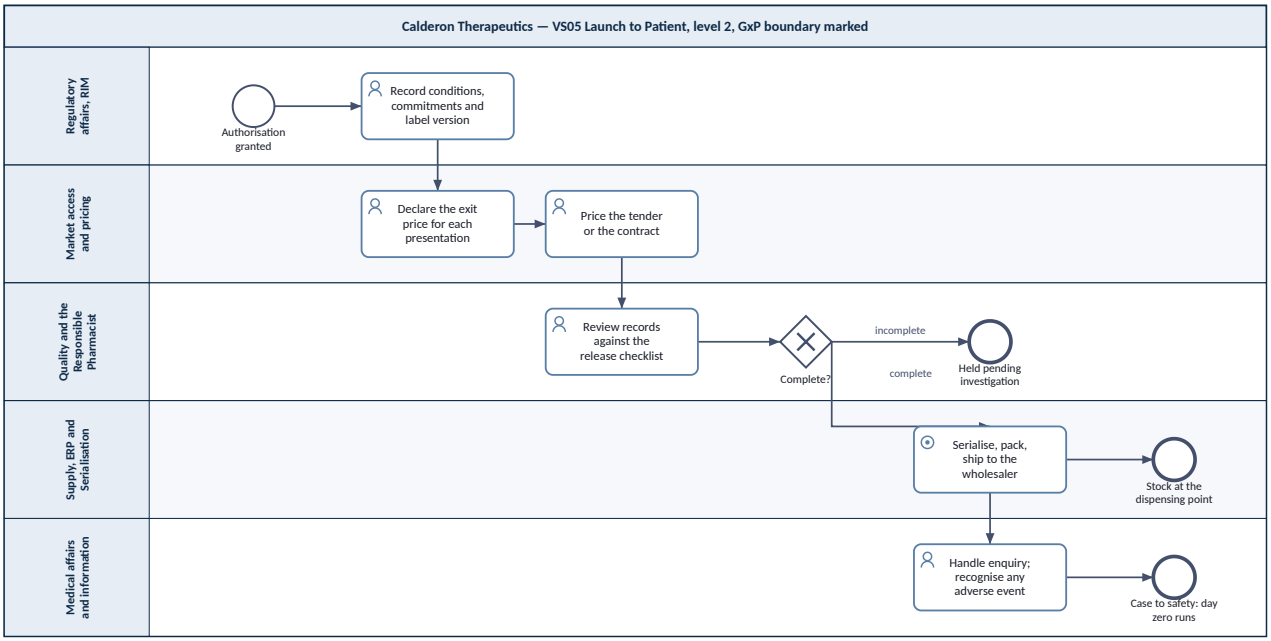


Figure 4.4 Step 4.11 drawn for `VS05 Launch to Patient`, which volume 1 does not model. Three lanes are regulated and two are not, and the last hand-off is the one that costs money: **day zero runs from the enquiry, not from the safety database**.

Step 4.11 — Repair the service, function, product and contract layer

Produce. An amended `O15 Business Service`, `O18 Product` and `O19 Contract`, and a decision on which of `O15`, `O17` and `O35` this company needs.

You need. `O15`, `O17` and `O35` rows 5–14; `O18` rows 5–8; `O19` rows 5–14; `R03`, `R04`, `R25`.

> **`O15 Business Service — Ref, Business service, Capability, Description, Exposure, Criticality`**
>
> Four sheets hold the same ten names. `O13`'s level-1 capabilities, `O15`'s services, `O17`'s functions and `O35`'s business building blocks are one list with four suffixes, joined one-to-one by `R03` [6]. That is not an architecture; it is one list copied.
>
> Two defects inside `O15`. Every `Description` reads "Externally consumable service realising ..." while `Exposure` reads `Internal` on seven of ten rows. And `Criticality` disagrees with the capability the service realises on **eight of ten rows** — `BSV-02 Preclinical Development Service` is `Medium` against `C02`'s `Critical` [6].
>
> `R04 Business Service supported` is positional too: `BSV-01` → `A01` through `BSV-10` → `A10`. So the Regulatory Affairs Service is supported by the Safety Database, Pharmacovigilance by `RIM`, Manufacturing & Supply by the `eTMF`, Quality Assurance & Control by `MES (GMP)`. `A11 Statistical Computing` and `A12 Medical Information` support no service at all [6].

How.

1. Keep only the services consumed by somebody outside the performing function. In a mid-size manufacturer that is three or four, not ten.
2. Fix `Exposure` against the description, then fix the description. Medical Information is the externally facing one, because healthcare professionals ring it; Preclinical Development is not externally consumable.
3. Derive `Criticality` from the capability through `R03`, or state that the two are scored independently and why. Eight disagreements of ten is a generator, not a judgement.
4. Repair `R04` by asking which application the service runs on. The answer is usually the one named after it, which is why the rotation is easy to miss.
5. Test `O18 Product`. Four rows — Prescription Medicine, Generic Portfolio, Biologic Therapy, Clinical Trial Supply — and not one is a registration, a presentation or a SKU [7]. A product here is four things with four cardinalities, and `O18` holds one of them.
6. Test `O19 Contract`. Ten rows, every one a vendor agreement named after an application, `Counterparty` holding an application's name rather than a legal entity [7]. No contract manufacturing agreement, no technical or quality agreement, no clinical trial agreement, no supply agreement — so the sheet cannot say who this company is bound to.

Signed by. `GB-01 Architecture Board` for the service layer; the general counsel for `O19`; the head of commercial for `O18`.

Done when. `O15` holds only services with an external consumer, no `Exposure` value contradicts its description, `R04` names the application that runs each service, and `O19.Counterparty` names legal entities.

Shows up in. `RPT13 Orphan Analysis`, once `A11` and `A12` are recorded as supporting nothing, and `RPT02 Application Portfolio`.

Goes wrong when. You keep all four sheets and reconcile them. Four one-to-one sheets over ten names cost four times the maintenance and carry the same information.

Look it up. Volume 1, chapter 9, *The object that breaks most models: "a product" is four things*; chapter 4, *Registered, priced, reimbursed: three different things*.

Step 4.12 — Take Phase B through G1 and record what you could not establish

Produce. A minuted `GT-1` decision, and a written list of what Phase B could not establish.

You need. Steps 4.1 to 4.11; `O58`; `O57`; `O31 Deliverable` rows 8–10.

> **`O31 Deliverable — Ref, Deliverable, Phase, Status`**
>
> Three Phase B rows: `DLV-04 Business Architecture`, `DLV-05 Capability Map`, `DLV-06 Value Stream Definition`, all marked `Approved` with no approver and no date. Set them to what your governance supports.
>
> `GT-1 G1 Design` tests that the design "meets principles, standards and reference architecture" [11]. Minute that `O05 Principle`'s twenty-four rows say nothing about validated state, data integrity, patient safety, retention or the regulator, so the test is weaker than it reads.

How.

1. Put the amended `O09`, `O10`, `O11`, `O13`, `O14`, `O15`, `O16` and `R22` to `GB-01` as one package. Piecemeal approval produces a business layer that does not join up.
2. Attach the step 4.9 count by regime. Phase D will be held to that number.
3. Write the "could not establish" list. From Calderon: how many stages `VS03` really has; who the Responsible Pharmacist's deputy is; whether quality is centralised or site-autonomous; which unit owns master data; whether Customer Service's 211 people include Medical Information.
4. Minute the absence of a quality body. `GB-01` has approved a business architecture with regulated steps in it, and no quality body exists to hold the regulated view.
5. Hand the GxP marking to Phase C as an input, not a reference. Classification of records is a GCP and POPIA obligation before it is an IT one, and the marking is where it starts.

Signed by. `GB-01 Architecture Board` for `GT-1`, with the quality assurance manager accepting the regime marking and the Responsible Pharmacist noting the release steps.

Done when. `GT-1` is minuted, the three `O31` Phase B rows carry a status your governance supports, and the list is attached to `DLV-04`.

Shows up in. `RPT06 Requirements Traceability`, where `REQ-022` to `REQ-027` — one per level-1 capability, each reading "exposes measures that the executive can act on" — can be tested against capabilities that mean something.

Goes wrong when. The gate passes on the capability map alone. That was the part already nearly right; the organisation, the roles and the processes are the work.

Look it up. Volume 1, chapter 12, *What is actually installed*.

What Phase B hands to Phase C

Artefact	Where it lives	Who owns it
Organisation units for the five regimes	`O09`, amended	Chief operating officer
Actors and roles that differ from each other	`O10`, `O11`, `R21`	Chief operating officer
Regulated roles with names and deputies	`O70 Regulated Role`	Quality assurance manager
Three branches decomposed, ten rows re-scored	`O13`, `R01`, `RPT03`	Function heads, `GB-01`
Capability ownership that is not positional	`R22`, amended	Chief operating officer
Level-1 process model for two value streams	`O16`, amended	Process owners
Every step marked GxP and by regime	`O16`, two new columns	Quality assurance manager
Count of steps by regime	One page, attached to `DLV-04`	Chief enterprise architect

The last two rows are what Phase C is built on. Classification, lineage, retention and every application's validation state follow from which steps are regulated — and no later phase can establish that, because by then the systems are being specified.

THE CHAPTER IN TEN LINES

- Nobody in the repository does a pharmaceutical job. That is the whole of Phase B.
- O13's ten level-1 capabilities and O14's five value streams are genuinely pharmaceutical. Keep both and replace what sits under them.
- O09 Organization Unit has twelve units and no Quality, Regulatory Affairs, Pharmacovigilance, Manufacturing or Clinical Development. Legal has 240 people and Technology 116.
- Six of twelve O09 heads appear in no row of O10 Actor.
- O10 and O11 hold the same eighteen titles, R21 maps them one-to-one, and Performed by reads Person on every row.
- The only pharmaceutical job titles in the repository are seven free-text values in the Owner column of O20 Data Entity, and they resolve to nothing.
- R22 Org Unit owns Capability is a positional join: ORG-01 to C01 through ORG-10 to C10. Human Resources owns Regulatory Affairs and Technology owns Pharmacovigilance.
- R04 Business Service supported by Application is positional too, which puts Pharmacovigilance on RIM and Quality Assurance on MES.
- O14 claims twenty-four stages. O16 holds twenty, four per stream. Four stages have no process row.
- O16's twenty rows are named for their stage number and described by their stage number. There is no process model; your work starts here.
- RPT03 Capability Heatmap gives every one of the ten capabilities a gap of exactly 1 and a disposition of Sustain, so it ranks nothing.
- O16 has no GxP column and no regime column. Add both, and name the regime rather than the flag, because the regime decides the inspection and the retention clock.
- Mark the unregulated steps explicitly. A blank cell is read as regulated by quality and unregulated by delivery.

SOURCES FOR THIS CHAPTER

- **1.** Pharmaceuticals_TOGAF_Metamodel.xlsx, sheet O09 Organization Unit, rows 5-16 (headers row 4); twelve units with Head and Headcount, including Customer Service at 211 and Legal at 240 against Technology at 116
- **2.** Pharmaceuticals_TOGAF_Metamodel.xlsx, sheets O10 Actor rows 5-22, O11 Role rows 5-22 and R21 Role performed by Actor rows 5-22; the same eighteen titles in both object sheets, one-to-one in R21, Performed by reading Person on all eighteen and Status reading Filled on all eighteen
- **3.** Pharmaceuticals_TOGAF_Metamodel.xlsx, sheets O13 Business Capability rows 5-104 and R01 Capability decomposition rows 5-94; ten level-1, thirty level-2 and sixty level-3 rows, ninety parent-child links, level-2 names formed from a pool of ten suffixes and level-3 names from Define and Operate
- **4.** Pharmaceuticals_TOGAF_Metamodel.xlsx, sheets O14 Value Stream rows 5-9 and R02 Value stream uses Capabilit... rows 5-19; O14's Stages column totals twenty-four, and O14.Capabilities disagrees with R02 on VS04 and VS05
- **5.** Pharmaceuticals_TOGAF_Metamodel.xlsx, sheet O16 Business Process, rows 5-24; twenty rows, four per value stream, each named for its stage number and described as Stage N of the value stream, with no GxP column and no regime column
- **6.** Pharmaceuticals_TOGAF_Metamodel.xlsx, sheets O15 Business Service rows 5-14, O17 Business Function rows 5-14, O35 Business Building Block rows 5-14, R03 Capability realised by Busi... rows 5-14 and R04 Business Service supported rows 5-14; four sheets over the same ten names, joined one-to-one, with R04 mapping BSV-01 to A01 through BSV-10 to A10
- **7.** Pharmaceuticals_TOGAF_Metamodel.xlsx, sheets O18 Product rows 5-8 (four rows, not six), O19 Contract rows 5-14 (ten vendor agreements whose Counterparty column holds application names) and R25 Application supports Produc... rows 5-8
- **8.** Pharmaceuticals_TOGAF_Metamodel.xlsx, sheet R22 Org Unit owns Capability, rows 5-14; ORG-01 to C01 through ORG-10 to C10 in sequence, with ORG-11 Internal Audit and ORG-12 Legal owning nothing
- **9.** Pharmaceuticals_TOGAF_Metamodel.xlsx, sheet O20 Data Entity, rows 5-18, Owner column; eight distinct owners of which only Data Steward appears in O10 Actor
- **10.** Pharmaceuticals_TOGAF_Metamodel.xlsx, sheet RPT03 Capability Heatmap, rows 5-14; Gap computed as target minus current, equal to 1 on all ten rows, Disposition Sustain on all ten
- **11.** Pharmaceuticals_TOGAF_Metamodel.xlsx, sheets O57 Governance Body rows 5-9 and O58 Governance Gate rows 5-8; five bodies, none of them a quality body, and four gates
- **12.** SAHPRA, SAHPGL-INSP-02 v10, Guidelines for Good Manufacturing Practice of Medicines, 13 April 2026; assigns batch-release duties and the duty that manufacturing and testing processes remain in a validated state to the Responsible Pharmacist. Cited as volume 1 cited it; re-check the version before quoting it to a client https://www.spotlightnsp.co.za/wp-content/uploads/2026/05/final-gmp-guideline-for-publication_20260424.pdf
- **13.** Pharmacy Act 53 of 1974, establishing the South African Pharmacy Council, with which the Responsible Pharmacist is separately registered; identified by title, as volume 1 identified it https://pharmcouncil.co.za/Media/Default/Pharmacy%20Act/Pharmacy%20Act%2053%20of%201974.pdf
- **14.** SAHPRA, SAHPGL-CEM-PV-04 v10, Guideline on Post-Marketing Reporting of Adverse Drug Reactions to Human Medicines in South Africa, 22 September 2025; the responsible person for pharmacovigilance and the back-up should reside in South Africa, and expedited clocks run from first awareness by anyone in the company https://www.sahpra.org.za/wp-content/uploads/2025/09/SAHPGL-CEM-PV-04_v10-Guideline-on-Post-Marketing-Reporting-of-ADRs-to-Human-Medicines-in-South-Africa.pdf
- **15.** The Open Group, TOGAF Standard, Architecture Development Method phase descriptions, public library landing page.org was robots-disallowed for the research behind volume 1, so the phase content in this volume comes from volume 1 and from the repository. Identified, not retrieved: The Open Group's public pages are robots-disallowed to automated retrieval, so no edition or wording is claimed from it.

CHAPTER 5

Data Architecture

Fourteen entities, nothing blank, and the column that says which system is the master is wrong on eight of them.

Before you open a single data sheet

Phase C in the ADM delivers the target data and application architectures and the gap between them and the baseline [21]; this item covers the data half. Phase C data is the phase this repository looks best in and is worst at. Fourteen entities, sixty attributes, twelve logical components, twelve physical components, ten mastering rules, eleven relationships, a metadata sheet and a published ownership report. Nothing is blank. **And the single most important column in the domain is wrong on eight of fourteen rows** [1].

A repository can be internally consistent, fully populated and completely untrue. `ZZ Checks` will not catch it. Referential integrity will not catch it — every reference resolves. No data-quality rule will catch it, because no rule knows that a safety database does not hold molecules. What catches it is a named person who owns the entity reading the row out loud and saying no. Your job in Phase C is to arrange that conversation fourteen times and write down what was said.

Calderon Therapeutics, the company in the repository, is an invention; so is Veldtkruin Group, volume 1's worked example [22]. Keep their data apart.

> **The GxP overlay for Phase C, data.** Three things make this phase different from the same phase in any other industry.
>
> **Classification is a good clinical practice and a POPIA obligation before it is an IT one.** The `Classification` column on `O20 Data Entity` is not a label for an access-control tool to consume. For subject data it decides lawful basis, cross-border position, breach impact and whether the processing needed prior authorisation at all [11][12].
>
> **Retention runs from an event, not from a calendar date.** Batch documentation runs from batch expiry or certification; trial essential documents from final closure or formal discontinuation of development; traceability records from trade-item expiry [13][15][16]. A retention period here is a computation over operational data held in other systems, which is why it gets its own sheet in step 5.6 rather than a cell in a policy document.
>
> **The dictionary version is part of the datum.** A coded adverse event without the dictionary and the version it was coded under is not a record, it is a string. `O37 Data Attribute` has sixty rows and not one of them is a dictionary version [2].

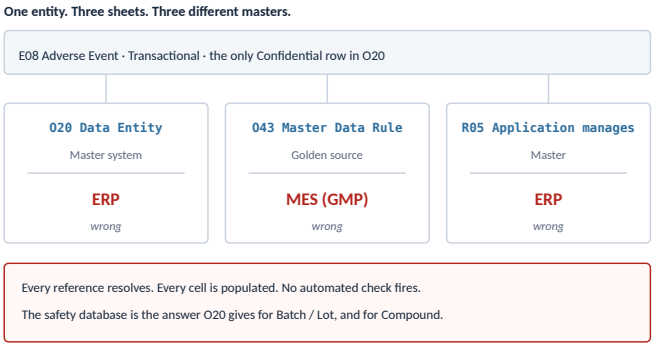


Figure 5.1 One entity, three sheets, three masters. `O20`, `O43` and `R05` each answer the question for `E08 Adverse Event`, and no two agree. Every reference resolves, so no automated check fires.

BEFORE YOU TOUCH `O20`, ANSWER THESE FOUR

Which system is the master for a compound, and who told you so?

Who signs off the classification of subject data, and under which instrument?

What is the record class of a batch record, and what event starts its clock?

Which of the twelve applications holds GxP records, and where is that written down?

Then count the domain yourself, because `00 Index` cannot be trusted to. Headers are on row 4 and the first record on row 5 on every sheet. `O20` has fourteen rows. `O37` has sixty, which is **ten entities at six attributes each** — so `E11 Deviation / CAPA`, `E12 Material`, `E13 Serialised Unit` and `E14 Site / Facility` have no attributes at all [1][2]. `O21` has twelve rows and `O44` ten, and `O36` carries eleven relationships forming a single chain `E01` → `E02` →…→ `E12` [4][5]. The last two entities have no logical component, no building block and no relationship. Write the line down — *fourteen entities, ten with attributes, twelve with a logical component, ten with a building block* — and take it into the first workshop.

Step 5.1 — Create the entity for the validated system

Produce. A fifteenth row in `O20 Data Entity` for the GxP computerised system, and the entry in `O67 Validated System` it points at.

You need. The counting above. `EXT Industry Extension`. The validation lead or, failing that, the head of quality assurance. Volume 1 chapter 9.

> **`O20 Data Entity` — add one row**
>
> `Ref`: `E15`. `Data entity`: `GxP Computerised System`. `Class`: `Master`. `Definition`: a computerised system whose fitness for a defined regulated purpose has been documented, tested, approved and maintained under change control. `Classification`: `Internal`. `Owner`: a named person, not a unit. `Master system`: the validated-system register itself, which is `O67`. `Lineage`: `Absent` until you build it, and say so.

How.

1. Read `EXT Industry Extension`. It declares six object types this sector needs beyond the TOGAF core — `X01 Compound`, `X02 Study / Trial`, `X03 Batch / Lot`, `X04 Regulatory Submission`, `X05 GxP Computerised System`, `X06 Adverse Event` — and holds **no instances of any of them** [8].
2. Five of the six have an echo in `O20` as entities — `E01`, `E02`, `E09`, `E06`, `E08`. `X05` has none. **There is no entity anywhere in the repository for the validated system** [1].
3. Add `E15` with the columns above, and cross-reference it to `O67 Validated System`, the sheet item 2 told you to create. `E15` is the entity; `O67` is the register of its instances. Do not duplicate the columns.
4. Add the entity to `O21` as `LDC-13 Validated system register` and to `O44` as `DBB-11 Validated system register`, with a testable acceptance criterion: *every application in `O23` appears exactly once in `O67` with a validation state and a next review date.*
5. Tell the quality function. They almost certainly hold a list already, in a spreadsheet, and the two lists will disagree. That disagreement is a finding and it is yours.

Signed by. The head of quality assurance owns the content. The repository has no quality body at all — `O57 Governance Body` has five bodies and none of them is a change control board, a validation committee or a data integrity forum [8] — so until item 2's regulated-role work lands, the `Architecture Board` signs that the entity exists and the quality function signs what is in it.

Done when. `E15` exists and the count of applications in `O67` equals the count in `O23`.

Shows up in. `RPT05 Data Ownership` gains a fifteenth row, and `RPT13 Orphan Analysis` should not list it, because `DBB-11` gives it a building block.

Goes wrong when. You create a separate quality-owned register instead of extending the portfolio. Two inventories and no answer is the near-universal state, and the disagreement gets discovered by an inspector rather than by you.

Look it up. Volume 1 chapter 9, *The object most architects leave out: the validated system itself*, has the grain and the attribute set; volume 1 chapter 13, *Which GxP, and for which record*, has why the record and not the system is the unit of scope.

Step 5.2 — Reclassify every entity, with the instrument named

Produce. A corrected `Classification` column on all fifteen rows of `O20`, each with the instrument that compels it recorded alongside.

You need. Step 5.1. The information officer. Clinical development. Volume 1 chapters 9 and 16.

> **`O20 Data Entity`, column `Classification`, plus the column you add beside it**
>
> `O20` has no column for the basis of a classification. Add one — `Instrument` — and populate it. An assigned classification with no reasoning fails an audit, and here the reasoning is usually a named section of a named Act or guideline.

How.

1. Read the column as it stands. `E03 Subject` is **`Internal`**. `E07 Marketing Authorisation`, `E12 Material` and `E14 Site / Facility` are **`Public`**. `E08 Adverse Event` is the only `Confidential` row in the sheet [1].
2. Deal with `E03` first. A pseudonymised trial participant record is health data and therefore special personal information; processing it is prohibited unless a ground in the Act applies, and the research ground carries its own tests and its own sufficient-guarantees condition [11][12]. It is also the record good clinical practice source-data rules exist to protect [13]. `Internal` is not a defensible value. Set it to the highest class the organisation operates.
3. Deal with the three `Public` rows. A marketing authorisation's existence is public; its conditions, approved sites and label history are not. A material master carries the approved-supplier qualification, and the registered site list is a regulatory object.
4. Open `O37 Data Attribute`. The compound identifier `ATR-001` is `Internal`; **the compound name `ATR-002` is `Public` and marked `Key: Yes`** [2]. The name of a compound in development is usually the most commercially sensitive string in the company.
5. Record the instrument for every row. Where you cannot name one, write `No instrument identified` rather than guessing — that is a finding for the information officer, not a gap for you to fill.

Signed by. The information officer, for every row touching personal information; the data owner for the rest. The repository names neither. `O20.Owner` holds eight job titles — Head of Supply Chain, Head of Commercial, Head of Clinical Development, Data Steward, Head of Preclinical Development, Head of Discovery Research, Head of Pharmacovigilance, Head of Regulatory Affairs — and **only `Data Steward` exists in `O10 Actor` or `O11 Role`** [1][8]. The sheet names the industry's jobs in exactly one place, and in that place they resolve to nobody.

Done when. Every row has a classification and an instrument, and the information officer has signed the subject row specifically, by name.

Shows up in. `RPT05 Data Ownership`, whose `Classification` column is a straight copy of `O20` [7]. Until you fix `O20`, the published report tells the organisation that trial participants' records are internal and registered sites are public.

Goes wrong when. You treat classification as an input to a tool rather than to a lawful basis.

Look it up. Volume 1 chapter 9, *Subject — the most sensitive record the company holds*, for why pseudonymisation is not a compliance exit; volume 1 chapter 16, *POPIA and health data*, for the instruments.

Form 5-A — Entity classification and ownership

Feeds `O20 Data Entity`, columns `Ref`, `Data entity`, `Class`, `Classification`, `Owner`. The `Definition` is typed straight into Excel; `Master system` and `Lineage` are Form 5-B. A first pass is fifteen rows — the fourteen you inherited plus `E15`. Signed by the information officer for every row carrying personal information, and by the named data owner for the rest.

Ref	Data entity	Class	Classification	Owner (named person)	Instrument
E03	EXAMPLE — delete this row	Master	Special / highest class	A. Mokoena, Head of Clinical Ops	POPIA s 26, s 27; SA GCP §5.10

Step 5.3 — Correct the master system column, entity by entity

Produce. A corrected `Master system` column on `O20`, with the person who confirmed each value named against it.

You need. Step 5.2. The data owners from `O09`-successor units, the application owners from `O23`, and ninety minutes with the four heads whose systems master most of the estate.

> **`O20 Data Entity`, column `Master system`**
>
> One system per entity. Not "ERP and MES". If two systems both hold it, one masters and the other consumes, and `R05 Application manages Data Entity` is where you record which is which with its `Role` column set to `Master` or `Consumer`.

How.

1. Read the column against the entity and stop at the first row. `E01 Compound / Molecule` is mastered by **Safety Database** [1]. The safety database holds events, not molecules.
2. Read the rest. `E02 Study / Trial` → ERP. `E03 Subject` → Statistical Computing. `E05 Clinical Data Point` → MES (GMP). `E06 Regulatory Submission` → Serialisation. `E08 Adverse Event` → ERP. `E09 Batch / Lot` → Safety Database. `E14 Site / Facility` → CTMS [1]. **Eight of fourteen are wrong, and they are wrong by rotation** — the column has been shifted against the rows, not guessed. An analysis environment is downstream of the master; a manufacturing execution system does not hold subject data; a serialisation platform does not hold a dossier.
3. Check whether it propagates. It does: `R05` repeats the same fourteen assignments exactly [6]. One wrong column, two sheets.
4. Note while you are there that `R05`'s fourteen `Master` rows carry **application names where every other relationship sheet carries a reference**, and that its fourteen `Consumer` rows cycle `A01`, `A04`, `A07`, `A10` through all fourteen entities [6]. `00 Index` claims every relationship row names both ends by reference and by name. Half of `R05` does not.
5. Fill the column in a workshop, out loud, one row at a time, writing the name of the person who gave each answer in the margin. Fourteen rows takes under an hour and it is the highest-value hour in Phase C. Where the answer is contested, record both candidates and raise a decision rather than a gap.

Signed by. The data owner per entity, countersigned by the application owner of the system named. Use `GB-03 Data Governance Council`, whose mandate is to designate data owners and resolve disputes [8], and note that `O60 Governance Decision` already carries `DEC-04 Designate data owners` as `Outstanding`.

Done when. Every row names one system and one person.

Shows up in. `RPT05 Data Ownership`, which republishes the column unchanged [7], and `R05`, which must be re-cut from `O20` rather than edited alongside it.

Goes wrong when. You correct `O20` and leave `R05`. Within a month someone runs the relationship sheet instead of the object sheet and the rotation is back.

Look it up. Volume 1 appendix 20, *System class to what it masters*, is the cross-map to argue from; volume 1 chapter 10, *Master data governance, and why it is a compliance matter*, is why this is a compliance conversation and not a tidiness one.

Form 5-B — Master system and lineage

Feeds `O20 Data Entity`, columns `Ref`, `Data entity`, `Master system` and `Lineage`. The `Confirmed by` column does not exist in `O20` — add it, because a master system nobody will put their name to is a guess. Fifteen rows for a first pass. Signed by the data owner per row, countersigned by the application owner of the system named.

Ref	Data entity	Master system	Lineage	Confirmed by (name)	Date
E01	EXAMPLE — delete this row	ELN / LIMS	Partial	T. Naidoo, Research Informatics	

Step 5.4 — Reconcile `O43` against `O20` and retire one of them

Produce. One authoritative statement of golden source per entity, and a decision recorded about which sheet holds it.

You need. Step 5.3 complete. `GB-03 Data Governance Council`.

> **`O43 Master Data Rule`, columns `Entity`, `Rule type`, `Golden source`**
>
> `O43` holds ten rows for `E01`–`E10` and names a `Golden source` for each. `O20` names a `Master system` for all fourteen. The two columns mean the same thing and must not both be authoritative.

How.

1. Put the two columns side by side for the ten entities both sheets cover. **They disagree on every one of the ten** [1][3]. `MDR-01` puts Compound in `ELN / LIMS` where `O20` says Safety Database. `MDR-03` puts Subject in `CTMS` where `O20` says Statistical Computing. `MDR-08` puts Adverse Event in `MES (GMP)` where `O20` says ERP.
2. Notice that `O43`'s answers are not better, just different — Adverse Event in a manufacturing execution system is as wrong as Adverse Event in an ERP. Two sheets, two answers, neither right.
3. Decide which sheet is authoritative. The defensible answer is `O20.Master system`, because every report and every relationship sheet already resolves against it, and `O43` becomes the sheet that holds the *rule* — match, survivorship, validation — and not the source.
4. Then fix `O43`'s `Rule type` column, which is pattern-assigned: four golden-source rules, four match rules, one survivorship rule and one validation rule distributed across the entities in a rotation that has nothing to do with the entities [3]. Subject does not need a survivorship rule; it needs a single enrolling system and a prohibition on duplicates.
5. Delete the `Golden source` column from `O43` or rename it `Source (from O20)` and make it a lookup. Two authorities is the defect; one authority plus a reference is the fix.
6. Record the decision in `O60 Governance Decision` and accept that `O60` **has no date column**, so nothing in it can be overdue [8]. Put the date in the decision text.

Signed by. `GB-03 Data Governance Council`, which is the only body in `O57` with a mandate that covers this.

Done when. One sheet names a source, the other names a rule, and a data owner has read both.

Shows up in. Nowhere — no report in the thirteen reconciles `O20` against `O43`. Write the check yourself and run it monthly.

Goes wrong when. Both sheets stay authoritative because deleting a column feels destructive. A repository with two answers to one question is worse than one with none, because it is quotable.

The eight wrong rows of O20.Master system, and the shift that made them

Entity		Master system recorded	Should be
E01	Compound / Molecule	Safety Database	A registry or ELN
E02	Study / Trial	ERP	CTMS
E03	Subject	Statistical Computing	EDC
E05	Clinical Data Point	MES (GMP)	EDC
E06	Regulatory Submission	Serialisation	RIM
E08	Adverse Event	ERP	Safety Database
E09	Batch / Lot	Safety Database	MES or ERP
E14	Site / Facility	CTMS	RIM or ERP

Not guesses. The column is offset against the rows, so the sheet reads as **a consistent, fully populated, completely untrue answer.**

Figure 5.2 The eight wrong rows of `O20.Master system`. The column is offset against the rows rather than guessed, which is why the sheet reads as internally consistent.

Step 5.5 — Redraw the entity relationships

Produce. A `O36 Data Entity Relationship` sheet that describes the industry's objects rather than the row order of `O20`.

You need. Step 5.3. Volume 1 chapter 9 open in front of you. Clinical data management and quality assurance in the room.

> **`O36 Data Entity Relationship`, columns `From entity`, `Relationship`, `To entity`, `Cardinality`**
>
> `R19 Data Entity relates to Data Entity` is the relationship sheet for the same eleven rows and carries identical content [4]. Edit `O36` and re-cut `R19`; do not maintain both.

How.

1. Read the eleven rows in order. `Compound / Molecule` *is composed of* `Study / Trial`; `Study / Trial` *is composed of* `Subject`; `Subject` *is derived from* `Protocol`; `Protocol` *references* `Clinical Data Point`; `Marketing Authorisation` *is composed of* `Adverse Event`; `Deviation / CAPA` *is composed of* `Material` [4]. The shape is a chain, row *n* to row *n+1*, down the `O20` list: the sheet order with verbs added.
2. Check the cardinalities. `REL-01` is a composition at **N:M**, which is not a composition. `REL-09 Batch / Lot has Specification N:M` inverts the real direction — a specification governs many batches and a batch is judged against one approved specification at its release date.
3. Rebuild from the conceptual model. A subject belongs to one study. A clinical data point belongs to one subject and one visit. An adverse event references a subject and may reference a batch. A batch is made to one specification version and belongs to one site. A marketing authorisation is granted against one submission, one authority, one market.
4. Add the two relationships the repository cannot express at all: the dated link from a serialised unit's pack back to the marketing authorisation that made it saleable, and the link from any record to the validated system that produced it, which is `E15`. `E13` and `E14` must each appear at least once.

Signed by. The data owners of both ends of each relationship. Where the two owners disagree on cardinality, that is an architectural finding and goes to `GB-03`.

Done when. No entity is absent, no composition is N:M, and you can read any row aloud to a clinical data manager without being corrected.

Shows up in. Nothing — no report in the thirteen publishes the data model, which is why the chain survived. Draw it as a view and put the view in the pack.

Goes wrong when. You accept the chain because it validates. Every reference in it resolves, and referential integrity is not meaning.

Look it up. Volume 1 chapter 9, *The object that breaks most models: "a product" is four things*, for the cardinality argument; volume 1 chapter 17, *The four identities of a product*, for how to draw it.

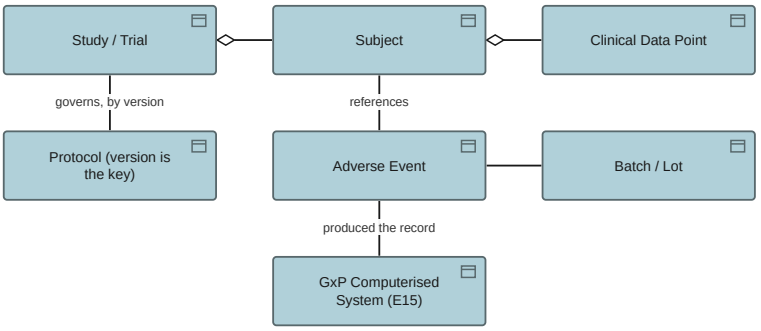


Figure 5.3 The clinical core of the data model, redrawn. `O36` 's eleven rows form a chain down the sheet order; this is the shape a clinical data manager would recognise, with `E15` added so every record names the validated system that produced it.

Step 5.6 — Define `O69 Retention Clock` and populate it from what is sourced

Produce. A new sheet, `O69 Retention Clock`, one row per record class, with every entity in `O20` mapped to exactly one of its rows.

You need. Steps 5.2 and 5.3. The records manager if there is one, the information officer, the head of quality assurance, and legal. Volume 1 chapters 10 and 16.

> **`O69 Retention Clock` — create it. Columns, in this order**
>
> `Ref` · `Record class` · `Period` · `Clock starts at` · `Instrument` · `System of record` · `Disposal approved by`.
>
> `Record class` is the unit, not the system and not the company. `Period` may be a computation — "1 year after batch expiry, or 5 years after certification, whichever is longer" is a period. `Clock starts at` names the **event**, and the system named in `System of record` must be able to supply that event's date. `Instrument` is the named section of the named instrument; where none states a period, the cell reads `No period stated` and the row still exists. **This sheet is defined here, once, and referenced by every other item in this volume.**

How.

1. Create the sheet with those seven columns, headers on row 4 to match the rest of the file.
2. Transcribe the record classes volume 1 verified. They are in the two tables below. Use what volume 1 printed and nothing more.
3. For each row, name the system that holds the trigger date. Batch certification and expiry sit in the manufacturing execution and quality systems; trial closure in the trial management system; trade-item expiry in the serialisation and ERP pair. If no system holds the event, the period is unimplementable and that is the finding.
4. Map every `O20` entity to exactly one record class. `E09 Batch / Lot` to batch documentation. `E03 Subject`, `E04 Protocol` and `E05 Clinical Data Point` to clinical trial essential documents. `E13 Serialised Unit` to traceability records.
5. Where the instrument states no period, write `No period stated` and assign a working period from the longest adjacent obligation **with the basis recorded**. Volume 1 is explicit: pharmacovigilance records have no stated period in the operative guideline [14], and the same is true of destruction records and import documents. "We chose seven years" with no reasoning is the answer that fails an audit.
6. Add the column `O42` has only partly: the date the retention classification was last reviewed. Guideline versions move faster than their own stated validity.

Signed by. Legal signs the instrument column, the head of quality assurance the GMP rows, the information officer the rows carrying personal information. Disposal of a GMP record is approved by a named quality role with a legal duty — the Responsible Pharmacist in a licensed South African operation — which **appears nowhere in `O10 Actor` or `O11 Role`** [8], so until item 2's `O70 Regulated Role` is populated, record the name in `Disposal approved by` and the vacancy as a risk.

Done when. Every entity in `O20` maps to exactly one `O69` row, every `O69` row has an instrument or an explicit `No period stated`, and every event in `Clock starts at` has a system that can supply it.

Shows up in. `O42 Metadata Element`, which carries `MTD-06 Retention rule` at coverage `Partial` [5]; once `O69` exists and is mapped, that cell can honestly read `Full`, and this is the only route by which it ever can.

Goes wrong when. Somebody sets a single corporate retention period. In an estate whose clocks run from three years to forty, one number is either unlawfully short for the longest clock or needlessly expensive for everything else — and either way it cannot be evaluated, because you cannot compute "one year after batch expiry" without the batch expiry date.

Look it up. Volume 1 chapter 10, *Retention as a data-model constraint*, for the modelling consequence; volume 1 chapter 16, *The retention clocks*, for the verified periods and their instruments; volume 1 appendix 20, *Retention clock by record class*, for the cross-map.

The verified record classes, first half — the periods volume 1 printed, with their instruments. Anything absent is absent because volume 1 could not source it.

Ref	Record class	Period	Clock starts at	Instrument
RC-01	Batch documentation	1 year after batch expiry, or 5 years after certification, whichever is longer	Batch expiry, or certification	Adopted manufacturing guide, Part I §4.11 [13]
RC-02	Clinical trial essential documents, investigator and sponsor	At least 10 years after final closure, or at least 2 years after formal discontinuation of development	Trial closure, or discontinuation	Good clinical practice §5.10.5, §6.4.6 [15]
RC-03	Research ethics committee trial records	At least 3 years after trial completion	Trial completion	Good clinical practice §4.3.6 [15]
RC-04	Traceability and unique-identifier records	Up to 5 years after the expiry of a trade item	Trade-item expiry	Traceability guideline §3.1.2 [16]
RC-05	Prescriptions, Schedules 2 to 6, at the last seller	At least 5 years	Date of sale	Medicines Act s 22A(6)(o) [17]
RC-06	Specified Schedule 5 and Schedule 6 register	**No period stated**; balanced quarterly and completed within 14 days	Quarter end	Medicines Act s 22A(6)(n) [17]

Second half, same columns.

Ref	Record class	Period	Clock starts at	Instrument
RC-07	Pharmacovigilance records	**No period stated** in the operative guideline — a finding, not a licence	—	Post-marketing reporting guideline [14]
RC-08	Safety surveillance obligation, which is not a retention period	Until 6 months after the expiry date of the last marketed batch	Last marketed batch expiry	Post-marketing reporting guideline [14]
RC-09	Marketing-code records and financial-transaction agreements	At least 5 years	—	Marketing code v19 §§3.7.1, 3.3.13.1.2 [9]
RC-10	Chemical exposure assessments and air monitoring	Minimum 30 years	—	Hazardous Chemical Agents Regulations 2021, reg 9(e) [18]
RC-11	Noise monitoring, audiometry and medical surveillance including baseline audiograms	40 years	—	Noise-induced hearing loss regulations, reg 11(e)–(f) [19]
RC-12	Employee name, occupation, time worked, remuneration	3 years from the last entry	Last entry	Basic Conditions of Employment Act 75 of 1997, s 31 [10]
RC-13	Company and accounting records	7 years	—	Companies Act 71 of 2008, s 24 [20]

> **One reading of that table to carry into every later phase.** The longest clocks sit on the least glamorous data — forty years for audiometry against ten for trial documents — so the obligation outlives the application four or five times over, and the requirement is an **export format** carrying the audit trail and the dictionary version, readable without the source application.

Form 5-C — Retention clock

Feeds `O69 Retention Clock`, a sheet the repository does **not** have and that you create in step 5.6. This form covers `Ref`, `Record class`, `Period`, `Clock starts at`, `Instrument` and `System of record`; type `Disposal approved by` straight into Excel once `O70 Regulated Role` is populated. A first pass is the thirteen classes volume 1 verified plus any the company's own legal register adds. Signed by legal for the instrument, the head of quality assurance for GMP rows and the information officer for rows carrying personal information.

Ref	Record class	Period	Clock starts at	Instrument	System of record
RC-01	EXAMPLE — delete this row	1 yr after expiry or 5 yrs after certification, longer	Batch expiry or certification	Mfg guide Part I §4.11	MES (GMP)



Figure 5.4 Retention as a computation. A single corporate period cannot be evaluated: the governing clock for batch documentation runs from certification or from batch expiry, whichever is longer, and both dates live in operational systems.

Step 5.7 — Put the attributes the industry needs into `O37`

Produce. An attribute set per entity that a clinical data manager and a quality assurance auditor would both recognise.

You need. Steps 5.2 and 5.6. Clinical data management, pharmacovigilance, quality control.

> **`O37 Data Attribute`, columns `Ref`, `Entity`, `Attribute`, `Type`, `Key`, `Classification`**
>
> Six columns, so the whole sheet transcribes from one form. Keep `Ref` contiguous — `ATR-061` onward — and do not renumber the sixty rows you inherited.

How.

1. Read the sixty rows. Ten entities, and the same six attributes against each: `Identifier`, `Name`, `Status`, `Effective date`, `Owner`, `Classification` [2]. It is a template, not an analysis.
2. Read the classifications across a single entity and watch them contradict the entity's own. `Batch / Lot` is `Restricted` in `O20`; four of its six attributes are `Public` in `O37`. `Subject` is `Internal` in `O20`; its `Name` attribute is `Confidential` and its `Owner` attribute `Restricted` [1][2]. No rule anywhere constrains the two sheets against each other.
3. Add the attributes that make a coded record a record. For `E08 Adverse Event`: the coding dictionary, **the dictionary version**, the coding date, the coder and the coding method. For `E05 Clinical Data Point`: the observation time as distinct from the entry

time, the originating instrument and the audit-trail reference. For `E09 Batch / Lot`: the specification version it was judged against, the certification date and the expiry date — because `O69` cannot compute without them.

4. Add attributes for `E13` and `E14`, which have none: for a serialised unit, its identifier scheme, aggregation parent and commissioning state; for a site, its licence reference, the activities licensed and the dated validity of each.
5. Set `Key` honestly. `ATR-002 Compound / Molecule · Name` is marked `Key: Yes` [2]; a name is not a key here, because the same molecule carries several across its life. And keep every added row's classification no more permissive than its entity's.

Signed by. The data owner of the entity. For the dictionary-version attributes, the pharmacovigilance lead — a role the repository does not carry.

Done when. Every entity in `O20` has at least one attribute, no attribute is more permissive than its entity, and every coded attribute has a version attribute beside it.

Shows up in. `O42 Metadata Element`, where `MTD-10 Primary key` sits at `Partial` [5] — the consequence of a `Key` column filled by pattern.

Goes wrong when. The dictionary version is left out as reference data. It is not reference data. A report produced in 2023 must be reproducible in 2029, and between the two a term can be demoted, merged or moved to a different class — so a coded value without its version cannot be interpreted, and a trend line across a release boundary steps for no clinical reason.

Look it up. Volume 1 chapter 9, *The dictionary-version problem*, is the whole argument; volume 1 chapter 9, *Controlled vocabularies as architectural dependencies*, is why the dictionary is a stored versioned asset and not a library call; volume 1 appendix 20, *Dictionary and vocabulary to its versioning and licensing*, is the cross-map.

The four metadata elements that are compliance elements, and three of them are absent. `O42 Metadata Element` has fourteen rows. In its `Governance` block, `MTD-05 Sensitivity` reads `Full`, `MTD-06 Retention rule` reads `Partial`, and **`MTD-07 Consent basis` and `MTD-08 Lineage` both read `Absent`** — the lawful basis for processing, and the path from source to report, in an estate holding special personal information under a ten-year clock [5]. `MTD-03 Steward` is `Absent` too, which sits oddly beside an `O20.Owner` column naming eight stewards who do not exist as actors. And two rows, `MTD-11 Physical location` and `MTD-13 Freshness`, carry the coverage value **`Recorded, not held`**, which is not a coverage: either the metadata is held and has a coverage, or it is not held and the coverage is `Absent`.

Fix `MTD-08` first, because `ZZ Checks` reports four entities with no lineage and `O20.Lineage` names them — `E04`, `E06`, `E12`, `E14`, all `Absent`, two of them regulatory objects [1][8]. Fix `MTD-07` second; it is filled by the information officer against each processing operation, and the architecture's job is to carry it as an attribute on the entity rather than as a document reference. Then decide the two `Recorded, not held` rows and re-state the coverage figures in the Phase C pack. A metadata sheet that honestly reports three `Absent` rows is more use than one that reports none. **Look it up.** Volume 1 chapter 10, *Data integrity as a design constraint, not a policy*, for what attribution and completeness oblige of the design.

Step 5.8 — Reject the generated sheets: migration, mapping and data sets

Produce. An emptied and rebuilt `O39`, `O40` and `O38`, or a written statement that all three are placeholder and are not to be used.

You need. Step 5.3. The application owners of both ends of each pairing. Commercial, clinical development and the information officer in one room — which is the hard part.

> **`O39 Data Migration`, `O40 Source To Target Mapping`, `O38 Data Set`**
> Six migration rows, twelve mapping rows, six data sets. All three are generated, and all three will be quoted at you in Phase E if you leave them.

How.

1. Read `O39`. `MIG-01` moves `Compound / Molecule` from CTMS to Safety Database; `MIG-03` moves `Subject` from Safety Database to eTMF [5]. **Neither pairing is possible.** Read the six rows together and the generator is visible: the from-system of each row is the to-system of the row above, walked down the application list.
2. Read `O40`. `MAP-01` maps a `CTMS field` to an `EDC field` for `Compound / Molecule`, rule `Direct` [5]. No field name, no type, no unit, no transformation — and a direct mapping between two systems that do not hold the entity is not a mapping.
3. Empty both, or mark every row `Placeholder — do not use`; `O29 Gap` already shows the file tolerates an empty sheet. Then fix `O40`'s shape before its content: a mapping row in a validated estate names the source field, its type, its unit, its precision, the target field, the transformation and the verification performed at the boundary.
4. Record that a migration is a **validation activity**. Value and **meaning** must both be verified, the audit trail must travel with the record, and the dictionary version with any coded value. A load that reconciles row counts has verified nothing. Hand the plan to Phase E and F.

5. Turn to `O38`. Four of the six are plausible. `DST-05 Risk measurement set` is described as "exposure and position data", which is not this industry's vocabulary, and **`DST-02 Customer analytics set`** is a "curated customer view for analysis" [5]. In a pharmaceutical company *customer* covers four different parties — prescriber, dispenser, funder, patient — so one curated view across them is four lawful bases in one table.
6. Rewrite each description to name its purpose, its lawful basis and the record classes from `O69` it contains. A data set that cannot name its lawful basis is not approved.

> **Two joins the architecture prohibits, and you say so in the first workshop.** Trial data and commercial data must not be joined — a single view spanning both is a consent and competition finding, not a feature. And pharmacovigilance assessment must not be reachable by commercial interest. Write both into `O06 Requirement`, not into a note: ArchiMate 4 has no Constraint element, so when item 16 draws this, `O07 Constraint`'s rows become `Requirement` elements.

Signed by. The application owner of the receiving system for each migration, the head of quality assurance for any GxP record class in `O69`, and the information officer for each data set's lawful basis. Where a set spans the firewall, nobody signs it.

Done when. No row asserts a pairing nobody has confirmed, every surviving mapping row names a unit and a verification, and every data set names a purpose and a lawful basis.

Shows up in. `RPT08 Roadmap`, through `WP-02 Data ownership and mastering`, which is `Approval: Not started` and `Delivery: Past end date` [8] — the work package that would have done this is already late and unapproved.

Goes wrong when. The rows survive into a business case. Six impossible migrations costed at a day rate is a number somebody will defend.

Look it up. Volume 1 chapter 10, *Data migration in a validated estate*, for the verification obligation; volume 1 chapter 10, *Integration at the information-model level*, for what a mapping row carries; volume 1 chapter 9, *Two boundaries the model must carry*, for both prohibitions.

Step 5.9 — Make the data building blocks and the physical components testable

Produce. Acceptance criteria on `O44` that test the system rather than the spreadsheet, and a corrected `Type` column on `O22`.

You need. Steps 5.1 and 5.3. The application owners. `O58 Governance Gate`.

> **`O44 Data Building Block ( What it must do , Acceptance criteria ) and O22 Physical Data Component ( Application , Type )`**
>
> `O55 Building Block` is the union sheet, carrying `DBB-01` – `DBB-10` alongside the business, application and technology blocks with a `Domain` column. `R09` and `R13` resolve against **`O55`**, not against `O44` — so a block edited only in `O44` will not move the traceability report.

How.

1. Read the ten acceptance criteria in `O44`. Every one reads *"No <entity> record without a named owner."* [5] That is a test of the repository, not of the estate, and `O20` already fails it on seven of fourteen rows because the owners named do not exist as actors.
2. Rewrite each as something a gate can test. For `DBB-03 Subject mastering`: *one enrolling system holds the subject identifier; no other system mints one; the re-identification key is not held by the sponsor; every downstream copy carries the originating system and extraction time.* That is four testable conditions, and `G2 Build complete` can evidence each.
3. Add `DBB-11` for the validated-system register from step 5.1, and blocks for `E11` – `E14`, which currently have none.
4. Re-cut `O55` from the four domain sheets after every change and check `R09` and `R13` still resolve. `R13` has thirty-seven rows against `O55`'s thirty-seven blocks [8]; adding blocks without adding delivery rows leaves orphans.
5. Turn to `O22`. `PDC-01 ELN / LIMS database` has `Type: File` [4] — a laboratory system's results store is not a file. The `Type` column cycles `File`, `Relational`, `Time series`, `Document` in a pattern unrelated to the applications. Correct it from the application owners, and add the one column a validated estate needs on a physical store: whether its contents are a GxP record, which drives backup, restore evidence and the retention row from `O69`.

Signed by. `GB-01 Architecture Board`, which holds `G1` and `G2`, signs the criteria. The application owner signs the physical type.

Done when. Every acceptance criterion names a condition a gate can evidence, `O55` reconciles to the four domain sheets, and no `O22` row is typed by pattern.

Shows up in. `RPT06 Requirements Traceability`, whose `Reaches a block` column resolves through `R09` to `O55`. It has thirty-three data rows for thirty-two requirements, **the last being a sentence** [7].

Goes wrong when. Criteria are written as statements of intent. `O59 Compliance Assessment` would hold the verdicts and it is **empty**, along with `R24`, so nothing has ever been tested against these criteria and nobody has noticed.

Step 5.10 — Republish `RPT05` and read its last row

Produce. A re-cut `RPT05 Data Ownership`, and a note on what its headline numbers should read once Phase C is done.

You need. Steps 5.2, 5.3 and 5.6 complete.

> **`RPT05 Data Ownership`, columns `Ref`, `Entity`, `Class`, `Classification`, `Owner`, `Master system`, `Lineage`**
>
> A derived report: every column is a copy from `O20` with `Definition` dropped. It computes nothing, so it cannot be wrong on its own — it can only republish.

How.

1. Re-cut the report from the corrected `O20`. Fifteen rows now, not fourteen.
2. Read its last row before you hand it to anyone. It is not a record: the string `4 of 14 entities have no lineage from report back to source.` sits in the **`Ref` column** [7], so anything loading this report with pandas gets a fifteenth entity called a sentence.
3. Check the other twelve reports for the same defect. **Nine of the thirteen end with a sentence in the `Ref` column** — `RPT01`, `RPT04`, `RPT05`, `RPT06`, `RPT08`, `RPT09`, `RPT10`, `RPT11` and `RPT13` [7][8], so `RPT13 Orphan Analysis` holds **two** records and a sentence, not three. Write the rule down for whoever automates against this file: skip any row whose `Ref` contains a space.
4. State what the numbers should read. Entities with no lineage: zero, or a named plan per entity with a date. Entities classified `Public`: zero, unless signed. Owners resolving to a person in `O10 Actor`: fifteen of fifteen.
5. Say what the report cannot show. It has no column for the record class, so it cannot show retention; none for the instrument, so it cannot show basis; and none for the validated system that produced the data. Add the three, or accept that `RPT05` is an ownership report and not a data governance report. Then put it in front of `GB-03` with the fourteen signatures from step 5.3 attached: the report is not the deliverable, the signatures are.

Signed by. `GB-03 Data Governance Council`, countersigned by the information officer on every row carrying personal information.

Done when. The report reconciles row for row to `O20`, the sentence row is documented as a known artefact, and every owner resolves to a named person.

Shows up in. `RPT01 Dashboard` and `ZZ Checks`, which both claim figures this phase moves. They are honest about nine things and silently wrong about three: object types populated reads 65 against at most 63, records reads 850 against 814, and *gaps overdue* reads zero, which is meaningless because `O29 Gap` is empty [8].

Goes wrong when. The report is circulated as evidence of data governance. It is evidence of a copy. The governance is in the fourteen conversations and the fifteen signatures.

Look it up. Volume 1 chapter 10, *The reporting layer, and the reconciliations that are always wrong*, for why a derived report that computes nothing still misleads.

What leaves Phase C data

Five artefacts, and a sixth that is the real one.

Artefact	Where it lives	Signed by
Entity classification with instruments	`O20`, new `Instrument` column	Information officer, data owners
Master system and golden source, entity by entity	`O20`, re-cut `R05`, `O43` demoted to rules	Data owners, `GB-03`
The validated-system entity	`O20` row `E15`, register in `O67`	Head of quality assurance
The retention clock	New sheet `O69`, mapped to every entity	Legal, quality, information officer
Testable data building blocks	`O44`, re-cut into `O55`	`GB-01 Architecture Board`

The sixth is the list of what you could not establish: which system genuinely masters a compound, what period applies to pharmacovigilance records, and who the Responsible Pharmacist is. Hand it over as a deliverable, with a name against each line. An honest open list is worth more than a complete sheet nobody believes.

THE CHAPTER IN TEN LINES

- Count the data domain yourself: fourteen entities, ten with attributes, twelve with a logical component, ten with a building block.
- There is no entity anywhere in the repository for the validated system. Create E15 and point it at O67.
- O20.Master system is wrong on eight of fourteen rows, and wrong by rotation, so the sheet reads as consistent.
- R05 repeats the same eight errors, so correcting O20 alone leaves the defect live in a relationship sheet.
- O43 names a golden source for ten entities and disagrees with O20 on every one. One sheet holds the source, the other the rule.
- Subject data classified Internal is the one classification error that changes a lawful basis. The information officer signs that row by name.
- Seven of the eight owners named in O20 do not exist in O10 Actor or O11 Role.
- O69 Retention Clock is defined here: record class, period, the event the clock starts at, the instrument, the system of record, who approves disposal.
- Retention runs from an event, so a period is a join to operational data and not a setting on a records platform.
- Where an instrument states no period, write No period stated and record the basis for whatever you assign.
- O42 reports consent basis and lineage as Absent, which are the two metadata elements this estate most needs.
- O39 and O40 pair systems that cannot hold the entity being moved. Empty them rather than inherit them.
- Nine of the thirteen reports end with a sentence in the Ref column, so skip any row whose Ref contains a space.

SOURCES FOR THIS CHAPTER

- **1.** Pharmaceuticals_TOGAF_Metamodel.xlsx, sheet O20 Data Entity, rows 5-18 (Classification, Owner, Master system and Lineage columns as populated)
- **2.** Pharmaceuticals_TOGAF_Metamodel.xlsx, sheet O37 Data Attribute, rows 5-64 (sixty rows across ten entities; ATR-001 and ATR-002)
- **3.** Pharmaceuticals_TOGAF_Metamodel.xlsx, sheet O43 Master Data Rule, rows 5-14 (Rule type and Golden source columns)
- **4.** Pharmaceuticals_TOGAF_Metamodel.xlsx, sheets O21 Logical Data Component rows 5-16, O22 Physical Data Component rows 5-16, O36 Data Entity Relationship rows 5-15 and R19 Data Entity relates to Data Entity rows 5-15
- **5.** Pharmaceuticals_TOGAF_Metamodel.xlsx, sheets O38 Data Set rows 5-10, O39 Data Migration rows 5-10, O40 Source To Target Mapping rows 5-16, O42 Metadata Element rows 5-18 and O44 Data Building Block rows 5-14
- **6.** Pharmaceuticals_TOGAF_Metamodel.xlsx, sheet R05 Application manages Data Entity, rows 5-32 (fourteen Master rows and fourteen Consumer rows)
- **7.** Pharmaceuticals_TOGAF_Metamodel.xlsx, report sheets RPT01 Dashboard, RPT04 Interface Catalogue, RPT05 Data Ownership, RPT06 Requirements Traceability, RPT08 Roadmap, RPT09 Regulatory Coverage, RPT10 Technology Currency, RPT11 Security Position and RPT13 Orphan Analysis - each ending in a sentence in the Ref column
- **8.** Pharmaceuticals_TOGAF_Metamodel.xlsx, sheets EXT Industry Extension rows 5-10, ZZ Checks, 00 Index, O10 Actor, O11 Role, O29 Gap, O30 Work Package, O55 Building Block, O57 Governance Body, O59 Compliance Assessment, O60 Governance Decision and R13 Work Package delivers Building Block
- **9.** Marketing Code Authority, The South African Code of Marketing Practice for Health Products, v19, SS 3.7.1 and 3.3.13.1.2 (five-year retention of marketing-code records and financial-transaction agreements), as printed in volume 1 chapter 16 https://www.marketingcode.co.za/
- **10.** Basic Conditions of Employment Act 75 of 1997, s 31 - employee name, occupation, time worked and remuneration, three years from the last entry, as printed in volume 1 chapter 16 https://www.gov.za/documents/basic-conditions-employment-act
- **11.** Protection of Personal Information Act 4 of 2013, s 26 - prohibition on processing of special personal information https://popia.co.za/section-26-prohibition-on-processing-of-special-personal-information/
- **12.** Protection of Personal Information Act 4 of 2013, s 27 - general authorisation concerning special personal information, including the research ground and its sufficient-guarantees condition https://popia.co.za/section-27-general-authorisation-concerning-special-personal-information/
- **13.** PIC/S, PE 009-17 (Part I), Guide to Good Manufacturing Practice for Medicinal Products, 25 August 2023 - S 4.11 batch documentation retention; the edition volume 1 read https://picscheme.org/en/publications
- **14.** SAHPRA, SAHPGL-CEM-PV-04 v10, Guideline on Post-Marketing Reporting of Adverse Drug Reactions to Human Medicines in South Africa, 22 September 2025 - no retention period stated; surveillance until six months after the expiry of the last marketed batch https://www.sahpra.org.za/wp-content/uploads/2025/09/SAHPGL-CEM-PV-04_v10-Guideline-on-Post-Marketing-Reporting-of-ADRs-to-Human-Medicines-in-South-Africa.pdf
- **15.** Department of Health, South African Good Clinical Practice: Clinical Trial Guidelines, 3rd edition, 2020 - SS 5.10.5 and 6.4.6 ten-year essential-document retention; S 4.3.6 three-year ethics committee retention; S 5.10.1.2 ALCOA-CCEA for source documents https://www.sahpra.org.za/wp-content/uploads/2021/06/SA-GCP-2020_Final.pdf
- **16.** SAHPRA, SAHPGL-INSP-RC-04 v1, Guideline on Traceability of Health Products, effective October 2025 - S 3.1.2, five-year retention after trade-item expiry https://www.sahpra.org.za/wp-content/uploads/2025/10/SAHPGL-INSP-RC-04_v1-Guideline-on-Traceability-of-Health-Products.pdf
- **17.** SAHPRA, SAHPGL-CEM-NS-02 v3, Guideline to the Scheduling of Substances and Medicines, effective 30 May 2022 - Medicines Act s 22A(6)(o) five-year prescription retention at the last seller and s 22A(6)(n) quarterly balancing of the Schedule 5 and 6 register within 14 days https://www.sahpra.org.za/wp-content/uploads/2022/05/SAHPGL-CEM-NS-02_v3-Guideline_Scheduling-of-Substances-and-Medicines.pdf
- **18.** Regulations for Hazardous Chemical Agents, GN R.280 in GG 44348, 29 March 2021, made under the Occupational Health and Safety Act 85 of 1993 - reg 9(e), exposure assessment and air-monitoring records, minimum thirty years https://www.gov.za/sites/default/files/gcis_document/202103/44348rg11263gon280.pdf
- **19.** Noise-Induced Hearing Loss Regulations under the Occupational Health and Safety Act 85 of 1993, reg 11(e)-(f) - noise monitoring and medical surveillance records including baseline audiograms, forty years
- **20.** Companies Act 71 of 2008, s 24 - company and accounting records, seven years https://www.gov.za/documents/companies-act
- **21.** The Open Group, TOGAF Standard, Architecture Development Method, Phase C: Information Systems Architectures. Identified, not retrieved: The Open Group's public pages are robots-disallowed to automated retrieval, so no edition or wording is claimed from it.
- **22.** Pharmaceuticals - a starter pack for enterprise architects (volume 1), chapters 9, 10, 13, 16 and 17 and appendix 20, Thotnr Consulting, October 2026

CHAPTER 6

Application Architecture

Twelve applications, most of them regulated, and not one column anywhere that says which.

What this half of Phase C is for

Phase C's application half delivers the target application architecture and the gap against the baseline [13]. In this repository the baseline is twelve applications, twelve services, ten building blocks, thirty-six interfaces and a published portfolio report, and **most of them are GxP systems while nothing anywhere records which**.

Read `O23 Application Component`'s column headers once and the problem is visible before you read a single row: `Ref`, `Application`, `Type`, `Description`, `Disposition`, `Criticality`, `Health`, `In service`, `Users`, `Owner` [1]. **No validation state. No last-periodic-review date. No GxP flag.** The repository cannot answer the first question an inspector asks — *which of your systems hold GxP records* [14]. Everything in this item is downstream of that absence.

Then read four rows together and the second problem appears. `A10 Serialisation` is `Critical`, `Adequate` and `Retire`. `A08 QMS` is `Critical`, `Healthy` and `Replace`. `A12 Medical Information` is `Critical`, `At risk` and `Consolidate`. `A06 eTMF` is `High`, `Poor` and `Invest` — the only one of the four that reads like a decision somebody took [1].

Calderon Therapeutics, the company in this repository, is an invention. So is Veldtkruin Group, volume 1's worked example [12]. Keep their data apart.

> **The GxP overlay for Phase C, application.** Every application in a pharmaceutical estate carries three facts that a generic portfolio does not hold: a **validation state**, a **last-periodic-review date** and a **named system owner who is a person**. The repository has no column for any of them.
>
> That is not a cosmetic gap. Validation is a state you maintain, not a thing you did, and periodic review is the mechanism that keeps it true [12] [14]. A disposition of `Replace` on a validated system is a validation project with a migration inside it, not a procurement. A disposition of `Retire` is a decommissioning plan with a records-readability obligation that outlives the system [12]. And an interface into a validated system sits inside that system's validation scope, which makes `R06 Application interfaces with` — nineteen rows unmanaged and thirty-one unlogged — the single largest unassessed compliance surface in the file [5].
>
> So the first substantive act of this phase is to create `O67 Validated System`, the sheet item 2 defines, and reconcile it row for row against `O23`. Every step below assumes you have.



Figure 6.1 Four rows of `O23`, and the columns the sheet has not got. Three critical applications carry dispositions that reduce them with no successor named; `A06 eTMF` is the only one of the four that reads like a decision somebody took.

Step 6.1 — Count the application domain and name the two orphans

Produce. A coverage count of the application domain, and the list of applications that fall out of each sheet.

You need. The repository. Thirty minutes. No stakeholders.

> **Read, do not write:** `O23`, `O24 Information System Service`, `O45 Application Building Block`, `R04 Business Service supported by Application`, `RPT13 Orphan Analysis`
>
> Headers on row 4, first record row 5. Match sheets on the `O`, `R` or `RPT` prefix and never on the full phrase — twenty-three tab names are cut at Excel's thirty-one characters, and `'R04 Business Service supported '` ends in a space.

How.

1. Count `O23` and `O24`. Twelve rows each, `A01` – `A12` and `ISS-01` – `ISS-12`, one service per application [1][2].
2. Count `O45`. **Ten rows, `ABB-01` – `ABB-10`, against `A01` – `A10`** [3]. `A11 Statistical Computing` and `A12 Medical Information` have no application building block, so nothing in the target architecture delivers them.
3. Count `R04`. **Ten rows**, pairing `BSV-01` – `BSV-10` with `A01` – `A10` [4]. The same two applications support no business service.
4. Open `RPT13 Orphan Analysis` and confirm it: `A11` and `A12`, both *"No business service maps to it"* [10]. Note that the sheet has three data rows and the third is the sentence *"An orphan is not always wrong. It is always worth a question."* in the `Ref` column — so it holds **two** records, not three.
5. Count `R25 Application supports Product`. Four rows, pairing `PRD-01` – `PRD-04` with `A01`, `A04`, `A07`, `A10` — every third application in sheet order [9]. Eight support no product.
6. Write the line down: *twelve applications, ten with a building block, ten supporting a business service, four supporting a product, two orphaned both ways.* One orphan, `A12`, is `Critical` and `At risk`.

Signed by. Nobody. Your own count, which you keep and quote.

Done when. You can name the two orphans and say which sheets they are absent from.

Shows up in. `RPT02 Application Portfolio`, which reports all twelve and so hides the fact that two of them connect to nothing, and `RPT13`, which reports it and is four rows long.

Goes wrong when. You accept `ZZ Checks`. It claims 850 records against 814 counted, and the 36-record difference is exactly the row count of `R06` — a relationship sheet counted as records [11]. A headline wrong by a whole sheet is not a safe basis for a portfolio count.

Look it up. Volume 1 chapter 12, *What is actually installed*, for what a real estate of this size looks like, and why thirty to sixty things that are plausibly systems is the expectation.

Step 6.2 — Add the three columns, in `O67`, and reconcile it to `O23`

Produce. `O67 Validated System` populated for every application in `O23`, with the GxP determination signed by quality assurance.

You need. Step 6.1. The head of quality assurance. The validation lead. Every application owner once you have found the real ones in step 6.3. Volume 1 chapters 11 and 13.

> **`O67 Validated System` — created in item 2. Columns**
>
> `Ref` · `Application` · `GxP` · `Validation state` · `Last periodic review` · `Next review due` · `System owner` · `Process owner` · `Validation lead` · `URS reference` · `Audit trail reviewed` · `Decommissioning plan`.
>
> The grain is **one system instance, in one environment, for one defined use** — not one row per application. A productive instance and a validation instance are different rows, and the same product validated for two purposes has two scopes. Make this the application portfolio with extra columns, not a separate quality-owned spreadsheet.

How.

1. Create one `O67` row per `O23` row to begin with, then split any application that has more than one validated use. Twelve rows becomes fifteen or twenty in a real estate.
2. Set `GxP` by asking **which GxP, and for which record** — manufacturing, clinical, laboratory, pharmacovigilance, distribution, several, or none. It is a family of regimes, and the same product carries different obligations in a GLP laboratory and a GMP one.
3. Determine scope on the **record**, not the system: a system is in scope because records in it are GxP records and decisions depend on them. On that test most of the twelve applications in `O23` are in scope — `A05 RIM`, `A09 ERP` and `A12 Medical Information` are the arguable three — and the determination is quality assurance's to sign, not yours to assert.
4. Fill `Validation state` from the validation lead's files, not the vendor's. Expect at least one system in service since 2010 or 2011 — `A01`, `A05` and `A07` all are [1] — whose validation package is on a departed engineer's shared drive.
5. Fill `Last periodic review` and compute `Next review due` from a risk-justified frequency. A blank `Last periodic review` on a fifteen-year-old GMP system is the finding, and worth more than anything else you produce this phase.
6. Reconcile. Any application in `O23` with no `O67` row, and any `O67` row with no `O23` ref, is a defect. The count must match exactly and be re-checked at every gate.

Signed by. The head of quality assurance signs the `GxP` and `Validation state` columns. The named system owner signs their own row. The repository has **no quality governance body at all** — `O57 Governance Body` holds five bodies and none of them is a change control board, a validation committee or a data integrity forum [11] — so record that absence as a finding and route the signature through the quality function directly until item 2's work lands.

Done when. `O67` and `O23` reconcile row for row, every row has a `GxP` value signed by quality assurance, and no GxP row has a blank `Next review due`.

Shows up in. `RPT02 Application Portfolio`, which must be re-cut to carry the three columns, and in `RPT09 Regulatory Coverage`, which is the report an inspector would ask for and the one that would embarrass you.

Goes wrong when. Quality already holds a validated-system list in a spreadsheet and you build a second one. Two inventories and no answer is the near-universal state, and the disagreement gets found by an inspector rather than by you.

Look it up. Volume 1 chapter 11, *The capability no other industry has: the register of validated systems*, for the full attribute set; volume 1 chapter 13, *Which GxP, and for which record*, for the two scoping errors that cost more than everything else.

Form 6-A — Validated system register, identify

Feeds `O67 Validated System`, **a sheet the repository does not have** and that item 2 tells you to create. This form covers `Ref`, `Application`, `GxP`, `Validation state`, `System owner` and `Process owner`. Type `Validation lead`, `URS reference`, `Audit trail reviewed` and `Decommissioning plan` straight into Excel; `Last periodic review` and `Next review due` are Form 6-B. One row per system instance per defined use — a first pass on this estate is fifteen to twenty rows, not twelve. Signed by the head of quality assurance for the `GxP` and `Validation state` columns, and by each named system owner for their own row.

Ref	Application	GxP	Validation state	System owner (person)	Process owner (person)
VAL-01	EXAMPLE — delete this row	GMP, GDP	Validated, 2019	N. Dlamini	R. Pillay, QA

Form 6-B — Periodic review position

Feeds `O67 Validated System`, columns `Ref`, `Application`, `Last periodic review` and `Next review due`, and seeds `O68 Periodic Review Schedule`, which item 11 defines. `Frequency` must be risk-justified and the justification recorded; a blank `Last review` is itself the finding and the row still gets filled in. Same row count as Form 6-A. Signed by the validation lead.

Ref	Application	Last periodic review	Frequency	Next review due	Open actions
VAL-01	EXAMPLE — delete this row	None recorded	2 years	Overdue	4

Step 6.3 — Replace the `Owner` column with people

Produce. An `Owner` column on `O23` holding named individuals, and the organisation unit moved to a column of its own.

You need. Step 6.1. The chief information officer, and an hour of somebody's political capital. Volume 1 chapter 12.

> **`O23 Application Component`, column `Owner`**
>
> Add `System owner` for the person and keep `Owner` for the unit, or rename. One name per application, with a deputy, because a single-named owner who leaves takes the row's meaning with them.

How.

1. Read the column as it stands. It holds **organisation units, not people**, and the units are wrong: `A04 Safety Database` → **Legal**. `A07 MES (GMP)` → **Internal Audit**. `A10 Serialisation` → **Human Resources**. `A01 ELN / LIMS` → **Executive**. `A03 EDC` → **Supply & Logistics** [1].
2. Notice the pattern rather than the individual errors. `A02 CTMS` is owned by `Customer Service`, and `O09 Organization Unit`'s twelve units include Customer Service at a headcount of 211 against Technology at 116, with **no Quality, no Regulatory Affairs, no Pharmacovigilance, no Manufacturing and no Clinical Development** [11]. The owner column is wrong because the organisation sheet it draws on describes a generic corporate.
3. Fill it with people. An application owner is a person who can be rung up and who can stop a change. Internal Audit cannot own a manufacturing execution system — an assurance function owning the system it audits is a segregation finding on its own.
4. For each GxP application name three people, not one: the **system owner**, accountable for the system; the **process owner**, accountable for the regulated process it supports; and the **validation lead**. They are different jobs, frequently held by one person, which is itself worth recording.
5. Copy the names into `O67` in the same sitting, so the two sheets never diverge. Where no name exists — and here at least two will not — write `Vacant` and raise it. `ZZ Checks` reports vacant architecture roles as zero and is correct about `O10 Actor`, which is not the same as the estate having owners [11].

Signed by. The chief information officer signs the list. Each named owner signs their own row, which is the point of the exercise and the part that takes the time.

Done when. Every application names a person, every GxP application names three, and no row names an assurance function as the owner of a system it assures.

Shows up in. `RPT02 Application Portfolio`, whose `Owner` column is a straight copy of `O23` [10]. Until you fix `O23`, the published portfolio tells the organisation that Human Resources owns pack-level serialisation.

Goes wrong when. You record the unit because it is faster and nobody will argue with it. A unit owner is how a validated system reaches a periodic review with nobody to sign the report.

Look it up. Volume 1 chapter 12, *What is actually installed*, for the second list — the systems that are not on the validated inventory — which is where the risk sits and which only a named owner will admit to.

Step 6.4 — Resolve the four self-contradictions, each with a dated decision

Produce. Four governance decisions, each with a date, a body and a named proposer, and four corrected rows.

You need. Steps 6.2 and 6.3. `GB-01 Architecture Board` and `GB-02 Investment Committee`.

> **`O23`, columns `Disposition`, `Criticality`, `Health`, and `O60 Governance Decision`**
>
> `O60` has columns `Ref`, `Decision`, `Body`, `Status` and **no date column** [11], so nothing in it can be overdue. Put the date in the decision text until the column exists.

How.

1. Take `A08 QMS` first, because it is the clearest. `Critical`, `Healthy`, `Replace`, 64 users, in service since 2020 [1]. Replacing a critical, healthy, five-year-old system needs a recorded reason — a change to a validated system is a change-control record before it is a backlog item [15] — and there is none anywhere in the file. Either the reason exists and belongs in `O60`, or the disposition is wrong.
2. Take `A12 Medical Information`. `Critical`, `At risk`, `Consolidate` [1]. A critical system at risk needs stabilising before it is merged into something, and the something is not named. It is also one of the two orphans from step 6.1 and has no building block in `O45`.
3. Take `A10 Serialisation` and hand it to step 6.5, because retiring a critical system with no successor is a different kind of problem.
4. Take `A06 eTMF`. `High`, `Poor`, `Invest` [1] — the only one of the four that is internally coherent. Say so out loud in the workshop: this row is the template for the other three.
5. For each, record the decision in `O60` with a date in the text, and check it against `O54 Architecture Alternative`, where `ALT-B Foundations first` at $46m is `Recommended` while `ALT-E Compliance led` at $31m is `Partly adopted` [11]. Two live answers upstream means your four dispositions have no settled strategy to hang on.
6. Re-cut `RPT02` afterwards and read the four rows again. If any still shows a critical system with a disposition nobody can explain, the step is not done.

Signed by. `GB-01 Architecture Board` for the disposition; `GB-02 Investment Committee` for anything with money attached. Note that `O60` already carries `DEC-01 Adopt the recommended architecture alternative` as `Outstanding` [11], so the body that should settle this has not settled the question above it.

Done when. Each of the four rows has a disposition, a recorded reason, a date and a named proposer, and no `Critical` row has a disposition that reduces the system without a named successor.

Shows up in. `RPT02 Application Portfolio` and `RPT08 Roadmap`. A healthy portfolio shows no `Critical` application with a `Retire` or `Consolidate` disposition and no named successor; this one shows three.

Goes wrong when. The four rows are treated as data-quality defects and corrected quietly. They are unmade decisions, and correcting the cell without taking the decision just moves the contradiction somewhere nobody is looking.

Step 6.5 — Decide what actually happens to the critical system you are retiring

Produce. A decommissioning plan for `A10 Serialisation`, or a reversal of its disposition.

You need. Step 6.4. The head of quality assurance. Supply chain. Regulatory affairs. Legal.

> **`O23` row `A10`, `O67` column `Decommissioning plan`, `O69 Retention Clock`**
>
> `O69`, defined in item 5, carries `RC-04 Traceability and unique-identifier records` at up to five years after the expiry of a trade item [17]. That clock, not the system's end of life, sets how long the records must stay readable.

How.

1. State the position plainly. `A10` is `Critical`, `Adequate` and `Retire`, with 170 users, in service since 2012 [1]. **No successor is named anywhere in the repository** — not in `O45`, not in `O54`, not in `O30 Work Package`.

2. Check what it carries. Serialisation and traceability are a statutory obligation, and `O63 Control Objective` records `CO-03 Falsified Medicines / DSCSA` as **`In breach`** [11]. You are retiring the system that implements an obligation already in breach.
3. Decide, in one meeting, between three answers: retire with a named successor and a dated cutover; retain and invest; or reverse the disposition and record why. There is no fourth answer and "retire" on its own is not one of the three.
4. If retirement stands, write the decommissioning plan before the cutover date, not after. It needs: the record classes the system holds, from `O69`; the export format carrying the audit trail; the verification that the export is readable without the source application; and the named person who approves disposal when the clock expires.
5. Budget it. Decommissioning a validated system is a validation activity and nobody budgets it [12]; the cost sits in no work package in `O30`, while `O52 Implementation Factor` carries `IF-03 Peak trading freeze` as a `Constraint` — a retail freeze that does not exist here, in place of the two that do: the annual product quality review cycle and the inspection window.
6. Record the outcome in `O67.Decommissioning plan` as a document reference, not a status.

Signed by. The head of quality assurance signs the decommissioning plan. The release signatory — a named person with a legal duty, which in a licensed South African operation is the Responsible Pharmacist and which the repository does not carry [11] — signs that the records remain available for the retention period.

Done when. `A10` has either a named successor with a dated cutover or a reversed disposition, and `O67.Decommissioning plan` points at a document somebody has approved.

Shows up in. `RPT09 Regulatory Coverage`, where `CO-03` should move off `In breach` before the system that implements it is switched off, and `RPT08 Roadmap`, which should carry the decommissioning work package it currently does not.

Goes wrong when. The system is switched off and the data is kept "in a database somewhere". A retention obligation that outlives the application by four or five times needs an export format with its reference data, cheap to build at the start and impossible to reconstruct later.

Look it up. Volume 1 chapter 13, *Decommissioning, which nobody budgets*, for the obligation; volume 1 chapter 16, *Serialisation and traceability*, for what the obligation actually is.

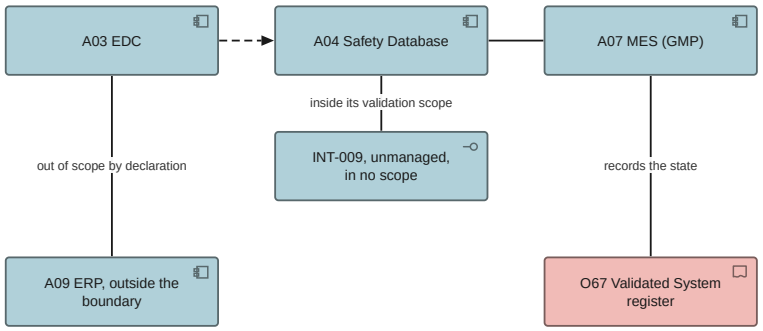


Figure 6.2 The validated boundary, drawn on the repository's own refs. The interface between two validated systems is inside the scope of the one it writes to, and is almost always nobody's row. Declaring `A09 ERP` out of scope is a valid architectural act, provided the declaration is honest and enforced.

Step 6.6 — Re-pair the business services to the applications

Produce. A `R04` that says which application supports which business service, confirmed by the service owner.

You need. Steps 6.1 and 6.3. The ten business service owners from `O15 Business Service`.

> **`R04 Business Service supported by Application`, columns `Business service`, `Service name`, `Application`, `Application name`**
>
> The tab name ends in a space. Match on the `R04` prefix.

How.

1. Read the ten rows in order and the pairing is visible: `BSV-01` → `A01`, `BSV-02` → `A02`, down to `BSV-10` → `A10` [4]. It is paired by index, not by fact.
2. Read what that produces. **`BSV-04 Regulatory Affairs Service` is supported by `A04 Safety Database`. `BSV-05 Pharmacovigilance Service` by `A05 RIM`. `BSV-06 Manufacturing & Supply Service` by `A06 eTMF`. `BSV-08 Commercial & Market Access Service` by `A08 QMS`** [4]. Regulatory affairs runs on the safety database and manufacturing runs on the trial master file.

3. Note the one row that lands: `BSV-07 Quality Assurance & Control Service` → `A07 MES (GMP)`. One in ten, by coincidence of position.
4. Re-pair it in a workshop with the service owners, allowing many-to-many. A business service here is almost never supported by exactly one application: pharmacovigilance needs the safety database, the case-intake channel, the dictionary and the submission gateway.
5. Place `A11` and `A12`, or record explicitly that they support no business service and belong in `O54` as candidates for removal. Then re-cut `R03 Capability realised by Business Service` the same day, because it carries the same index pairing and feeds the same reports.

Signed by. The business service owner per row. Where the service has no owner — and in this repository none of the ten does, because `O15` has no owner column — raise that as the finding before you fill the sheet.

Done when. Every row is confirmed by a named service owner and no row survives only because its reference numbers match.

Shows up in. `RPT13 Orphan Analysis`, which should shrink or grow for reasons you can explain, and `RPT03 Capability Heatmap`, which resolves through `R03`.

Goes wrong when. You fix `R04` and leave `R03`. Index pairing is a property of how this repository was generated, so wherever you find one instance, look for the others: `R05`, `R07`, `R18`, `R20` and `R25` all carry it.

Look it up. Volume 1 chapter 11, *One: what is the system of record for the compound, the study, the batch and the submission?*, for the four pairings that have to be right before the rest matter.

Step 6.7 — Catalogue the interfaces and put each one inside a validation scope

Produce. A corrected `R06`, and a named validation scope for every interface that touches a GxP system.

You need. Steps 6.2 and 6.3. The integration lead. The validation lead. Volume 1 chapter 12.

> **`R06 Application interfaces with`, columns `Ref`, `Source`, `Source name`, `Target`, `Target name`, `Type`, `Frequency`, `Managed`, `Logged`**
>
> Thirty-six rows, `INT-001` – `INT-036`. Add two columns: `Validation scope`, naming the system whose scope the interface sits in, and `Interface specification`, naming the document.

How.

1. Count it before you read it. **Nineteen of thirty-six are `Unmanaged`. Thirty-one carry no logging. Sixteen are both. Only two of thirty-six are managed and logged. Six are `Direct DB read`, and all six are unlogged** [5].
2. Read the six direct database reads by name: `INT-001` ELN / LIMS to RIM, `INT-002` ELN / LIMS to eTMF, `INT-012` Safety Database to RIM, `INT-023` QMS to CTMS, `INT-024` QMS to MES (GMP), `INT-027` ERP to Statistical Computing [5]. Three of them are marked `Managed`, which cannot be true of a read that bypasses the target's application layer and its audit trail.
3. Read `INT-009`, the one that matters most: EDC to Safety Database, type `File`, hourly, **`Unmanaged`** [5]. That is the path a serious adverse event takes from the trial to the reporting clock.
4. Check type against frequency. Five rows are type `Batch` at frequency `Real time` and one is `Event` at `Nightly` [5]. A batch interface is not real time; one of the two columns is wrong on six rows.
5. Assign a validation scope to every row touching a GxP system. This is the most common audit gap in the industry, and it is structural: each endpoint is validated by the function that owns it, and the interface belongs to an integration team outside the GxP organisation.
6. Specify the three things that close it. An **interface specification that is a validation deliverable**, naming every field, its type, its unit and its transformation. **Verification at the boundary**, recorded. And a **provenance marker on every record that crossed** — source system, extraction time, originating user — cheap at design time, impossible to retrofit.

Signed by. The validation lead signs the scope assignment. The system owners at both ends sign the interface specification. `GB-01 Architecture Board` holds `G1 Design`, which is where an unmanaged interface into a validated system should have been stopped.

Done when. Every row has a validation scope, every GxP row has an interface specification reference, and no row is both `Managed` and a direct database read.

Shows up in. `RPT04 Interface Catalogue`. A healthy estate shows zero unmanaged, zero unlogged and zero direct database reads; this one shows nineteen, thirty-one and six. Read the report's last row before you circulate it: the string `19 of 36 interfaces are unmanaged and 31 carry no logging.` sits in the **`Ref` column**, so anything loading it gets a thirty-seventh interface called a sentence [10].

Goes wrong when. Attribution is lost in transit. A named person creates the record, the interface posts it onward as the integration service account, and the receiving audit trail says the service account created it — which breaks the first letter of the data-integrity set

[16].

Look it up. Volume 1 chapter 12, *Interfaces are routinely in nobody's validation scope*, for what gets corrupted and the three-part pattern that works.



Figure 6.3 `INT-009`**, and the gap it sits in.** Both endpoints are validated by the function that owns them. The interface belongs to an integration team outside the GxP organisation, so the place the data actually gets corrupted is the place nobody assessed.

Step 6.8 — Publish the services, or stop calling them services

Produce. An `O24` that distinguishes a published, contracted service from an integration point with a name.

You need. Step 6.7. The integration lead.

> **`O24 Information System Service`, columns `Information system service`, `Application`, `Style`, `Published`**
> Twelve rows. `Style` takes `API`, `Batch`, `Event`, `File`; `Published` takes `Published` or `Unpublished`.

How.

1. Read the names. Every row is *"<application> Service"*, described as *"Service exposed by <application>"* [2] — twelve rows, one per application, which is how you can tell nobody derived them from what the applications expose.
2. Read `Published`. Five of twelve are `Unpublished`, including `ISS-01 ELN / LIMS Service` [2], while `R06` shows three interfaces leaving `A01` [5]. An unpublished service with three consumers is a point-to-point integration with a service name on it.
3. Read `Style` against `R06.Type` for the same application. `ISS-02 CTMS Service` is `Batch`, while the interfaces out of `A02` are `API`, `Event` and `Event` [2][5]. Two sheets, same traffic, different answers.
4. Decide what a service means here, in one sentence: a published interface with a contract, a version, a named owner and a consumer list. Anything else is an integration point, and calling it a service hides the fact that nobody owns its contract. Then re-derive `O24` from `R06` rather than from `O23` — a service exists because something consumes it.
5. For every service into or out of a GxP system, record the contract version alongside the validation scope from step 6.7. A contract change there is a change-control record before it is a backlog item.

Signed by. The integration lead signs the service list; the system owner signs each contract.

Done when. Every `Published` row has a contract reference and a consumer list, and no `Unpublished` row has a consumer in `R06`.

Shows up in. `RPT04 Interface Catalogue`, where a published service should account for the managed rows and an integration point for the unmanaged ones.

Goes wrong when. The sheet is left as twelve rows of furniture and then quoted in a target architecture as evidence that the estate is service-oriented. `O60`'s `DEC-07 Grant dispensation for point-to-point transfer`, already `Approved` [11], is the real state of affairs.

Step 6.9 — Make the application building blocks testable

Produce. Acceptance criteria on `O45` that a gate can evidence, and building blocks for the two applications that have none.

You need. Steps 6.1 and 6.7. `O58 Governance Gate`.

> **`O45 Application Building Block`, columns `What it must do`, `Acceptance criteria`**
> `O55 Building Block` is the union sheet, carrying `ABB-01` – `ABB-10` with the business, data and technology blocks and a `Domain` column. **`R09` and `R13` resolve against `O55`**, not against `O45`, so a block edited only in `O45` will not move the traceability report.

How.

1. Read the ten `What it must do` cells. Every one reads *"<application> at target disposition"* [3], which restates the disposition column and obliges nothing.
2. Read the ten acceptance criteria. Every one reads *"No unmanaged interface to or from <application>."* [3] That is a genuinely good criterion — and it is currently **false for every application in the estate**, because nineteen of thirty-six interfaces are unmanaged [5]. Keep the criterion and say that out loud.
3. Add the criteria a GxP application needs and none of the ten carries: the system appears in `O67` with a current validation state; its periodic review is not overdue; its audit trail has a named reviewer and a recorded review; every interface into it has a validation scope and a specification.
4. Add `ABB-11` and `ABB-12` for `A11` and `A12`, the two orphans. A critical, at-risk application with no building block cannot be delivered by any work package.
5. Re-cut `O55` from the four domain sheets after every change and check `R09` and `R13` still resolve. `R13` carries thirty-seven rows against `O55`'s thirty-seven blocks [3]; adding blocks without adding delivery rows leaves orphans in the traceability report.
6. Set each criterion against a gate. `G1 Design` tests design against principles and standards; `G2 Build complete` tests that every acceptance criterion is evidenced [11]. A criterion no gate can test is a sentence.

Signed by. `GB-01 Architecture Board`, which holds `G1` and `G2`.

Done when. Every building block names a condition a gate can evidence, `O55` reconciles to the four domain sheets, and `A11` and `A12` each have a block.

Shows up in. `RPT06 Requirements Traceability`, whose `Reaches a block` column resolves through `R09` to `O55` — and whose thirty-third data row is a sentence, not a requirement [10].

Goes wrong when. Nothing ever tests them. `O59 Compliance Assessment` would hold the verdicts and it is **empty**, along with `R24 Compliance Assessment tests` [11], so no acceptance criterion in this repository has ever been assessed and the four gates have never bitten.

Step 6.10 — Apply a standard to the systems that carry the obligations

Produce. An `R18` that covers all twelve applications, and a GxP standard set in `O32` where there is currently none.

You need. Step 6.2. The chief architect. The validation lead.

> **`R18 Standard applies to Application and O32 Standard`**
>
> `R18` has sixteen rows; `O32` has sixteen standards. `Conformance` takes `Conformant`, `Partial`, `Breach` or `Not assessed`.

How.

1. Read `R18`'s `Application` column. It cycles `A01`, `A03`, `A05`, `A07`, `A09`, `A11` — the odd-numbered applications, in order, three times [7]. **Six of the twelve applications have no standard applying to them at all**, and those six include `A04 Safety Database`, `A08 QMS`, `A10 Serialisation` and `A12 Medical Information` — which is **all three `Critical` applications** [1][7].
2. Read `O32`'s sixteen standards. They are generic IT standards, and **not one of them is a GxP standard** [7]: no computerised-system validation standard, no electronic records standard, no data-integrity standard, no audit-trail-review standard.
3. Read two rows that cannot both be true. `STD-02 Event schema standard` is `Deprecated` and `Conformant`; `STD-07 Logging and monitoring standard` is `Deprecated` and in `Breach` [7]. A deprecated standard cannot be conformed to or breached — it has been withdrawn.
4. Add the GxP standards the estate needs, in `O32`'s own columns — `Ref`, `Standard`, `Category`, `Status`, `Conformance` — so they transcribe straight in. Item 19 carries the full starter set; here, add at least the validation and audit-trail-review standards, because steps 6.7 and 6.9 depend on them.
5. Re-cut `R18` to cover all twelve applications and set `Conformance` honestly. `Not assessed` is the only honest value for something nobody has looked at.
6. Never state a standard's version from memory. Volume 1 carries the version facts with retrieval dates.

Signed by. `GB-01 Architecture Board` approves standards. The validation lead signs the GxP ones.

Done when. Every application appears in `R18`, no `Deprecated` standard carries a conformance verdict, and `O32` contains at least one standard about validated state.

Shows up in. `RPT12 Standards Conformance`, which currently reports sixteen standards against six applications and reads as coverage.

Goes wrong when. The three critical applications stay uncovered, because they are the ones nobody wants assessed, and absence of assessment reads as conformance on every dashboard.

Look it up. Volume 1 chapter 15, *What goes stale first, in order*, for why a version stated from memory is the fastest way to lose a room.

Step 6.11 — Settle where each application runs, and on what

Produce. One answer per application for location and platform, reconciled across three sheets.

You need. Step 6.3. Infrastructure. Volume 1 chapter 13 for the qualification-against-validation distinction, which item 7 works through.

> **`R20 Application located at Location`, `R07 Application deployed on Technology Component`, `O27 Physical Technology Component`, `O48 Environment`**
>
> Three sheets answer the question and they are not reconciled anywhere. Run the join yourself: `R20` gives the application's location directly; `R07` plus `O27.Location` gives it by way of the platform component.

How.

1. Run the join. **Nine of the twelve applications are in two places at once** [6][8]. `A01 ELN / LIMS` is at Head Office by `R20` and at the Distribution Hub by way of `PTC-01`. `A04 Safety Database` is at the API Manufacturing Site and at the Quality Control Laboratory. Three agree — `A07`, `A09`, `A10` — by coincidence of index.
2. Read `R07`'s pairings, which are index-paired like everything else: `A01` on `PTC-01 GMP MES servers`, `A02 CTMS` on `PTC-02 LIMS application tier`, `A04 Safety Database` on `PTC-04 Clinical EDC platform`, `A08 QMS` on `PTC-08 Statistical computing grid` [6]. Every row names a component built for a different application.
3. Find the two that are past vendor support and see which applications sit on them. `PTC-07 Laboratory instruments network` ended support in 2026-08 and carries `A07 MES (GMP)`, a GMP system. `PTC-11 Enterprise storage array` ended 2026-01 and carries `A11 Statistical Computing` [6][8]. Two production systems, one of them GMP, on unsupported platforms today.
4. Read `R07`'s `Environment` column: **all twelve rows read `Production`** [6]. So `O48`'s other three environments — `ENV-02 Pre-production`, `ENV-03 Test`, `ENV-04 Development` — are referenced by nothing, and the repository cannot show where a validation instance runs. Item 7 takes `O48`'s inverted personal-data column, which is a POPIA problem rather than an application one.
5. Correct `R20` and `R07` from infrastructure's own records, then re-run the join as a standing check. It is four lines of code and the fastest honest finding in the repository.
6. Record each platform component's qualification state separately from the validation state of the application on it. Infrastructure is qualified; the computerised system on top of it is validated; different owners, different evidence.

Signed by. Infrastructure signs the platform rows; the system owner signs the location. Platform currency sits with the chief information officer.

Done when. `R20` and the `R07`-to-`O27` join give the same location for all twelve, and no GxP application sits on a component past vendor support without a dated remediation.

Shows up in. `RPT10 Technology Currency`, which correctly reports two components past support and, like the others, ends in a sentence in the `Ref` column [10].

Goes wrong when. The mismatch is treated as a data problem. Two locations for one application means two different people believe two different things about where the regulated records are held, and one of them answers the inspector.

Look it up. Volume 1 chapter 12, *The platform consolidation story, and why the framing every deck uses has broken*, before anybody proposes a single platform for the estate.

Nine of twelve applications are in two places at once

Application	R20 says	R07 + O27.Location says	
A01 ELN / LIMS	Head Office	Distribution Hub	*differs*
A02 CTMS	Research Campus	Head Office	*differs*
A03 EDC	Clinical Ops Centre	Research Campus	*differs*
A04 Safety Database	API Manufacturing	Quality Control Lab	*differs*
A06 eTMF	Quality Control Lab	Clinical Ops Centre	*differs*
A07 MES (GMP)	Distribution Hub	Distribution Hub	*agrees*
A08 QMS	Validated Cloud Region	Fill and Finish Site	*differs*
A11 Statistical Comp.	Clinical Ops Centre	Quality Control Lab	*differs*

A07 and A11 run on components past vendor support today — PTC-07 and PTC-11.

Figure 6.4 `R20` against the `R07`-to-`O27` join. Nine of twelve applications have two locations; the three that agree do so by coincidence of index. The join is four lines of code and the fastest honest finding in the repository.

Step 6.12 — Republish `RPT02` and `RPT04`, and read their last rows

Produce. A re-cut portfolio report carrying the three new columns, and a re-cut interface catalogue with validation scope on every row.

You need. Every step above.

> **`RPT02 Application Portfolio` and `RPT04 Interface Catalogue`**
>
> `RPT02` is `O23` with `Description` dropped — nine columns, twelve rows, no computation [10]. `RPT04` is `R06` with the `Ref`-and-name pairs collapsed, thirty-seven data rows for thirty-six interfaces.

How.

1. Re-cut `RPT02` from the corrected `O23` and add `GxP`, `Validation state` and `Next review due` from `O67`. A portfolio report here that cannot show validation state is not one.
2. Re-cut `RPT04` from the corrected `R06` and add `Validation scope`. State the four numbers on the face of the report: interfaces, unmanaged, unlogged, direct database reads. The healthy figure for the last three is zero.
3. Read both last rows before circulating either. `RPT04`'s thirty-seventh row is a sentence in the `Ref` column, and **nine of the thirteen reports end the same way** — `RPT01`, `RPT04`, `RPT05`, `RPT06`, `RPT08`, `RPT09`, `RPT10`, `RPT11` and `RPT13` [10]. Write the rule down for whoever automates against this file: skip any row whose `Ref` contains a space.
4. Say which of the thirteen an inspector would ask to see: `RPT02`, once it carries validation state, and `RPT09 Regulatory Coverage`, which is the one that would embarrass you — `CO-03`, `CO-04` and `CO-05` all read `In breach` while `O59`, the sheet that would hold the evidence, is empty [11].
5. Put the portfolio in front of `GB-01` with the signatures from steps 6.2 and 6.3 attached, and keep both reports in step with `O67` from now on. The moment they diverge, the organisation has two inventories again.

Signed by. `GB-01 Architecture Board` for the portfolio; the head of quality assurance for the `GxP` and `Validation state` columns; the validation lead for the interface catalogue's scope column.

Done when. Both reports reconcile to their source sheets, both carry the new columns, and the sentence rows are documented as known artefacts rather than discovered by a loader.

Shows up in. `RPT01 Dashboard` and `ZZ Checks`, which are honest about nine figures and silently wrong about three — object types populated reads 65 against at most 63, records reads 850 against 814, and *gaps overdue* reads zero, which is meaningless because `O29 Gap` is empty [11].

Goes wrong when. The portfolio is circulated as evidence of control. It is evidence of a copy. The control is in `O67`, in the named owners, and in the periodic reviews that are not overdue.

Look it up. Volume 1 chapter 16, *Who inspects, and what they ask for*, for the twelve-row table of who comes through the door and what they open first.

What leaves Phase C application

Artefact	Where it lives	Signed by
Validated-system register, reconciled to the portfolio	`O67`, re-cut `RPT02`	Head of quality assurance
Named system, process and validation owners	`O23`, `O67`	Chief information officer, each owner
Four dated dispositions for the four contradictions	`O23`, `O60`	`GB-01`, `GB-02`
A decommissioning plan or a reversal for `A10`	`O67`, `O69`	Quality assurance, release signatory
Interfaces with a validation scope and a specification	`R06`, re-cut `RPT04`	Validation lead, both system owners
Testable application building blocks	`O45`, re-cut into `O55`	`GB-01 Architecture Board`

And the list of what you could not establish: which systems have a current validation package, when each was last reviewed, who owns the thirty-six interfaces, and what replaces `A10`. Hand it over with a name against each line. In this estate the absences are the finding.

THE CHAPTER IN TEN LINES

- O23 has no validation state, no last-periodic-review date and no GxP flag, and nine of the twelve applications are GxP systems.
- Create O67 Validated System and reconcile it to O23 row for row before anything else in this phase.
- The grain of O67 is one system instance, in one environment, for one defined use, not one row per application.
- Four rows of O23 contradict themselves. Three are critical systems being reduced with no successor named; A06 eTMF is the honest one.
- The Owner column holds organisation units, and the units are wrong. Human Resources owns serialisation and Internal Audit owns the GMP execution system.
- A11 Statistical Computing and A12 Medical Information are orphans in O45, R04 and R25, and A12 is critical and at risk.
- R04 pairs business services to applications by index, so regulatory affairs runs on the safety database and manufacturing on the trial master file.
- R06: nineteen of thirty-six interfaces unmanaged, thirty-one unlogged, sixteen both, only two managed and logged, and all six direct database reads unlogged.
- INT-009 carries serious adverse events from EDC to the safety database as an unmanaged hourly file.
- Six of twelve applications have no standard applying to them, and those six include all three Critical ones.
- O32 holds sixteen generic IT standards and not one GxP standard. Two deprecated standards still carry conformance verdicts.
- Nine of twelve applications are in two places at once, and two production systems sit on components past vendor support today.
- RPT04's last row is a sentence in the Ref column, and so is the last row of nine of the thirteen reports.

SOURCES FOR THIS CHAPTER

- **1.** Pharmaceuticals_TOGAF_Metamodel.xlsx, sheet O23 Application Component, rows 5-16 (Disposition, Criticality, Health, In service and Owner columns; A06, A08, A10 and A12)
- **2.** Pharmaceuticals_TOGAF_Metamodel.xlsx, sheet O24 Information System Service, rows 5-16 (Style and Published columns)
- **3.** Pharmaceuticals_TOGAF_Metamodel.xlsx, sheets O45 Application Building Block rows 5-14, O55 Building Block rows 5-41 and R13 Work Package delivers Building Block rows 5-41
- **4.** Pharmaceuticals_TOGAF_Metamodel.xlsx, sheet 'R04 Business Service supported ' (tab name ends in a space), rows 5-14, with O15 Business Service rows 5-14
- **5.** Pharmaceuticals_TOGAF_Metamodel.xlsx, sheet R06 Application interfaces with, rows 5-40 - 36 interfaces, Type, Frequency, Managed and Logged columns counted directly
- **6.** Pharmaceuticals_TOGAF_Metamodel.xlsx, sheets R07 Application deployed on Technology Component rows 5-16 and R20 Application located at Location rows 5-16
- **7.** Pharmaceuticals_TOGAF_Metamodel.xlsx, sheets R18 Standard applies to Application rows 5-20 and O32 Standard rows 5-20
- **8.** Pharmaceuticals_TOGAF_Metamodel.xlsx, sheets O27 Physical Technology Compone rows 5-16 and O12 Location rows 5-12
- **9.** Pharmaceuticals_TOGAF_Metamodel.xlsx, sheet R25 Application supports Product, rows 5-8, with O18 Product rows 5-8
- **10.** Pharmaceuticals_TOGAF_Metamodel.xlsx, report sheets RPT02 Application Portfolio, RPT04 Interface Catalogue, RPT06 Requirements Traceability, RPT09 Regulatory Coverage, RPT10 Technology Currency and RPT13 Orphan Analysis
- **11.** Pharmaceuticals_TOGAF_Metamodel.xlsx, sheets EXT Industry Extension, ZZ Checks, 00 Index, O09 Organization Unit, O10 Actor, O29 Gap, O30 Work Package, O48 Environment, O52 Implementation Factor, O54 Architecture Alternative, O57 Governance Body, O58 Governance Gate, O59 Compliance Assessment, O60 Governance Decision, O63 Control Objective and R24 Compliance Assessment tests
- **12.** Pharmaceuticals - a starter pack for enterprise architects (volume 1), chapters 11, 12, 13, 15 and 16 and appendix 20, Thotnr Consulting, October 2026
- **13.** The Open Group, TOGAF Standard, Architecture Development Method, Phase C: Information Systems Architectures. Identified, not retrieved: The Open Group's public pages are robots-disallowed to automated retrieval, so no edition or wording is claimed from it.
- **14.** European Commission, EudraLex Volume 4 - Annex 11 Computerised Systems, revision January 2011 (in force; it expects an up-to-date listing of all relevant systems and their GMP functionality). A draft revision was consulted jointly with PIC/S from 7 July to 7 October 2025 and no final text had been published as at the date volume 1 was written https://health.ec.europa.eu/medicinal-products/eudralex/eudralex-volume-4_en
- **15.** PIC/S, PE 009-17 (Part I), Guide to Good Manufacturing Practice for Medicinal Products, 25 August 2023 - the edition volume 1 read, for change control, deviation and documentation expectations on computerised systems https://picscheme.org/en/publications
- **16.** Department of Health, South African Good Clinical Practice: Clinical Trial Guidelines, 3rd edition, 2020 - SS 5.10.3.1-5.10.3.2 audit trails and no deletion of entered data; S 5.10.1.2 ALCOA-CCEA for source documents https://www.sahpra.org.za/wp-content/uploads/2021/06/SA-GCP-2020_Final.pdf
- **17.** SAHPRA, SAHPGL-INSP-RC-04 v1, Guideline on Traceability of Health Products, effective October 2025 - obligations on manufacturers and registration holders, and five-year retention after trade-item expiry https://www.sahpra.org.za/wp-content/uploads/2025/10/SAHPGL-INSP-RC-04_v1-Guideline-on-Traceability-of-Health-Products.pdf

CHAPTER 7

Technology Architecture and the validated estate

Infrastructure is qualified; the computerised system on top of it is validated. Two activities, two owners, two evidence sets - and the repository you were handed has a column for neither.

What this phase is for, and the one distinction it turns on

Phase D turns the application architecture into the platforms, services and locations that carry it, and in this industry it does one more thing that no sibling volume has to do: it separates **qualification** from **validation** and gives each of them an owner.

The distinction is not pedantry. **Infrastructure is qualified. The computerised system that runs on it is validated.** Different activity, different document set, different signatory, different review cycle. Volume 1 chapter 13 says it in one line — the words are not interchangeable, and using them interchangeably tells a room of quality people that you have not done this before [9]. A qualified platform is an asset you cite, once, from every application validation that sits on it, instead of re-proving the operating-system build twelve times. Where that citation chain is written down, validation cost falls. Where it is not, every project qualifies the platform again in its own binder.

`Pharmaceuticals_TOGAF_Metamodel.xlsx` gives you nine technology sheets and two reports, and not one column anywhere records a qualification state or a qualification reference. You will add them as columns on `O27 Physical Technology Compone`, not as a sheet: this volume has agreed that `O67` – `O70` are the only new sheets.

> **The repository's technology domain is internally consistent and largely untrue.** Four components sit in buildings they cannot be in, a cloud region is staffed by eighty-two people, the environment sheet says personal data is absent from production and present in development, and the deployment relationship pairs application *n* with component *n* for all twelve rows. None of it fails a referential-integrity check. All of it fails a walk round the building.

Two invented organisations appear in this volume and their data must never be mixed. **Calderon Therapeutics** is the company in the repository you were handed. **Veldtkruin Group** is volume 1's worked example. Both are fictional.

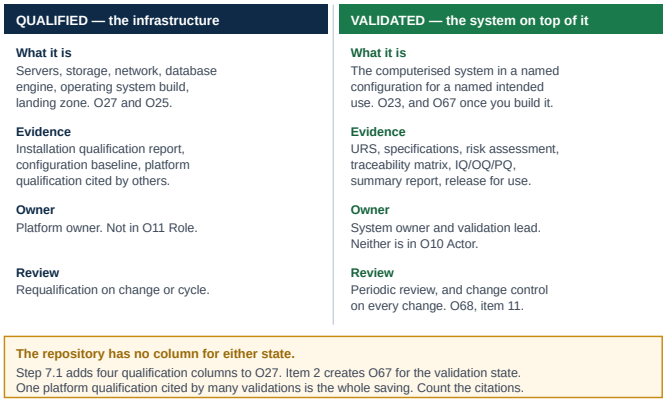


Figure 7.1 Qualification against validation, with the owner and the evidence on each side. **The saving is in the citation**: one platform qualification cited by many application validations, rather than the same operating-system build proved twelve times in twelve binders.

Before you open a sheet

ANSWER THESE FROM MEMORY, THEN CHECK THEM

Who signs a platform qualification here, and who signs an application validation? If it is the same person, which hat are they wearing?

Which platform qualification is cited by more than one application validation? Name it.

Does non-production hold trial participants' data today? Who would know, and how would they prove it either way?

When a recovery test last ran, what evidence was produced, and where is it filed?

The steps

Step 7.1 — Split the estate into what is qualified and what is validated

Produce. A one-page scope split: every row of `O27 Physical Technology Compone` marked as infrastructure to be qualified, and every row of `O23 Application Component` marked as a computerised system to be validated, each with a named owner and the document that carries the evidence.

You need. The application list from item 6. The validation lead — who is not in the repository. The platform owner. Volume 1 chapter 13 for the deliverable set.

Where it lands.

> **`O27 Physical Technology Compone` — four new columns, not a new sheet**
>
> Add `Qualification state`, `Qualification reference`, `Qualified by` and `Requalification due` to the right of `Location`. `Qualification state` takes one of: `Not qualified`, `Under qualification`, `Qualified`, `Qualified with open deviations`, `Requalification due`. `Qualification reference` is the installation-qualification document number, not a folder path. `Qualified by` is a person. `Requalification due` is a date, and a blank in it is the finding.

How.

1. Print `O27` and `O23` side by side. Twelve rows each, which is the whole estate.
2. For each `O27` row, write which applications cite its qualification. Nine of twelve rows read `Vendor` in the `Vendor` column, so start by finding out who the supplier is — supplier assessment is part of the validation obligation and you cannot assess a supplier called `Vendor`.
3. For each `O23` row, write the validation state and the document reference. These go to `O67 Validated System`, defined in item 2.
4. Mark every row where the same person signs the qualification and the validation. That is a segregation question, not a staffing one.
5. Count the citations. If no platform qualification is cited twice, the estate is paying for the same qualification twelve times.

Signed by. The qualification side by the platform owner; in `O57 Governance Body` the nearest body is `GB-05 Architecture Function`, which is not a quality body and cannot approve a qualification. The validation side is signed by the validation lead and Quality Assurance, neither of which exists in `O10 Actor` or `O11 Role`. Until item 2's `O70 Regulated Role` is populated, record the Head of Service Management as the interim platform owner and say in the minute that the validation signatory is vacant.

Done when. Every `O27` row carries a qualification state and a named person, and every `O23` row carries a validation state, and the two lists do not share a signatory for the same system.

Shows up in. `RPT10 Technology Currency`, which today repeats `O27` verbatim and adds nothing. After this step it carries a qualification column and the count of rows with no qualification reference — which on this repository starts at twelve.

Goes wrong when. The architect writes "validated" against a server. A server is qualified. The sentence is wrong in a way that costs you the room.

Look it up. Volume 1, chapter 3, "Validation, qualification, verification"; and chapter 13, "Computerised system validation, as it is actually practised".

Step 7.2 — Put the physical components in the buildings they are actually in

Produce. A corrected `Location` column on `O27`, with a dated note recording what each value was before you changed it.

You need. `O12 Location`'s eight sites. The site master file or the licence for each GMP site. The person who can walk you into the room.

Where it lands.

> **`O27 Physical Technology Compone`**
>
> `Location` must hold a value that appears in `O12 Location`, column `Location`. Four rows do not survive contact with the building: `PTC-01 GMP MES servers` is recorded at `Distribution Hub`, `PTC-02 LIMS application tier` at `Head Office`, `PTC-04 Clinical EDC platform` at the `Quality Control Laboratory`, and `PTC-08 Statistical computing grid` at the `Fill and Finish Site`. A manufacturing execution system does not run from a warehouse; a laboratory information management tier does not run from head office; clinical data capture does not run from a quality-control laboratory [1][2].

How.

1. Join `O27.Location` to `O12.Location` and confirm every value resolves. All twelve do, which is why no check catches this.
2. For each of the four rows above, ask the site to confirm the rack. Write the answer and the date.
3. Do not reason from the two cross-references agreeing. `R20 Application located at Loca` pairs `A01` with `LOC-01` down to `A08` with `LOC-08`, then wraps; `R07 Application deployed on Tec` pairs `A01` with `PTC-01` down to `A12` with `PTC-12`. Both are position joins, so where they agree they agree by arithmetic [7].
4. Record the licensed activity of each GMP site next to its name. A sterile fill-and-finish site and an oral solid dose site are different regulated objects with different inspection scopes.
5. Where a component is in two places — a cluster split across sites — add a row rather than a compound string, and renumber from `PTC-13`.

Signed by. The site quality head for each GMP location, countersigned by the platform owner. The repository has no site quality role; record the `Head of Operations` and flag the gap.

Done when. Every `O27` row's `Location` has been confirmed by somebody who has been in the room, with a date against the confirmation.

Shows up in. `RPT10 Technology Currency`, whose `Location` column is a straight copy of `O27`. Nothing recomputes, so the corrected value appears only once you rebuild the report.

Goes wrong when. You correct the sheet and not the drawing. The location error has usually been copied into a slide that three people have presented.

Look it up. Volume 1, chapter 7, "A registered manufacturing site is a regulated object" and "What each site is licensed to do".

Step 7.3 — Decide what the validated cloud region is, and stop staffing it

Produce. A corrected `O12 Location` row for `LOC-08`, plus a one-paragraph statement of the data residency position for every record class that lands there.

You need. The hosting contract. The shared-responsibility split. The POPIA position on health data leaving the Republic.

Where it lands.

> **`O12 Location`**
>
> `LOC-08 Validated Cloud Region` is typed `Technology` with a `Scale` of **82 people**. A region is not staffed. Either `Scale` means something other than headcount on this row, in which case the column has two meanings and is unusable, or the cell was filled by pattern. Replace the value with the unit the row actually measures — availability zones, or the count of qualified environments — and say in the cell which it is.

How.

1. Add the eight `Scale` values. They total 1,215 people, of which 82 sit in a region. `O09 Organization Unit` totals 1,879, so 664 people have no location — or 746 once you take the region's 82 out, which you should, because nobody sits in a region [1][2].
2. Decide what `LOC-08` is for. `O27` already carries `PTC-09 Validated cloud landing zone` at `Head Office`, which is a second answer to the same question.
3. List the record classes hosted there. For each, record the lawful basis for the transfer. Health information may not leave South Africa without a lawful basis and either adequate protection in the destination or POPIA-aligned contractual obligations [11].
4. Get the five cloud conditions in writing: the responsibility split, change notification with a deferral window, access to the provider's audit evidence, a tested export format, and the cross-border basis [9].
5. Where any of the five is missing, write that the system is not in a maintainable validated state in that hosting model. Do not soften it.

Signed by. The Information Officer for the transfer basis and the platform owner for the qualification. The repository has neither role; `GB-03 Data Governance Council` is the nearest body and its mandate covers data owners, not transfers.

Done when. `LOC-08` carries a unit that is not people, and every record class hosted there has a named lawful basis and a named signatory.

Shows up in. `RPT10 Technology Currency`, where the cloud rows stop reading as sites. The residency position does not show up anywhere, because the repository has no column for it; keep it with the phase deliverable `DLV-11 Technology Architecture`.

Goes wrong when. The region is treated as qualified because the contract says "validated cloud". A supplier's marketing adjective is not a qualification reference.

Look it up. Volume 1, chapter 15, "Cloud regions, and the check this book teaches instead of a table" and "Health-data residency under POPIA"; chapter 16, "Section 72, and the adequacy burden POPIA puts on you".

Form 7-A — Physical component qualification register

Six of the seven columns of `O27 Physical Technology Compone`, in sheet order, plus the four qualification columns step 7.1 adds. `Vendor` is typed straight into Excel, because nine of the twelve rows read `Vendor` today and the real answer is a procurement enquiry. One pass needs twelve rows — the whole estate. Signed by the platform owner.

Ref	Physical technology component	Support end	Position	Location	Qualification state
PTC-07	EXAMPLE — delete. Laboratory instruments network	2026-08	Past support	Distribution Hub	Not qualified

Second pass, same `Ref`: `Qualification reference`, `Qualified by`, `Requalification due`, and the date somebody confirmed the location. Type those four straight in.

Step 7.4 — Reconcile the three sheets that name the same seven building blocks

Produce. One status per technology building block, agreed, and a note of which sheet is now authoritative.

You need. `O26 Logical Technology Componen`, `O46 Technology Building Block`, `O47 Reference Model Entry`. The reference architecture from item 6.

Where it lands.

> **`O26 Logical Technology Componen` and `O47 Reference Model Entry`**
>
> The same seven entries appear in all three sheets — GxP-validated cloud, 21 CFR Part 11 controls, CDISC standards, IDMP data model, cold chain telemetry, scientific data lake, and federated learning or secure enclave. These are genuinely pharmaceutical building blocks and the best thing in the technology domain. The status columns contradict each other on **all seven**: `O26` calls GxP-validated cloud `Current` while `O47` calls it `Candidate`; `O26` calls 21 CFR Part 11 controls `Emerging` while `O47` calls them `Adopted`; and so on down the list [5].

How.

1. Lay the three sheets side by side on the building-block name, not on the reference.
2. Agree one status vocabulary. `O26` uses `Current` and `Emerging`; `O47` uses `Adopted` and `Candidate`. Pick one, write it into both, record the mapping.
3. Note the second `Ref` column on `O46 Technology Building Block`. The sheet has **two columns headed `Ref`**; the second holds `T01` – `T07`. Call it `Ref (T)` in writing. Nothing else in the workbook uses that key space, and no relationship sheet resolves against it — `R09` and `R13` resolve their `Building block` column against `O55 Building Block`, the 37-row union sheet with a `Domain` column [5].
4. Decide whether `Ref (T)` survives. An unreferenced key space is how a repository grows a second identity for every object.
5. For each of the seven, write what "adopted" would mean here: in production, for a named service, with its qualification or validation evidence filed.

Signed by. `GB-01 Architecture Board`, which holds G1 and G2 under `O58 Governance Gate` and is the right body for a standards decision.

Done when. The three sheets give one answer per building block and the status vocabulary is written down once.

Shows up in. Nowhere automatically — there is no technology-building-block report among the thirteen. Carry the reconciled list in `DLV-12 Technology Standards`, which `O31 Deliverable` already marks `Approved`, which it should not be.

Goes wrong when. Somebody treats `O47.Status = Adopted` as evidence of adoption. It is a statement of intent with no acceptance criterion behind it, which is step 7.5.

Look it up. Volume 1, chapter 14, "The validation and quality layer"; chapter 11, "The capability no other industry has: the register of validated systems".

Step 7.5 — Rewrite the acceptance criteria so a gate can fail them

Produce. Seven acceptance criteria that a reviewer at `GT-2 G2 Build complete` can mark pass or fail from evidence.

You need. The reconciled list from step 7.4. `O58 Governance Gate`. Volume 1 chapter 13's deliverable set.

Where it lands.

> **`O46 Technology Building Block`**
>
> `Acceptance criteria` currently restates the block's own name. `TBB-01 GxP-validated cloud` is accepted when there is "GxP-validated cloud in production for at least one service"; `TBB-03 CDISC standards` when there are "CDISC standards in production for at least one service" [5]. Every one of the seven is the name plus eleven words. No reviewer can fail that, which is the same as having no criterion. Replace each with a condition that names the evidence, the signatory and the date.

How.

1. For each block, write the criterion as a sentence containing a document reference. "A platform qualification report for the landing zone, approved by Quality Assurance, cited by at least two application validations" is testable. "In production for at least one service" is not.
2. Add the negative condition. For 21 CFR Part 11 controls: the audit trail is switched on in production, is not editable, and a review record exists for the last period. Volume 1's characteristic failure is that the capability ships switched off [9].
3. For CDISC standards and the IDMP data model, name the version, and take it from the submission the company actually made rather than from memory. The version is part of the datum.
4. For cold chain telemetry, write the criterion against the excursion, not the sensor: an excursion produces an alarm, an investigation record and a disposition decision.
5. Mirror all seven into `O55 Building Block`, rows `TBB-01` – `TBB-07`, because that is where `R09` and `R13` read them. Mark every block with no evidence and no owner as a gap for item 8.

Signed by. `GB-01 Architecture Board` for the criteria; the validation lead countersigns the two that touch electronic records, because those criteria become validation acceptance tests.

Done when. Each of the seven criteria names a document, a signatory and a period, and a reviewer who has never met you could mark it.

Shows up in. `O59 Compliance Assessment`, which is empty and whose columns — `Building block`, `Acceptance criterion`, `Gate`, `Verdict`, `Evidence` — are exactly the shape of this step's output. Item 10 populates it.

Goes wrong when. The criteria are written as a count. "Nine of twelve systems compliant" is a project measure, not an acceptance criterion, and it hides which three.

Look it up. Volume 1, chapter 13, "The validation master plan and the deliverable set".

`O48 Environment` says the company holds no personal data in production and holds it in development. `ENV-01 Production` records `Personal data present: No`. `ENV-02 Pre-production` and `ENV-04 Development` record `Yes` [3]. Read literally that is impossible. Read with `O41 Security Control`, where `SEC-07 Non-production masking` is `Partial`, and with `O65 Enforcement Point`, where `EP-05 Non-production provisioning` reads `Most refreshes are unmasked`, the live reading is the second one: **non-production holds unmasked trial participants' data** [3][6]. That is a POPIA matter and a Good Clinical Practice matter before it is an IT hygiene matter, and the subject record is the most protected class in the estate. Do not file it as a data-quality defect on an environment sheet.

Step 7.6 — Correct the environment sheet and bring non-production inside the boundary

Produce. A corrected `O48 Environment`, an environment-by-record-class statement, and a dated remediation commitment for every environment that holds subject data it should not.

You need. The classification work from item 5. `SEC-07`. `EP-05`. The Information Officer.

Where it lands.

> **`O48 Environment`**
>
> Four rows, one column of consequence: `Personal data present`. Fill it from a query against the environment, not from memory. Then add two columns the sheet does not have: `Masking method` and `Last refresh verified`. Without the second, `Personal data present: No` is an assertion rather than a finding.

How.

1. Run the query. Count records in each non-production environment that carry a subject identifier, a date of birth, or free text from a case narrative.
2. Record the number, not the verdict. "Two hundred and eleven subject records in pre-production, counted on this date" survives a challenge; "masked" does not.
3. Treat every environment holding GxP records as inside the validation scope of the system above it. A pre-production instance used for release validation produces evidence, so it is in scope.
4. For each environment, name the person who approves a refresh. `EP-05` is in no policy relationship at all — `R17 Policy enforced at Enforcem` pairs `POL-01` to `EP-01` down to `POL-04` to `EP-04` and leaves `EP-05` out [6] — so the policy protecting personal information across production and non-production is enforced at the design review gate rather than at the point that copies the data.
5. Write the remediation as two dated items: stop the next unmasked refresh, then remove what is already there. They are different pieces of work with different approvals.

Signed by. The Information Officer, with the validation lead for the scope decision. Neither is in the repository. `GB-03 Data Governance Council` is the nearest body and should take the minute.

Done when. Every row of `O48` carries a counted value with a date, `EP-05` is mapped to `POL-01` in `R17`, and a dated commitment exists for each environment that fails.

Shows up in. `RPT11 Security Position`, which lists `SEC-07` as `Partial` and ends with a sentence row reading that two controls are absent. The report does not mention masking coverage at all; add it.

Goes wrong when. The sheet is corrected and the pipeline is not. A corrected `No` in `ENV-04` with an unchanged refresh job is a worse state than the original error, because now the sheet defends the practice.

Look it up. Volume 1, chapter 13, "Data integrity"; chapter 16, "POPIA and health data".

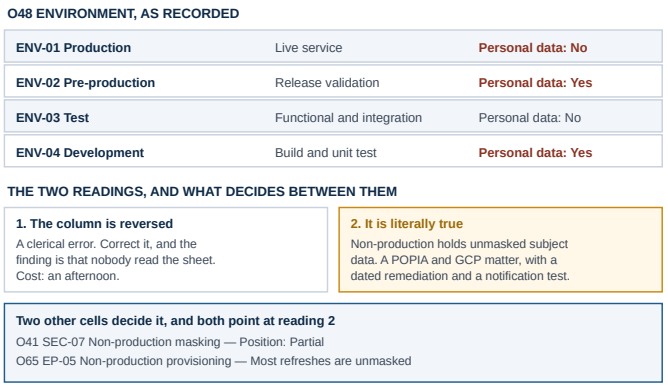


Figure 7.2 The environment sheet inverted, and the two cells elsewhere in the repository that settle which reading is live. **Both point the same way.** Treat it as a personal-information finding with a dated remediation, not as a data-quality defect on a technology sheet.

Step 7.7 — Put the deployment relationship back on facts

Produce. A rebuilt `R07 Application deployed on Tec` with one row per application, per component, per environment — and the count of rows it gains.

You need. The corrected `O27`. The corrected `O48`. Whoever runs the platform.

Where it lands.

> **`R07 Application deployed on Tec`**
>
> Columns `Application`, `Application name`, `Platform component`, `Component name`, `Environment`. Twelve rows today: `A01` on `PTC-01`, `A02` on `PTC-02`, down to `A12` on `PTC-12`, and every `Environment` cell reads `Production` [7]. The sheet therefore asserts that `ELN / LIMS` runs on `GMP MES servers`, that the `Safety Database` runs on the `Clinical EDC platform`, and that `QMS` runs on the `Statistical computing grid`. It also asserts twelve independent silos: no two applications share a database cluster, a storage array or a network.

How.

1. Delete the diagonal. Rebuild from the platform inventory, one row per deployment.
2. Add the non-production rows. `O48` records four environments and `R07` records one, so expect the row count to triple at least.
3. Flag the two rows where a validated application sits on a component that is already past vendor support: `A07 MES (GMP)` on `PTC-07 Laboratory instruments network`, and `A11 Statistical Computing` on `PTC-11 Enterprise storage array` [1][7]. Whether or not the pairing is real, go and check it, because if it is real it is a periodic-review finding today.
4. Where a component serves several applications, write the qualification reference once and cite it from each. That citation is the payoff from step 7.1.

Signed by. The platform owner. The validation lead countersigns the environment column, because it determines which instances are inside a validation scope.

Done when. No `Environment` cell is blank, non-production deployments appear, and every application-to-component pairing has been confirmed by someone who can name the host.

Shows up in. Nothing computes over `R07` among the thirteen reports, which is part of why it has never been questioned. After this step it is the join that lets `RPT10 Technology Currency` answer "which validated systems are running on unsupported infrastructure" — the question an inspector asks in the first hour.

Goes wrong when. The rebuild is done from a configuration management database that is itself a position join. Check three rows against the host before you trust three hundred.

Look it up. Volume 1, chapter 12, "Interfaces are routinely in nobody's validation scope"; chapter 13, "Periodic review".

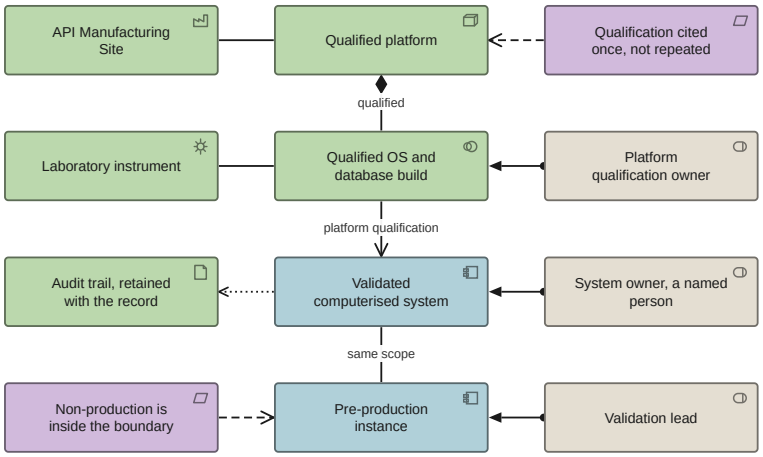


Figure 7.3 The deployment relationship drawn properly: a qualified platform serving a validated system, with two owners and two evidence sets. **The pre-production instance is inside the same validation scope**, because it produces release-validation evidence — which is why `O48 Environment` is a Phase D problem and not an environments problem.

Step 7.8 — Re-tier resilience against the regulatory clock, not the label

Produce. A rebuilt `O49 Resilience Tier` where the recovery objective is derived from an obligation with a clock, and a test date against every row.

You need. The reporting clocks from volume 1 chapter 16. The release process. The business continuity clause of the computerised-systems annex.

Where it lands.

> **`O49 Resilience Tier`**
>
> Twelve rows, columns `Application`, `Tier`, `Recovery objective`, `Tested`. **Nine of twelve read `Never tested`** — only `RES-03 EDC`, `RES-05 RIM` and `RES-08 QMS` have been tested. Tier and objective do not track each other: `RES-08 QMS` is Tier 2 with a **15-minute** objective and tested, while `RES-04 Safety Database` is Tier 3 with **4 hours** and never tested. Tier 2 holds objectives ranging from fifteen minutes to twenty-four hours, which is a ninety-six-fold spread inside one tier [4]. Add one column: `Clock it serves`.

How.

1. For each application, write the obligation with the shortest clock that depends on it. The safety database carries the expedited reporting clock, and day zero runs from any employee's first knowledge of a case. It is the lowest tier in the estate.
2. Derive the recovery objective from that clock and a margin, then set the tier from the objective. Doing it in the other order is how `RES-08` got fifteen minutes.
3. Mark the three applications whose loss stops batch release. Those are the ones where recovery time is a supply decision, not an availability metric.
4. Book a test for each of the nine untested rows and write the date in. The evidence wanted is a tested *restore*, not a completed backup [9].
5. Correct `MSR-13 Recovery objectives tested` in `O04 Measure`. It records a baseline of **53** in unit `count` against a population of twelve applications, of which three are tested [8]. A count of 53 out of 12 is not a rounding problem.

Signed by. The Head of Operations for the tier, the Responsible Pharmacist or the release signatory for any row that gates release — a role the repository does not carry. Record the Head of Operations and name the vacancy.

Done when. Every row has a clock, a derived objective, a tier consistent with the objective and a test date in the future or the past. No row says `Never tested` without a booked date.

Shows up in. `RPT01 Dashboard` has no resilience line at all, which is itself worth saying out loud. `O04 Measure`'s `MSR-13` and `MSR-14` are where this becomes visible; after this step `MSR-13` reads 3 of 12 and rises.

Goes wrong when. The tier is set from criticality in `O23 Application Component`. That column is already contradictory — `A10 Serialisation` is `Critical` and dispositioned `Retire` — so inheriting from it propagates the contradiction into the recovery plan.

Look it up. Volume 1, chapter 16, "Pharmacovigilance" and "The retention clocks".

Form 7-B — Resilience re-tier worksheet

The five real columns of `049 Resilience Tier` in sheet order, plus the `Clock it serves` column step 7.8 adds. Twelve rows is the whole estate; a first pass fits on one page. Signed by the Head of Operations and the release signatory.

Ref	Application	Clock it serves	Recovery objective	Tier	Tested
RES-04	EXAMPLE — delete. Safety Database	Expedited report, day zero from first knowledge			Never tested

Step 7.9 — Take the two past-support components to a decision

Produce. A decision per component — renew, replace, or accept with a dated compensating control — each with the revalidation consequence costed.

You need. `RPT10 Technology Currency`. The vendor's support statement. The validation lead's estimate of what the change costs in revalidation.

Where it lands.

> **`027 Physical Technology Compone`**
>
> Two rows are `Past support` today: `PTC-07 Laboratory instruments network`, support ended **2026-08**, and `PTC-11 Enterprise storage array`, support ended **2026-01** [1]. Both sit under GxP systems. Three more cross within two years — `PTC-01 GMP MES servers` at 2027-12, `PTC-06 Document management vault` at 2028-04 — so the sheet needs a `Decision` and a `Decision date`, added beside the qualification columns from step 7.1.

How.

1. Write two sentences for each: what breaks if it fails, and what the fix costs in revalidation. The second number is the one the committee has never seen.
2. Treat the change as a regulatory project with an information technology component, not the reverse. A platform upgrade under a validated application triggers an impact assessment, a risk evaluation, re-testing proportionate to the risk, Quality Assurance approval and a validation documentation update [9].
3. Where the answer is "accept for now", write the compensating control and its review date. Accepting an unsupported component is a defensible position; accepting it silently is not.
4. Correct `MSR-07 Components within vendor support` in `O04 Measure`. It records a baseline of **22%** while `027` shows ten of twelve supported, which is 83% [1][8]. One of those two numbers is going to a board.
5. Put the three forward-dated crossings into the phase E gap list with their dates, so they arrive as planning input rather than as an incident.

Signed by. `GB-02 Investment Committee` for the money, `GB-01 Architecture Board` for the standard, and the validation lead for the revalidation estimate.

Done when. Each past-support row carries a decision, a date and a revalidation estimate, and `MSR-07` agrees with `027`.

Shows up in. `RPT10 Technology Currency`, whose last row is a sentence rather than a record — the string "2 components are running past vendor support today." sits in the `Ref` column, exactly as it does in `RPT04 Interface Catalogue`. Anything loading the report with a dataframe library gets a thirteenth component called a sentence [8].

Goes wrong when. The replacement is scheduled into a quarter the quality function cannot staff. That is item 8's constraint and it binds here first.

Look it up. Volume 1, chapter 15, "What goes stale first, in order"; chapter 13, "Decommissioning, which nobody budgets".

Step 7.10 — Close the controls the regime names, and stop pretending about the gateway

Produce. A corrected `O41 Security Control` and `O65 Enforcement Point`, with a named owner and a date against each control that is `Absent` or `Partial`.

You need. `RPT11 Security Position`. The identity provider's owner. Volume 1 chapter 13 on the electronic-records obligations.

Where it lands.

> **`O41 Security Control` and `O65 Enforcement Point`**
>
> Two controls are `Absent`: `SEC-01 Identity lifecycle` and `SEC-03 Privileged access management`. Four are `Partial`, including `SEC-04 Attributable access`, whose description reads "Every access traceable to a person" [6]. In a validated estate that is not a security preference — unique individual accounts, no shared logins and attribution are requirements of the electronic-records regime [9][10]. `O65` is blunter than `O41`: `EP-02 Identity provider` reads `Enforces authentication, not attribution`, and `EP-03 API gateway` reads `Product not yet selected` — while `O28 Platform Service` lists `PSV-02 API gateway` as a platform service and `SEC-05 Interface logging` says it is enforced at the API gateway.

How.

1. Write the chain out: a policy enforced at a point that is a product nobody has bought. `POL-03 Material activity is logged and observable` is `Breached` in `O64 Security Policy` and is mapped by `R17` to `EP-03`, the unselected gateway [6].
2. Separate the two absences. Identity lifecycle is a joiner-mover-leaver process and is mostly procedural. Privileged access management is a product and a procedure, and it is the control an inspector reaches for when asking who could have changed a record.
3. For `SEC-04`, enumerate the systems with shared or generic accounts. Name them. This is the single most common electronic-records finding in the industry and it is always discovered by counting accounts, never by reading a policy.
4. Decide the gateway. `DEC-02 Select the integration product` in `O60 Governance Decision` is `Outstanding`, and `O62 Delivery Risk` carries the same thing as `RSK-01` at likelihood `Certain`. It is one decision with three homes.
5. Re-map `R17` so each policy points at the point that actually enforces it, and add the missing `EP-05` row from step 7.6.

Signed by. The Security Architect in `O11 Role` for the controls, `GB-01 Architecture Board` for the standard, and the validation lead for anything touching audit trails or attribution.

Done when. No control is `Absent` without an owner and a date, `EP-03` names a product or the decision is dated, and every policy in `O64` maps to a point that exists.

Shows up in. `RPT11 Security Position`, which also ends in a sentence row rather than a record: "2 controls are absent. A control that is absent is not a control." [8] Two absent and four partial is the honest count; the report's own closing line is the best sentence in the workbook.

Goes wrong when. Attributable access is logged as a technology task. Its hard part is operational — the shared account on the laboratory workstation that four analysts use because the instrument software was configured that way in 2014.

Look it up. Volume 1, chapter 13, "21 CFR Part 11 and EU GMP Annex 11" and "Audit-trail review, the most under-resourced obligation in the regime".

Step 7.11 — Fix the network sheet, then ask it the only question that matters

Produce. A corrected `O50 Network Link` and a single statement per GMP site: what stops if the link stops.

You need. The network owner. The corrected `O12`. The licensed activity of each site.

Where it lands.

> **`O50 Network Link`**
>
> Ten rows, columns `From`, `To`, `Bandwidth`, `Diverse path`. The rows describe a ring through all eight locations and then repeat two of the hops. `NWL-01` and `NWL-09` are the identical pair Research Campus to Clinical Operations Centre at 10 Gbps with a diverse path; `NWL-02` and `NWL-10` are the same pair Clinical Operations Centre to API Manufacturing Site with **two different answers** — 1 Gbps with a diverse path, and 40 Gbps without one [7]. Six of the ten rows read `Diverse path: No`, including both hops between the GMP sites and the quality-control laboratory.

How.

1. Deduplicate. Eight distinct pairs across ten rows, and one pair that contradicts itself.
2. Replace the ring. A ring through `Validated Cloud Region` between the `Distribution Hub` and `Head Office` is a pattern, not a topology.
3. For each GMP site, write what stops. A quality-control laboratory that cannot reach the laboratory information management tier stops releasing; a fill-and-finish site cut off from the manufacturing execution system falls back to paper, which is legitimate only if the hybrid is declared in advance.
4. Where the answer is paper, name the procedure and confirm it has been exercised. An undeclared fallback is a data-integrity finding waiting to happen.
5. Record support hours against the link, not only bandwidth. A twenty-four-hour plant and a global trial are not served by business-hours support.

Signed by. The network owner, with the site quality head for the fallback procedure.

Done when. No duplicate pairs remain, no pair carries two answers, and each GMP site has a written statement of what stops and what the declared fallback is.

Shows up in. No report reads `O50`, so this one is invisible until something fails. Carry it in `DLV-11 Technology Architecture` and in the resilience rows of step 7.8.

Goes wrong when. Diversity is recorded from the order form rather than the route. Two circuits from one supplier down one duct is one path with two invoices.

Look it up. Volume 1, chapter 15, "Support hours against a twenty-four-hour plant and a global trial"; chapter 13, "Seven pieces of folklore, punctured" for the hybrid rule.

Step 7.12 — Populate `O67 Validated System` from the technology side and sign the phase out

Produce. The technology half of `O67 Validated System`, the phase D deliverables, and the list of gaps this phase hands to item 8.

You need. Everything from steps 7.1 to 7.11. `O31 Deliverable`. `O55 Building Block`.

Where it lands.

> **`O67 Validated System` — the sheet you created in item 2**
>
> Columns `Ref`, `Application`, `GxP`, `Validation state`, `Last periodic review`, `Next review due`, `System owner`, `Process owner`, `Validation lead`, `URS reference`, `Audit trail reviewed`, `Decommissioning plan`. Phase D fills `GxP`, `Validation state`, `System owner` and `Validation lead` for all twelve applications, and sets `Audit trail reviewed` to a date or to blank — never to `Yes`. The two review-date columns belong to item 11, which defines `O68 Periodic Review Schedule`.

How.

1. One row per application per environment per defined use. Twelve applications is not twelve rows.
2. Fill `GxP` from the record, not the system. The test is which records a system holds and what decisions depend on them.
3. Fill `System owner` with a person who can be telephoned. `O23 Application Component`'s `Owner` column holds organisation units, and the wrong ones — `A04 Safety Database` is owned by `Legal`.
4. Write the gap list for item 8 as you go: twelve rows with no qualification reference, nine rows never tested, two components past support, two controls absent, four environments with no verified refresh, seven building blocks with no real acceptance criterion.
5. Mark `DLV-11 Technology Architecture` and `DLV-12 Technology Standards` honestly. `O31 Deliverable` has both as `Approved` today, which cannot be true of a technology architecture whose location column was wrong in four rows this morning.

Signed by. `GB-01 Architecture Board` for the architecture, the validation lead for `O67`, and `GB-02 Investment Committee` for anything with a cost. The validation lead is the signatory the repository does not have; item 2's `O70 Regulated Role` is where the vacancy is recorded.

Done when. `O67` has a row for every GxP system in every environment, with a named owner and a named validation lead, and the gap list is written down rather than remembered.

Shows up in. `RPT10 Technology Currency` and `RPT11 Security Position` both become answerable against a question an inspector asks. `ZZ Checks` still claims `Object types populated: 65`; after this phase the repository carries a sixty-sixth object type the check does not know about.

Goes wrong when. `O67` is created and then maintained by the architecture function. It is a quality record. If nobody in the quality system owns it, it is a spreadsheet that will disagree with the real register within a quarter.

Look it up. Volume 1, chapter 17, "The validated system itself" and "The validated boundary and the clinical/commercial firewall"; chapter 19, "Week one: find out what is validated".

Form 7-C — Validated system register, technology half

Six of the twelve columns of `O67 Validated System`, in sheet order. **This sheet does not exist in the repository you were handed** — item 2 creates it as `O67`. `Process owner`, `URS reference`, `Audit trail reviewed` and `Decommissioning plan` are typed straight into Excel; the two review dates belong to `O68 Periodic Review Schedule` in item 11. One row per system per environment, so budget thirty rows for twelve applications; continue on a second copy. Signed by the validation lead.

Ref	Application	GxP	Validation state	System owner	Validation lead
VAL-01	EXAMPLE — delete. MES (GMP), production	GMP	Validated with open deviations		

What this phase hands to phase E

Six numbers, and every one of them becomes a row in `O29 Gap`, which is empty today.

From	The difference	Count
`027`	Rows with no qualification reference	12
`027`	Components past vendor support	2
`049`	Applications never recovery-tested	9
`041`	Controls absent	2
`048`	Environments with no verified refresh	4
`046`	Blocks with no testable acceptance criterion	7

Phase E's first act is to write those down. The gap sheet, the gap relationship and the gap report are all empty, and `ZZ Checks` reports `Gaps overdue: 0` over the top of it [8].

THE CHAPTER IN TEN LINES

- Infrastructure is qualified; the system on top of it is validated. Different owner, different evidence, different review.
- The repository has no column for either state. Add four qualification columns to O27; the validation state lives in O67.
- One platform qualification cited by many application validations is where the saving is. Count the citations.
- Four components are recorded in buildings they cannot be in, and both location joins agree only by arithmetic.
- LOC-08 Validated Cloud Region has a Scale of 82 people. A region is not staffed, and 746 employees have no location at all.
- O48 Environment says production holds no personal data and development does. With SEC-07 Partial and EP-05 unmasked, read it literally.
- R07 pairs application n with component n for all twelve rows and records only one of four environments.
- Nine of twelve applications have never been recovery-tested, and Tier 2 holds objectives from 15 minutes to 24 hours.
- The Safety Database carries the expedited reporting clock and is the lowest tier in the estate.
- MSR-07 says 22 per cent of components are within support; O27 says 83 per cent. MSR-13 counts 53 tests across 12 applications.
- SEC-01 and SEC-03 are absent, POL-03 is breached, and the enforcement point for the breached policy is a product nobody has bought.
- O46 has two columns headed Ref; the second holds T01 to T07, a key space nothing in the workbook resolves against.
- Phase D hands Phase E six counted differences. They are the first six rows of O29 Gap, which is empty.

SOURCES FOR THIS CHAPTER

- **1.** Pharmaceuticals_TOGAF_Metamodel.xlsx, sheet O27 Physical Technology Compone, headers row 4, records rows 5-16 (twelve components; Location, Support end, Position and Vendor columns as cited)
- **2.** Pharmaceuticals_TOGAF_Metamodel.xlsx, sheet O12 Location, rows 5-12 (eight sites; LOC-08 Validated Cloud Region, Scale 82 people) and sheet O09 Organization Unit, rows 5-16 (headcount column, twelve units)
- **3.** Pharmaceuticals_TOGAF_Metamodel.xlsx, sheet O48 Environment, rows 5-8 (four environments; Personal data present column)
- **4.** Pharmaceuticals_TOGAF_Metamodel.xlsx, sheet O49 Resilience Tier, rows 5-16 (twelve applications; Tier, Recovery objective and Tested columns)
- **5.** Pharmaceuticals_TOGAF_Metamodel.xlsx, sheets O25 Technology Service rows 5-16, O26 Logical Technology Componen rows 5-11, O28 Platform Service rows 5-12, O46 Technology Building Block rows 5-11 (two columns headed Ref, the second holding T01-T07) and O47 Reference Model Entry rows 5-15
- **6.** Pharmaceuticals_TOGAF_Metamodel.xlsx, sheets O41 Security Control rows 5-12, O64 Security Policy rows 5-8, O65 Enforcement Point rows 5-9, and R17 Policy enforced at Enforcem rows 5-8
- **7.** Pharmaceuticals_TOGAF_Metamodel.xlsx, sheets O50 Network Link rows 5-14, R07 Application deployed on Tec rows 5-16 and R20 Application located at Loca rows 5-16
- **8.** Pharmaceuticals_TOGAF_Metamodel.xlsx, reports RPT10 Technology Currency and RPT11 Security Position (each ending in a sentence row in the Ref column), RPT01 Dashboard, ZZ Checks, and sheet O04 Measure rows 5-20
- **9.** Pharmaceuticals - a starter pack for enterprise architects (volume 1), chapter 13, GxP, validation and data integrity - the qualification and validation distinction, the validation deliverable set, periodic review and the restore evidence, the five cloud conditions, and the capability that ships switched off
- **10.** European Commission, EudraLex Volume 4 - EU GMP guidelines index (Annex 11 Computerised Systems, revision January 2011, with no entry for a revision in force), as cited in volume 1 chapter 13 https://health.ec.europa.eu/medicinal-products/eudralex/eudralex-volume-4_en
- **11.** goLegal, New POPIA Regulations: Processing of health information, 10 April 2026 - legal-commentary secondary source for health information as special personal information and the section 72 restriction on cross-border transfer, as cited in volume 1 chapter 13 https://www.golegal.co.za/popia-regulations-health/

CHAPTER 8

Opportunities and Solutions

The gap sheet is empty, the gap relationship is empty, the gap report is empty - and the dashboard reports no gaps overdue. Your first act in this phase is to write down the differences the repository has already recorded everywhere else.

What this phase is for, and the sheet that is empty

Phase E turns a baseline and a target into a set of options with costs, and then into work packages and transition states somebody can fund [16]. In this repository it starts from a hole.

`O29 Gap` has seven column headings — `Ref`, `Gap`, `Domain`, `Building block`, `Severity`, `Status`, `Due` — and **no rows**. `R12 Gap closed by Work Package` has five headings and no rows. `RPT07 Gap Analysis` therefore computes over nothing and shows nothing. And `ZZ Checks` reports `Gaps overdue: 0` against a threshold of `0`, which reads as a clean bill of health and means precisely nothing: no gap can be overdue when no gap exists [1][2].

Meanwhile `O31 Deliverable` marks `DLV-13 Consolidated Gaps` as **`Approved`** [11]. Somebody signed off a gap analysis with no gaps in it. That is the first thing to say out loud in this phase, and it is not a joke at anybody's expense — it is the commonest state of a repository that was populated from a template rather than from an estate.

> **You do not discover gaps in Phase E. You collect them.** Every gap is a difference *already recorded* somewhere in the repository — a `Current` against a `Target`, a `Position: Absent`, a `Tested: Never tested`, a `Lineage: Absent`, a `Managed: Unmanaged`, an `Evidence: In breach`. Phase E's work is to transcribe those differences into one sheet, give each a severity that means something and a date that can expire, and then decide what closes it. If you find yourself inventing a gap, you have left the evidence behind.

Two invented organisations appear in this volume. **Calderon Therapeutics** is the company in the repository you were handed; **Veldtkruin Group** is volume 1's worked example. Both are fictional and their data must never be mixed.

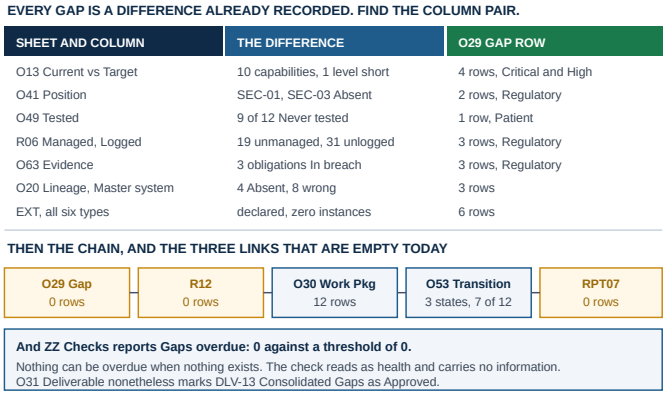


Figure 8.1 The derivation. **A gap is a column pair, not a judgement** — a `Current` against a `Target`, an `Absent` against a policy, a `Never tested` against a clock. Three links in the chain are empty today, and the dashboard reports the emptiness as a clean result.

Before you open a sheet

ANSWER THESE BEFORE THE FIRST WORKSHOP

How many systems can the quality function revalidate in one quarter, alongside everything else it owes? Who told you, and when?

Which two of the five alternatives in `O54` are currently live, and who thinks the matter is settled?

What is the real change-freeze window in this company, and what date does it start?

The steps

Step 8.1 — Derive the gap set from differences already recorded

Produce. A single list of gaps, each one traceable to a cell in another sheet.

You need. The hand-offs from items 4 to 7. No workshop. This step is a desk exercise and it should take a day.

Where it lands.

> **`O29 Gap`**
>
> `Ref` takes `GAP-001` upward. `Gap` is one sentence naming the difference, not the remedy. `Domain` takes `Business`, `Data`, `Application` or `Technology`, matching `O55 Building Block`'s `Domain` column. `Building block` resolves against **`O55`**, the 37-row union sheet — not against `O35`, `O44`, `O45` or `O46`, which is where `R09` and `R13` already point. `Severity` and `Due` come in steps 8.2 and 8.3.

How. Work the table below, sheet by sheet, and write one row per difference.

Sheet	The difference already recorded	Gaps it yields
`O13` / `RPT03`	Every one of the ten level-1 capabilities is one maturity level short, and every `Disposition` reads `Sustain` [8]	4 — the `Critical` and `High` capabilities sitting at level 2
`O09`	No Quality, Regulatory Affairs, Pharmacovigilance, Manufacturing or Clinical Development unit	5
`O10` / `O11`	Eighteen generic corporate titles, no regulated role, every `Performed by` reading `Person`	1, against `O70`
`O16`	Twenty processes named `<value stream> — stage N`, under five genuinely good value streams	1 per value stream, 5
`O20` / `O43`	Eight `Master system` values wrong, four `Lineage: Absent`, and `O43`'s `Golden source` disagreeing on every entity it names	3
`EXT`	Six industry object types declared, zero instances of any of them	6
`R06`	19 of 36 interfaces `Unmanaged`, 31 unlogged, 6 direct database reads [10]	3
`O23`	Four rows whose `Criticality`, `Health` and `Disposition` contradict each other; `Owner` holding organisation units	2
`O27`	Two components past vendor support, twelve with no qualification reference	2
`O49`	Nine of twelve applications `Never tested`	1
`O41` / `O65`	`SEC-01` and `SEC-03` `Absent`; `EP-03 API gateway` reading `Product not yet selected`	3
`O32` / `RPT12`	Five standards in `Breach`, four `Not assessed`, seven `Deprecated` — four of those seven recorded `Conformant` or `Partial` — and not one GxP standard in the sheet [10]	2

Signed by. Nobody yet. This is an architect's desk output and signing it before step 8.2 invites a severity argument you cannot win.

Done when. Every row of `O29` names the sheet and column it came from, in the `Gap` text itself. Forty-odd rows is a normal first pass on this repository.

Shows up in. `RPT07 Gap Analysis`, which goes from zero rows to the number you wrote.

Goes wrong when. A gap is written as a solution. "Implement an integration platform" is a work package; "19 of 36 interfaces are unmanaged and 31 carry no logging" is a gap.

Look it up. Volume 1, chapter 19, "The checks that take an afternoon".

Step 8.2 — Give each gap a severity that a regulator would recognise

Produce. A severity against every `O29` row, using a scale defined in writing before you apply it.

You need. The gap list from 8.1. `O63 Control Objective`. The inspection-response obligation.

Where it lands.

> **`O29 Gap`**
>
> `Severity` is a free-text column, so you are defining the vocabulary. Use four values and write them down: `Regulatory` — an obligation is unevidenced or in breach today; `Patient` — the difference could reach a patient or a subject; `Operational` — it costs money or time; `Hygiene` — it is untidy. Do not use `High`, `Medium` and `Low`. They compress exactly the distinction this industry needs.

How.

1. Mark `Regulatory` first. `O63` records `CO-03`, `CO-04` and `CO-05` as `In breach` while `O59 Compliance Assessment` — the sheet that would hold the evidence — is empty [10].
2. Mark `Patient` second. The unmasked non-production environments, the untested safety database recovery, and the subject record classified `Internal` are all in this class.
3. Check `O04 Measure` against the sheets. `MSR-01 Obligations with evidence` records a baseline of **54%** while `O63` shows two of seven evidenced, which is 29% [10]. Two numbers, one measure. The discrepancy is itself a `Regulatory` gap, because the 54% is what reaches a board.
4. Count the severities and publish the counts. A gap list that is 80% `Hygiene` has not been read.
5. Resist ranking. Severity is a class, not an order; the sequencing argument belongs in step 8.6 where capacity decides it.

Signed by. `GB-01 Architecture Board` for the scale, and the quality function for every `Regulatory` row. The repository has no quality body: `O57 Governance Body`'s five bodies are the Architecture Board, the Investment Committee, the Data Governance Council, the Executive Committee and the Architecture Function. None of them can accept a regulatory severity [11].

Done when. Every row has one of the four values, the scale is written into the phase deliverable, and the `Regulatory` rows have been read by somebody from quality.

Shows up in. `RPT07 Gap Analysis`, column `Severity`. A healthy first pass on this repository shows roughly a dozen `Regulatory`, a handful of `Patient`, and the rest split.

Goes wrong when. `Regulatory` is used for anything touching a regulated system. It means the obligation is unevidenced *today*, which is a much smaller and much more useful set.

Look it up. Volume 1, chapter 16, "Inspection response" and "Who inspects, and what they ask for".

Form 8-A — Gap register, first pass

Six of the seven columns of `O29 Gap`, in sheet order. `Status` is set to `Open` on entry and typed straight into Excel. Every `Gap` cell must name the sheet and column the difference came from. A first pass on this repository needs about forty rows; continue on a second copy of this form. Signed by the Chief Enterprise Architect, with quality countersigning the `Regulatory` rows.

Ref	Gap	Domain	Building block	Severity	Due
GAP-001	EXAMPLE — delete. O49: nine of twelve applications never recovery-tested	Technology	TBB-01	Patient	

Step 8.3 — Put a date in `Due` that can actually expire

Produce. A date against every gap, derived from an external obligation wherever one exists.

You need. The regulatory commitment calendar. The inspection history. `O52 Implementation Factor`.

Where it lands.

> **`O29 Gap`**
>
> `Due` is a date column with no rows in it. Fill it from the obligation, not from the plan. A gap against `CO-04 Pharmacovigilance reporting` is due before the next periodic safety report, not when a work package happens to land. Note while you are here that `O60 Governance Decision` has **no date column at all**, so no decision in this repository can be overdue either [3].

How.

1. For each `Regulatory` gap, find the external date: a renewal, a periodic report, a commitment, an inspection window. Registration renewal in South Africa is due six months before expiry, and variations cannot be in flight during the renewal, so a renewal date blocks a change window as well as setting one [15].
2. For `Patient` gaps, set the date from exposure, not convenience. An unmasked non-production refresh runs on a cycle; the date is before the next refresh.
3. For `Operational` and `Hygiene`, set a date you intend to move and say so.
4. Re-run the `ZZ Checks` line. `Gaps overdue` is now a number that can be non-zero, which is the point of the check and the first time it has meant anything [2].
5. Add a `Date set by` note in the `Gap` text where the date came from an obligation. In six months somebody will ask why.

Signed by. `GB-01 Architecture Board`, with the regulatory affairs manager for every date derived from an obligation — a role `O11 Role` does not carry.

Done when. No `Due` cell is blank, and at least the `Regulatory` rows cite the obligation the date came from.

Shows up in. `ZZ Checks`, line `Gaps overdue`, and `RPT07 Gap Analysis`, column `Due`. If the number is still zero after this step, either the dates are all in the future — possible — or they were all set to the end of the programme, which is the usual answer.

Goes wrong when. The dates are set from the roadmap. Then the gap sheet cannot ever embarrass the plan, which is the only thing it is for.

Look it up. Volume 1, chapter 7, "The regulatory commitment calendar" and "Medicine registration renewal".

Step 8.4 — Delete the trading peak and put the real windows in

Produce. A corrected `O52 Implementation Factor` naming the windows this company actually has.

You need. The quality calendar. The inspection history. The regulatory commitment calendar.

Where it lands.

> **`O52 Implementation Factor`**
>
> Eight rows, columns `Ref`, `Implementation factor`, `Category`, `Deduction`. `IF-03 Peak trading freeze` is categorised `Constraint` with the deduction `No change in the peak window` [4]. **That is retail vocabulary in a pharmaceutical repository, and it is load-bearing** — a `Constraint` shapes the plan. A reader who inherits it plans around a trading peak this company does not have, and does not plan around the two windows it does: the **annual product quality review cycle**, which runs all year and consumes the same quality people who would validate a new system, and the **inspection window**, which is not a date you choose. `O07 Constraint` repeats the error at `CON-06 Change freeze windows` — "Peak periods are closed to change" [4].

How.

1. Replace `IF-03` with two rows. The first is the product quality review cycle: reviews are normally annual per product, and the obligation sits with production and quality control jointly [14]. Count the products, multiply by the quality person-days each review takes, and divide by the year.
2. The second is the inspection window. Record the last inspection date, the deficiency response deadline the company works to, and the open commitments. An inspection response has a hard deadline and it consumes the same people.
3. Correct `CON-06` in `O07 Constraint` to match, and note that `O07` has no ArchiMate home — when item 16 draws the repository, those eight rows become `Requirement` elements.
4. Keep `IF-04 Regulatory deadline` and `IF-08 Site access windows`. Both are real: a fixed external date, and physical work limited to plant shutdowns.
5. Add the mass data-change event. A Single Exit Price adjustment is a change across every pack with a hard effective date, and 2026 carried two of them — this sector's equivalent of a tax-rate change [13].

Signed by. The quality head for the review cycle, the Responsible Pharmacist for anything touching release, and `GB-02 Investment Committee` for the plan consequence. The first two roles are not in the repository.

Done when. No row of `O52` or `O07` refers to a trading peak, and both real windows carry a date range and a person-day figure.

Shows up in. Nothing computes over `O52` among the thirteen reports, which is why the error has survived. It shows up in the roadmap — or it does not, and that is the failure.

Goes wrong when. The freeze is recorded as a date range with no owner. A window with no owner is negotiated away in the first steering meeting.

Look it up. Volume 1, chapter 7, "The Annual Product Quality Review"; chapter 4, "The adjustment is an event, and 2026 had two of them".

Form 8-B — Implementation factors, corrected

All four real columns of `O52 Implementation Factor`, in sheet order, plus the window dates you must record somewhere and this sheet has no column for — put them in the `Deduction` cell. Eight to twelve rows is a first pass. Signed by the quality head and the Investment Committee.

Ref	Implementation factor	Category	Deduction, with dates
IF-03	EXAMPLE — delete and replace. Peak trading freeze	Constraint	No change in the peak window
IF-03a	Annual product quality review cycle	Constraint	

Step 8.5 — Record quality assurance capacity as the binding constraint

Produce. One number: how many computerised-system validations or revalidations the quality function can absorb in a quarter, alongside everything else it owes.

You need. The quality head. The validation lead. The product quality review workload from step 8.4. `O62 Delivery Risk`.

Where it lands.

> **`O52 Implementation Factor`**
>
> Replace `IF-06 Concurrent capacity unknown`, categorised `Assumption` with the deduction "Plan assumes capacity that is unrecorded" [4], with a measured figure. An assumption is what you write before you have asked. Once asked, it is a `Constraint`, and it is the one that decides the shape of every option in step 8.6.

How.

1. Ask the quality head for the figure directly. Expect a small number — two or three systems a quarter in a mid-size estate — and expect it to be lower than anyone on the programme assumes.
2. Write the arithmetic out loud, because the point is not the number but why it is the number: the people who would write the user requirements, execute the qualification scripts, review the audit trails and approve the change records are the same people assembling the product quality reviews and answering the inspection [12].
3. Convert it into a plan rule: at most *n* validation events per quarter, and a revalidation counts as one.
4. Test the rule against the estate. `O23 Application Component` holds four systems marked `Replace` or `Retire` and nine GxP systems in all. An option that revalidates nine systems in one quarter is not an option.
5. Record the figure, the date it was given and the person who gave it, in the `Deduction` cell. Capacity changes; an undated figure is defended for years.

Signed by. The quality head. Nobody else can give this number, and no architect should estimate it.

Done when. `IF-06` carries a number, a date and a name, and the number has been repeated back to the Investment Committee in a sentence beginning "the year allows".

Shows up in. `O62 Delivery Risk`, where `RSK-03 Delivery capacity below plan` is recorded as `Likely` with impact `High` [7]. After this step it stops being a risk and becomes an input.

Goes wrong when. The figure is taken from the delivery function. Delivery capacity and quality assurance capacity are different numbers, and in this industry the second one is smaller and binds first.

Look it up. Volume 1, chapter 18, "6. The roadmap and what the year allows"; chapter 19, "The ninety-day shape".

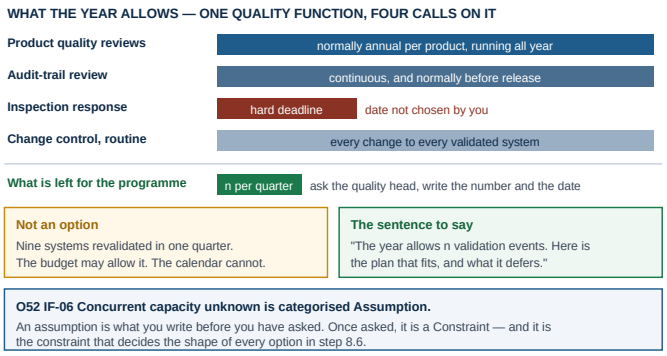


Figure 8.2 Quality assurance capacity, not budget, is what the year allows. **The people who would validate a new system are the same people assembling the product quality reviews**, reviewing audit trails and answering the inspection — so the programme gets what is left, and the number is small.

Step 8.6 — Score the five alternatives against revalidation load, not against cost

Produce. A scored comparison of the five rows of `O54 Architecture Alternative`, with a revalidation-load column the sheet does not have.

You need. The capacity figure from 8.5. The gap list. `O54`. `O66 Architecture Increment`.

Where it lands.

> **`O54 Architecture Alternative` — three new columns, not a new sheet**
>
> Five rows, columns `Ref`, `Alternative`, `Description`, `Cost`, `Recommendation`. Add `Systems revalidated`, `Quality person-days` and `Quarters required` to the right of `Cost`. Without them the sheet compares options on money alone, and money is not the binding constraint.

How.

1. Fill the three columns for each alternative. `ALT-D Big bang replacement` at $88m is already `Rejected`; after this step it is rejected for the right reason, which is that it cannot be staffed rather than that it is dear.
2. Look hard at `ALT-E Compliance led` at **$31m**, marked `Partly adopted`. Compliance-led options look cheap because they count mechanisms and not validation events. Price the revalidation and the number moves.
3. Compare `ALT-B Foundations first` at **$46m**, marked `Recommended`, against the sum of the work packages. `O30 Work Package`'s twelve rows total **$84.3m** [6], which is nearly twice the recommended option and close to the rejected big-bang figure. The roadmap in the repository costs an alternative nobody chose.
4. Divide the quality person-days by the quarterly capacity and read off the duration. That number, not the cost, is what you present.
5. Write the recommendation as one sentence with the constraint in it: the option is chosen because it fits the validation capacity of the year, and it would change if the capacity changed.

Signed by. `GB-02 Investment Committee`, which holds G0 funding under `O58 Governance Gate`, with the quality head countersigning the revalidation load.

Done when. All five rows carry the three new columns, and the recommended option's quarters figure has been stated to the committee.

Shows up in. Nothing computes over `O54`. Carry it in `DLV-14 Architecture Roadmap`, which `O31 Deliverable` correctly marks `Draft` [11].

Goes wrong when. The comparison is run on cost and the quality function sees it for the first time at the steering committee. Then the option is re-opened after funding, which is the most expensive sequence available.

Look it up. Volume 1, chapter 13, "Seven pieces of folklore, punctured"; chapter 18, "4. What it costs to be wrong".

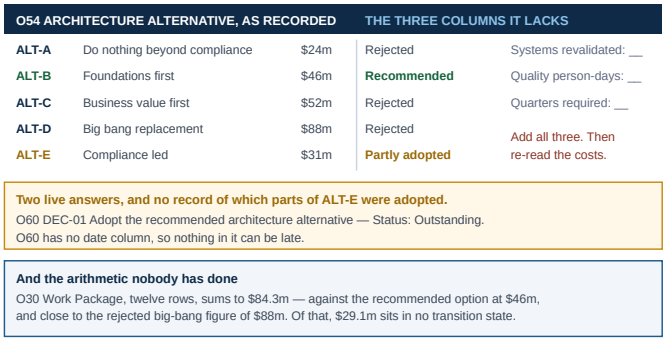


Figure 8.3 The five alternatives as the repository records them, the three columns that would make the comparison real, and the arithmetic underneath. **The roadmap in this repository costs an alternative nobody chose.**

Form 8-C — Alternative scoring, with the three columns `O54` lacks

`O54 Architecture Alternative`'s `Ref` and `Cost`, plus the three columns step 8.6 adds. These are **new columns on `O54`, not a new sheet** — this volume has agreed that `O67`–`O70` are the only sheets it creates. Five rows is the whole sheet. Signed by the Investment Committee and the quality head.

Ref	Alternative	Cost	Systems revalidated	Quality person-days	Quarters required
ALT-E	EXAMPLE — delete. Compliance led	$31m			

Step 8.7 — Close `DEC-01`, and remove the second live answer

Produce. A dated decision adopting one alternative, and a corrected `Recommendation` column with exactly one live value.

You need. Step 8.6's scoring. `O60 Governance Decision`. `GB-02 Investment Committee`.

Where it lands.

> **`O54 Architecture Alternative` and `O60 Governance Decision`**
>
> `O54` marks `ALT-B Foundations first` `Recommended` **and** `ALT-E Compliance led` `Partly adopted`. Two live answers, and no record of which parts of `ALT-E` were adopted or by whom. `O60 Governance Decision` then records `DEC-01 Adopt the recommended architecture alternative` as **`Outstanding`**, together with `DEC-02 Select the integration product`, `DEC-03 Fund compliance mechanisms` and `DEC-04 Designate data owners` [3]. Four outstanding decisions, and the sheet has no date column, so none of them is late.

How.

1. Add `Date raised`, `Date due` and `Date taken` to `O60`. Three columns, and they turn a list into a register.
2. Resolve `ALT-E` explicitly. Either the compliance-led parts are folded into `ALT-B` and `ALT-E` becomes `Rejected`, or they are a separate increment with their own name. `Partly adopted` with no list of the parts is not a decision.
3. Take `DEC-01` to the Investment Committee with the quarters figure, not the cost figure.
4. Note that `DEC-02 Select the integration product` appears three times in the repository — as a decision here, as `IF-01 Vendor selection not made` in `O52`, and as `RSK-01` in `O62` at likelihood `Certain` [3][4][7]. One decision, three homes, and it has been outstanding long enough to become all three.
5. Write the consequence of deferring each outstanding decision in one sentence. That sentence is what gets the decision taken.

Signed by. `GB-02 Investment Committee` for `DEC-01` to `DEC-03`, `GB-03 Data Governance Council` for `DEC-04`.

Done when. `O54` has one live recommendation, `O60` has three date columns, and every outstanding decision has a due date and a named consequence.

Shows up in. No report reads `O60`, which is part of why four decisions have sat outstanding. Put the four into the `DLV-14 Architecture Roadmap` front page as a dated decisions table.

Goes wrong when. The architect adopts an option by writing it into the roadmap. The adoption is a governance act; a roadmap is evidence that somebody drew it.

Look it up. Volume 1, chapter 18, "5. The target and the transition".

Step 8.8 — Re-derive the work packages from the gaps, and populate `R12`

Produce. A work-package set where every package closes a named gap, and `R12 Gap closed by Work Package` with a row for every gap.

You need. The gap list. `O30 Work Package`. `O55 Building Block`. `R13`.

Where it lands.

> **`R12 Gap closed by Work Package`**
>
> Columns `Gap`, `Gap name`, `Severity`, `Work package`, `Package name`. Empty today. One row per gap, and a gap with no work package is the most useful row in the sheet — it is the thing nobody has funded [1].

How.

1. Check `R13 Work Package delivers Build` before you trust it. All 37 building blocks are allocated, three or four to each of the twelve packages, by position: `WP-01` gets `BBB-01`, `DBB-03`, `ABB-05` and `TBB-07`; `WP-04 Technology currency` gets `DBB-06 Regulatory Submission mastering` and `ABB-08 QMS target` [5]. It is arithmetic, not a plan.
2. Check `R09 Requirement satisfied by Bu` the same way. Thirty-two requirements map one-to-one onto the first thirty-two rows of `O55`, which leaves `TBB-03` to `TBB-07` — CDISC standards, the IDMP data model, cold chain telemetry, the scientific data lake and the secure enclave — satisfied by no requirement at all. And the `Traced` column reads `No` on exactly every fourth row, including `REQ-001 Evidence GMP / GCP / GLP / GVP`, which is the right first requirement in the sheet [5].
3. Rebuild both from the gap list. A building block with no gap behind it does not need a work package.
4. Leave the honest blanks. Where a `Regulatory` gap has no package, write the gap reference in `R12` with the work package cell empty and count those rows.
5. Reconcile `O30 Work Package` with `O56 Project`. They are the same twelve items twice, same names, same costs, same dates — and `WP-08 Process automation` is `Approval: Not started` with `Delivery: Delivered` [6].

Signed by. `GB-01 Architecture Board` for the mapping, `GB-02 Investment Committee` for the packages that carry money.

Done when. `R12` has one row per gap, the rows with no package are counted, and no work package exists without a gap behind it.

Shows up in. `RPT07 Gap Analysis`, whose `Work package` column fills in, and `RPT06 Requirements Traceability`, where the eight requirements reaching no building block stop being a modulo pattern and become eight real questions.

Goes wrong when. The existing twelve packages are kept and the gaps are mapped onto them. Then the plan is unchanged and the gap sheet is decoration.

Look it up. Volume 1, chapter 18, "3. The findings".

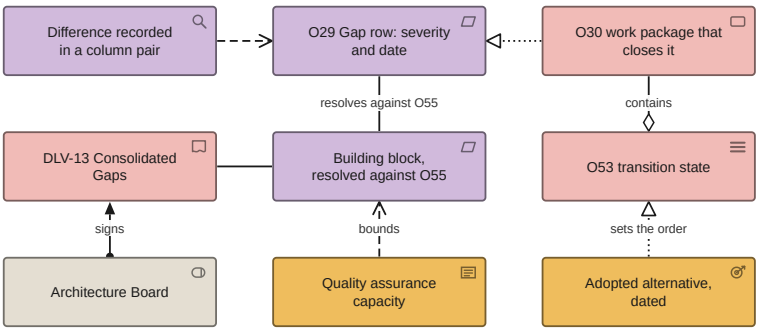


Figure 8.4 The Phase E chain, with the one element the repository has no sheet for. **Quality assurance capacity bounds the chain**, and there is no column anywhere in the workbook to record it — which is why step 8.5 puts it in `O52 Implementation Factor` as a measured constraint rather than an assumption.

Step 8.9 — Reconcile the roadmap cost and allocate every package to a transition state

Produce. A transition-state set whose work packages sum to the cost of the alternative you chose, with nothing unallocated.

You need. `O53 Transition Architecture`. `R15`. `O66 Architecture Increment`. The adopted alternative.

Where it lands.

> **`O53 Transition Architecture`**
>
> Three states, and the `Exit criteria` column reads **`5 exit criteria`**, `6 exit criteria`, `7 exit criteria` — a count, not criteria [6]. Replace each with the criteria themselves, written so a reviewer at `GT-2 G2 Build complete` can mark them. `R15 Transition contains Work Pa` allocates seven of the twelve packages; `WP-06`, `WP-08`, `WP-09`, `WP-10` and `WP-12` are in **no** transition state, and they total **$29.1m** [6].

How.

1. Write the exit criteria out. Eighteen sentences across three states, each naming evidence and a signatory.
2. Allocate the five orphaned packages or delete them. `$29.1m` with no transition state is either a plan nobody drew or a budget nobody questioned.
3. Reconcile the totals. Twelve packages at **$84.3m** against `ALT-B` at **$46m**; `TA-1` alone carries $32.3m [6]. Say which figure is going to the committee and why.
4. Map the three transition states onto the six rows of `O66 Architecture Increment`. There is no mapping today: three states, six increments, and `INC-3 Compliant by mechanism` and `INC-5 Analytics and evidence` appear in no state at all [6].
5. Check each state against the capacity rule from 8.5. A state whose packages imply more validation events than the quarters allow is not a state; split it.

Signed by. `GB-02 Investment Committee` for the money, `GB-01 Architecture Board` for the exit criteria, the quality head for the validation load per state.

Done when. Every package sits in exactly one state, every state has written exit criteria, the increments map onto the states, and the sum reconciles to the adopted alternative or the difference is explained in one sentence.

Shows up in. `RPT08 Roadmap`, which today repeats `O30` and ends with the sentence "3 work packages are past their end date." sitting in the `Ref` column [9]. Three of twelve packages are past their end date before the programme has been approved.

Goes wrong when. The transition states are renamed increments. They are different objects: a state is a stable configuration the business can run in, an increment is what a delivery adds.

Look it up. Volume 1, chapter 18, "5. The target and the transition".

Step 8.10 — Move the certainties out of the risk sheet

Produce. A corrected `O62 Delivery Risk` carrying only things that might happen, with the certainties moved to `O52` as factors.

You need. `O62`. `O52`. The decisions register from 8.7.

Where it lands.

> **`O62 Delivery Risk`**
>
> Eight rows, columns `Ref`, `Risk`, `Likelihood`, `Impact`, `Owner`. **Three of the eight read likelihood `Certain`**: `RSK-01 Integration product not selected`, `RSK-07 Benefit not measurable at G3` and `RSK-08 Unmanaged interfaces proliferate` [7]. A certainty is not a risk. `RSK-01` is already recorded as `IF-01 Vendor selection not made` in `O52` and as `DEC-02` in `O60`, so it has three entries and no owner who can close it.

How.

1. Move all three `Certain` rows to `O52 Implementation Factor` with category `Dependency` or `Constraint`, and delete them from `O62`.
2. For `RSK-07 Benefit not measurable at G3`, note what it actually says: the governance gate `GT-3 G3 Benefit` tests a stated benefit against a date, and the programme already knows it cannot produce one. That is a measure-definition gap, and it belongs in the gap sheet.
3. For `RSK-08 Unmanaged interfaces proliferate`, the certainty is a consequence of `EP-03` having no product. One cause, two symptoms.

4. Re-rate what is left. Five rows, of which `RSK-04 Legacy platform failure before renewal` is `Possible` with impact `Critical` and is the only one that maps to a dated external fact — the two components already past vendor support.
5. Add the risk the sheet does not carry: that a validation or revalidation slips and a transition state cannot exit. Owner, the quality head.

Signed by. `GB-01 Architecture Board`, with each row's named owner accepting it in the minute. Five of the eight owners in `O62` are executives who have not been asked.

Done when. No row of `O62` reads `Certain`, every remaining row has an owner who has accepted it, and the validation-slippage risk is on the sheet.

Shows up in. No report reads `O62`. Put the five surviving rows on the roadmap front page with their owners, because a risk nobody reads is not managed.

Goes wrong when. The certainties are re-rated to `Likely` to keep them on the risk sheet. That is a presentational fix and it leaves the issue unowned.

Look it up. Volume 1, chapter 19, "What ninety days will not achieve".

Step 8.11 — Consolidate, mark the deliverables honestly, and hand over

Produce. The three Phase E deliverables, a one-page honest statement of what the year allows, and a corrected `O31 Deliverable`.

You need. Everything from 8.1 to 8.10.

Where it lands.

> **`O31 Deliverable` and `O33 Architecture Contract`**
>
> `DLV-13 Consolidated Gaps` is `Approved` while `O29` was empty this morning; change it to `Draft` until the gap list is signed. `DLV-14 Architecture Roadmap` and `DLV-15 Work Packages` are already `Draft`, correctly. `O33 Architecture Contract` has eight rows against twelve work packages, so `WP-09` to `WP-12` have no contract, and six of the eight read `Not yet assessable` while `ACO-02` and `ACO-04` read `Non-compliant` [11].

How.

1. Publish the gap list, the alternative scoring and the transition states as one document. Three sheets, one narrative.
2. Write the honest year in a paragraph: this many validation events, these windows closed, these four decisions outstanding, this is what the year allows.
3. Correct `O31` to match reality. A deliverable marked `Approved` that nobody can produce is worse than one marked `Draft`.
4. Create the four missing architecture contracts, or say which packages are proceeding without one.
5. Check the orphans before you finish. `RPT13 Orphan Analysis` carries **two** records — `A11 Statistical Computing` and `A12 Medical Information`, both with no business service mapped — and then a closing sentence in the `Ref` column [9]. Two orphans in a twelve-application estate is plausible; a report that is two rows and an aphorism is not an analysis.
6. Count your own repository before quoting its headline. `ZZ Checks` and `RPT01 Dashboard` both claim **850 records**; the sum across the 64 `O` sheets plus `EXT` is **814**, and the 36 difference is exactly the row count of `R06 Application interfaces with` [2].

Signed by. `GB-01 Architecture Board` for the architecture, `GB-02 Investment Committee` for the roadmap, the quality head for the capacity statement.

Done when. `O29`, `R12`, `O52`, `O53`, `O54` and `O60` are all populated and consistent, `O31` reflects what exists, and the capacity sentence has been said in a room with the Investment Committee in it.

Shows up in. `RPT07 Gap Analysis` and `RPT08 Roadmap`, which stop being decoration, and `ZZ Checks`, whose `Gaps overdue` line finally computes over something.

Goes wrong when. The gap list is handed over without the capacity figure. Then Phase F plans the gaps in the order of their severity, which is the one ordering the quality calendar cannot support.

Look it up. Volume 1, chapter 19, "How to run the workshops" and "What to say when you do not know".

What this phase hands to phase F

Four things, and none of them is a Gantt chart: a populated `O29 Gap` with dates that can expire; one live alternative in `O54` with a dated `DEC-01` behind it; three transition states with written exit criteria and every work package allocated; and one number from the quality head that bounds the year. Phase F sequences those. It does not revisit them.

THE CHAPTER IN TEN LINES

- O29 Gap, R12 and RPT07 are all empty, and ZZ Checks reports Gaps overdue: 0 over the top of them.
- O31 Deliverable marks DLV-13 Consolidated Gaps as Approved. Somebody signed a gap analysis with no gaps in it.
- You do not discover gaps in this phase. Every gap is a difference already recorded in a column pair somewhere else.
- Use Regulatory, Patient, Operational and Hygiene as severities. High, Medium and Low compress the one distinction that matters here.
- IF-03 Peak trading freeze is retail vocabulary and it is load-bearing. Replace it with the product quality review cycle and the inspection window.
- IF-06 records concurrent capacity as an Assumption. Ask the quality head, and it becomes the binding constraint.
- An option that revalidates nine systems in one quarter is not an option. Budget is not what the year allows.
- O54 has two live answers, ALT-B Recommended and ALT-E Partly adopted, and DEC-01 to adopt one of them is Outstanding.
- O60 Governance Decision has no date column, so none of its four outstanding decisions can be late.
- The twelve work packages sum to $84.3m against a recommended option of $46m, and $29.1m sits in no transition state.
- R13 allocates all 37 building blocks to work packages by position, and R09 leaves TBB-03 to TBB-07 satisfied by no requirement.
- Three of the eight rows in O62 read likelihood Certain. A certainty is an issue, and RSK-01 already has two other homes.
- O53 records 5, 6 and 7 exit criteria as counts rather than criteria, and six increments map onto three states not at all.

SOURCES FOR THIS CHAPTER

- **1.** Pharmaceuticals_TOGAF_Metamodel.xlsx, sheets O29 Gap (seven headers at row 4, no records) and R12 Gap closed by Work Package (five headers, no records), and report RPT07 Gap Analysis (seven headers, no records)
- **2.** Pharmaceuticals_TOGAF_Metamodel.xlsx, ZZ Checks and RPT01 Dashboard (Gaps overdue 0 against a threshold of 0; Records 850 against 814 counted across the 64 O sheets plus EXT, a difference equal to the 36 rows of R06 Application interfaces with)
- **3.** Pharmaceuticals_TOGAF_Metamodel.xlsx, sheets O54 Architecture Alternative rows 5-9 and O60 Governance Decision rows 5-11 (four decisions Outstanding; no date column on the sheet)
- **4.** Pharmaceuticals_TOGAF_Metamodel.xlsx, sheets O52 Implementation Factor rows 5-12 (IF-03 Peak trading freeze, IF-06 Concurrent capacity unknown) and O07 Constraint rows 5-12 (CON-06 Change freeze windows)
- **5.** Pharmaceuticals_TOGAF_Metamodel.xlsx, sheets O55 Building Block rows 5-41 (37 rows with a Domain column), R09 Requirement satisfied by Bu rows 5-36 (a one-to-one position join; Traced reads No on every fourth row) and R13 Work Package delivers Build rows 5-41
- **6.** Pharmaceuticals_TOGAF_Metamodel.xlsx, sheets O30 Work Package rows 5-16 (twelve packages summing to $84.3m), O56 Project rows 5-16, O53 Transition Architecture rows 5-7, R15 Transition contains Work Pa rows 5-11 (seven of twelve packages allocated; the five unallocated summing to $29.1m) and O66 Architecture Increment rows 5-10
- **7.** Pharmaceuticals_TOGAF_Metamodel.xlsx, sheet O62 Delivery Risk rows 5-12 (three rows at likelihood Certain: RSK-01, RSK-07 and RSK-08)
- **8.** Pharmaceuticals_TOGAF_Metamodel.xlsx, sheet O13 Business Capability rows 5-104 and report RPT03 Capability Heatmap rows 5-14 (ten level-1 capabilities, every Gap value 1, every Disposition Sustain)
- **9.** Pharmaceuticals_TOGAF_Metamodel.xlsx, reports RPT08 Roadmap, RPT13 Orphan Analysis (two records, A11 and A12) and RPT06 Requirements Traceability - nine of the thirteen reports close with a prose sentence in the first data column
- **10.** Pharmaceuticals_TOGAF_Metamodel.xlsx, sheets O04 Measure rows 5-20 (MSR-01 at 54 per cent, MSR-08 at 31 per cent), O63 Control Objective rows 5-11, O59 Compliance Assessment (empty), O32 Standard rows 5-20 with RPT12 Standards Conformance, and R06 Application interfaces with rows 5-40
- **11.** Pharmaceuticals_TOGAF_Metamodel.xlsx, sheets O31 Deliverable rows 5-24, O33 Architecture Contract rows 5-12, O57 Governance Body rows 5-9 and O58 Governance Gate rows 5-8
- **12.** Pharmaceuticals - a starter pack for enterprise architects (volume 1), chapters 18 and 19 - quality assurance capacity rather than budget as the binding constraint, and the arithmetic of what the year allows
- **13.** Pharmaceuticals - a starter pack for enterprise architects (volume 1), chapters 4 and 7 - the Single Exit Price adjustment as a dated mass data-change event, and the portfolio and regulatory cadences
- **14.** PIC/S, PE 009-17 (Part I), Guide to Good Manufacturing Practice for Medicinal Products, 25 August 2023 - the product quality review as normally annual at section 1.10, as read and cited in volume 1 chapter 13. PE 009-18 came into force 24 September 2026 and was not read https://picscheme.org/docview/11332
- **15.** SAHPRA, Communication to Stakeholders HPA-RNW01-2025/26, 21 August 2025, Medicines Certificate of Registration Renewal Implementation Framework (renewal due six months before expiry; variations may not be in flight during the renewal), as cited in volume 1 chapter 16 https://www.sahpra.org.za/wp-content/uploads/2025/08/HPA-RNW01-2025_26-Medicines-Certificate-of-Registration-Renewal-Implementation-Framework.pdf
- **16.** The Open Group, TOGAF Standard - Architecture Development Method, public online edition. Identified, not retrieved: The Open Group's public pages are robots-disallowed to automated retrieval, so no edition or wording is claimed from it.

CHAPTER 9

Migration Planning

Three transition states, six increments, eighteen exit criteria nobody wrote down, and a validation calendar that decides all of it.

What Phase F is for, and what this repository says about it

Phase E decided what to do. Phase F decides **in what order, inside what the year allows, and with whose signature on the sequence.** The standard puts it plainly enough: the phase finalises the implementation and migration plan and assures that it is coordinated with the way the enterprise manages change [21]. It is the phase that turns a set of work packages into a dated plan somebody can be held to.

Read the sequencing sheets in order and the plan looks finished. `O53 Transition Architecture` holds three states — `TA-1 Governed and supportable`, `TA-2 Integrable`, `TA-3 Business visible` — each with a description and a work-package list [1]. `O30 Work Package` holds twelve packages with costs, start and end dates, an `Approval` state and a `Delivery` state [2]. `O56 Project` holds twelve projects with a lead. `O66 Architecture Increment` holds six increments, each with a `Lands` month. `R15`, `R13`, `R14` and `R23` wire them together, and `RPT08 Roadmap` prints the whole thing with a footer [3].

Now read it as an auditor would.

`O53`'s `Exit criteria` column reads **`5 exit criteria`**, `6 exit criteria`, `7 exit criteria`. Not the criteria. The count of them. Eighteen criteria are asserted and none is written down [1]. The column is the most consequential one in the sheet — it is the test that says a transition state has arrived — and it has been filled with its own cardinality.

`O30 Work Package` and `O56 Project` are **the same twelve items twice**: same names, same costs, same dates, with a `Lead` added [2]. `R14 Project delivers Work Package` then maps `PRJ-01` to `WP-01`, `PRJ-02` to `WP-02` and so on down to twelve, so the relationship sheet is an identity function over a duplicated list. A work package is a unit of architectural scope; a project is a funded vehicle with a manager, a budget and a steering line. Twelve of one against twelve of the other, in the same order, means nobody has yet decided how the work is actually to be bought.

Three dispositions in `O30` are impossible as written. `WP-08 Process automation` reads `Approval: Not started` and `Delivery: Delivered` — delivered without being approved [2]. `WP-07 Core platform renewal` reads `Approval: In flight` and `Delivery: Delivered`. And `WP-02`, `WP-04` and `WP-10` read `Delivery: Past end date`, which `ZZ Checks` counts as three and `RPT08`'s footer repeats as a finding [3].

> **CHECK THE ARITHMETIC ON "PAST END DATE" BEFORE YOU REPEAT IT**
>
> The earliest `End` date in `O30` is **`2029-02`** and the earliest `Start` is **`2027-02`** [2]. On a reading in late 2026, **nothing in this roadmap has reached its start date, let alone its end date.** `Past end date`, `In flight` and `Delivered` are strings somebody typed into a `Delivery` column, not computed facts — which means the dashboard's "3 work packages past end date" is right about the three rows and wrong about what those rows mean. Say this in the first steering pack. Your credibility for the rest of Phase F rests on being the person who checked.

Add the costs up. The twelve packages in `O30` total **$84.3m** [2]. `O54 Architecture Alternative` marks `ALT-B Foundations first` at **$46m** as `Recommended` and `ALT-E Compliance led` at **$31m** as `Partly adopted` [4]. The plan the sequencing sheets describe costs 1.8 times the alternative the investment committee was told it was buying, and `O60 Governance Decision` still records `DEC-01 Adopt the recommended architecture alternative` as `Outstanding` [4]. You are sequencing a programme that has not been chosen.

Calderon Therapeutics, the pharmaceutical company in this repository, is an invention. So is Veldtkruin Group in volume 1. Their data never mixes.

The overlay: a migration is a validation activity

This is the phase where the regulated estate stops being background and starts setting the plan.

Three sentences carry the whole chapter. **A migration is a validation activity, and the obligation is to verify that data are not altered in value and in meaning** [5]. Meaning, not bytes. A coded adverse event moved from one system to another without its dictionary version has changed what it means, because the term it carries was defined by a release that the target does not record [6]. And **a transition state that revalidates nine systems in one quarter is not a plan**, because the people who would write the user requirements, execute the qualification scripts, review the audit trails and approve the change records are the same small group already committed to the annual product quality review cycle. **Quality assurance capacity, not budget, is what the year allows.**

Nothing in the 105 sheets records that capacity. There is no column for validation effort, no column for validation state, no column for a periodic-review due date. `O52 Implementation Factor`, the sheet whose whole job is to record what constrains the plan, holds `IF-03 Peak trading freeze` as a `Constraint` with the deduction `No change in the peak window` [7]. That is retail vocabulary. A reader who inherits

that row plans around a trading peak this industry does not have, and does not plan around the two windows it does: the review cycle and the inspection window.

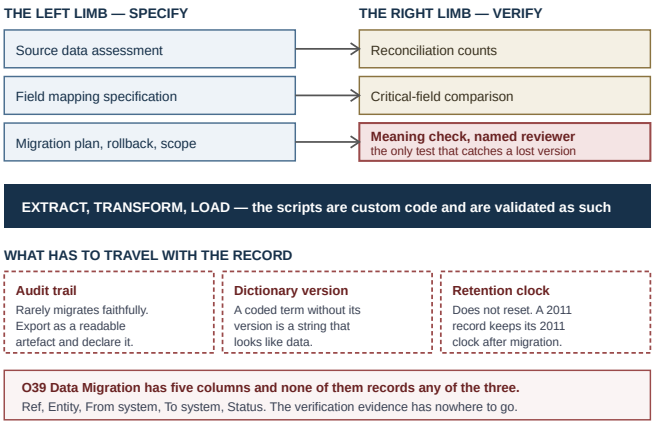


Figure 9.1 A migration is verified on **value and on meaning**. The three riders at the foot — audit trail, dictionary version, retention clock — are what carry meaning, and none of them survives a naive extract.

Look it up. Volume 1 chapter 10, *Data migration in a validated estate*, has the deliverable set and the six problems specific to this estate. Chapter 9's *The dictionary-version problem* has the five mechanisms by which a coded value changes meaning without anyone touching it. Chapter 13, *Decommissioning, which nobody budgets*, has the cost you are sequencing when you decide to leave a system read-only. Chapter 18, *6. The roadmap and what the year allows*, does the capacity arithmetic out loud.

BEFORE YOU SEQUENCE ANYTHING, ANSWER THESE

What is the current open deviation and change-control backlog, and who told you the number?

How many person-days a month does the quality function already owe to the review cycle?

When is the next inspection window, and which system would you least like to show?

Who signs a migration verification report in this company, by name?

The twelve steps

Step 9.1 — Separate the work packages from the projects, and price the plan honestly

Produce. A reconciled register in which `O30 Work Package` holds architectural scope and `O56 Project` holds funded vehicles, plus a one-page cost note.

You need. `O30`, `O56`, `R14`, `O54 Architecture Alternative`, `O33 Architecture Contract`, and thirty minutes with whoever holds the investment case.

Where it lands.

> **`O30 Work Package` and `O56 Project` — all columns, rows 5 onward**
>
> In `O30`, keep `Ref`, `Work package`, `Domains`, `Cost`, `Start`, `End`, `Approval`, `Delivery`, and treat `Cost` as architectural estimate, not budget. In `O56`, `Project` stops being a copy of the work-package name: a project may deliver several work packages or part of one, and `Work package` becomes a list. `Lead` must be a person who exists in `O10 Actor` — today the leads include `Internal Auditor` on `PRJ-11`, which is a role that cannot run a delivery.

How.

1. Add the twelve `Cost` values. The sum is $84.3m. Put it next to `ALT-B`'s $46m and `ALT-E`'s $31m and ask the investment committee which number is the plan.
2. Rewrite the `Delivery` column as of today's date, computed from `Start` and `End`, not typed. Every row becomes `Not started`, because every start date is in the future.
3. Resolve `WP-08` and `WP-07`: a package cannot be delivered ahead of its own approval. Decide which of the two cells is wrong and record the correction.
4. Re-map `R14` so it stops being an identity function. Where one project really does deliver one package, say so and leave it; where it does not, split the row.
5. Note the four work packages — `WP-09` to `WP-12` — that have no row in `O33`, because `GT-0 G0 Charter` tests that a contract exists [8].

Signed by. `GB-02 Investment Committee` for the cost reconciliation; `GB-01 Architecture Board` for the scope split. Neither body is a quality body, so neither can sign the validation effort — see step 9.7.

Done when. The cost total in your note matches the sum of `O30.Cost`, and the architecture board minute names which alternative in `O54` the plan implements.

Shows up in. `RPT08 Roadmap` reprints `O30` verbatim, so the corrected `Delivery` column appears there the same day. The footer should stop saying three packages are past their end date.

Goes wrong when. You accept the duplication as a modelling convention and carry twelve phantom projects into Phase G, where each one needs a contract, a gate and an assessment it will never get.

Step 9.2 — Write the eighteen exit criteria the sheet only counts

Produce. Written, testable exit criteria for each transition state, replacing the counts in `O53.Exit criteria`.

You need. `O53`, `R15 Transition contains Work Package`, `O55 Building Block` for the acceptance criteria of the blocks each state's packages deliver, `O04 Measure` for anything you intend to measure, and Form F-1.

Where it lands.

> **`O53 Transition Architecture — Ref`, `Transition architecture`, `Description`, `Work packages`, `Exit criteria`**
>
> `Exit criteria` becomes the criteria, one per line in the cell, numbered. Keep the count as the first line if you want the old value preserved. A criterion names the sheet and column that tests it: *`R06.Managed` reads `Managed` on every interface into `A08 QMS`* is a criterion. *Integration working well* is not.

How.

1. For each state, list the work packages from `R15` — `TA-1` has `WP-01`, `WP-02`, `WP-04`, `WP-05`; `TA-2` has `WP-03`; `TA-3` has `WP-07` and `WP-11` [1].
2. For each package, pull the building blocks from `R13` and their `Acceptance criteria` from `O55`.
3. Turn each acceptance criterion into an exit criterion by adding the test: which sheet, which column, what value.
4. Add one criterion per state that no building block gives you: **the validated systems touched by this state are in a validated state on the day the state closes**, evidenced from `O67 Validated System`.
5. Where a criterion cannot be tested from any sheet, write it anyway and mark it `Manual`. An untestable criterion you can see beats a count you cannot.

Signed by. `GB-01 Architecture Board` publishes the criteria; the validation lead signs the validated-state criterion, because no body in `O57 Governance Body` has standing to.

Done when. `O53.Exit criteria` contains eighteen or more sentences and no integers standing alone, and each one names a sheet, a column and a value or is marked `Manual`.

Shows up in. Nowhere automatically — no report reads `O53`. That is itself worth reporting: the test that closes a transition state is invisible to all thirteen reports.

Goes wrong when. You write the criteria as benefits. A benefit is tested at `GT-3 G3 Benefit` after operation. An exit criterion is tested on the day the state closes, by somebody looking at a cell.

Form F-1 — Transition state exit criteria

Feeds `O53 Transition Architecture`, columns in sheet order. A first pass needs one row per transition state, so three rows today and up to six if step 9.3 makes you add states. Write the criteria on the back of the sheet if the cell runs out. Signed by the chair of `GB-01 Architecture Board`, countersigned by the validation lead.

Ref	Transition architecture	Description	Work packages	Exit criteria
TA-1	Governed and supportable	EXAMPLE — delete this row	WP-01, WP-02, WP-04, WP-05	1. R06.Managed reads Managed on all 7 ELN / LIMS interfaces. 2. …

Step 9.3 — Map the six increments onto the three transition states

Produce. A decision, minuted, on whether this programme has three transition states or six.

You need. `O53`, `O66 Architecture Increment`, `R15`, `R23 Increment delivered by Project`.

Where it lands.

> **`O66 Architecture Increment` — `Ref`, `Increment`, `Lands`, `What is added`**
>
> Add nothing to the column set; change the values. Either `Increment` names collapse onto the three `O53` states, or `O53` grows to six rows and each increment gets one. `Lands` must stop being a bare month offset: convert it to `YYYY-MM` against a stated month zero, or the column cannot be read against `O30.Start`.

How.

1. Match on name. `INC-1 Governed and supportable`, `INC-2 Integrable` and `INC-4 Business visible` carry the same names as `TA-1`, `TA-2` and `TA-3` [9]. Three of six map by identity; the brief that there is no mapping is half true.
2. The other three — `INC-3 Compliant by mechanism`, `INC-5 Analytics and evidence`, `INC-6 Core renewal` — belong to no state. Decide: three new states, or fold them in.
3. Fix `R23`. It maps each increment to exactly one project, and only to the odd-numbered ones: `PRJ-01`, `PRJ-03`, `PRJ-05`, `PRJ-07`, `PRJ-09`, `PRJ-11` [9]. The six even-numbered projects — including `PRJ-02 Data ownership and mastering` and `PRJ-10 Compliance mechanisms` — deliver no increment at all.
4. Read the names across `R23` and three of six disagree: `INC-3 Compliant by mechanism` is delivered by `PRJ-05 Access and security`, `INC-5 Analytics and evidence` by `PRJ-09 Resilience uplift`, `INC-6 Core renewal` by `PRJ-11 Customer experience`.
5. Take the earliest `O30.Start` of `2027-02` as month 1 and place the six `Lands` values on the same axis. `INC-1` lands eleven months before its only project starts; `INC-6` lands thirty-one months after its only project ends. Three of six fall outside the delivering project's window and the other three land while it is still running.

Signed by. `GB-01 Architecture Board`. The chair owns the increment-to-state decision because it decides what gets gated when.

Done when. Every row of `O66` resolves to exactly one row of `O53`, and every `Lands` value is a calendar month derived from a stated month zero that appears somewhere in the repository.

Shows up in. `RPT08 Roadmap` prints `O30` only, so increments never reach a report. Until step 9.12 that is a gap you carry.

Goes wrong when. You present both the three states and the six increments to a steering committee without reconciling them, and the committee picks whichever has the earlier dates.

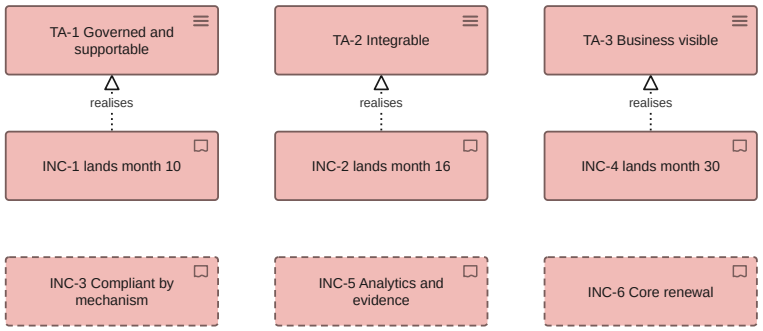


Figure 9.2 Three transition states against six increments. Three map by name identity. The three dashed increments — `INC-3`, `INC-5`, `INC-6` — realise no state in `O53`, and the decision in step 9.3 is whether to add three states or fold them in.

Step 9.4 — Place the five work packages that belong to no transition state

Produce. Either five new rows in `R15`, or a written decision that the five sit outside the transition sequence and why.

You need. `R15`, `O30`, `O53`, `O62 Delivery Risk`.

Where it lands.

> **`R15 Transition contains Work Package — Transition, Name, Work package, Package name`**
> Seven rows today. Add rows until every one of the twelve `O30` refs appears at least once, or record the exclusion. The sheet name is cut at Excel's 31 characters; match on the `R15` prefix.

How.

1. List the orphans: `WP-06 Analytics and reporting`, `WP-08 Process automation`, `WP-09 Resilience uplift`, `WP-10 Compliance mechanisms`, `WP-12 Supplier and partner integration` [1].
2. Add their costs: **$29.1m of $84.3m sits in no transition state** — a little over a third of the programme.
3. Look at which one is missing. `WP-10 Compliance mechanisms` is the compliance work package, and it is in no state, has no increment through `R23`, and reads `Delivery: Past end date`.
4. `WP-09 Resilience uplift` is also outside, while `O49 Resilience Tier` records nine of twelve applications as `Never tested`, including `RES-04 Safety Database` at Tier 3 with a four-hour objective [10].
5. Assign each orphan to a state or write the exclusion into the state's `Description`. An exclusion with a reason is a plan. An omission is not.

Signed by. `GB-01 Architecture Board`, with `GB-02 Investment Committee` noting the $29.1m.

Done when. `R15` has at least twelve rows, or `O53.Description` names every excluded package.

Shows up in. Nowhere. No report joins `O30` to `O53`. Build the join yourself in the comparison tool and keep it as a standing check.

Goes wrong when. The orphans turn out to be the packages nobody funded, and you discover it at `GT-0` when four of them have no contract in `O33` either.

Step 9.5 — Build the plan against the calendars that actually bind

Produce. A one-page delivery-year calendar showing what the year allows, and a corrected `O52 Implementation Factor`.

You need. `O52`, the regulatory commitment calendar, the review cycle, the inspection history, and volume 1 chapter 18.

Where it lands.

> **`O52 Implementation Factor` — `Ref`, `Implementation factor`, `Category`, `Deduction`**
>
> Delete or rewrite `IF-03 Peak trading freeze`. Add rows for the windows this industry has: the annual product quality review cycle, the inspection window, registration renewal submission dates, the variation notification clocks, and the price adjustment effective date. `Category` takes `Constraint`; `Deduction` says what the plan may not do and when.

How.

1. Lay the review cycle across the year. Each review consumes quality person-days, and they run all year, not in a season [11].
2. Add the registration renewal dates. Registration is valid five years and the renewal goes in six months before expiry, and no variations may be in flight during renewal review [12].
3. Add the price adjustment as a mass data-change event with a hard effective date, the way another industry treats a tax-rate change [11].
4. Add the inspection window. You cannot plan it, so plan the freeze around the response obligation it creates.
5. Rewrite `IF-06 Concurrent capacity unknown` from an `Assumption` to a `Constraint` with a number in it, once step 9.7 gives you the number.

Signed by. The head of quality assurance and the head of regulatory affairs — neither of whom appears in `O09 Organization Unit`, `O10 Actor` or `O11 Role`. Record their names in `O70 Regulated Role` and in the plan's signature block.

Done when. `O52` holds no retail vocabulary and every `Constraint` row carries a date or a recurring window.

Shows up in. `O52` reaches no report. Put the calendar in the migration plan deliverable `DLV-16` instead, where a steering committee will read it.

Goes wrong when. You plan around budget-year quarters. The year has about forty-four usable weeks and the quality function is already committed in most of them.

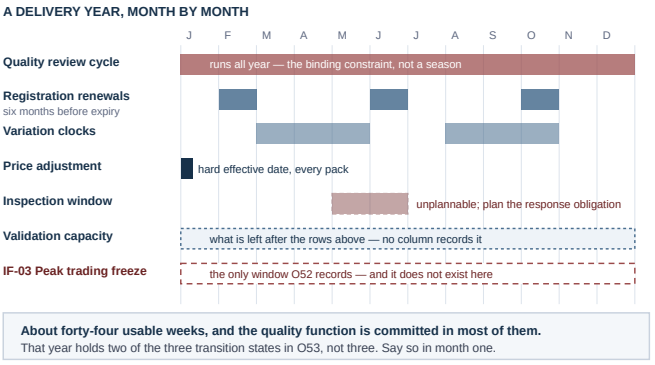


Figure 9.3 The three calendars that bind a delivery year in this industry, and the one `O52 Implementation Factor` records instead. Draw this for your own company before you commit to a transition-state date.

Step 9.6 — Size the validation effort, and let it set the sequence

Produce. A validation-effort estimate per system touched, in quality person-days, and a sequence that fits it.

You need. `O23 Application Component` for the twelve applications, `O67 Validated System` for which are GxP and in what state, `R13` for which blocks each package delivers, the validation lead.

Where it lands.

> **`O67 Validated System` — `Ref`, `Application`, `GxP`, `Validation state`, `Last periodic review`, `Next review due`, `System owner`, `Process owner`, `Validation lead`, `URS reference`, `Audit trail reviewed`, `Decommissioning plan`**
>
> Defined in item 2 of this workbook. Phase F reads it and writes nothing to it. If it does not exist yet, stop and create it: you cannot sequence a programme in a regulated estate without knowing which systems carry regulated records, and no sheet in the repository carries that fact.

How.

1. Count the GxP systems in scope per transition state. Nine of the twelve applications in `O23` are GxP systems [13].
2. For each, ask the validation lead for a person-day estimate at the right rigour: a configuration change, a version upgrade and a replacement are three different numbers.
3. Sum per quarter and compare with the quality function's uncommitted capacity. The difference is your sequence constraint.

4. Where the sum exceeds capacity, move scope, not dates. Moving the date and keeping the scope is how a transition state arrives with no evidence behind it.
5. Record the estimate in the migration plan and the constraint in `O52`. Neither has a column in the repository, and saying so is part of the deliverable.

Signed by. The validation lead, countersigned by the head of quality assurance. The architecture board cannot sign this; it has no quality membership [14].

Done when. Every quarter of the plan has a validation person-day total below the capacity number the quality function gave you in writing.

Shows up in. Nothing computes it. Carry it as a column in your own copy of `RPT08 Roadmap` and show it at every gate.

Goes wrong when. Nobody asks, the plan assumes validation is a line item inside each project, and `TA-1` arrives needing four systems revalidated in the same quarter as the review cycle peak.

THE ONE SENTENCE TO PUT IN FRONT OF THE STEERING COMMITTEE

Nine systems cannot be revalidated in one quarter, and no amount of money changes that. The constraint is a small number of named people with a statutory duty, and two of the three transition states in `O53` are scoped as though it did not exist. A plan that ignores this does not fail at the end. It fails at the first gate that asks for evidence, which is `GT-2 G2 Build complete`, and by then the money is spent.

Step 9.7 — Rewrite `O39 Data Migration`, because every row is impossible

Produce. Six corrected migration rows, each with a named source of record and a named target.

You need. `O39`, `O20 Data Entity` for the recorded master system, `O43 Master Data Rule` for the recorded golden source, `R05`, and the data owners designated in Phase C.

Where it lands.

> **`O39 Data Migration — Ref, Entity, From system, To system, Status`**
>
> `From system` is the system that holds the authoritative record today, which is a question you ask a data owner, not a column you copy. `To system` is the target in the approved architecture. `Status` takes `Not started`, `Planned`, `In progress`, `Verified`. Note that `Verified` is not in the sheet's current value set and must be, because `In progress` and the end of a validated migration are not the same state.

How.

1. Read the six rows. `MIG-01` moves Compound from `CTMS` to `Safety Database`. `MIG-03` moves Subject from `Safety Database` to `eTMF`. `MIG-05` moves Clinical Data Point from `eTMF` to `QMS` [15]. None of those pairings is possible.
2. Check the pattern before you correct one row at a time. The six `From system` values are the application list rotated by one position and the six `To system` values rotated by three. The column was generated, not recorded.
3. Reconcile against `O20.Master system` and `O43.Golden source`, which disagree with each other on every entity they share — `O20` puts Compound in `Safety Database`, `MDR-01` puts it in `ELN / LIMS` [16]. Two sheets, two answers, and `O39` agrees with neither.
4. Get a data owner to say, in writing, which system holds the authoritative record for each entity. Their answer is `From system`.
5. Add the six entities `O39` omits. `O40` carries twelve; `O39` carries six. Either six entities migrate with no plan or six mappings exist for migrations nobody intends.

Signed by. The designated data owner per entity, from `O20.Owner` once that column holds people. Today it holds job titles, so the signature line is empty.

Done when. No row of `O39` names a source system that a data owner has not confirmed in writing, and `O39` and `O40` cover the same entity set.

Shows up in. `RPT05 Data Ownership` reprints `O20`, so a corrected master system shows there. `O39` itself reaches no report.

Goes wrong when. You migrate from the system the repository names rather than the system that holds the record, and reconcile successfully against the wrong source.

Step 9.8 — Replace the placeholder field mapping with a specification

Produce. A field-level mapping specification per entity, with transformation rules and the handling of values that have no target.

You need. `O40` , `O37 Data Attribute` , `O36 Data Entity Relationship` , the source system's data dictionary, Form F-2.

Where it lands.

> **`O40 Source To Target Mapping — Ref , Entity , Source field , Target field , Rule`**
>
> `Source field` and `Target field` take real field names. `Rule` takes `Direct` , `Transform` , `Lookup` , and two values the sheet needs and does not have: `No target` and `Carry version` . One row per field, so twelve rows becomes several hundred, which is the correct order of magnitude.

How.

1. Read `MAP-01` : entity Compound / Molecule, source field `CTMS field` , target field `EDC field` , rule `Direct` [17]. The field names are the application names with the word "field" after them. All twelve rows read that way.
2. Compare `O40` 's target against `O39` 's. For all six entities they share, the source agrees and the target never does: `O40` 's target is one position further round the same rotation. Two generated columns, generated differently.
3. Expand to real fields. `O37 Data Attribute` has sixty attributes across the entities and is the place to start, though it carries its own defects — `ATR-002 Compound / Molecule · Name` is classified `Public` and marked `Key: Yes` .
4. For every coded value, add a row for the dictionary and a row for its version, with rule `Carry version` . A coded term without its version is a string that looks like data [6].
5. For every field with no target, write the row with rule `No target` and a disposition: archived, left in the legacy system, or lost with a signature against it.

Signed by. The data owner for the entity and the validation lead jointly. The mapping specification is a validation deliverable, not a design note [18].

Done when. Every field in the source extract appears exactly once in `O40` with a rule, and no cell contains the word "field" as a placeholder.

Shows up in. No report reads `O40` . Its output appears in the migration verification report, which is the artefact an inspector asks for.

Goes wrong when. The mapping is done in the migration tool and never written down, so the verification protocol has nothing to verify against and the tool becomes the specification.

Form F-2 — Field mapping specification

Feeds `O40 Source To Target Mapping` , columns in sheet order. A first pass on one entity needs twenty to eighty rows; continue on a second copy of this form. Signed by the data owner and the validation lead. Note the two `Rule` values the sheet needs and does not have: `No target` and `Carry version` .

Ref	Entity	Source field	Target field	Rule
MAP-13	EXAMPLE — delete	AE_TERM_CODE	ae_coded_term	Direct
MAP-14	EXAMPLE — delete	AE_DICT_VERSION	ae_dict_version	Carry version

Step 9.9 — Decide migrate, archive or leave read-only, per record class

Produce. A disposition decision per record class, with the retention clock that forces it.

You need. `O69 Retention Clock`, `O20`, `O39`, `O23.In service` for how old each system is, and volume 1 chapter 16's retention clocks.

Where it lands.

> **`O69 Retention Clock — Ref`, `Record class`, `Period`, `Clock starts at`, `Instrument`, `System of record`, `Disposal approved by`**
>
> Defined in item 5 of this workbook. Phase F reads it to decide what has to come across. `Clock starts at` is the column that decides the plan: a clock that runs from batch expiry or from trial closure cannot be evaluated without the date that triggers it, which usually lives in a different system from the record.

How.

1. Classify every record set in scope against `O69`. Periods in this estate run from a few years to forty, and three of the longest are event-triggered.
2. Apply the default: **migrate the minimum, archive the rest, decide at design time** [5].
3. For each archive decision, specify the export format and the reference data that makes it readable without the source application — including the dictionary version and the audit trail, neither of which survives a naive extract.
4. Remember the clocks do not reset. A record created in 2011 and migrated in 2027 keeps its 2011-based clock, and a target computing retention from its own creation date gets it wrong by sixteen years.
5. For anything left read-only, write the decommissioning plan now and put its cost in `O30`. It is not in the $84.3m.

Signed by. The record class's disposal approver from `O69`, and the Responsible Pharmacist for anything in the batch-release chain.

Done when. Every record set in `O39` has a disposition, and every archive decision names a format and a rendering mechanism.

Shows up in. Nothing. `O69` does not exist in the delivered repository and no report reads it. That absence is a finding to carry into Phase G.

Goes wrong when. Everything is migrated because deciding is hard, the migration becomes more than half the programme cost, and nobody has read the 2011 records since 2012.

Look it up. Volume 1 chapter 10, *Retention as a data-model constraint*, has the verified periods and their triggers. Chapter 16, *The retention clocks*, has the instruments behind them.

Step 9.10 — Write the migration verification protocol before the first extract

Produce. A verification protocol with reconciliation counts, critical-field comparison and a meaning check with a named reviewer.

You need. `O40` as corrected, `O55` acceptance criteria for the data blocks, the validation lead, and a user who can read a record and say whether it is right.

Where it lands.

> **`O59 Compliance Assessment — Ref`, `Building block`, `Block name`, `Acceptance criterion`, `Gate`, `Verdict`, `Evidence`**
>
> The verification outcome per data building block belongs here, at `GT-2 G2 Build complete`. The sheet is empty today and item 10 of this workbook fills it. Phase F's job is to write the criterion so that Phase G has something to assess.

How.

1. Write the reconciliation: record counts in, out, rejected, and the disposition of every rejection. A migration with no record of its exceptions cannot assert its own completeness.
2. Write the critical-field comparison: full comparison, not sampled, on the fields that carry regulatory meaning.
3. Write the meaning check. A named person reads a representative set of records in the target and confirms they read correctly — which is the only test that catches a coded term whose dictionary version did not travel.
4. Decide the audit trail. Migrating it faithfully is usually impossible; exporting it as a readable artefact and retaining it alongside is acceptable if declared.
5. Decide the signatures. A compliant electronic signature is bound to its record and cannot be recreated in a new system, so the honest option is to retain the signed rendering as a verified true copy and declare it.

Signed by. The validation lead approves the protocol; the business reviewer signs the meaning check; the system owner from `O67` accepts the result.

Done when. The protocol names, for each entity, the reconciliation query, the critical-field set, the meaning reviewer and the rejection disposition rule.

Shows up in. `O59`, once Phase G starts populating it. Your protocol is what makes the `Evidence` column fillable.

Goes wrong when. Verification is a row count. Row counts match on migrations that destroyed meaning, which is exactly the failure the obligation was written to catch [5].

Step 9.11 — Move the certainties out of the risk register

Produce. A risk register where every row is uncertain, and an issue list where the certainties live.

You need. `O62 Delivery Risk`, `O52 Implementation Factor`, `O29 Gap`, `R12`, `O60`.

Where it lands.

> **`O62 Delivery Risk` — `Ref`, `Risk`, `Likelihood`, `Impact`, `Owner`**
>
> `Likelihood` must not take the value `Certain`. Three rows do: `RSK-01 Integration product not selected`, `RSK-07 Benefit not measurable at G3` and `RSK-08 Unmanaged interfaces proliferate` [19]. A certainty is an issue. Move it to `O52` as a `Dependency` or a `Constraint`, where `IF-01 Vendor selection not made` already records the first of the three.

How.

1. Move the three `Certain` rows to `O52` and renumber. `O62` drops to five rows and becomes readable.
2. Note what the move reveals: `RSK-07 Benefit not measurable at G3` is certain, and `GT-3 G3 Benefit` tests "stated benefit confirmed against a date". No sheet carries a benefit date. `O61` records `CR-G04 Record benefit realisation dates per package` as `Approved` and nothing has been recorded.
3. Add the risks this plan actually carries and `O62` omits: quality capacity below the validation demand, a migration that cannot evidence meaning, and an inspection inside the transition window.
4. Look at `O29 Gap`. It is empty, `R12 Gap closed by Work Package` is empty in consequence, and `RPT07 Gap Analysis` prints nothing [20]. `ZZ Checks` reports "Gaps overdue: 0", which is true and meaningless.
5. `O61` already records `CR-G03 Raise unfunded building blocks as gaps` as `Raised`. Close it by populating `O29` from the building blocks no work package delivers.

Signed by. The named `Owner` on each `O62` row. Three of the current owners — `Head of Procurement`, `Head of Delivery`, `Chief Enterprise Architect` — exist in `O10 Actor`; check each one does before you leave the sheet.

Done when. No row of `O62` reads `Likelihood: Certain`, and `O29 Gap` has rows.

Shows up in. `RPT07 Gap Analysis`, which goes from zero rows to a populated report — the single most visible improvement available in Phase F.

Goes wrong when. The register keeps the certainties because removing them makes the risk count look better, and the steering pack reports eight risks of which three have already happened.

Step 9.12 — Publish the plan and hand it to Phase G

Produce. `DLV-16 Migration Plan` and `DLV-17 Project Charters` moved from `Draft` to `Approved`, with the roadmap report reprinted.

You need. Everything above, `O31 Deliverable`, `O33 Architecture Contract`, `RPT08 Roadmap`.

Where it lands.

> **`O31 Deliverable` — `Ref`, `Deliverable`, `Phase`, `Status`**
>
> `DLV-16` and `DLV-17` are the two Phase F deliverables and both read `Draft`. They move to `Approved` only when a body with the mandate has minuted the approval. Note that `DLV-13 Consolidated Gaps` already reads `Approved` while `O29 Gap` is empty — an approved deliverable with no content, and the precedent you are refusing to repeat.

How.

1. Assemble the plan: the sequence, the exit criteria, the calendar, the validation capacity number, the migration dispositions, the verification protocol.

2. Issue the missing architecture contracts so `O33` covers all twelve work packages, because `GT-0 G0 Charter` tests that a contract exists.
3. Reprint `RPT08 Roadmap` from the corrected `O30`. Twelve rows, a computed `Delivery` column, and a footer that states the cost total and the validation capacity constraint.
4. Minute the approval at `GB-01 Architecture Board` and the funding at `GB-02 Investment Committee`, and resolve `DEC-01` before either.
5. Hand Phase G a list: which blocks are assessable, at which gate, against which criterion. That list is the input to `O59`.

Signed by. `GB-01 Architecture Board` for the plan, `GB-02 Investment Committee` for the money, and the head of quality assurance for the sequence. The third signature has no body in `O57` and item 10 creates one.

Done when. `DLV-16` and `DLV-17` read `Approved` against a dated minute, and `RPT08` reconciles to `O30` row for row.

Shows up in. `RPT08 Roadmap` and `RPT06 Requirements Traceability`, where a requirement that now reaches a work package with a date stops being a requirement nothing will deliver.

Goes wrong when. The plan is approved and the exit criteria are not, so `TA-1` closes on the chair's judgement rather than on a test, and Phase G inherits a transition state nobody can audit.

What you hand to Phase G

Four things, and only four. **A sequence with dates that a validation calendar allows. Eighteen written exit criteria that name a sheet, a column and a value. A migration plan whose verification protocol tests meaning, not row counts.** And **a number for quality assurance capacity per quarter**, in writing, from the person who owns it.

If you hand over three of the four, hand over the fourth as a gap in `O29` rather than as an assumption in `O52`. A gap gets a work package. An assumption gets discovered in month nine.

THE CHAPTER IN TEN LINES

- O53's Exit criteria column holds a count, not criteria. Eighteen criteria are asserted and none is written down.
- O30 Work Package and O56 Project are the same twelve items twice, and R14 maps them one to one.
- The earliest start date in O30 is 2027-02 and the earliest end date 2029-02, so nothing is past its end date. Past end date, In flight and Delivered are typed strings.
- The twelve work packages total 84.3m dollars against a recommended alternative of 46m, and DEC-01 to adopt an alternative is still Outstanding.
- R15 places seven of twelve work packages. Five, worth 29.1m dollars, sit in no transition state — including the compliance package.
- Three of the six increments share a name with a transition state. R23 maps increments only to the odd-numbered projects, so six projects deliver no increment at all.
- O39 Data Migration moves Compound from CTMS to Safety Database and Subject from Safety Database to eTMF. The From and To columns are the application list rotated.
- O40 maps CTMS field to EDC field with rule Direct. Its target is always one position further round the rotation than O39's.
- A migration is a validation activity and the obligation is to verify value and meaning. A row count matches on migrations that destroyed meaning.
- A coded term migrated without its dictionary version has changed what it means.
- IF-03 Peak trading freeze is retail vocabulary, and it is a Constraint on the plan. The windows that bind here are the review cycle and the inspection.
- Quality assurance capacity, not budget, is what the delivery year allows. No column in 105 sheets records it.
- Three rows of O62 read Likelihood: Certain. A certainty is an issue and belongs in O52.

SOURCES FOR THIS CHAPTER

- **1.** Pharmaceuticals_TOGAF_Metamodel.xlsx, Calderon Therapeutics — sheets O53 Transition Architecture (rows 5-7) and R15 Transition contains Work Package (rows 5-11)
- **2.** Pharmaceuticals_TOGAF_Metamodel.xlsx — sheets O30 Work Package (rows 5-16), O56 Project (rows 5-16) and R14 Project delivers Work Package (rows 5-16)
- **3.** Pharmaceuticals_TOGAF_Metamodel.xlsx — sheets RPT08 Roadmap (13 rows, the last a sentence in the Ref column) and ZZ Checks
- **4.** Pharmaceuticals_TOGAF_Metamodel.xlsx — sheets O54 Architecture Alternative (rows 5-9) and O60 Governance Decision (rows 5-11)
- **5.** European Commission, EudraLex Volume 4 — Annex 11 Computerised Systems (migration checks that data are not altered in value and/or meaning); as cited in volume 1 chapter 10 https://health.ec.europa.eu/medicinal-products/eudralex/eudralex-volume-4_en
- **6.** MedDRA MSSO, What's New with MedDRA Version 29.0 and the MSSO, March 2026, and Version 29.1, August 2026 — twice-yearly releases with term demotion, merger and primary-class change; as cited in volume 1 chapter 9 https://files.meddra.org/www/Training%20Materials/2026/Materials/001363_Whats_New_with_MedDRA_V29_0_and_the_MSSO.pdf
- **7.** Pharmaceuticals_TOGAF_Metamodel.xlsx — sheet O52 Implementation Factor (rows 5-12)
- **8.** Pharmaceuticals_TOGAF_Metamodel.xlsx — sheets O33 Architecture Contract (rows 5-12) and O58 Governance Gate (rows 5-8)
- **9.** Pharmaceuticals_TOGAF_Metamodel.xlsx — sheets O66 Architecture Increment (rows 5-10) and R23 Increment delivered by Project (rows 5-10)
- **10.** Pharmaceuticals_TOGAF_Metamodel.xlsx — sheet O49 Resilience Tier (rows 5-16)
- **11.** Volume 1, Pharmaceuticals — a starter pack for enterprise architects, chapter 18 section 6 and chapter 7, on the annual product quality review cycle, the regulatory commitment calendar and the price adjustment as a dated mass-change event
- **12.** SAHPRA, Communication to Stakeholders HPA-RNW01-2025/26, 21 August 2025 — Medicines Certificate of Registration renewal: five-year validity, application six months before expiry, no variations in flight during renewal review https://www.sahpra.org.za/wp-content/uploads/2025/08/HPA-RNW01-2025_26-Medicines-Certificate-of-Registration-Renewal.pdf
- **13.** Pharmaceuticals_TOGAF_Metamodel.xlsx — sheet O23 Application Component (rows 5-16), which carries no validation-state, GxP or periodic-review column
- **14.** Pharmaceuticals_TOGAF_Metamodel.xlsx — sheet O57 Governance Body (rows 5-9)
- **15.** Pharmaceuticals_TOGAF_Metamodel.xlsx — sheet O39 Data Migration (rows 5-10)
- **16.** Pharmaceuticals_TOGAF_Metamodel.xlsx — sheets O20 Data Entity (rows 5-18) and O43 Master Data Rule (rows 5-14)
- **17.** Pharmaceuticals_TOGAF_Metamodel.xlsx — sheets O40 Source To Target Mapping (rows 5-16) and O37 Data Attribute (rows 5-64)
- **18.** Volume 1, chapter 13, on the validation deliverable set and the migration plan and report as validation deliverables
- **19.** Pharmaceuticals_TOGAF_Metamodel.xlsx — sheet O62 Delivery Risk (rows 5-12)
- **20.** Pharmaceuticals_TOGAF_Metamodel.xlsx — sheets O29 Gap (0 rows), R12 Gap closed by Work Package (0 rows) and RPT07 Gap Analysis (0 rows)
- **21.** The Open Group, TOGAF Standard, Architecture Development Method — Phase F, Migration Planning. Identified, not retrieved: The Open Group's public pages are robots-disallowed to automated retrieval, so no edition or wording is claimed from it.

CHAPTER 10

Implementation Governance, validation and release

Four sound gates, five governance bodies, zero compliance assessments — and a release signature the repository has no row for.

What Phase G is for, and what this repository says about it

Phase G is where the architecture stops describing and starts testing. Something is built, somebody says whether it conforms, and the verdict is written down with the evidence behind it — conformance assessed against the architecture contract [1].

Read the governance sheets and the regime looks complete. `O58 Governance Gate` holds four rows — `GT-0 G0 Charter` before work starts, `GT-1 G1 Design` before build, `GT-2 G2 Build complete` before release, `GT-3 G3 Benefit` after operation — each with a `When` and a `What it tests` [2]. These four gates are the best-written thing in the repository. Do not redesign them. `O57 Governance Body` holds five bodies with a cadence, a membership and a mandate, and records `GB-01 Architecture Board` as holding G1 and G2 and approving dispensations with an expiry [3]. `O33 Architecture Contract` holds eight contracts. `O60 Governance Decision` holds seven decisions. `O61 Governance Change Request` holds five requests, one of them a dispensation with an expiry date.

`O59 Compliance Assessment` holds **nothing**. Header row only — `Ref`, `Building block`, `Block name`, `Acceptance criterion`, `Gate`, `Verdict`, `Evidence` — and `R24 Compliance Assessment tests` is empty in consequence [4]. Meanwhile `O63 Control Objective` records `CO-01 GMP / GCP / GLP / GVP` as `Evidence: Evidenced` [5]. The sheet that would hold the evidence has no rows.

So the gates are designed, the bodies named, the contracts issued — and the gate has produced **zero records**, while the compliance sheet asserts that the largest obligation in the industry is evidenced. That is not a gap to tidy up. It is the chapter.

Two measures in `O04 Measure` make the same point from the other end. `MSR-15 Architecture compliance rate` carries a baseline of **59%** and `MSR-16 Initiatives through a design gate` carries **44%** [6]. `O59` is the only sheet either could be computed from, and it is empty.

Calderon Therapeutics, the pharmaceutical company in this repository, is an invention. So is Veldtkruin Group in volume 1. Their data never mixes.

The overlay: the gate that matters is release, and a named person signs it

In a pharmaceutical company the governance gate that carries legal weight is **release** — of a batch for sale, and of a computerised system for use. The architecture board's gates govern a model. Release governs whether product reaches a patient, and the signature on it belongs to a named individual with a statutory duty.

In South Africa that individual is the Responsible Pharmacist, and this is not a naming difference from the European qualified person. The holder appoints one, named on the manufacturing licence, not more than one per licence, registered with the South African Pharmacy Council, and accountable in two directions — to the regulator under the Medicines Act and to the Council under the Pharmacy Act, which means two professional bodies and two disciplinary routes [7][8]. The duties include certifying, before release for sale, that each batch meets its registration conditions, good manufacturing practice and legal requirements, taking account of storage and transport conditions, sampling and quality-control results, **and ensuring that manufacturing and testing processes remain in a validated state** [7].

Read that last duty again, because it is the line that reaches the architecture. **Keeping systems in a validated state is explicitly a Responsible Pharmacist duty here, not an IT matter.** It makes validation status a release-relevant fact, and gives the signer a legitimate interest in the validated-system register, the periodic-review schedule and the open change records. A release view that does not surface those three is incomplete against the duty as written. And because the role requires Council registration, **the release signature cannot be delegated offshore** — a pattern that works in other industries and does not work here.

The repository has no row for this person. `O10 Actor` and `O11 Role` hold the same eighteen generic corporate titles, and `O57 Governance Body` has five bodies and not one quality body: no change control board, no validation committee, no data integrity forum [3]. The signatory who matters most in Phase G is in no sheet that could name them.

Look it up. Volume 1 chapter 13, *The Responsible Pharmacist*, has the four ways the role differs from the European equivalent. Chapter 5, *Quality and manufacturing*, has the surrounding roles and why the validation lead is the most useful first interview in the building. Chapter 8, *Batch manufacture to release*, has the release flow and the join across six systems. Chapter 16, *Licensing, and the Responsible Pharmacist*, has the licence conditions and the dated obligations.

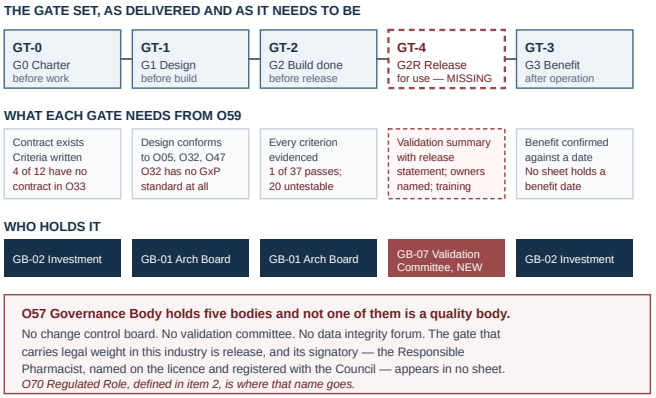


Figure 10.1 The four gates in `058` are sound, and none of them is release. **`GT-2` ends where release begins.** The fifth column is the gate step 10.2 adds; the bottom band is the body that holds each one once step 10.1 has created the quality bodies.

BEFORE YOU OPEN THE GOVERNANCE SHEETS

Who is the Responsible Pharmacist named on each manufacturing licence, and what is the Council registration number?

Who signed the last release for use of a computerised system, and where is that signature held?

`059` is empty. Who stopped a gate sitting, and what did they fear it would find?

Which body in this company holds change control for a validated system, and does it know it does?

The thirteen steps

Step 10.1 — Create the quality bodies `057` does not have

Produce. Three new rows in `057 Governance Body`, and a decision on which of them holds which gate.

You need. `057`, `058`, `010 Actor`, `070 Regulated Role`, the head of quality assurance and the validation lead.

Where it lands.

> **`057 Governance Body — Ref, Governance body, Cadence, Membership, Mandate`**
>
> Add `GB-06 Change Control Board`, `GB-07 Validation Committee` and `GB-08 Data Integrity Forum`. `Mandate` must name the gate each one holds and the thing it can refuse. `Membership` names roles from `011 Role` and `070 Regulated Role` — and `070` is where the Responsible Pharmacist, the QA release officer and the validation lead live, because none of them exists in `010` or `011`.

How.

1. Read the five existing bodies. `GB-01 Architecture Board` holds G1 and G2; `GB-02 Investment Committee` holds G0 funding and G3 benefit; `GB-03 Data Governance Council` designates data owners; `GB-04 Executive Committee` escalates; `GB-05 Architecture Function` publishes criteria and maintains the repository [3].
2. Note what no body does: approve a change to a validated system, approve a validation plan, or accept a system for release for use.
3. Add the change control board. In most pharmaceutical companies it already exists inside the quality system — your job is to record it, not to invent it, and to connect it to the gates.
4. Give the validation committee the release-for-use decision and the periodic-review outcome.

5. Write the escalation path. Where the architecture board and the change control board disagree about a GxP system, the quality body wins, and `O57.Mandate` should say so in those words.

Signed by. The head of quality assurance, with `GB-04 Executive Committee` ratifying the mandate split. The architecture board cannot grant itself authority over a quality decision.

Done when. Every gate in `O58` has a named holder in `O57`, and every GxP change has a body that can refuse it.

Shows up in. No report reads `O57`. The visible effect is that `O60 Governance Decision` acquires a `Body` value that is a quality body, which is the first time that has happened in this file.

Goes wrong when. You create an architecture-owned validation committee in parallel with the quality function's existing one, and the company ends up with two registers that disagree — the near-universal state, and the thing an inspector finds rather than you.

Step 10.2 — Add the gate the four-gate set does not have

Produce. A fifth row in `O58 Governance Gate` for release for use, with its holder and its evidence set.

You need. `O58`, `O67 Validated System`, `O57` as extended in step 10.1.

Where it lands.

> **`O58 Governance Gate — Ref, Gate, When, What it tests`**
>
> Add `GT-4`, gate name `G2R Release for use`, `When` = *after `G2`, before the system is used on regulated records*, `What it tests` = *the validation summary report carries a release-for-use statement, the system owner and process owner are named, training records are complete, and the periodic-review date is set*. Keep the `GT-` numbering sequential though the gate sits between G2 and G3: `Ref` records order of creation, the `Gate` name order of use.

How.

1. Read `GT-2 G2 Build complete`: `When` = *before release*, `What it tests` = *every acceptance criterion evidenced* [2]. The gate ends where release begins and nothing governs the step itself.
2. Decide what release for use means here. It is the point at which a validated system may hold regulated records, and it is evidenced by a validation summary report with a release statement.
3. Give the gate to `GB-07 Validation Committee`, not to the architecture board.
4. Name the evidence: validation summary report, traceability matrix, executed qualification protocols with their deviations closed, training records, the operating procedures for audit-trail review and periodic review.
5. Write the refusal condition. A gate that cannot say no is a milestone, not a gate.

Signed by. `GB-07 Validation Committee`, with the system owner from `O67` accepting the system. For a batch, the Responsible Pharmacist; for a system, the validation lead and the system owner.

Done when. `O58` has five rows, `GT-4` has a holder, and no system in `O67` reads `Validation state: Validated` without a `GT-4` record behind it.

Shows up in. `O59 Compliance Assessment`, whose `Gate` column can now take `GT-4` — the value that lets a validation verdict be recorded at all.

Goes wrong when. Release for use is treated as an event inside the project rather than a gate, so there is no record of who accepted the system and the first periodic review has no baseline.

Step 10.3 — Fill `O59 Compliance Assessment`, and derive every row rather than inventing it

Produce. The first populated rows of `O59`, derived from `O55 Building Block` and the gate set, plus a note recording that before them the sheet was empty.

You need. `O55` for the 37 refs and their `Acceptance criteria`; `O58` as extended; `R13 Work Package delivers Building Block` for which package owns each block; `O33` for the contract; Forms G-1 and G-2.

Where it lands.

> **`O59 Compliance Assessment` — `Ref`, `Building block`, `Block name`, `Acceptance criterion`, `Gate`, `Verdict`, `Evidence`**
>
> `Ref` takes `CA-001` onward; the sheet has no reference space because it has no rows, so you are setting the convention. `Building block` takes a ref from **`O55`**, the union sheet, not from the four domain sheets — `BBB-`, `DBB-`, `ABB-` or `TBB-`. `Acceptance criterion` is copied from `O55` verbatim, not paraphrased. `Gate` takes `GT-0` to `GT-4`. `Verdict` takes `Conformant`, `Conformant with dispensation`, `Non-conformant` or `Not assessable`. `Evidence` names the artefact, its date and where it is held — never the word "yes".

How.

1. Count first. `O55` holds **37 rows**: ten `BBB-`, ten `DBB-`, ten `ABB-` and seven `TBB-`, each with one acceptance criterion [9]. Thirty-seven criteria is your row count for a first pass, one per block, at the gate where the criterion can be tested.
2. Assign the gate. A criterion about design meets `GT-1`; a criterion about something built meets `GT-2`; a criterion about a system holding regulated records meets `GT-4`; a criterion about a benefit meets `GT-3`.
3. Copy the criterion verbatim, then test it as written. Steps 10.4 to 10.6 do this by domain and the answers are not encouraging.
4. Where a criterion is not testable as written, record `Not assessable` with an `Evidence` cell that says why. That is a true verdict and it is far more useful than a blank.
5. Populate `R24 Compliance Assessment tests` from the rows you just wrote, so the assessment is reachable from the building block as well as from the gate.

Signed by. `GB-05 Architecture Function` prepares the assessment; `GB-01 Architecture Board` signs `GT-1` and `GT-2` verdicts; `GB-07 Validation Committee` signs `GT-4`.

Done when. `O59` has at least 37 rows, `R24` has the same count, and no `Evidence` cell is blank on a row whose `Verdict` is `Conformant`.

Shows up in. `MSR-15 Architecture compliance rate` and `MSR-16 Initiatives through a design gate` acquire a source for the first time. Expect the real numbers to be well below the typed 59% and 44%.

Goes wrong when. You write 37 rows all reading `Conformant` because the project said so. An assessment with no artefact behind it is worse than an empty sheet, because the empty sheet was honest.

Form G-1 — Compliance assessment, identify

Feeds `O59 Compliance Assessment`, first four columns in sheet order. Form G-2 covers the rest. A first pass needs 37 rows, one per `O55` building block — so four copies of this form. Signed by the domain architect who prepared it.

Ref	Building block	Block name	Acceptance criterion
CA-001	EXAMPLE — delete	ABB-08 QMS target	No unmanaged interface to or from QMS

Form G-2 — Compliance assessment, verdict

Feeds `O59 Compliance Assessment`, last three columns plus `Ref` to tie it to Form G-1. Signed by the body that holds the gate: `GB-01 Architecture Board` for `GT-1` and `GT-2`, `GB-07 Validation Committee` for `GT-4`, `GB-02 Investment Committee` for `GT-3`.

Ref	Gate	Verdict	Evidence
CA-001	GT-2	Conformant	R06 extract, 3 QMS rows, all Managed, dated __

Step 10.4 — Assess the ten application criteria, because they are the only testable ones

Produce. Ten `O59` rows with real verdicts and real evidence, from one query.

You need. `O55` rows `ABB-01` to `ABB-10`, `R06 Application interfaces with`, `O23 Application Component`.

Where it lands.

> **`O59 Compliance Assessment` — the ten `ABB-` rows, `Gate` = `GT-2`**
>
> `Evidence` cites the extract: sheet `R06`, the rows for that application, the `Managed` column value on each, and the date you pulled it. This is the one place in the repository where a verdict can be computed rather than asserted.

How.

1. Read the criterion. Every `ABB-` row reads *No unmanaged interface to or from <application>* [9]. That is testable against `R06.Managed`, which takes `Managed` or `Unmanaged`.
2. Count per application. `R06` has 36 rows and **19 read `Unmanaged`** [10]. Counting both ends of every interface: `ELN / LIMS` touches five unmanaged, `RIM` five, `Safety Database` four, `EDC` four, `eTMF`, `Serialisation`, `CTMS`, `Medical Information`, `MES (GMP)` and `Statistical Computing` three each, `ERP` two.
3. **`QMS` touches none.** Three interfaces, all `Managed`. `ABB-08 QMS target` is the only one of the ten that passes, and nine are `Non-conformant` on the day you look.
4. Then read the other half of the `ABB-08` row. `What it must do` is *QMS at target disposition*, and `O23` records `A08 QMS` as `Disposition: Replace`, `Criticality: Critical`, `Health: Healthy` [11]. The criterion passes and the block is not delivered, because the criterion does not test the thing the block must do. Record `Conformant` against the criterion and raise the mismatch as a change request in step 10.10.
5. Add the pharmaceutical reading. `R06` records six `Direct DB read` interfaces, two of them `QMS` reading `CTMS` and `MES (GMP)` [10]. A direct database read into or out of a validated system sits inside that system's validation scope and is routinely in nobody's. Five of 36 interfaces are logged.

Signed by. `GB-01 Architecture Board` at `GT-2`, with the integration owner present.

Done when. Ten `O59` rows carry a verdict computed from a dated `R06` extract, and the one `Conformant` row carries the disposition mismatch as a linked change request.

Shows up in. `RPT04 Interface Catalogue`, which reprints `R06` and adds a footer — *19 of 36 interfaces are unmanaged and 31 carry no logging.* Note that its last row is that sentence sitting in the `Ref` column, so anything loading the report programmatically gets a 37th interface called a sentence [12].

Goes wrong when. You report "one of ten conformant" without the second reading, and the board concludes `QMS` is in good shape three months before it is replaced.

Step 10.5 — Assess the twenty criteria that cannot be tested as written, and rewrite them

Produce. Twenty `O59` rows reading `Not assessable`, each with the rewrite that would make it assessable.

You need. `O55` rows `BBB-01` to `BBB-10` and `DBB-01` to `DBB-10`, `O20 Data Entity`, `O43 Master Data Rule`, `O04 Measure`, `O13 Business Capability`.

Where it lands.

> **`O55 Building Block — Ref`, `Domain`, `Building block`, `What it must do`, `Acceptance criteria`**
>
> The rewrite goes here, in `Acceptance criteria`, and the verdict goes in `O59`. A criterion is testable when it names a sheet, a column and a value. *Discovery Research measured and owned* names none of the three.

How.

1. Read the ten `DBB-` criteria. Each reads *No <entity> record without a named owner* [9]. The test is `O20.Owner`, and that column holds job titles — `Head of Supply Chain`, `Head of Commercial`, `Data Steward` — not named people [13]. Verdict `Not assessable`, and the rewrite is *every row of `O20` names a person who appears in `O70 Regulated Role` or `O10 Actor`.*
2. Note the second problem on the same ten rows. `O20.Master system` and `O43.Golden source` disagree on every entity they share: `E01 Compound / Molecule` is mastered in `Safety Database` per `O20` and in `ELN / LIMS` per `MDR-01` [13]. A mastering block cannot be conformant while two sheets name two golden sources.
3. Read the ten `BBB-` criteria. Each reads *<capability> measured and owned*. No measure in `O04` is mapped to a capability and `R22 Org Unit owns Capability` maps capabilities to organisation units that include `Customer Service` and exclude Quality, Regulatory Affairs and Pharmacovigilance [6].
4. Rewrite each `BBB-` criterion to name the `MSR-` ref and the target value from `O13.Target`. `C01 Discovery Research` has `Current: 2`, `Target: 3` — so the criterion is *`O13.Current` reads 3 or better for `C01`, measured by `MSR-xx`, owned by a named person.*
5. Record all twenty as `Not assessable` with the rewrite in the `Evidence` column, and raise one change request against `O55` rather than twenty.

Signed by. `GB-03 Data Governance Council` for the `DBB-` rewrites; `GB-01 Architecture Board` for the `BBB-` rewrites.

Done when. No `O55` acceptance criterion lacks a sheet, a column and a value, and all twenty `O59` rows carry a rewrite rather than a blank.

Shows up in. `RPT05 Data Ownership`, which reprints `O20` with its owner and master-system columns and adds *4 of 14 entities have no lineage from report back to source.*

Goes wrong when. The rewrite is done in a document rather than in `O55`, so Phase G next quarter assesses against the original wording and gets the same twenty blanks.

Step 10.6 — Assess the seven technology criteria, and define what "in production" means

Produce. Seven `O59` rows, and a definition of production that a verdict can rest on.

You need. `O55` rows `TBB-01` to `TBB-07`, `R07 Application deployed on Technology`, `O48 Environment`, `O27 Physical Technology Compone`, `O46 Technology Building Block`.

Where it lands.

> **`R07 Application deployed on Tec — Application`, `Application name`, `Platform component`, `Component name`, `Environment`**
>
> Twelve rows, every one reading `Environment: Production`. `O48 Environment` holds four environments, so three of them have no application deployed in them anywhere in the repository. Until `R07` carries non-production rows, no criterion of the form "in production" can be distinguished from "exists".

How.

1. Read the criterion. Every `TBB-` row reads *<building block> in production for at least one service* [9]. The test needs a sheet that records what is in production, and `R07` is the only candidate.
2. Check `R07` before you rely on it. `A01` is deployed on `PTC-01`, `A02` on `PTC-02`, and so on to `A12` on `PTC-12` [14]. The mapping is the row index. Three of the twelve are diagnostic: `A04 Safety Database` is deployed on `PTC-04 Clinical EDC platform` while `PTC-05` is literally named `Safety database`; `A07 MES (GMP)` on `PTC-07 Laboratory instruments network` while `PTC-01` is `GMP MES servers`; `A03 EDC` on `PTC-03 Validated database cluster`.

3. Note the consequence. By that rotation, `MES (GMP)` runs on `PTC-07` and `Statistical Computing` on `PTC-11` — **the two components `O27` records as past vendor support**, 2026-08 and 2026-01 [15]. A `GT-1` verdict on technology standards cannot be derived from a sheet generated this way.
4. Record all seven `TBB-` rows as `Not assessable`, `Evidence` naming `R07`'s rotation as the reason. Note that `O46 Technology Building Block` carries two columns headed `Ref`, the second holding `T01` to `T07`; call it `Ref (T)`.
5. Write the definition. *In production* means: deployed in the environment `O48` designates as production, serving at least one `O24 Information System Service` row marked `Published`, on a platform component whose `O27.Position` reads `Supported`. Three conditions, three sheets, all checkable.

Signed by. `GB-01 Architecture Board` at `GT-1`, with the technology domain architect and the qualification engineer. Infrastructure is qualified; the system on top is validated. Keep the two signatures apart.

Done when. The definition of production is written into `O58.What it tests` for `GT-1`, and `R07` either carries non-production rows or is marked as unreliable in your own notes.

Shows up in. `RPT10 Technology Currency`, which reprints `O27` and adds *2 components are running past vendor support today.*

Goes wrong when. You accept `R07` at face value, certify that the validated cloud block is in production, and discover at the next periodic review that the component underneath it went out of support ten months ago.

WHY `O59` BEING EMPTY IS THE FINDING, NOT THE TASK

An inspector's first request in a computerised-systems inspection is the validation master plan, because it tells them what to hold the company to [16]. The second is a list of the systems holding regulated records. The third is the evidence for one of them, picked at random. `O59` is the sheet where the third answer lives, and it has no rows while `O63` asserts that the good-practice regimes are evidenced. **The gap between those two cells is the audit finding.** Fill `O59` with honest `Non-conformant` and `Not assessable` verdicts and the gap closes, because a documented absence is a defensible position and an undocumented assertion is not.

Step 10.7 — Replace `O63`'s tautologies with control objectives

Produce. Seven rewritten `Control objective` cells, each stating something testable about the design.

You need. `O63`, `O41 Security Control`, `O64 Security Policy`, `O06 Requirement`, Form G-3.

Where it lands.

> **`O63 Control Objective – Ref, Obligation, Source, Control objective, Evidence`**
>
> `Obligation` is what the law or guideline requires. `Control objective` is what must be true of the design for the obligation to be met — a different sentence, testable, naming an artefact. `Evidence` takes `Evidenced`, `Partial` or `In breach` and must resolve to an `O59` row.

How.

1. Read the column. Every one of the seven cells reads *Control that evidences <the obligation restated>* [5]. `CO-01` reads *Control that evidences GMP / GCP / GLP / GVP.* That is the obligation with four words in front of it.
2. Write the first one properly. For `CO-02 21 CFR Part 11 and Annex 11` the objective is not "evidence Part 11". It is: *every system in `O67` holding regulated electronic records has an audit trail that cannot be disabled by a user, a signature carrying signer, date, time and the meaning of signing, and a documented audit-trail review at a stated frequency.* Three testable clauses, each with an evidence artefact.
3. Do the same for the other six. The test is whether a reviewer could say no.
4. Reconcile the measure. `MSR-01 Obligations with evidence` carries a baseline of **54%** while `O63` shows two of seven `Evidenced` — **29%** [6]. Two sheets, two numbers, one measure. Pick `O63`, because it can be recounted.
5. Add the evidence join. An `Evidence` value of `Evidenced` is only legitimate when a row of `O59` names the artefact. Today none can, because `O59` is empty.

Signed by. `GB-05 Architecture Function` drafts; the head of regulatory affairs and the head of quality assurance approve. Neither appears in `O09 Organization Unit`.

Done when. No cell in `O63.Control objective` begins with the words "Control that evidences", and every `Evidenced` row cites an `O59` ref.

Shows up in. `RPT09 Regulatory Coverage`, which reprints `O63` verbatim and adds a footer.

Goes wrong when. You rewrite the column into something longer but still untestable. The test is not length. It is whether a reviewer with the artefact in front of them could record a verdict.

Step 10.8 — Add the South African obligations, because there are none in the sheet

Produce. New `O63` rows for the obligations that actually bind a South African pharmaceutical company, with their control objectives.

You need. `O63`, Form G-3, volume 1 chapters 16 and 20.

Where it lands.

> **`O63 Control Objective — Ref, Obligation, Source, Control objective, Evidence`**
>
> Add rows from `CO-08` upward. `Source` names the instrument, not a category: the Act and section, the guideline and its reference code, the regulation and its gazette number. An obligation with no named instrument cannot be evidenced, because nobody can check what it requires.

How.

1. Read what is there. `CO-03 Falsified Medicines / DSCSA` and `CO-02 21 CFR Part 11 and Annex 11` are European and United States framings. `CO-06 Data protection and health data` names no instrument at all [5]. **There is no South African obligation anywhere in the sheet.**
2. Add the registration duties: registration is valid five years, renewal six months before expiry, and no variations in flight during renewal review [17]. The control objective is a register with the derived renewal-due date and a computed alert.
3. Add the licensing duties: a manufacturing licence with renewal every five years at least 180 days before expiry, particulars changes within 30 days, and a named Responsible Pharmacist with a Council number [7][8]. The objective is a licence register whose named-pharmacist field cannot be null.
4. Add the pricing duty: one gazetted single exit price per pack, the only price at which a manufacturer or importer may sell to a private buyer, adjusted on a hard effective date [18]. The control objective is that no private-sector transaction can price off anything else, and that the adjustment is a controlled mass data change.
5. Add scheduling under the Act, pharmacy licensing under the Pharmacy Act, and data protection by name — special personal information, the sponsor's lawful ground, the prior-authorisation notification where it applies, and the transfer determination the responsible party makes itself and must date and reason [19]. Five more rows, each with an instrument.

Signed by. The head of regulatory affairs for the first four; the information officer for the data-protection rows; the Responsible Pharmacist countersigns the licensing row because the licence is in their name.

Done when. `O63` holds at least twelve rows, every `Source` cell names an instrument with a version or a date, and no obligation in the sheet applies only outside South Africa without being marked as such.

Shows up in. `RPT09 Regulatory Coverage`, which goes from seven rows to twelve and stops being a report about somebody else's jurisdiction.

Goes wrong when. You delete the European and United States rows. Keep them — a company that exports is bound by them too — but mark the jurisdiction, because an obligation without a jurisdiction is how the wrong control gets built.

Look it up. Volume 1 chapter 16 has the obligation set in full, section by section — *Registration, variation and renewal*, *Single Exit Price*, *Scheduling*, *Schedules 0 to 8*, *Pharmacy law* and *POPIA and health data*. Appendix 20, *South African instrument to its EU or US analogue*, is the table that stops an imported control set doing damage: read its last column, where the analogue does not exist.

Form G-3 — Control objective

Feeds `O63 Control Objective`, columns in sheet order. A first pass needs twelve to sixteen rows: the seven already in the sheet, rewritten, plus five or more South African obligations. Signed by the head of regulatory affairs, countersigned by the head of quality assurance.

Ref	Obligation	Source	Control objective	Evidence
CO-08	EXAMPLE — delete	Medicines Act 101 of 1965, s 22C and regs 23-24	Licence register holds a named Responsible Pharmacist with a Council number per licence; field cannot be null	In breach

Step 10.9 — Break the one-control-per-obligation mapping in `R16`

Produce. A corrected `R16`, and a note on what the sheet was doing instead of reading `O41`.

You need. `R16 Control Objective met by Co`, `O41 Security Control`, `O63`, `RPT11 Security Position`.

Where it lands.

> **`R16 Control Objective met by Co — Control objective`, `Obligation`, `Control`, `Control name`, `Position`**
> One row per control that contributes to an objective, so seven rows becomes thirty or more. `Position` must be read from `O41.Position`, not copied from `O63.Evidence`. The sheet name is cut at Excel's 31 characters; match on the `R16` prefix.

How.

1. Read the mapping. `CO-01` maps to `SEC-01`, `CO-02` to `SEC-02`, down to `CO-07` to `SEC-07` [20]. Seven objectives, seven controls, strictly positional. `SEC-08 Data-at-rest encryption` is mapped to nothing.
2. Then check the `Position` column against `O41`. `R16` records `CO-01` as `Evidenced` through `SEC-01 Identity lifecycle`, and `O41` records `SEC-01.Position` as **`Absent`** [20][21]. The largest obligation in the industry is evidenced by a control that does not exist.
3. Compare all seven. `R16.Position` is a verbatim copy of `O63.Evidence` and **disagrees with `O41.Position` on four of the seven** — `CO-01`, `CO-02`, `CO-04` and `CO-05`. The relationship sheet never reads the control sheet.
4. Say the structural thing out loud: **mapping each obligation to exactly one access control is how a compliance regime gets reduced to an identity problem.** Good manufacturing practice is not met by joiner-mover-leaver processing. Expand to the real control set — procedural controls, system configuration, audit-trail review, training, qualification — most of which have no sheet here.
5. Rebuild `RPT11 Security Position` from `O41` and check the footer. Two controls are absent: `SEC-01 Identity lifecycle` and `SEC-03 Privileged access management` [21].

Signed by. The security architect from `O11 Role` for the control positions; the head of quality assurance for whether the mapped set actually meets the obligation.

Done when. No row of `R16` carries a `Position` that differs from the `O41.Position` of the control it names, and no obligation in `O63` is met by exactly one control.

Shows up in. `RPT11 Security Position` and `RPT09 Regulatory Coverage`, which will disagree with each other until `R16` is fixed. That disagreement is the thing to show the board.

Goes wrong when. Somebody rebuilds `R16` by copying `O63.Evidence` again, because it is the column that makes the report look finished.

Step 10.10 — Date `O60`, and put an expiry on every dispensation in `O61`

Produce. `O60` with dates, and `O61` with an expiry on every `Granted` row.

You need. `O60 Governance Decision`, `O61 Governance Change Request`, `O54`, `O29 Gap`.

Where it lands.

> **`O60 Governance Decision — Ref, Decision, Body, Status`**
>
> Four columns and **no date column**, so nothing in the sheet can be overdue [22]. Add `Raised`, `Due`, `Decided` and `Expires`. A decision with a status and no date is a position, not a decision.

How.

1. Read the four `Outstanding` rows: `DEC-01 Adopt the recommended architecture alternative`, `DEC-02 Select the integration product`, `DEC-03 Fund compliance mechanisms`, `DEC-04 Designate data owners` [22]. Those four are what the entire roadmap waits on.
2. Note the contradiction on `DEC-01`. `O54` already marks `ALT-B Foundations first` as `Recommended` and `ALT-E Compliance led` as `Partly adopted` — so something has been adopted while the decision to adopt is outstanding.
3. Note the one that governs a validated estate. `DEC-07 Grant dispensation for point-to-point transfer` reads `Approved` in `O60` with no expiry, while the same dispensation appears in `O61` as `CR-G01` against `WP-04` expiring `2029-06` [23]. A reader of `O60` alone sees a permanent dispensation. Add the `Expires` column and copy the date across.
4. Work the other `O61` rows. `CR-G05 Refuse or re-baseline the integration package` reads `Board decision required` with no date. `CR-G04 Record benefit realisation dates per package` reads `Approved` and nothing has been recorded anywhere, which is why `GT-3 G3 Benefit` — *stated benefit confirmed against a date* — cannot be passed by any work package in the file. `CR-G03 Raise unfunded building blocks as gaps` reads `Raised`, and `O29 Gap` is still empty.
5. Set the rule: **no dispensation without an expiry, and no expiry beyond the next periodic review of the system it affects.** A dispensation that outlives a periodic review becomes the configuration.

Signed by. `GB-01 Architecture Board` for the dispensation expiries; `GB-02 Investment Committee` for the four outstanding decisions; `GB-07 Validation Committee` for any dispensation touching a GxP system.

Done when. Every `O60` row has a date in `Decided` or a date in `Due`, and every `Granted` row in `O61` has a value in `Expires`.

Shows up in. No report reads `O60` or `O61`. Add the open-decision list to the board pack yourself; four outstanding decisions on an $84.3m programme is the single most governable fact in the repository.

Goes wrong when. The dispensation is granted, the expiry passes unnoticed because nothing computes it, and the point-to-point transfer into a validated system becomes the architecture.

Step 10.11 — Complete `O33`, and make a contract something a verdict can rest on

Produce. Twelve architecture contracts, and a defensible basis for every `Compliance status`.

You need. `O33 Architecture Contract`, `O30 Work Package`, `O55`, `O59` as populated, `O58`.

Where it lands.

> **`O33 Architecture Contract — Ref, Architecture contract, Work package, Compliance status`**
>
> Eight rows for twelve work packages: `ACO-01` to `ACO-08` cover `WP-01` to `WP-08`, and `WP-09` to `WP-12` have no contract [24]. `Compliance status` takes `Not yet assessable`, `Compliant`, `Non-compliant` or `Compliant with dispensation`, and each value must resolve to `O59` rows.

How.

1. Issue `ACO-09` to `ACO-12`. `GT-0 G0 Charter` tests that scope, acceptance criteria and a contract exist, so four work packages cannot pass their first gate [2].
2. Look at the two `Non-compliant` rows: `ACO-02` against `WP-02 Data ownership and mastering` and `ACO-04` against `WP-04 Technology currency` [24]. Both are packages whose `Delivery` column reads `Past end date`. Neither has a single assessment row anywhere in the file behind the verdict, because `O59` is empty.
3. Decide what a contract contains here. The acceptance criteria from `R13`'s building blocks, the gate each is tested at, the dispensations in force with their expiry, and — for any GxP system — the validation deliverable set the supplier will produce.

4. Add the supplier clause. Where a vendor's documentation will be relied on as the functional specification, the contract must say so and must require the supplier assessment to be available [16].
5. Re-derive `Compliance status` from `O59`. A contract is `Non-compliant` when a named criterion has a `Non-conformant` verdict with evidence, not when somebody felt the project was behind.

Signed by. `GB-01 Architecture Board` issues the contract; the project lead from `O56.Lead` accepts it. Check that lead exists in `O10 Actor` — `PRJ-11` reads `Internal Auditor`, which cannot accept a delivery contract.

Done when. `O33` has twelve rows and every `Compliance status` other than `Not yet assessable` cites at least one `O59` ref.

Shows up in. Nothing reads `O33`. Put the contract register in the board pack beside the gate calendar.

Goes wrong when. The four missing contracts are issued as templates with the acceptance criteria left blank, and `GT-0` passes on the existence of a document rather than on its content.

Step 10.12 — Build the release decision record, and name who signs it

Produce. A release decision record with its named inputs, and the `O67` columns that feed it.

You need. `O67 Validated System`, `O68 Periodic Review Schedule`, `O70 Regulated Role`, the release checklist, Form G-4.

Where it lands.

> **`O67 Validated System — Ref`, `Application`, `GxP`, `Validation state`, `Last periodic review`, `Next review due`, `System owner`, `Process owner`, `Validation lead`, `URS reference`, `Audit trail reviewed`, `Decommissioning plan`**
>
> Defined in item 2 of this workbook. Phase G reads four of its columns at every release: `GxP`, `Validation state`, `Next review due` and `Audit trail reviewed`. **The release decision record itself is not a repository object.** It lives in the quality system at `(batch, market)` grain for a batch, and in the validation summary report for a system. What the repository owes it is those four columns and the named roles in `O70`.

How.

1. Write the input list, and keep it a list rather than a narrative, because the guideline treats the release checklist as a named artefact [7]. For a batch: authorisation conformity, good manufacturing practice conformity, legal requirements, storage and transport conditions, sampling, quality-control results, transmissible-agent status where relevant.
2. Add the three the duty implies and the checklist does not spell out: the **validation state** of every system that contributed a record to the batch, the **open change records** against those systems, and the **periodic-review state** from `O68`. The validated-state duty makes all three release-relevant [7].
3. Name the signer. One Responsible Pharmacist per licence, named on the licence, Council-registered, resident. Record the name, the Council number and the deputy in `O70 Regulated Role`.
4. Record the signature with its meaning. A compliant electronic signature carries the signer's printed name, the date and time, and the meaning of the signing. A signature without a declared meaning is a timestamp.
5. Build the view. Six systems contribute to a release and the join key is the batch number, which is not formatted the same way in any two of them. That join is the architecture deliverable the signer will actually use.

Signed by. The Responsible Pharmacist for a batch. For a system, `GB-07 Validation Committee` at `GT-4` with the system owner accepting. Neither signature can be delegated offshore, because the first requires Council registration and the second requires accountability inside the quality system.

Done when. Every GxP system in `O67` has a `Validation state`, a `Next review due` and a named `System owner`, and the release view shows all three to the signer without anybody assembling it by hand.

Shows up in. `RPT02 Application Portfolio` reprints `O23` with its disposition, criticality, health and owner — and no validation column. Extend the report once `O67` exists, because the portfolio and the validated-system register must be the same register.

Goes wrong when. The validated-system register lives in the quality function's spreadsheet and the application portfolio lives in the repository, the two disagree, and the disagreement is discovered by an inspector rather than by you.

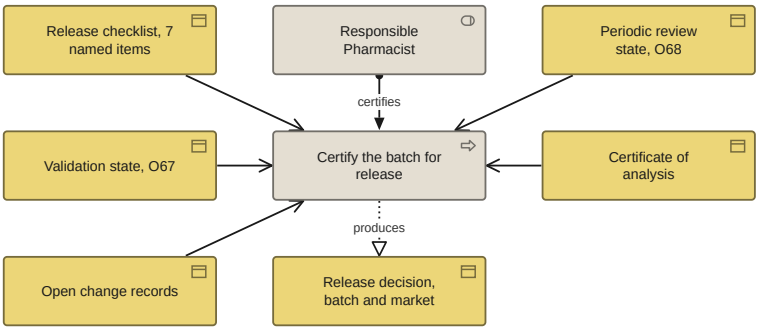


Figure 10.2 The release decision and its named inputs. The record itself is not a repository object — it lives in the quality system at batch and market grain. What the repository owes it is four columns of `O67 Validated System` and a named person in `O70 Regulated Role`.

Form G-4 — Release-relevant validation state

Feeds `O67 Validated System`, six of its twelve columns — the four Phase G reads at every release plus the two that identify the row. **`O67` is a sheet the repository does not have**; item 2 of this workbook defines it in full and the remaining six columns are typed straight into Excel. A first pass needs one row per GxP system instance, which is nine or more for the twelve applications in `O23`. Signed by the validation lead; accepted by the system owner.

Ref	Application	GxP	Validation state	Next review due	Audit trail reviewed
VAL-01	EXAMPLE — delete	Yes	Validated	2027-03	2026-09, risk-targeted

Step 10.13 — Publish `RPT09` honestly, and rehearse the inspection

Produce. A `RPT09 Regulatory Coverage` that computes rather than copies, and a two-hour inspection rehearsal.

You need. `RPT09`, `O63` as rewritten, `O59` as populated, `R16` as corrected, and the eleven bodies in volume 1 chapter 16.

Where it lands.

> **`RPT09 Regulatory Coverage` — `Ref`, `Obligation`, `Source`, `Control objective`, `Evidence`**
>
> Today it is `O63` reprinted verbatim with a typed footer — *3 obligations have no mechanism or a mechanism in breach* [25]. Add two computed columns: the count of `O59` rows evidencing each objective, and the date of the most recent one. An obligation with zero evidence rows reads zero, whatever `O63.Evidence` says.

How.

1. Recount the footer. Three rows read `In breach`, and two more read `Partial` — so five of seven are not evidenced, not three. And the two reading `Evidenced` have no evidence anywhere, because `O59` is empty. The honest number today is **seven of seven** [5] [25].
2. Note the report's shape. Nine of the thirteen `RPT` sheets end in a typed sentence sitting in the `Ref` column, `RPT09` among them [12][25]. Those footers are commentary, not computation.

3. Decide which reports an inspector would ask for. The validated-system register first, then the list of GxP systems, then evidence for one of them. `RPT02 Application Portfolio` is the nearest thing the repository has and it carries no validation column.
4. Decide which would embarrass you. **`RPT09 Regulatory Coverage` is the one** — a report titled regulatory coverage with no South African obligation in it, three breaches, and an evidence column with nothing behind it.
5. Rehearse. Two hours, one domain architect playing the inspector, three questions: which systems hold regulated records, show me the evidence for this criterion, who signed the last release. Any answer taking more than ten minutes to assemble is your next work package.

Signed by. The head of quality assurance owns inspection readiness. `GB-05 Architecture Function` owns the report.

Done when. `RPT09` computes its evidence count from `O59`, its footer states the real number, and the rehearsal has produced a dated action list.

Shows up in. `RPT01 Dashboard`, whose `Control objectives without evidence` line reads 3 against a true figure of 7 until `O59` and `O63` are both fixed.

Goes wrong when. The report is corrected and the footer is left alone, so the one line a busy executive reads still says three.

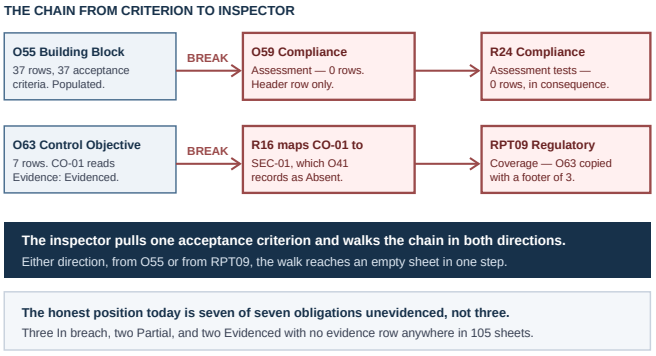


Figure 10.3 The evidence chain an inspector walks, and the two places it is broken in this repository. `O59` has no rows, so `R24` has none either — and `RPT09` reports an evidence position it computes from nothing.

What you hand to Phase H

Four things. **A populated `O59` with honest verdicts**, including the twenty that read `Not assessable` and why. **A gate set with five gates**, one of which is release for use, each with a named holder in `O57`. **An `O63` that names South African instruments** and a `R16` that reads `O41` instead of copying `O63`. And **a release decision whose inputs are a list**, with the named signatory recorded in `O70 Regulated Role`.

Phase H then has one job Phase G cannot do for it: keeping all of that true. A change to a validated system is a change-control record before it is a backlog item, and periodic review is what keeps the validated state from quietly expiring [26]. Hand over the dispensation expiry dates first. They go stale while nobody is computing them.

THE CHAPTER IN TEN LINES

- O58's four gates are the best-written thing in the repository. Do not redesign them. None of them is release.
- O59 Compliance Assessment has no rows, so R24 has none either, and the gate has produced zero records.
- O63 records CO-01 GMP / GCP / GLP / GVP as Evidenced while the sheet that would hold the evidence is empty.
- O57 has five governance bodies and not one quality body: no change control board, no validation committee, no data integrity forum.
- O55 holds 37 acceptance criteria. One of them passes today: ABB-08 QMS target, because QMS has no unmanaged interface.
- The ABB-08 criterion passes while the block is not delivered, because the criterion does not test what the block must do.
- Twenty of the 37 criteria are not testable as written. Ten test an owner column that holds job titles; ten name no measure.
- Every Control objective cell in O63 reads Control that evidences the obligation restated. That is the obligation with four words in front of it.
- There is no South African obligation anywhere in O63. No SAHPRA registration duty, no single exit price, no POPIA by name, no Medicines Act scheduling, no Pharmacy Act licensing.
- R16 maps each obligation to exactly one SEC control, its Position column copies O63.Evidence, and CO-01 is evidenced through SEC-01 Identity lifecycle, which O41 records as Absent.
- O60 Governance Decision has no date column, and DEC-07 grants a dispensation with no expiry while CR-G01 records the same one expiring 2029-06.
- The release signatory in South Africa is the Responsible Pharmacist: one per licence, Council-registered, with a duty to keep processes in a validated state.
- RPT09 Regulatory Coverage is O63 copied with a footer saying three. The honest number is seven of seven.

SOURCES FOR THIS CHAPTER

- **1.** The Open Group, TOGAF Standard, Architecture Development Method — Phase G, Implementation Governance. Identified, not retrieved: The Open Group's public pages are robots-disallowed to automated retrieval, so no edition or wording is claimed from it.
- **2.** Pharmaceuticals_TOGAF_Metamodel.xlsx, Calderon Therapeutics — sheet O58 Governance Gate (rows 5-8)
- **3.** Pharmaceuticals_TOGAF_Metamodel.xlsx — sheets O57 Governance Body (rows 5-9), O10 Actor (rows 5-22) and O11 Role (rows 5-22)
- **4.** Pharmaceuticals_TOGAF_Metamodel.xlsx — sheets O59 Compliance Assessment (0 rows, header row 4 only) and R24 Compliance Assessment tests (0 rows)
- **5.** Pharmaceuticals_TOGAF_Metamodel.xlsx — sheet O63 Control Objective (rows 5-11)
- **6.** Pharmaceuticals_TOGAF_Metamodel.xlsx — sheets O04 Measure (rows 5-20), R22 Org Unit owns Capability and O09 Organization Unit (rows 5-16)
- **7.** SAHPRA, SAHPGL-LIC-01 v3, Guide on How to Apply for a Licence to Manufacture, June 2022 — one Responsible Pharmacist named per licence, Council registration, batch certification before release, the release checklist as a named artefact, and the duty that processes remain in a validated state https://www.sahpra.org.za/wp-content/uploads/2022/06/SAHPGL-LIC-01-Guide-On-How-to-Apply-for-A-Licence-To-Manufacture.pdf
- **8.** South African Pharmacy Council, Responsible Pharmacist — Overview, under the Pharmacy Act 53 of 1974 (dual accountability to the Council and to the medicines regulator) https://pharmcouncil.co.za/Responsible_Overview-2
- **9.** Pharmaceuticals_TOGAF_Metamodel.xlsx — sheet O55 Building Block (37 rows, 5-41: ten BBB, ten DBB, ten ABB, seven TBB)
- **10.** Pharmaceuticals_TOGAF_Metamodel.xlsx — sheet R06 Application interfaces with (36 rows, 19 Unmanaged, 5 Logged, 6 Direct DB read)
- **11.** Pharmaceuticals_TOGAF_Metamodel.xlsx — sheet O23 Application Component (rows 5-16)
- **12.** Pharmaceuticals_TOGAF_Metamodel.xlsx — sheets RPT04 Interface Catalogue (37 data rows, the last a sentence in the Ref column) and the nine RPT sheets carrying a typed footer sentence: RPT01, RPT04, RPT05, RPT06, RPT08, RPT09, RPT10, RPT11, RPT13
- **13.** Pharmaceuticals_TOGAF_Metamodel.xlsx — sheets O20 Data Entity (rows 5-18) and O43 Master Data Rule (rows 5-14)
- **14.** Pharmaceuticals_TOGAF_Metamodel.xlsx — sheets R07 Application deployed on Technology (rows 5-16, all Environment: Production) and O48 Environment (rows 5-8)
- **15.** Pharmaceuticals_TOGAF_Metamodel.xlsx — sheet O27 Physical Technology Compone (rows 5-16; PTC-07 support end 2026-08 and PTC-11 support end 2026-01, both Past support)
- **16.** Volume 1, Pharmaceuticals — a starter pack for enterprise architects, chapter 13, on the validation master plan as the first document an inspector asks for, the per-system deliverable set, and supplier documentation as the functional specification
- **17.** SAHPRA, Communication to Stakeholders HPA-RNW01-2025/26, 21 August 2025 — registration validity of five years, renewal six months before expiry, and no variations in flight during renewal review https://www.sahpra.org.za/wp-content/uploads/2025/08/HPA-RNW01-2025_26-Medicines-Certificate-of-Registration-Renewal.pdf
- **18.** Volume 1, chapter 16 and appendix 20, on the single exit price as the only price at which a manufacturer or importer may sell to a private buyer, and the 2026 adjustments with their effective dates
- **19.** Protection of Personal Information Act 4 of 2013 — s 26 and s 27 on special personal information, s 32 on health and sex life, s 57 on prior authorisation and s 72 on transborder flows; with the Information Regulator's Guidance Note on Applications for Prior Authorisation https://inforegulator.org.za/wp-content/uploads/2020/07/Revised-Guidance-Note-on-applications-for-Prior-Authorisation.pdf
- **20.** Pharmaceuticals_TOGAF_Metamodel.xlsx — sheet R16 Control Objective met by Co (rows 5-11)
- **21.** Pharmaceuticals_TOGAF_Metamodel.xlsx — sheets O41 Security Control (rows 5-12; SEC-01 and SEC-03 Absent) and RPT11 Security Position
- **22.** Pharmaceuticals_TOGAF_Metamodel.xlsx — sheets O60 Governance Decision (rows 5-11, four columns, no date column) and O54 Architecture Alternative (rows 5-9)
- **23.** Pharmaceuticals_TOGAF_Metamodel.xlsx — sheet O61 Governance Change Request (rows 5-9)
- **24.** Pharmaceuticals_TOGAF_Metamodel.xlsx — sheets O33 Architecture Contract (rows 5-12), O30 Work Package (rows 5-16) and O56 Project (rows 5-16)
- **25.** Pharmaceuticals_TOGAF_Metamodel.xlsx — sheet RPT09 Regulatory Coverage (7 data rows reprinting O63, plus a typed footer sentence in the Ref column)
- **26.** Volume 1, chapter 13, Periodic review — the documented check that the system is still in a validated state, with its review items and risk-justified frequency

CHAPTER 11

Architecture Change Management and the validated state

Validation is a state you maintain, not a thing you did. Periodic review is the mechanism that keeps it true, and the repository has no column for its due date.

What Phase H is for, and why it is different here

Phase H decides whether a change to the estate is absorbed by the architecture you already have or forces a new turn of the ADM [1]. In the sibling volumes of this series that is a question about cost and scope. Here it is a regulated question, and the wrong answer is an inspection finding.

The reason takes one sentence, and volume 1 gives it a section of its own: **validation is a maintained state, not a project phase** [15]. A system was validated on a date, by a summary report a named person signed, and everything afterwards either keeps that statement true or quietly makes it false. Two mechanisms keep it true: **change control**, which catches the deliberate changes, and **periodic review**, which catches everything else — drift, configuration creep, a departed administrator, an intended use that moved while the validation package stood still [15].

Calderon Therapeutics, the company in this repository, and volume 1's **Veldtkruin Group** are both inventions. Never mix their data.

> **A change to a validated system is a change-control record before it is a backlog item.**
>
> Not after, not alongside. Before. The sequence that gets companies into trouble is the ordinary one: somebody raises a ticket, a team sizes it, it enters a sprint, and a change-control record is written afterwards so the paperwork matches. In a validated estate that order is the defect, because the record is where the four impact assessments live — quality, validation, regulatory and data integrity — and at least one of them can stop the change [16].
>
> The regulatory assessment is the one most often skipped, by someone who does not have the registered state in front of them; the consequence arrives years later as a running process that is not the registered process [16]. Your contribution is specific: make `O61 Governance Change Request` the first record, not the last.

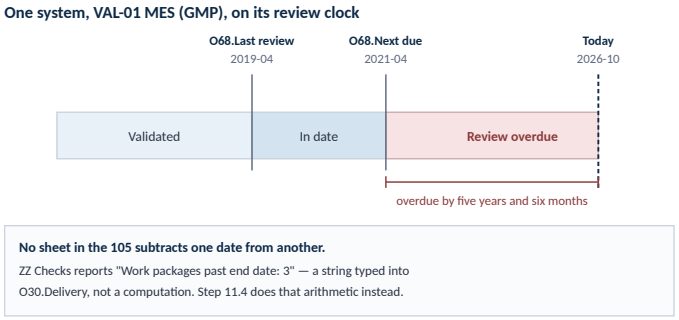


Figure 11.1 The clock nothing computes. `O68.Last review` plus `O68.Frequency` gives `O68.Next due`; the distance between that date and today is the only number Phase H really has. The repository types its overdue counts in as strings instead.

Before you open a sheet

ANSWER THESE BEFORE YOU TOUCH THE GOVERNANCE SHEETS

Who owns the architecture repository as a document — not who edits it, who owns it? What version is it at today?

For the oldest GMP system in the estate, when was the last periodic review, and who holds the report?

Who decides whether a change needs revalidation, and where is that decision written down?

The steps

Step 11.1 — Make the architecture repository a controlled document

Produce. A front-sheet entry on `00 Index` naming the repository's owner, its version, its approval date and the rule for changing a row — plus a change log.

You need. `00 Index`, `O57 Governance Body`, `O31 Deliverable`, and item 2's decision on whether an architecture decision is a quality record. Half a day.

Where it lands.

> **`00 Index`, rows 1 to 10, and a new sheet `01 Change Log`**
>
> Rows 5 to 10 of `00 Index` carry prose answers to *What this is*, *Structure*, *Referential integrity* and three more [12]. Add four rows in the same shape: `Owner`, `Version`, `Approved`, `Change rule`. The change log is not a numbered object sheet — it is a log, with `Date`, `Sheet`, `Ref`, `Was`, `Is`, `Raised by`, `Approved by`.

How.

1. Name the owner as a person, not `GB-05 Architecture Function`. `O57`'s mandate for `GB-05` reads *Publishes criteria, assesses compliance, maintains the repository* [3] — a mandate, not an owner. Write the Chief Enterprise Architect's name in the cell.
2. Set the version. If the file has never carried one it is 1.0 as handed over and 2.0 after items 2 to 10. Write the change rule in one sentence and make it restrictive: a row changes on a change-log entry naming the body that approved it. Nobody edits a cell because it looks wrong.
3. Decide, from item 2, whether the repository is inside the quality system; if it is not, say so on the front sheet.
4. Add `DLV-21 Repository baseline and change log` to `O31 Deliverable` with `Phase: H`. `O31` has twenty rows and **no Phase H deliverable at all**, no approver column and no date column [9].

Look it up. Volume 1, chapter 13, section *What the architect is actually expected to produce*, and chapter 16, section *Inspection response*.

Signed by. `GB-01 Architecture Board` approves the baseline; the owner signs their own name into the cell. Where item 2 put the repository inside the quality system, the Head of Quality counter-signs: an uncontrolled quality record is worse than no quality record.

Done when. `00 Index` names an owner, a version and an approval date, and `01 Change Log` has at least one row — the row recording its own creation.

Shows up in. Nothing computes over it. A controlled document is checked by a person on a cadence, and step 11.12 sets the cadence.

Goes wrong when. The repository is versioned once, at hand-over, then edited by four people for eighteen months. The version on the front page is the one an auditor quotes back at you.

Step 11.2 — Define `O68 Periodic Review Schedule`

Produce. A new sheet, `O68 Periodic Review Schedule`, eight columns, no rows yet.

You need. Item 2's `O67 Validated System` and its `VAL-` key space. Items 7 and 10 both defer the review dates to this sheet by name [3]. Thirty minutes in Excel.

Where it lands.

> **`O68 Periodic Review Schedule` — the second sheet the repository does not have**
>
> Create it after `O67 Validated System`, on the five-row convention: title, subtitle, blank, headers on row 4, first record on row 5. Eight columns, in order:
>
> | Column | What goes in it |
> |---|---|
> | `Ref` | `PRS-01` onward. `PRS-` is used nowhere in the 105 sheets |
> | `System` | The `VAL-` ref from `O67`, then the application name. Never the `A0n` ref: a validated system is not always an application |
> | `Frequency` | A period with a justification reference: `2 years (RA-07)`. Never a bare number |
> | `Last review` | `YYYY-MM-DD`, or `Never`. `Never` is legitimate and common |
> | `Next due` | `YYYY-MM-DD`, computed from `Last review` plus `Frequency`. A date, never a word |
> | `Reviewer` | A named person, independent of the system's day-to-day administration |
> | `Outcome` | One of `Validated state confirmed`, `Confirmed with actions`, `Not confirmed`, `Not performed` |
> | `Actions open` | An integer, and a reference to where the actions live |

How.

1. Type the eight headers into row 4 exactly as above; item 13 and item 18's forms quote them.
2. Key on `O67`, not on `O23 Application Component`. `O67`'s grain is one system instance in one environment for one defined use, which is the grain a review is performed at.
3. Restrict `Outcome` to the four values. `Not performed` is the one people leave out, and the one that makes the sheet honest.
4. Keep `Frequency` here and nowhere else. `O67` holds `Last periodic review` and `Next review due` as dates and no frequency — item 2 pinned that, and a frequency in two sheets diverges. Add the sheet to `00 Index` with `0 records`, and a change-log row from step 11.1.

Look it up. Volume 1, chapter 13, section *Periodic review* — the twelve review items and the risk-justified frequency [15].

Signed by. `GB-05 Architecture Function` for the structure; `GB-07 Validation Committee`, added to `O57` by item 10, owns the content from the first row [3].

Done when. Eight headers sit on row 4, `Outcome` is restricted to four values, and `00 Index` carries the sheet.

Shows up in. No report yet. Step 11.4 builds the only view that matters.

Goes wrong when. `O68` is created as a copy of `O67` with two columns added. It is a schedule, not a register: one system appears in it many times over the years, and the history is the evidence.

Form 11-A — Periodic review schedule, part one: schedule it

Feeds `O68 Periodic Review Schedule`, a sheet the repository does not have — create it first, per Step 11.2. Six of the eight columns. A first pass needs one row per `O67` row, so twelve to twenty rows. `Outcome` and `Actions open` are Form 11-C, filled after the review. Signed by the validation lead and the system owner jointly.

Ref	System	Frequency	Last review	Next due	Reviewer
PRS-01	EXAMPLE — delete this row. VAL-01 MES (GMP)	1 year (RA-02)	Never	Overdue	

Step 11.3 — Populate `O68`, and make every frequency risk-justified

Produce. One `O68` row per validated system, with a frequency somebody can defend and a `Next due` date computed rather than typed.

You need. Form 11-A printed, `O67` as items 2, 6 and 7 filled it, the validation lead, the system owners, and the risk assessments behind the frequencies. Two hours, then a week of chasing dates.

Where it lands.

> **`O68 Periodic Review Schedule`, rows 5 onward**
>
> Fill `Ref`, `System`, `Frequency` and `Reviewer` in the room. `Last review` comes from documents, not from memory; where no report can be produced, the cell reads `Never`, whatever anybody remembers. `Next due` is a formula on the other two.

How.

1. Set the frequency from risk: the regimes in `O67.GxP`, the criticality in `O23`, the record class, and how much the system changes. A GMP batch-record system changing monthly is annual; a stable GLP instrument system is two or three years.
2. Record the justification reference in the cell. `2 years (RA-07)` resolves to a risk assessment; a bare `2 years` is a preference, and a preference is what an inspector picks at.
3. Write `Never` where no report exists. `O23` has three applications in service since 2010 or 2011 — `A01 ELN / LIMS`, `A05 RIM` and `A07 MES (GMP)` [8]. A fifteen-year-old GMP system with `Last review: Never` is the most valuable row you will write this phase.
4. Compute `Next due`; do not type it. The arithmetic is the control, and a typed date drifts from the frequency within two cycles.
5. Name a reviewer independent of the system's administration. `O11 Role` holds eighteen generic titles and no validation engineer, so for most rows the reviewer is a vacancy — record it in item 2's `O70 Regulated Role` rather than leaving the cell blank. Then count the rows: `Never`, and `Next due` in the past, are the two numbers you take to the board.

Look it up. Volume 1, chapter 19, section *Week one: find out what is validated*; chapter 13, section *Computerised system validation, as it is actually practised*.

Signed by. The validation lead signs the frequencies and `GB-07 Validation Committee` the schedule. Where the real signatory is the Responsible Pharmacist — a statutory duty the repository has no row for — `O70 Regulated Role` records who it actually is [19].

Done when. Every `O67` row has an `O68` row, every `Frequency` a justification reference, and `Next due` is computed everywhere.

Shows up in. Item 14's validated-estate view. `ZZ Checks` still reports `Object types populated: 65` and knows nothing about `O67` or `O68` [12].

Goes wrong when. Every system gets two years because two years is what the last company did. Uniform frequency is the tell that no risk assessment happened anywhere.

Step 11.4 — Compute the overdue list, and publish it every month

Produce. A one-page overdue list: systems whose `Next due` is in the past, ordered by how far.

You need. `O68` populated. An hour a month, for ever.

Where it lands.

> **`A view over O68 Periodic Review Schedule`**
>
> Not a new sheet. A filter on `Next due` against today, with `System`, `Next due`, `Reviewer`, `Outcome` and `Actions open`. Publish it as a page in the board pack, not as a tab somebody has to open.

How.

1. Write the arithmetic down once: days between `Next due` and today. **Nothing in the 105 sheets subtracts one date from another.** `ZZ Checks` reports `Work packages past end date: 3`, and that string is typed into `O30.Delivery`, not computed from `O30.End` [13]. Do not repeat the pattern.
2. Sort descending by days overdue. The order is the agenda.
3. Add two counts underneath: rows reading `Never`, and rows reading `Not performed`.
4. Take it to `GB-07 Validation Committee` monthly and `GB-01 Architecture Board` quarterly. The board sees the trend; the committee sees the rows.
5. When a row goes overdue by a full frequency period, set `O67.Validation state` to `Validated, review overdue` — item 2 put that value in the list for this [3].

Look it up. Volume 1, chapter 16, section *Inspection response* — what the first hour of an inspection asks for.

Signed by. Nobody signs a list. `GB-07 Validation Committee` accepts it and the acceptance is minuted: an accepted overdue list is a managed risk, an unaccepted one a surprise.

Done when. The list exists, has a date on it, and was tabled at a meeting with a minute.

Shows up in. The board pack. No `RPT` sheet computes it, and there is no point adding one until somebody owns the cadence.

Goes wrong when. The list is built once, for the inspection, from the schedule rather than the reports. The schedule says a review happened; the report is what the inspector asks to see.

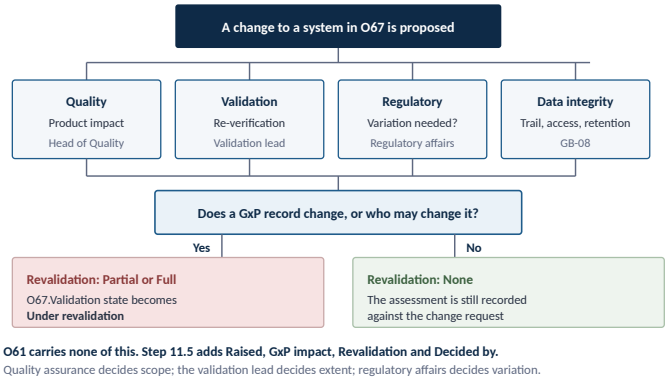


Figure 11.2 The triage, and the four columns `O61` has not got. Four assessments run in parallel and any one of them can stop the change. The architect owns the record, not the verdict: the test is on the **record**, never on the software.

Step 11.5 — Decide which changes trigger revalidation, and record who decides

Produce. A written triage rule, and a named decider for each class of change.

You need. The Head of Quality, the validation lead, the regulatory affairs manager, and whoever holds the registered state of the products. Two hours, and the hardest two of the phase.

Where it lands.

> **`O61 Governance Change Request` — four new columns**
>
> `O61` has `Ref`, `Change request`, `Against`, `Type`, `Status` and `Expires` [2]. Add `Raised`, `GxP impact`, `Revalidation` and `Decided by`. `GxP impact` takes the regimes affected or `None`; `Revalidation` takes `None`, `Partial`, `Full` or `Assessment pending`; `Decided by` takes a person, never a body.

How.

1. Say the finding out loud first. **No column anywhere in the 105 sheets would let you answer whether a change needs revalidation.** `O23 Application Component` has no validation state, no GxP flag and no review date [8]; `O61` records a change type — `Dispensation`, `Scope`, `Correction`, `Compliance` — and no quality impact at all [2].
2. Write the triage on the record, not the software. A change needs revalidation when it alters what a GxP record is, how it is created, who can change it, what the audit trail captures, or how it is retained. A change touching none of those does not, however large it looks.
3. Keep the four parallel assessments from volume 1's change-control flow: quality impact on product, validation impact and proportionate re-verification, **regulatory impact and whether a variation is required**, and data-integrity impact on trail, access, retention and migration [16].
4. Name the decider per class. Quality assurance decides GxP scope; the validation lead decides the extent of re-verification; regulatory affairs decides whether a variation is needed and where. **The architect decides none of these** and owns the record that a decision was taken.
5. Write down the three changes people get wrong: a vendor patch; an interface added to a validated system, which sits inside its validation scope [10]; and a dictionary or reference-data version change, which alters the meaning of records already held. Where quality and regulatory disagree, escalate to `GB-04 Executive Committee`, whose `O57` mandate is *Escalation point when bodies cannot agree* [3].

Look it up. Volume 1, chapter 8, section *Change control*; chapter 13, sections *Seven pieces of folklore, punctured* and *Data integrity*; chapter 3, section *Validation, qualification, verification*.

Signed by. The Head of Quality signs the triage rule. `GB-06 Change Control Board`, added to `O57` by item 10, operates it [3]. The repository had no quality body before that step, which is why this rule has never been written here.

Done when. The rule is one page, each class has a named decider, and `O61` carries the four new columns.

Shows up in. `O61`, in the falling proportion of rows reading `Assessment pending`.

Goes wrong when. The rule is written as a table of system types. It is a rule about records, and a table of types puts a validated spreadsheet outside the regime and a reporting tool inside it.

Form 11-B — Change triage record

Feeds `O61 Governance Change Request`, including the four columns Step 11.5 adds. One form per change to a system that appears in `O67 Validated System`. The four assessment verdicts go on the back of this sheet or into the change-control system; this form is the architecture record of them. Signed by the Head of Quality for `GxP impact` and by the validation lead for `Revalidation`.

Ref	Change request	Against	GxP impact	Revalidation	Decided by
CR-G06	EXAMPLE — delete this row. Vendor patch to MES	VAL-01	GMP, GDocP	Partial	

Step 11.6 — Widen `Against`, so a change request can be raised against a principle or a standard

Produce. An `O61` whose `Against` column resolves to any governed object, and the first batch of requests raised against `O05 Principle` and `O32 Standard`.

You need. `O61` , `O05 Principle` , `O32 Standard` , `O67` , `O55 Building Block` . A morning.

Where it lands.

> **`O61 Governance Change Request` , column `Against`**
>
> Today the five rows resolve to `WP-03` , `WP-04` and the string `Repository` [2]. Make the column take a ref from any of `O05` , `O32` , `O55` , `O67` , `O30` or `O68` , and write the ref, not a word. `Repository` becomes a change-log entry from step 11.1, not a change request.

How.

1. Classify the five existing rows. `CR-G01` and `CR-G05` are against work packages; `CR-G02` is a scope change; `CR-G03 Raise unfunded building blocks as gaps` and `CR-G04 Record benefit realisation dates per package` are corrections, not architecture changes [2].
2. Raise the standards requests. `O32` has `STD-02 Event schema standard` and `STD-15 Records management standard` both `Deprecated` and `Conformant` , and `STD-07 Logging and monitoring standard` `Deprecated` and in `Breach` [7]. A deprecated standard with a conformance verdict is a change request: retire it or reinstate it.
3. Raise the retention contradiction. `STD-04 Data retention standard` is `Current` and `Conformant` ; `STD-15 Records management standard` is `Deprecated` and `Conformant` [7]. Two standards over one subject, one of them dead, both conformant. In an industry whose retention clocks run from events rather than dates, that pair decides whether a batch record survives [18].
4. Raise requests against the principles. `O05` 's twenty-four are all owned by `Architecture Board` and none mentions validated state, data integrity, the patient's data or the regulator [6]. A principle nobody has ever raised a change request against is decoration.
5. Set the route: a request against a principle goes to `GB-01 Architecture Board` ; against a standard, to its owner; against a system in `O67` , to `GB-06 Change Control Board` with `GB-07 Validation Committee` consulted.

Look it up. Volume 1, chapter 16, section *The retention clocks*; appendix 20, section *Retention clock by record class*.

Signed by. `GB-01 Architecture Board` , whose `O57` mandate includes approving dispensations with an expiry [3].

Done when. Every `O61.Against` cell holds a ref, and at least three requests are open against `O32` rows.

Shows up in. `RPT12 Standards Conformance` , which reprints `O32` and will keep showing `Deprecated` beside `Conformant` until somebody closes those requests [7].

Goes wrong when. `Against` stays free text. Within a year it holds project names, team names and a sentence, and nothing joins.

Step 11.7 — Sweep the dispensations monthly, and bound every expiry by the next periodic review

Produce. A monthly dispensation sweep, and an expiry date on every dispensation in force.

You need. `O61` , `O60 Governance Decision` as item 10 dated it, `O68` from step 11.3. An hour a month.

Where it lands.

> **`O61 Governance Change Request` , column `Expires` , and `O60 Governance Decision`**
>
> `O61.Expires` is populated on **one of five rows** — `CR-G01 Dispensation for point-to-point file transfer` , expiring `2029-06` [2]. `O60` records the same dispensation as `DEC-07 Grant dispensation for point-to-point transfer` , status `Approved` , in a sheet with four columns and no date column at all [4]. Item 10 added `Raised` , `Due` , `Decided` and `Expires` to `O60` ; this step operates them.

How.

1. Read the pair. `O60` alone says a dispensation for point-to-point transfer was approved and never expires; `O61` says it expires in June 2029. A reader of either is misled, and neither sheet points at the other [2][4].
2. Copy the expiry across, then set the bounding rule: **no expiry beyond the next periodic review of the system the dispensation affects.** A dispensation that outlives a periodic review has become the configuration, and the review will record it as the documented state.
3. Test the rule on this dispensation. Point-to-point transfer into a validated system touches `R06` 's thirty-six interfaces, nineteen of them unmanaged [14]. If the affected system's `Next due` is 2027, a 2029 expiry is out of bounds and the date comes back.
4. Sweep monthly. Expiring within ninety days goes on the agenda; expired goes to `GB-06 Change Control Board` that week, as either a renewal with a fresh assessment or a dated withdrawal.
5. Record the withdrawal. A dispensation that lapses silently leaves the estate non-conformant with no record of when, which is the hardest state to explain afterwards.

Look it up. Volume 1, chapter 12, section *Interfaces are routinely in nobody's validation scope*.

Signed by. `GB-01 Architecture Board` grants and renews. `GB-07 Validation Committee` must agree any dispensation touching a system in `O67`.

Done when. Every `Granted` row in `O61` has an `Expires` date inside the affected system's next review, and `O60` and `O61` agree on `DEC-07` and `CR-G01`.

Shows up in. No report reads `O60` or `O61`. The sweep is a page you produce; nothing produces it for you.

Goes wrong when. The dispensation is renewed on the same assessment for six years. Three renewals without a fresh assessment is a decision to change the architecture, taken by nobody.

Step 11.8 — Classify every change by what it changes, and route it accordingly

Produce. A four-way classification of open changes, and the route each class takes.

You need. `O61`, `O66 Architecture Increment`, `O53 Transition Architecture`. Two hours.

Where it lands.

> **`O61 Governance Change Request`, `column Type`**
>
> `Type` holds `Dispensation`, `Scope`, `Correction` and `Compliance` across five rows [2]. Keep those values and add the class each belongs to: the Phase H change classes decide the route, not the local label [1].

How.

1. Use four classes. **Simplification** — the architecture absorbs the change with no new work. **Incremental** — the change is delivered inside an existing increment in `O66`. **Re-architecting** — the change starts a new ADM cycle. And the one this industry adds: **record correction** — the estate did not change, the repository was wrong.
2. Route record corrections to the change log from step 11.1 and nowhere near a backlog. `CR-G03` and `CR-G04` are both corrections and both sit in the change-request sheet as though somebody must build something [2].
3. Route incremental changes to an increment. `O66` holds six increments landing at months 10, 16, 22, 30, 42 and 56 [5], and item 9 mapped three of them onto the three transition states by name identity. An incremental change names its increment or it is not incremental.
4. Reserve re-architecting for the triggers in step 11.9: a change gets that class because a trigger fired, not because it is large.
5. Apply the overlay to the classification itself: **a change can be a simplification in architecture terms and a full revalidation in quality terms.** The two are independent, which is why Form 11-B carries both columns. Publish the counts by class each quarter.

Look it up. Volume 1, chapter 13, section *The system itself is regulated*.

Signed by. `GB-01 Architecture Board` classifies. `GB-02 Investment Committee` is told, because incremental and re-architecting classes both consume money it holds [3].

Done when. Every open change carries one of the four classes and, for the incremental ones, an `INC-` ref.

Shows up in. `RPT08 Roadmap`, which reprints `O30 Work Package` and shows new work only once incremental changes reach a package [13].

Goes wrong when. Everything is incremental, because incremental needs no decision. The estate changes continuously and the architecture is never re-examined.

Step 11.9 — Write the triggers that open a new ADM cycle, and date them

Produce. A written trigger list — six to ten lines — each with a named watcher and a review cadence.

You need. `O01 Driver`, `O66 Architecture Increment`, `O68`, and the business calendars item 9 built the plan against. Two hours.

Where it lands.

> **`O01 Driver` and a trigger list held with the change log**
>
> `O01` holds thirteen drivers, seven regulatory and six business, all owned by `Chief Compliance Officer` or `Group Executive`. A trigger is a driver with a threshold and a watcher. Record the thresholds beside the drivers; the list itself lives with step 11.1's change log.

How.

1. Write the regulated triggers first — they fire without warning: a new or revised obligation; an inspection finding; a product recall; a variation that changes a process; a site licence change; a dispensation refused.
2. Add the estate triggers: a system entering `Validated, review overdue`; a periodic review with `Outcome: Not confirmed`; a component passing vendor support, which `O27` already shows for `PTC-07` and `PTC-11` [14]; a resilience test failing.
3. Add the business triggers: an acquisition, a site opening or closing, a therapeutic area entering or leaving the portfolio, and a new archetype — a generics manufacturer that starts contract manufacturing is a different company in architecture terms.
4. Give each trigger a named watcher and a cadence.
5. Do the arithmetic that makes the cadence real. `O66`'s last increment lands at **month 56** [5]. On a two-year review cycle every system is reviewed twice before it arrives, so the review cycle is the faster clock and will surface the need for a new ADM turn long before month 56. Set the standing cadence from it: annual architecture review, quarterly trigger check, monthly overdue list.

Look it up. Volume 1, chapter 7, sections *The Annual Product Quality Review* and *Variation planning, and the correction that inverts a workflow*; chapter 2, section *How to tell which archetype you are in, in the first week*.

Signed by. `GB-01 Architecture Board` adopts the trigger list. Each watcher signs for their own trigger.

Done when. Every trigger has a watcher and a cadence, and the first quarterly check has a date in the diary.

Shows up in. `RPT01 Dashboard` has eighteen measures and none is a trigger state [12]. Put the list in the board pack, not the dashboard: a trigger is a judgement.

Goes wrong when. The list is written and filed. Four of these triggers fire in a normal year in this industry; none fires in a way that sends you an email.

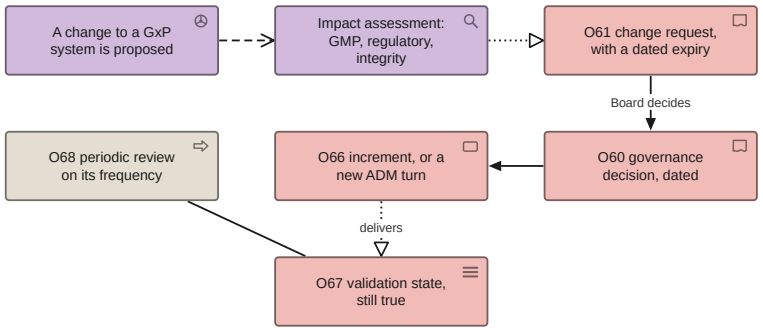


Figure 11.3 The Phase H chain, drawn on the repository's own refs. The change request realises the impact assessment and triggers the governance decision; the decision lands in an increment or opens a new ADM turn. Periodic review is the second, slower path to the same plateau — and the only one that fires when nobody raised anything.

Step 11.10 — Re-assess the things that decay, on a cadence

Produce. A quarterly re-assessment pack covering the sheets whose contents go stale.

You need. `O63 Control Objective` as item 10 rewrote it, `O62 Delivery Risk`, `O04 Measure`, `O41 Security Control`, `O49 Resilience Tier`. Half a day a quarter.

Where it lands.

> **`O63 Control Objective`, `O62 Delivery Risk`, `O04 Measure`**
>
> `O63`'s `Evidence` column decays fastest: three of seven read `In breach`, and `RPT09 Regulatory Coverage` reprints the sheet verbatim [11]. `O62` holds eight risks, three at `Likelihood: Certain` — items 8 and 9 moved those into `O52 Implementation Factor`; check they stayed moved [17].

How.

1. Re-read `O63.Evidence` against `O59 Compliance Assessment` as item 10 populated it. Evidence assessed eight months ago is not current evidence.
2. Check `R16 Control Objective met by Co`, a verbatim copy of `O63.Evidence` that never read `O41.Position` [11]. Item 10 broke the copy; verify a quarterly refresh has not restored it.
3. Re-score the measures that contradicted their own sheets. `MSR-01 Obligations with evidence` read 54% against `O63`'s two of seven [11]. A measure that disagrees with its source is worse than a missing one, because somebody quotes it.

4. Re-test what was never tested. `O49 Resilience Tier` had nine of twelve at `Never tested` [14]. Each test either changes the cell or confirms it; both are results.
5. Re-check the two absent controls. `SEC-01 Identity lifecycle` and `SEC-03 Privileged access management` both read `Absent`, and `SEC-01` is the control `CO-01` claimed its evidence through [11]. Keep the pack to one page per domain.

Look it up. Volume 1, chapter 13, section *Audit-trail review, the most under-resourced obligation in the regime*; chapter 16, section *Who inspects, and what they ask for*.

Signed by. `GB-08 Data Integrity Forum`, added to `O57` by item 10, for the audit-trail and control rows. `GB-01 Architecture Board` for the pack.

Done when. Each of the five sheets has a last-re-assessment date in the change log, none older than a quarter.

Shows up in. `RPT09 Regulatory Coverage`, `RPT11 Security Position` and `RPT12 Standards Conformance`. Republish all three after each pack, and drop the last row of each: nine of the thirteen reports end in a typed sentence rather than a record [12].

Goes wrong when. The pack is produced from the sheets instead of the estate. Re-reading a cell is not a re-assessment.

Form 11-C — Periodic review outcome

Feeds `O68 Periodic Review Schedule`, columns `Ref`, `Last review`, `Next due`, `Outcome` and `Actions open`, and updates `O67.Validation state` and `O67.Last periodic review`. One form per review performed. The review itself has twelve items and they do not fit here — this records its outcome, and the report is the evidence. Signed by the reviewer and the system owner.

Ref	System	Review date	Outcome	Actions open	New next due
PRS-01	EXAMPLE — delete this row. VAL-01 MES (GMP)	2026-11-14	Confirmed with actions	4	2027-11-14

Step 11.11 — Decide what happens to a system you switch off

Produce. A decommissioning position for every system with `Disposition: Retire` or `Replace`, and a `Decommissioning plan` value in `O67` for each.

You need. `O23 Application Component`, `O67 Validated System`, item 5's `O69 Retention Clock`. Half a day, and a budget conversation you will not win in one meeting.

Where it lands.

> **`O67 Validated System`, column `Decommissioning plan`**
>
> Takes `Yes` with a document reference, `No`, or `Not applicable`. A retiring system's records must stay available, readable and intelligible for their retention period; the periods are in `O69 Retention Clock` as `RC-nn` — cite the ref, do not restate it [18].

How.

1. List them. `O23` has `A10 Serialisation` at `Retire` with `Criticality: Critical`, and `A07 MES (GMP)` and `A08 QMS` at `Replace`, `A08` also `Critical` [8]. Three of twelve are on their way out; two are critical.
2. For each, answer the question that matters: how do the records come out, in what format, with what reference data, and what renders them in fifteen years? That decision is made at implementation or it is made expensively [15].
3. Write the retention position per record class, citing `RC-nn`. A batch record's clock runs from the batch, not from the switch-off date, and a clinical record's runs for decades [18].
4. Keep the system in `O68` until the records are migrated or archived. A system that is off and still holds GxP records is still in scope, and `O67.Validation state` has `Decommissioned, records retained` for that position [3]. Put the cost in front of `GB-02`

`Investment Committee` beside the replacement cost, in the same paper; separated, the decommissioning line gets cut.

Look it up. Volume 1, chapter 13, section *Decommissioning, which nobody budgets*; chapter 10, section *Retention as a data-model constraint*.

Signed by. The Head of Quality for the retention position, `GB-02 Investment Committee` for the cost, and the Responsible Pharmacist where a batch record is involved [19].

Done when. Every `Retire` and `Replace` row in `O23` has a `Decommissioning plan` value in `O67` that is not blank, and every `Yes` resolves to a document.

Shows up in. `RPT02 Application Portfolio`, which reprints `O23`'s disposition column and says nothing about records [12]. The gap between those two is the finding.

Goes wrong when. The successor is chosen, the old system is switched off at the end of the migration, and an auditor raises the records two years later.

Step 11.12 — Hold the annual architecture review, and re-baseline the repository

Produce. A dated annual review minute, a new repository version, and a decision on whether the next ADM cycle starts.

You need. Everything from steps 11.1 to 11.11. `O60 Governance Decision`, `O66`, the trigger list. A half-day meeting, once a year, in the same quarter every year.

Where it lands.

> **`00 Index` version row, `01 Change Log`, and `O60 Governance Decision`**
> The outcome is a dated decision in `O60` — item 10 gave the sheet its date columns. The version row on `00 Index` moves, and the change log carries the row count per sheet before and after.

How.

1. Open on the four `Outstanding` decisions: `DEC-01 Adopt the recommended architecture alternative`, `DEC-02 Select the integration product`, `DEC-03 Fund compliance mechanisms`, `DEC-04 Designate data owners` [4]. A year of Phase H that leaves them outstanding has governed nothing.
2. Read the trigger list from step 11.9 and say, on the record, whether a trigger fired. If one did, this meeting's output is a Phase A statement of architecture work, not a minute.
3. Take the overdue list and the dispensation sweep: together they say whether the validated state held.
4. Re-count the repository. Row 2 of `00 Index` claims 850 records; the per-sheet counts in rows 15 to 79 of the same sheet sum to 814 [12]. After items 2 to 11 the count is different again.
5. Re-baseline: new version, approval date, approver, and a change-log row naming every sheet whose row count changed. Then record what you could not establish — the honest list of unanswered questions is part of the deliverable, and in this estate most of its value.

Look it up. Volume 1, chapter 19, sections *What ninety days will not achieve* and *What to say when you do not know*; chapter 13, section *What the architect is actually expected to produce*.

Signed by. `GB-01 Architecture Board` and `GB-02 Investment Committee` jointly. The Head of Quality signs the validated-state section. Where item 2 placed the repository inside the quality system, the baseline is a controlled document and is approved as one.

Done when. `O60` carries a dated annual-review decision, `00 Index` a new version and date, and the change log reconciles the two.

Shows up in. Every report, because every one reprints a sheet you have changed. Republish all thirteen and drop each one's typed footer row [12].

Goes wrong when. The review happens in a different quarter each year and the comparison with last year becomes impossible. Fix the quarter, away from the annual product quality review.

What Phase H hands back

Three things, and they all go back to the Preliminary phase rather than forwards.

A repository with an owner, a version and a change log, so the next reader knows what they hold. **A validated state that is maintained rather than asserted** — `O67` current, `O68` populated, an overdue list with a cadence, a dispensation sweep with a bounding rule. And **a trigger list with watchers**, the only thing that will tell you the architecture you signed off last year has stopped describing the company.

What Phase H cannot hand over is the decision nobody has made. Four rows of `O60` read `Outstanding` and the whole $84.3m roadmap in `O30` waits on them [4][13]. Phase H either gets those four decided or writes down, in a minute with a date on it, that they were not.

THE CHAPTER IN TEN LINES

- Validation is a maintained state. Phase H is where it stays true or quietly stops being true.
- A change to a validated system is a change-control record before it is a backlog item.
- O68 Periodic Review Schedule is yours to create: eight columns, PRS-01 onward, keyed on O67.
- Frequency lives in O68 and nowhere else. O67 holds the two dates and no frequency.
- Nothing in the 105 sheets subtracts one date from another. The overdue list is arithmetic you do.
- Never is a legitimate value for Last review, and on a fifteen-year-old GMP system it is the finding.
- No column anywhere answers whether a change needs revalidation. Add four to O61, and a decider per class.
- DEC-07 grants a dispensation with no expiry; CR-G01 records the same one expiring 2029-06.
- No dispensation may expire later than the next periodic review of the system it affects.
- Four classes of change: simplification, incremental, re-architecting, and record correction.
- O31 Deliverable has no Phase H row at all, and no approver and no date column.
- The repository is a document. Give it an owner with a name, a version, and a change log.
- Four O60 decisions read Outstanding, and an $84.3m roadmap in O30 waits on them.

SOURCES FOR THIS CHAPTER

- **1.** The Open Group, TOGAF Standard, Architecture Development Method — Phase H, Architecture Change Management, and its simplification / incremental / re-architecting change classes (identified, not retrieved: the publisher's pages are unreachable from this environment)
- **2.** Pharmaceuticals_TOGAF_Metamodel.xlsx, Calderon Therapeutics — sheet O61 Governance Change Request, headers row 4, records rows 5–9 (Expires populated on CR-G01 only, of five rows)
- **3.** Pharmaceuticals_TOGAF_Metamodel.xlsx — sheet O57 Governance Body (rows 5–9), with GB-06 Change Control Board, GB-07 Validation Committee and GB-08 Data Integrity Forum added by item 10; sheet O11 Role (rows 5–22); and the O67 Validated System columns and state values fixed by item 2 and deferred to item 11 by items 6, 7 and 10
- **4.** Pharmaceuticals_TOGAF_Metamodel.xlsx — sheet O60 Governance Decision (rows 5–11: four columns, no date column; DEC-01 to DEC-04 Outstanding; DEC-07 Approved)
- **5.** Pharmaceuticals_TOGAF_Metamodel.xlsx — sheet O66 Architecture Increment (rows 5–10, landing at months 10, 16, 22, 30, 42 and 56) and sheet O53 Transition Architecture (rows 5–7)
- **6.** Pharmaceuticals_TOGAF_Metamodel.xlsx — sheet O05 Principle (rows 5–28: twenty-four principles, every Owner cell reading Architecture Board)
- **7.** Pharmaceuticals_TOGAF_Metamodel.xlsx — sheet O32 Standard (rows 5–20: STD-02 and STD-15 Deprecated and Conformant, STD-07 Deprecated and in Breach, STD-04 Current and Conformant) and report RPT12 Standards Conformance
- **8.** Pharmaceuticals_TOGAF_Metamodel.xlsx — sheet O23 Application Component (rows 5–16: no validation-state, GxP or periodic-review column; In service 2010 to 2023; A10 Retire and Critical, A07 and A08 Replace) and report RPT02 Application Portfolio
- **9.** Pharmaceuticals_TOGAF_Metamodel.xlsx — sheet O31 Deliverable (rows 5–24: twenty deliverables phased A to G and RM, no Preliminary row and no Phase H row, no approver column and no date column)
- **10.** Volume 1, Pharmaceuticals — a starter pack for enterprise architects, chapter 12, section "Interfaces are routinely in nobody's validation scope"
- **11.** Pharmaceuticals_TOGAF_Metamodel.xlsx — sheets O63 Control Objective (rows 5–11, three In breach), R16 Control Objective met by Co (rows 5–11), O41 Security Control (rows 5–12, SEC-01 and SEC-03 Absent) and O04 Measure (rows 5–20, MSR-01 at 54%); report RPT09 Regulatory Coverage
- **12.** Pharmaceuticals_TOGAF_Metamodel.xlsx — 00 Index (row 2 claiming 850 records against per-sheet counts in rows 15–79 summing to 814), ZZ Checks (15 checks), RPT01 Dashboard (18 measures), and the nine reports ending in a typed sentence in the first column: RPT01, RPT04, RPT05, RPT06, RPT08, RPT09, RPT10, RPT11, RPT13
- **13.** Pharmaceuticals_TOGAF_Metamodel.xlsx — sheet O30 Work Package (rows 5–16: twelve packages totalling $84.3m, Delivery typed rather than computed, earliest Start 2027-02) and report RPT08 Roadmap
- **14.** Pharmaceuticals_TOGAF_Metamodel.xlsx — sheets R06 Application interfaces with (36 rows, 19 Unmanaged), O27 Physical Technology Compone (PTC-07 support end 2026-08 and PTC-11 support end 2026-01, both Past support) and O49 Resilience Tier (rows 5–16, nine Never tested)
- **15.** Volume 1, chapter 13, GxP, validation and data integrity — sections "Periodic review" (the review items and the risk-justified frequency), "Seven pieces of folklore, punctured" (validation as a maintained state) and "Decommissioning, which nobody budgets"
- **16.** Volume 1, chapter 8, section "Change control" — the four parallel impact assessments, the regulatory assessment most often skipped, and the validated-system inventory and periodic-review baseline as the step nobody budgets
- **17.** Pharmaceuticals_TOGAF_Metamodel.xlsx — sheets O62 Delivery Risk (rows 5–12, three rows at Likelihood: Certain) and O52 Implementation Factor (rows 5–12)
- **18.** Volume 1, chapter 16, section "The retention clocks", and appendix 20, section "Retention clock by record class"; with item 5's O69 Retention Clock and its RC-01 to RC-13 key space
- **19.** SAHPRA, SAHPGL-LIC-01 v3, Guide On How to Apply for A Licence To Manufacture, June 2022 — one Responsible Pharmacist named per licence, batch certification before release, and the duty that processes remain in a validated state https://www.sahpra.org.za/wp-content/uploads/2022/06/SAHPGL-LIC-01-Guide-On-How-to-Apply-for-A-Licence-To-Manufacture_DN.pdf

CHAPTER 12

Requirements Management

The ADM phase that never closes, and the one place a pharmaceutical requirement differs from every other kind: it has to reach forward to a test and a signature, not only backward to a driver.

What this phase is for, and why it never closes

Requirements Management sits in the middle of the ADM wheel because every other phase both feeds it and reads from it [1]. It is the one activity with no start and no end, and the repository treats it that way: `O31 Deliverable` gives it a phase code of its own, `RM`, and exactly one deliverable, `DLV-20 Requirements Catalogue`, status `Draft` [2].

The sheets are `O06 Requirement` with thirty-two rows, `O07 Constraint` with eight, `O08 Assumption` with six, `R08 Requirement derives from Dr` and `R09 Requirement satisfied by Bu` with thirty-two rows each, and `RPT06 Requirements Traceability` [3][4][5]. Between them they describe a chain that looks complete and is, in two places, not a statement about anything.

Calderon Therapeutics, the company in this repository, and volume 1's **Veldtkruin Group** are both inventions. Never mix their data.

> **A user requirement specification is the first document in the V model, and it is the same artefact as a TOGAF requirement with the evidence burden attached.**
>
> In the sibling volumes a requirement is traced backward, to the driver that justifies it. Here it is traced **both ways**. Backward to a driver, as everywhere else. Forward to a specification, a design element, a test case, a test result and a signature — because the user requirement specification is where the left limb of the V begins, and performance qualification at the top of the right limb tests it [6].
>
> The mechanism that makes the V work is the traceability matrix, and an inspector will pull one requirement at random and walk the chain in both directions [6]. That is a harder test than any architecture board applies. It is also the reason the honest finding in this item is not that the requirements are badly written — plenty are serviceable — but that **the forward half of the chain does not exist in this repository at all.** `O59 Compliance Assessment` is empty and `R24 Compliance Assessment tests` is empty [7].
>
> One more thing the V model asks that the ADM does not. The user requirement specification is written **before** the product is chosen. Volume 1 is blunt about how often that is untrue, and about what it costs: requirements reverse-engineered from a product's capabilities so that the product can pass [6].

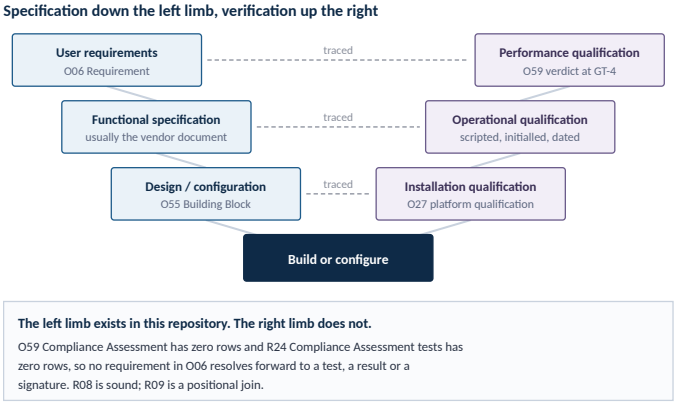


Figure 12.1 The V, drawn on the sheets that hold each limb. A user requirement specification is `O06 Requirement` with the evidence burden attached; performance qualification at the top of the right limb is what tests it. The dashed lines are the traceability an inspector walks in both directions.

Before you open a sheet

ANSWER THESE BEFORE YOU TOUCH `O06`

For one requirement of your choosing, can you name the test that proves it and the person who signed the result?

Which requirements in `O06` were written before a product was selected, and which after?

Who accepts a requirement on the business's behalf — not who wrote it, who accepts it?

The steps

Step 12.1 — Read `O06` before you add to it, and name the five families

Produce. A one-page reading of the requirement sheet: how many rows, how many families, which are usable as written.

You need. `O06 Requirement`. An hour at a desk.

Where it lands.

> **`O06 Requirement — Ref, Requirement, Statement, Category, Priority, Driver, Status`**
>
> Thirty-two rows, `REQ-001` to `REQ-032`, headers on row 4 [3]. Five generated families, and the `Category` column names four of them.

How.

1. Count the families. `REQ-001` to `REQ-007` are regulatory, one per obligation in `O63`; `REQ-008` to `REQ-015` are *mastered once*, one per data entity; `REQ-016` to `REQ-021` are *integration managed*, one per application; `REQ-022` to `REQ-027` are *measurable*, one per capability; `REQ-028` to `REQ-032` are *adopt*, one per technology building block [3].
2. Say what the repository gets right. **`REQ-001 Evidence GMP / GCP / GLP / GVP` is the right first requirement for this industry**, and the statement — *the architecture must produce evidence that GMP / GCP / GLP / GVP is met, without manual reconciliation* — is a real requirement with a real test behind it.
3. Mark the two families that are not requirements as written. The *measurable* six say a capability *exposes measures that the executive can act on*, which names no measure; the *adopt* five restate a building block's description [3].
4. Check `Status`. Twenty-one read `Approved` and eleven `Proposed`, and no column records who approved or when. A requirement approved by nobody on no date cannot be the baseline for a change.
5. Note what is absent, because in this estate the absences are the finding. There is **no requirement about validated state, audit trail, electronic signature, retention or decommissioning** anywhere in the thirty-two.

Look it up. Volume 1, chapter 13, section *What the architect is actually expected to produce*.

Signed by. Nobody. This is a reading, not a change. Take it to `GB-05 Architecture Function` before you start editing.

Done when. You can say the five family sizes without opening the file.

Shows up in. `RPT06 Requirements Traceability`, which reprints `O06` with one column added and ends in a typed sentence [5].

Goes wrong when. You start writing new requirements before you have read the thirty-two. Half of what you were about to write is already there in a weaker form, and two copies of a requirement is worse than one bad one.

Step 12.2 — Add the seven columns a specification needs and `O06` has not got

Produce. An `O06` whose columns can carry a user requirement specification.

You need. `O06`, the validation lead, item 2's `O67 Validated System`. A morning.

Where it lands.

> **`O06 Requirement`, seven new columns after `Status`**
>
> `Source`, the person or document the requirement came from. `GxP`, the regimes it bears on, or `None`. `Verification method`, one of `Test`, `Inspection`, `Demonstration`, `Analysis`. `Test reference`, the protocol and script that proves it. `Accepted by`, a named person. `Accepted date`. `Supersedes`, the ref of the requirement this one replaces.

How.

1. Add `Verification method` first. A requirement with no verification method cannot enter a validation plan, and four values cover everything: test it, look at it, show it working, or argue it from evidence.
2. Make `GxP` text, not a flag. A requirement can bear on GMP and not GCP, and the regime decides how much evidence the test has to produce.
3. Make `Accepted by` a person. `Status: Approved` with no name is the state twenty-one rows are in now, and it is the state an auditor asks about first.
4. Use `Supersedes` rather than editing in place. A requirement that changes after a system is validated is a change-control record, and the old text is evidence — step 12.11 operates it.
5. Leave `Test reference` blank until step 12.9 fills it from `O59 Compliance Assessment`. A test reference invented now is worse than an empty cell.

Look it up. Volume 1, chapter 13, sections *The V-model, and how it actually runs* and *The validation master plan and the deliverable set*.

Signed by. `GB-05 Architecture Function` for the structure; the validation lead for `Verification method`, because the word in that cell sets the evidence burden.

Done when. `O06` has fourteen columns, and `Verification method` is restricted to four values.

Shows up in. Nothing yet. `RPT06` reprints five of the original seven columns and will have to be rebuilt — step 12.6.

Goes wrong when. The columns are added and left empty for a year, so the sheet looks like a specification and is still a list.

Form 12-A — Requirement as a user requirement specification

Feeds `O06 Requirement`, six of its fourteen columns after Step 12.2. One row per requirement. A first pass over the regulatory family is seven rows; a full pass on a GxP system is forty to eighty. `Statement`, `Priority`, `Driver`, `Status`, `Test reference`, `Accepted date` and `Supersedes` are typed straight into Excel. Signed by the named acceptor, who must be in the business, not in technology.

Ref	Requirement	Category	GxP	Verification method	Accepted by
REQ-033	EXAMPLE — delete this row. Audit trail reviewable	Regulatory	GMP, GDocP	Test	

Step 12.3 — Rewrite `REQ-001` as something a test can prove

Produce. `REQ-001` decomposed into four or five requirements, each with a verification method.

You need. Step 12.2's columns, `O63 Control Objective` as item 10 rewrote it, the head of quality assurance. Two hours.

Where it lands.

> **`O06 Requirement , rows 5 onward`**
>
> `REQ-001` stays as the parent and keeps its text. The children take new refs from `REQ-033` onward, with `Supersedes` left blank — they do not supersede the parent, they decompose it. Each child's `Source` cell names `REQ-001` .

How.

1. Read the parent again: *the architecture must produce evidence that GMP / GCP / GLP / GVP is met, without manual reconciliation* [3]. It is the right ambition and it is four requirements in a coat.
2. Separate the regimes. GMP evidence is batch and manufacturing records; GCP evidence is subject and trial records; GLP is laboratory; GVP is safety. Each has a different system, a different retention clock and a different signatory.
3. Write each child so it names a system, a record class and an observable. *The batch record for any released batch can be produced from* `A07 MES (GMP)` *with its complete audit trail, in one operation, within one working day* is testable. *Produce evidence* is not.
4. Attach the verification method per child. Three of the four are `Test` — you can ask for the record. The fourth, *without manual reconciliation*, is `Inspection` : you watch somebody do it and count the steps.
5. Keep the parent. An inspector asks about the regime, not about your decomposition, and the parent is where the obligation is named.

Look it up. Volume 1, chapter 13, section *Which GxP, and for which record*; appendix 20, section *GxP regime to the record it compels*.

Signed by. The head of quality assurance accepts the children, because the regime scope is theirs. `GB-01 Architecture Board` notes them.

Done when. Four or five children exist, each names a system and a record class, and each has a verification method.

Shows up in. `RPT06` , with a longer requirement list and a higher proportion reaching a block.

Goes wrong when. The decomposition is written by the architecture function alone and quality first sees it at a gate. Then the regime scope is wrong and the whole branch is rework.

Step 12.4 — Check the derivation, and find the drivers nothing derives from

Produce. A corrected `R08` , and a short list of drivers no requirement answers.

You need. `O01 Driver` , `O06` , `R08 Requirement derives from Dr` . An hour.

Where it lands.

> **`R08 Requirement derives from Dr — Requirement , Requirement name , Driver , Driver name`**
>
> Thirty-two rows, one per requirement, and it agrees with `O06.Driver` on every row [3][4]. This relationship is sound. It is the only one of the two that is.

How.

1. Join `R08.Driver` against `O01` . Thirteen drivers exist; ten are used. **`DRV-09 Customer expectation` , `DRV-12 Talent scarcity` and `DRV-13 Sustainability reporting` derive no requirement at all** [8][4].
2. Decide each of the three. A driver nothing derives from is either not a driver here or a gap in the requirement set, and both are findings you write down.
3. Read the seven regulatory drivers. `DRV-01` to `DRV-07` are all `Origin: External` , `Pressure: High` , and all owned by `Chief Compliance Officer` [8]. One owner for every regulatory obligation in the company is a single point of failure in the governance, not a tidy sheet.
4. Add the drivers this industry has and this sheet lacks — a SAHPRA registration or variation duty, a single exit price declaration, POPIA as a named instrument. Item 3 rewrote the driver list; check your requirements reach the rows it added.
5. Re-check that `O06.Driver` and `R08` still agree after any edit. Two places holding one fact is how the rest of this repository went wrong.

Look it up. Volume 1, chapter 16, sections *Registration, variation and renewal* and *POPIA and health data*.

Signed by. `GB-01 Architecture Board` . Each driver's owner signs for their own rows.

Done when. Every driver in `O01` either derives a requirement or carries a written reason why not.

Shows up in. `RPT06` , whose `Driver` column is copied from `O06` and will show the new refs [5].

Goes wrong when. The three unused drivers are deleted to make the join tidy. Talent scarcity in a regulated estate is a real constraint on the plan; deleting it does not make it go away.

Step 12.5 — Break the positional join in `R09`, and re-trace by content

Produce. An `R09` whose rows say something about the world.

You need. `R09 Requirement satisfied by Bu`, `O55 Building Block`, `O06`. Half a day, and it is the most valuable half day in this item.

Where it lands.

> **`R09 Requirement satisfied by Bu — Requirement, Requirement name, Building block, Block name, Traced`**
>
> Thirty-two rows. `REQ-001` to `BBB-01`, `REQ-002` to `BBB-02`, and so on in index order through the thirty-seven blocks of `O55` until the refs run out at `TBB-02` [4][9].

How.

1. Prove the join to yourself before you argue it with anybody. `REQ-001 Evidence GMP / GCP / GLP / GVP` is matched to `BBB-01 Discovery Research capability uplift` [4]. A regulatory evidence requirement and a discovery-research capability. They have nothing to do with each other, and the only reason they are on the same row is that both are first in their sheet.
2. Check the tail. `REQ-032` lands on `TBB-02`, so **`TBB-03` to `TBB-07` satisfy no requirement** — CDISC standards, IDMP, cold chain telemetry, the scientific data lake and the secure enclave [4][9].
3. Re-trace the families that are easy. `REQ-008` to `REQ-015` are *mastered once* per entity and `DBB-01` to `DBB-10` are *mastering* per entity; match them by entity name, not by position. `REQ-016` to `REQ-021` match `ABB-01` to `ABB-06` by application name the same way.
4. Re-trace the families that are hard. The seven regulatory requirements reach no building block in `O55` at all, because no block in the sheet produces compliance evidence. That is not a tracing defect — it is a gap, and it belongs in `O29 Gap` from item 8.
5. Set `Traced` from the re-trace, not from the pattern. Step 12.6 says what the column currently holds.

Look it up. Volume 1, chapter 13, section *The V-model, and how it actually runs* — the traceability matrix and the two directions an inspector walks it.

Signed by. `GB-01 Architecture Board`, and the domain architect for each domain in `O55`.

Done when. No row of `R09` pairs a requirement with a block by index, and every unmatched requirement is a row in `O29 Gap`.

Shows up in. `RPT06.Reaches a block`, which is `R09.Traced` under another name [5].

Goes wrong when. The join is corrected and `R13 Work Package delivers Build` is left alone. `R13` cycles twelve packages round the same thirty-seven blocks by index, so the requirement now reaches a block that nothing delivers.

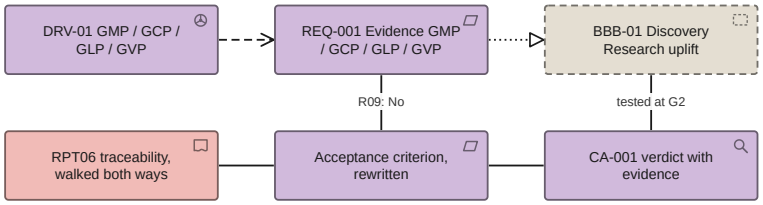


Figure 12.2 One requirement, traced. `DRV-01` to `REQ-001` is sound — `R08` agrees with `O06.Driver` on all thirty-two rows. `REQ-001` to `BBB-01` is `R09`, a positional join: a regulatory evidence requirement matched to a discovery-research capability because both are first in their sheet. The whole bottom row is absent: `O59` and `R24` have no rows at all.

Step 12.6 — Read `Traced: No` for what it is, and rebuild `RPT06`

Produce. A corrected `Traced` column, and a republished `RPT06` without its footer row.

You need. `R09`, `RPT06 Requirements Traceability`. Two hours.

Where it lands.

> **`R09.Traced` and `RPT06` Requirements Traceability**
>
> `Traced` holds `Yes` and `No`. `RPT06` has thirty-two data rows plus a typed sentence in the first column — *a requirement that reaches no building block is a requirement nothing will deliver* — and its last column, `Reaches a block`, is `R09.Traced` copied [4][5].

How.

1. List the eight `No` rows: `REQ-001`, `REQ-005`, `REQ-009`, `REQ-013`, `REQ-017`, `REQ-021`, `REQ-025`, `REQ-029` [4]. Every fourth row, starting at the first.
2. Say what that means. The column is not an assessment; it is a pattern, generated with the join it describes. A quarter of the rows read `No` because a quarter of the rows were made to read `No`.
3. Note which requirement heads the list. **`REQ-001`, the one requirement in the sheet that is right, is also the one the traceability report says reaches nothing** [4][5]. Nobody reading the report would check whether that was true.
4. Rebuild the column from step 12.5's re-trace and expect the honest number to be worse than eight. The seven regulatory requirements reach no block, and six of the technology *adopt* requirements reach a block that satisfies nothing else.
5. Drop the last row when you republish. Nine of the thirteen reports end in a typed sentence in the first column, `RPT06` among them; anything loading the report as data gets a requirement called a sentence [5].

Look it up. Volume 1, chapter 19, section *The checks that take an afternoon*.

Signed by. `GB-05 Architecture Function` publishes. `GB-01 Architecture Board` accepts the new number, which will be worse than the old one, and the minute should say so.

Done when. `RPT06` is regenerated from `R09`, has no footer row, and its count of `No` matches a re-trace somebody can show their working for.

Shows up in. `RPT06` itself, and in `MSR-15 Architecture compliance rate`, whose 59% has no source in any populated sheet [10].

Goes wrong when. The report is regenerated and the footer sentence is regenerated with it, because the generator writes it. Delete it in the generator, not in the output.

Step 12.7 — Score the thirty-seven acceptance criteria, and find the one that passes

Produce. A scored list of all thirty-seven `O55` acceptance criteria, with a verdict and a reason per row.

You need. `O55 Building Block`, `R06 Application interfaces with`, `O23 Application Component`, `O20 Data Entity`. A day.

Where it lands.

> **`O55 Building Block`, column `Acceptance criteria`**
>
> Thirty-seven rows: ten `BBB`, ten `DBB`, ten `ABB`, seven `TBB` [9]. Each criterion is one sentence generated from the block's name. Score each against the repository as it stands, and record the verdict in `O59 Compliance Assessment`, not in `O55`.

How.

1. Score the ten `ABB` criteria first, because they are the only ones testable as written: *no unmanaged interface to or from X*. `R06` has thirty-six interfaces and nineteen `Unmanaged`, so the test is a count [11].
2. Find the one that passes. **`ABB-08 QMS target` passes, and it is the only one of the thirty-seven that does**, because `A08 QMS` is the only application in `R06` with no unmanaged interface — three interfaces, all managed [11][9].
3. Then read why it passes. `O23` has `A08 QMS` at `Disposition: Replace`, `Criticality: Critical`, `Health: Healthy` [12]. The building block is *QMS at target disposition* and the criterion tests interface management instead. **The criterion passes while the block does not**, because the criterion does not test what the block must do.
4. Score the rest and keep the categories separate: nine non-conformant, twenty not testable as written, seven not assessable because nothing in the repository holds the fact they need.
5. Write the twenty down as rework, not as failures. A criterion that cannot be tested has no verdict, and recording `Fail` against it is as wrong as recording `Pass`.

Look it up. Volume 1, chapter 12, section *Interfaces are routinely in nobody's validation scope*.

Signed by. `GB-05 Architecture Function` scores. The domain architect for each domain signs their own rows. `GB-07 Validation Committee` sees any row touching a system in `O67`.

Done when. All thirty-seven have a verdict or an explicit `Not testable`, and the one pass has its reason written beside it.

Shows up in. `O59 Compliance Assessment` from step 12.9, and in item 10's gate assessments.

Goes wrong when. The one pass is reported as a pass. It is the best illustration in the file of a criterion that measures the wrong thing, and reporting it without the reason throws that away.

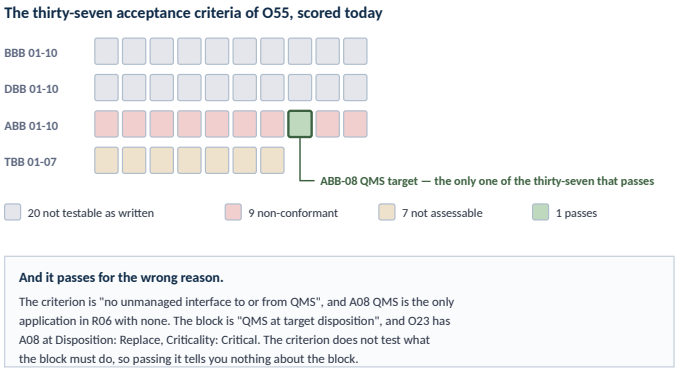


Figure 12.3 One of thirty-seven. The ten `ABB` criteria are the only ones testable as written, because `R06` can be counted. The twenty business and data criteria name no observable, and the seven technology criteria need a fact no sheet holds.

Step 12.8 — Write an acceptance criterion a gate can fail

Produce. The rule, and the first batch of rewritten criteria.

You need. Step 12.7's scoring. A day, spread over the four domain architects.

Where it lands.

> **`O55 Building Block`, column `Acceptance criteria`**
> One criterion per row, replacing the generated sentence. Items 7 and 10 applied this rule to the technology and the compliance-assessment halves; this step is where the rule itself is written down, so that item 13's reference and item 19's standards set can both point at it.

A criterion that a gate can fail has five parts. It names **the object** — a sheet and a ref, not a concept. It names **the observable** — a column, a count, a document. It names **the threshold** — a number, a date or a value, never "appropriate". It names **where the evidence is read from**, so two people get the same answer. And it names **who verifies it**, because `GT-2 G2 Build complete` tests that every acceptance criterion is evidenced and somebody has to hold the evidence [13].

Ref	Acceptance criterion as written	Rewritten so a gate can fail it
`BBB-05`	Pharmacovigilance measured and owned.	`C05 Pharmacovigilance` has a named person in `O70 Regulated Role`, and at least one `O04 Measure` row names a pharmacovigilance outcome with a baseline dated within 90 days. `O04` has none today.
`DBB-03`	No subject record without a named owner.	`O20.E03 Subject` names a person in `Owner`, `Classification` reads `Restricted` or above, and `O37` carries no `Public` attribute on `E03`.
`ABB-08`	No unmanaged interface to or from QMS.	Every `R06` row with `A08` in `Source` or `Target` reads `Managed` **and** `Logged: Yes`, and `O23.A08.Disposition` resolves to a dated decision.
`TBB-01`	GxP-validated cloud in production for at least one service.	One named application in `O23` is deployed on `PTC-09 Validated cloud landing zone` in `R07`, with an installation and operational qualification report referenced against its `O67` row.

Ref	Acceptance criterion as written	Rewritten so a gate can fail it
`DBB-09`	No batch / lot record without a named owner.	`O20.E09 Batch / Lot` names a person, `Master system` reads `MES (GMP)` or `ERP`, and `O43.MDR-09` names the same system.
`ABB-04`	No unmanaged interface to or from Safety Database.	`INT-009 EDC` to `Safety Database` reads `Managed`, and the interface is named inside the validation scope of both endpoints in `O67`.

How.

1. Start from the object, not the sentence. If you cannot name the sheet and the ref the criterion is about, the block itself is not defined well enough to have a criterion.
2. Replace every adjective with a cell. *Measured and owned* becomes a measure ref and an owner cell; *at target disposition* becomes a dated decision.
3. Put the threshold in. A criterion with no number cannot fail, and a criterion that cannot fail is not a gate, it is a wish.

4. Test the criterion backwards: write down what the repository would have to show for the answer to be `No`. If you cannot, rewrite it.
5. Keep one criterion per row. Two conditions joined by *and* give a verdict nobody can act on.

Look it up. Volume 1, chapter 9, section *The object most architects leave out: the validated system itself*.

Signed by. The domain architect writes; `GB-01 Architecture Board` approves the set. `GB-07 Validation Committee` approves any criterion that will be cited as validation evidence.

Done when. Every rewritten criterion names a sheet, a ref, an observable and a threshold, and the backwards test passes on each.

Shows up in. `O59 Compliance Assessment`, whose `Verdict` column becomes defensible, and at `GT-2` [13].

Goes wrong when. The criteria are rewritten by one person in one sitting. They read beautifully and the four domains do not recognise them, so nobody uses them at the gate.

Form 12-C — Acceptance criterion rewrite

Feeds `O55 Building Block`, column `Acceptance criteria`, and seeds `O59 Compliance Assessment`. One row per building block; thirty-seven rows in all, so use four copies, one per domain. `Verdict today` is from Step 12.7 and will usually read `Not testable`. Signed by the domain architect for the domain, and by `GB-01 Architecture Board` for the set.

Ref	Block	Verdict today	Object and ref	Observable and threshold	Verified by
ABB-08	EXAMPLE — delete. QMS target	Passes, wrongly	R06 rows on A08	All Managed and Logged	

Step 12.9 — Open `O59` as the forward half of the trace

Produce. A populated `O59 Compliance Assessment` and a populated `R24`, keyed from the rewritten criteria.

You need. Steps 12.7 and 12.8, item 10's `CA-001` reference space, `O58 Governance Gate` with the fifth gate item 10 added. Two days.

Where it lands.

> **`O59 Compliance Assessment` — `Ref`, `Building block`, `Block name`, `Acceptance criterion`, `Gate`, `Verdict`, `Evidence`**
>
> Seven columns, zero rows, header row 4 [7]. Item 10 set the reference space at `CA-001` onward; use it. `R24 Compliance Assessment tests` is the join from an assessment to the criterion it tests, and is also empty.

How.

1. Write one `O59` row per criterion per gate, not one per block. The same criterion assessed at `GT-1 G1 Design` and again at `GT-2 G2 Build complete` is two rows with two dates, and the pair is the evidence that anything changed.
2. Fill `Evidence` with a document reference, never with a sentence. *Assessed and found compliant* is what an empty evidence column looks like once somebody has tidied it.
3. Put the `Test reference` back into `O06` from step 12.2. That closes the loop: driver to requirement to block to criterion to assessment to evidence, and back.
4. Populate `R24` as you go. A row in `O59` with no row in `R24` is an assessment of nothing.
5. Use `GT-4 G2R Release for use` for anything touching a system in `O67 Validated System`. Item 10 added that gate because the four-gate set has no release gate, and release is the one a named person signs [13].

Look it up. Volume 1, chapter 16, section *Who inspects, and what they ask for*.

Signed by. `GB-05 Architecture Function` assesses; `GB-01 Architecture Board` holds `GT-1` and `GT-2`; the release signatory named in `O70 Regulated Role` holds `GT-4`.

Done when. `O59` has at least one row per building block, every `Evidence` cell resolves to a document, and `R24` has the same row count.

Shows up in. `RPT06`, and in `MSR-15` and `MSR-16`, whose only possible source is `O59` and which currently report 59% and 44% over an empty sheet [10].

Goes wrong when. `O59` is populated in one pass to clear the gate and never touched again. The verdicts age, and an eight-month-old verdict is not evidence.

Form 12-B — Traceability line

Feeds `R08`, `R09`, `O06` and `O59` together: one line per requirement, carrying the whole chain. Use it in a workshop, where arguing the chain out loud is the point. Thirty-two rows for the existing set, more as step 12.3 decomposes. Signed by the domain architect, and countersigned by the validation lead for any row whose `GxP` cell is not `None`.

Ref	Driver	Building block	Criterion, rewritten	Verification	Evidence ref
REQ-001	DRV-01	EXAMPLE — delete this row. None yet		Test	

Step 12.10 — Separate requirement from constraint and assumption

Produce. A cleaned `O07 Constraint` and `O08 Assumption`, with owners and expiry dates.

You need. `O07`, `O08`, volume 1 chapter 17's notation rule. Two hours.

Where it lands.

> **`O07 Constraint` and `O08 Assumption`**
>
> `O07` holds eight rows with `Ref`, `Constraint`, `Description`, `Type`; `O08` holds six with `Ref`, `Assumption`, `Status` [14]. Add `Owner` and `Expires` to both, and `Evidence` to `O08`.

How.

1. Know the notation rule before you draw any of this. **ArchiMate 4 has no Constraint element**, so the eight rows of `O07` become `Requirement` elements when item 16 draws the repository [15]. That is a modelling rule, not a reason to delete the sheet.
2. Test each constraint: is it a limit on the solution, or a requirement in disguise? `CON-05 Data residency` — *certain data may not leave the jurisdiction* — is a requirement, and belongs in `O06` with a verification method and the instrument named [14].
3. Test `CON-06 Change freeze windows` against item 8's work. The repository's freeze is `IF-03 Peak trading freeze`, which is retail vocabulary; the real windows here are the annual product quality review cycle and the inspection window.
4. Give every assumption an owner, an expiry and a test. `ASM-05 Regulatory interpretation remains stable` reads `Open`, has no owner and no date, and in a year when an obligation is under revision it is the assumption most likely to fail [14].
5. Convert the two at-risk assumptions. `ASM-03` and `ASM-04` read `At risk`, which is a risk, not an assumption; they belong in `O62 Delivery Risk` with an owner.

Look it up. Volume 1, chapter 17, section *What has no ArchiMate home at all*.

Signed by. `GB-01 Architecture Board`. Each named owner signs their own row.

Done when. Every `O07` row is a limit rather than a requirement, every `O08` row has an owner and an expiry, and the `At risk` rows have moved.

Shows up in. No report reads `O07` or `O08`. Add them to the architecture definition document by hand.

Goes wrong when. The sheets are left alone because nothing reads them. An assumption nobody tests is how a programme discovers in month twenty that it was wrong in month two.

Step 12.11 — Put a requirement change through change control

Produce. A rule binding a change to any requirement on a validated system into `O61 Governance Change Request`.

You need. Item 11's triage rule and `O61`'s four new columns, `O67 Validated System`. An hour to write, for ever to operate.

Where it lands.

> **`O06.Supersedes` and `O61 Governance Change Request`**
>
> A requirement change writes a new `O06` row with `Supersedes` pointing at the old ref, and the old row's `Status` becomes `Superseded`. The change request in `O61` carries the pair.

How.

1. Set the test. If the requirement bears on a system in `O67`, changing it is a change-control record before it is a backlog item, and the four impact assessments run [16].
2. Never edit a requirement in place once it has been accepted. The accepted text is what a test was written against, and overwriting it breaks the chain backwards.
3. Treat a protocol amendment as the worst case. It invalidates downstream structures — the study build, the edit checks, the analysis datasets — so one amendment can supersede a dozen requirements at once [17].
4. Treat a dictionary version change the same way. A new release of a coded terminology changes the meaning of records already held, and any requirement written against the old version needs re-reading, not re-testing [17].
5. Re-run the trace at every gate, not at the end. A requirement added in month nine with no building block is cheap to fix in month nine.

Look it up. Volume 1, chapter 9, sections *Protocol amendment, which invalidates downstream structures* and *The dictionary-version problem*.

Signed by. `GB-06 Change Control Board` for the change, the named acceptor for the new text, and the validation lead where a test has to be rewritten.

Done when. No accepted requirement has been edited in place since the rule was set, and every superseded row points at its successor.

Shows up in. `O61`, and in the `Supersedes` column of `O06`, which becomes the requirement history nobody kept before.

Goes wrong when. The rule is written for requirements and not for their acceptance criteria. A criterion changed quietly after a gate makes the gate's verdict meaningless.

Step 12.12 — Publish the catalogue and the matrix as one artefact

Produce. `DLV-20 Requirements Catalogue`, published, dated and owned, with the traceability matrix inside it rather than beside it.

You need. Everything from 12.1 to 12.11. `O31 Deliverable`. A day.

Where it lands.

> **`O31 Deliverable`, row `DLV-20`**
>
> `DLV-20 Requirements Catalogue`, phase `RM`, status `Draft` [2]. It is the only `RM` deliverable in the sheet, and `O31` has no approver column and no date column, so publishing it means publishing something the repository cannot timestamp. Add the two columns, and say in the text that you did.

How.

1. Publish them together. A catalogue without the matrix is a list, and a matrix without the catalogue is a join. The artefact an inspector asks for is the pair [6].

2. Carry the forward half visibly. Each row shows driver, requirement, building block, criterion, assessment and evidence, so the chain can be walked in both directions from any row.
3. Mark the gaps on the face of it. The seven regulatory requirements reaching no block, the five technology blocks satisfying no requirement, and the three drivers deriving nothing are the catalogue's most useful page.
4. Date it and name its owner. A requirements catalogue with no date is the artefact most likely to be quoted back at you from two baselines ago.
5. Set the re-publication cadence with item 11's annual review, so the catalogue, the repository baseline and the periodic-review position all move together.

Look it up. Volume 1, chapter 5, section *The regulator is a stakeholder with a data demand*.

Signed by. `GB-01 Architecture Board` approves. The named acceptors sign their own requirements. `GB-07 Validation Committee` signs where the catalogue is cited as a validation deliverable.

Done when. `DLV-20` is `Approved`, has a date and an approver in the columns you added, and the matrix is in the same document.

Shows up in. `RPT06`, and in item 14's report catalogue, where `RPT06` becomes one of the reports an inspector may legitimately be shown.

Goes wrong when. The catalogue is published as a spreadsheet export. An export has no date, no owner and no version, and within a quarter three versions are circulating.

What Requirements Management hands to every phase

One thing, in two directions. **Backward**, a requirement that resolves to a driver somebody owns. **Forward**, a requirement that resolves to a test and a signature — which is what makes it a user requirement specification rather than a line on a list.

The repository hands you the backward half intact and the forward half empty. `R08` is sound, `R09` is a positional join, `O59` and `R24` have no rows, and `RPT06` reports that `REQ-001` — the one requirement in the sheet that is right — reaches nothing [4][5][7]. Fix the forward half and most of Phase G stops being guesswork.

THE CHAPTER IN TEN LINES

- A user requirement specification is a TOGAF requirement with the evidence burden attached.
- Trace a requirement both ways: back to a driver, forward to a test, a result and a signature.
- O06 has thirty-two rows in five generated families. REQ-001 is the right first requirement.
- Nothing in O06 is about validated state, audit trail, electronic signature, retention or decommissioning.
- R08 is sound and agrees with O06.Driver on every row. R09 is a positional join.
- DRV-09, DRV-12 and DRV-13 derive no requirement at all.
- R09.Traced reads No on every fourth row. It is a pattern, not an assessment.
- RPT06 reports that REQ-001, the best requirement in the sheet, reaches nothing.
- Exactly one of the thirty-seven O55 acceptance criteria passes today, and it passes for the wrong reason.
- A criterion a gate can fail names an object, an observable, a threshold, a source and a verifier.
- O59 Compliance Assessment and R24 are both empty, so the forward half of every trace is missing.
- ArchiMate 4 has no Constraint element, so O07 becomes Requirement elements when you draw it.
- Never edit an accepted requirement in place. Supersede it, and send the change through change control.

SOURCES FOR THIS CHAPTER

- **1.** The Open Group, TOGAF Standard, Architecture Development Method — Requirements Management, the continuous activity at the centre of the ADM (identified, not retrieved: the publisher's pages are unreachable from this environment)
- **2.** Pharmaceuticals_TOGAF_Metamodel.xlsx, Calderon Therapeutics — sheet O31 Deliverable (rows 5–24: DLV-20 Requirements Catalogue, phase RM, status Draft; no approver column and no date column)
- **3.** Pharmaceuticals_TOGAF_Metamodel.xlsx — sheet O06 Requirement, headers row 4, records rows 5–36 (thirty-two rows in five generated families; twenty-one Approved and eleven Proposed; REQ-001 Evidence GMP / GCP / GLP / GVP)
- **4.** Pharmaceuticals_TOGAF_Metamodel.xlsx — sheets R08 Requirement derives from Dr (rows 5–36) and R09 Requirement satisfied by Bu (rows 5–36: REQ-001 to BBB-01 through REQ-032 to TBB-02; Traced reads No on REQ-001, 005, 009, 013, 017, 021, 025 and 029)
- **5.** Pharmaceuticals_TOGAF_Metamodel.xlsx — report RPT06 Requirements Traceability (thirty-two data rows plus a typed sentence in the first column; Reaches a block is R09.Traced copied), and the nine reports ending in such a sentence: RPT01, RPT04, RPT05, RPT06, RPT08, RPT09, RPT10, RPT11, RPT13
- **6.** Volume 1, Pharmaceuticals — a starter pack for enterprise architects, chapter 13, sections "The V-model, and how it actually runs" and "The validation master plan and the deliverable set" — the URS/PQ pairing, the traceability matrix walked in both directions, and the user requirement specification written after product selection
- **7.** Pharmaceuticals_TOGAF_Metamodel.xlsx — sheets O59 Compliance Assessment (seven headers on row 4, zero records) and R24 Compliance Assessment tests (five headers, zero records); the CA-001 reference space set by item 10
- **8.** Pharmaceuticals_TOGAF_Metamodel.xlsx — sheet O01 Driver (rows 5–17: thirteen drivers, DRV-01 to DRV-07 all Regulatory, External, High and owned by Chief Compliance Officer; DRV-09, DRV-12 and DRV-13 derive no requirement)
- **9.** Pharmaceuticals_TOGAF_Metamodel.xlsx — sheet O55 Building Block (37 rows, 5–41: ten BBB, ten DBB, ten ABB, seven TBB, with the Acceptance criteria column generated from each block name)
- **10.** Pharmaceuticals_TOGAF_Metamodel.xlsx — sheet O04 Measure (rows 5–20), where MSR-15 Architecture compliance rate reads 59% and MSR-16 Initiatives through a design gate reads 44%, with O59 as their only possible source
- **11.** Pharmaceuticals_TOGAF_Metamodel.xlsx — sheet R06 Application interfaces with (36 rows: 19 Unmanaged, 5 Logged, 6 Direct DB read; A08 QMS appears in three rows, all Managed)
- **12.** Pharmaceuticals_TOGAF_Metamodel.xlsx — sheet O23 Application Component (rows 5–16: A08 QMS at Disposition Replace, Criticality Critical, Health Healthy, in service 2020)
- **13.** Pharmaceuticals_TOGAF_Metamodel.xlsx — sheet O58 Governance Gate (rows 5–8: GT-0 to GT-3, GT-2 testing that every acceptance criterion is evidenced), with the fifth gate GT-4 G2R Release for use added by item 10
- **14.** Pharmaceuticals_TOGAF_Metamodel.xlsx — sheets O07 Constraint (rows 5–12) and O08 Assumption (rows 5–10: ASM-03 and ASM-04 At risk, no owner and no date column on either sheet)
- **15.** Volume 1, chapter 17, Modelling pharmaceuticals in ArchiMate, section "What has no ArchiMate home at all" — there is no Constraint element in ArchiMate 4, so a constraint is expressed as a Requirement
- **16.** Volume 1, chapter 8, section "Change control" — the four parallel impact assessments and the coupling between a technical change and a regulatory variation
- **17.** Volume 1, chapter 9, sections "Protocol amendment, which invalidates downstream structures" and "The dictionary-version problem"

APPENDIX A

The sheet-by-sheet reference

All 105 tabs in tab order: the exact name, the row count, the columns, what each one is for, which item of this workbook repairs it, and the one verified thing that is wrong with it.

How to read this reference

Keep this open beside Excel. All 105 tabs, in tab order, with the exact name, the row count, the columns in sheet order, what it is for, which item works on it, and the one verified thing wrong with it. Three conventions, then the tables.

Names are exact. Twenty-three tab names are cut at Excel's 31-character limit and are marked **[cut]**. `R04 Business Service supported` is cut *and* ends in a space, which no amount of squinting at the tab will show you. Match on the `O`, `R` or `RPT` prefix, never the full phrase [1].

Row counts are data rows, from row 5 down with blanks dropped. Row 1 is the title, row 2 a subtitle, row 3 blank, row 4 the header. A count of `0` means empty, and the finding says so in bold: two object sheets are, and two relationship sheets and one report in consequence [2][3].

The `Item` column names the item that works on the sheet, not every item that mentions it. Item 1 mentions nearly all of them and item 18 reprints the forms.

Key space	Where it resolves	Note
BBB- · DBB- · ABB- · TBB-	a domain sheet AND O55	R09 and R13 mean O55
T01 - T07	O46, second column headed Ref	resolved by nothing in the file
C01 · CO- · CON- · CTR-	O13 / O63 / O07 / O19	four sheets, one letter
A01 · ACT- · ACO- · ALT- · ATR-	O23 / O10 / O33 / O54 / O37	match on the full prefix
E01 · ENV- · EP-	O20 / O48 / O65	entity, environment, enforcement
REQ- · REL- · RES- · RME-	O06 / O36 / O49 / O47	requirement is not resilience
PRN- · PRD- · PRJ- · PRS-	O05 / O18 / O56 / O68 (new)	PRS- is the one you add
INT-001 - INT-036	R06 only, and RPT04	an interface has no object sheet
VS01 - VS05	O14 Value Stream	the new register uses VAL-, not VS-

Figure 13.1 Nine key spaces that collide. Match on the full prefix, never the first letter. Each of the four building-block prefixes appears twice, once on a domain sheet and once on the union sheet, and both relationship sheets that use them mean the union sheet.

> The prefix is not the sheet. `C01` is a capability in `O13`, `CO-01` an obligation in `O63`, `CON-01` a constraint in `O07` and `CTR-01` a contract in `O19`. The four building-block prefixes are worse: each appears on a domain sheet *and* on `O55`, and the relationship sheets mean `O55` [4].

The index sheet, and the number on it that is wrong

`00 Index` has no header row and no `Ref` column, so the row-4 rule does not apply. Rows 5 to 12 are eight prose statements. Row 13 is the header pair `Sheet` / `Contents`, and rows 14 to 78 are a per-sheet table of 65 rows, one per object sheet plus `EXT`, each reading `<n> records · <domain>`.

The index lists no relationship sheet and no report — 65 of 105 tabs. And **it contradicts itself inside one sheet**: row 2 claims 850 records, while the 65 counts below it sum to exactly **814**, which is what the sheets hold. The detail is right and the headline is wrong [1][3].

The domain labels below are the index's own. Two sit oddly: `O47 Reference Model Entry` under `Governance` rather than Technology, and `O35 Business Building Block` under `Business` rather than with the other building-block sheets. Both are part of why nobody reconciles them against their neighbours.

Motivation — eight sheets

Drivers, goals, objectives, measures, principles, requirements, constraints, assumptions. Phase A argues from these.

Sheet, exactly as the tab reads	Rows	Columns, in sheet order
`001 Driver`	13	`Ref` , `Driver` , `Type` , `Description` , `Origin` , `Owner` , `Pressure`
`002 Goal`	6	`Ref` , `Goal` , `Description` , `Owner` , `Status`
`003 Objective`	14	`Ref` , `Objective` , `Goal` , `Horizon` , `Status`
`004 Measure`	16	`Ref` , `Measure` , `Unit` , `Domain` , `Baseline` , `Cadence`
`005 Principle`	24	`Ref` , `Principle` , `Category` , `Statement` , `Owner` , `Status`
`006 Requirement`	32	`Ref` , `Requirement` , `Statement` , `Category` , `Priority` , `Driver` , `Status`
`007 Constraint`	8	`Ref` , `Constraint` , `Description` , `Type`
`008 Assumption`	6	`Ref` , `Assumption` , `Status`

Sheet	What it is for	Item	What is wrong with it
`001`	What pushes the company.	3	Seven regulatory drivers, all EU or US framed, all owned by the Chief Compliance Officer.
`002`	Six enterprise goals.	3	`GOL-01 Protect the licence to operate` is right; `GOL-03 Grow revenue from existing customers` is retail framing.
`003`	Fourteen objectives under the goals.	3	`OBJ-05 Single view of customer` and `OBJ-06 Cross-sell rate` belong to another sector.
`004`	Sixteen measures with a baseline.	3	`MSR-01` says 54% of obligations are evidenced; `063` shows two of seven. `MSR-13` reads 53 against twelve.
`005`	Twenty-four architecture principles.	19	Generic TOGAF, all owned by `Architecture Board` . None mentions validated state, data integrity, retention or the patient.
`006`	Thirty-two requirements with priority and driver.	12	`REQ-001 Evidence GMP / GCP / GLP / GVP` is the right first row. Twenty-five of the rest are generated per entity and capability.
`007`	Eight constraints on the work.	16	`CON-06 Change freeze windows` is a trading calendar. **ArchiMate 4 has no Constraint element**: draw these as `Requirement` .
`008`	Six assumptions, `Open` or `At risk` .	3	Honest as written. `ASM-03` and `ASM-04` are really `052` implementation factors.

Business — twelve sheets

The organisation, the capability tree, the value streams and what hangs under them. `035` is filed here, not with the building-block sheets.

Sheet, exactly as the tab reads	Rows	Columns, in sheet order
`009 Organization Unit`	12	`Ref` , `Organisation unit` , `Type` , `Head` , `Headcount`
`010 Actor`	18	`Ref` , `Actor` , `Type` , `Status`
`011 Role`	18	`Ref` , `Role` , `Type` , `Performed by`
`012 Location`	8	`Ref` , `Location` , `Type` , `Scale`
`013 Business Capability`	100	`Ref` , `Capability` , `Level` , `Parent` , `Description` , `Criticality` , `Current` , `Target` , `Disposition`
`014 Value Stream`	5	`Ref` , `Value stream` , `Scope` , `Stages` , `Capabilities`
`015 Business Service`	10	`Ref` , `Business service` , `Capability` , `Description` , `Exposure` , `Criticality`
`016 Business Process`	20	`Ref` , `Business process` , `Value stream` , `Description` , `Automation`
`017 Business Function`	10	`Ref` , `Business function` , `Capability` , `Description`
`018 Product`	4	`Ref` , `Product` , `Type` , `Description`
`019 Contract`	10	`Ref` , `Contract` , `Counterparty` , `Type` , `Expiry`
`035 Business Building Block`	10	`Ref` , `Business building block` , `Capability` , `What it must do` , `Acceptance criteria`

Sheet	What it is for	Item	What is wrong with it
O09	Twelve organisation units with a headcount.	4	No Quality, Regulatory Affairs, Pharmacovigilance, Manufacturing or Clinical Development. Customer Service 211 against Technology 116.
O10	Eighteen actors with a Status.	4	The same eighteen generic titles as O11, because R21 joins them by index. Every row reads Filled.
O11	Eighteen roles with a Performed by.	4	Every Performed by cell reads Person. No Responsible Pharmacist, qualified person, QA release officer or validation engineer.
O12	Eight sites with a Scale in people.	7	LOC-08 Validated Cloud Region is staffed at 82 people. Scale sums to 1,215 against O09's 1,879.
O13	One hundred capabilities, levels 1 to 3.	4	The ten level-1 names are good; below that they are generated. Scores do not roll up: C01 is Low, C01.01.02 Critical.
O14	Five value streams with stages and capabilities.	4	The best sheet in the file. Stages totals 24 against O16's 20 rows; Capabilities disagrees with R02 throughout.
O15	Ten business services, one per capability.	4	Described as externally consumable while Exposure is Internal on seven of ten, and Criticality disagrees with its capability on eight.
O16	Twenty business processes under the value streams.	4	There is no process model: rows named <value stream> — stage N, described as Stage N of <value stream>.
O17	Ten business functions, one per capability.	4	Every description is the same sentence. The sheet adds nothing O13 does not carry.
O18	Four product types.	4	Defensible as a type list, but not the four identities of a product. No registration or pack exists anywhere.
O19	Ten contracts with a counterparty and expiry.	4	Counterparty holds application names, not legal entities. No quality, contract manufacturing, clinical trial or supply agreement.
O35	Ten business building blocks with criteria.	4	Each criterion reads "<capability> measured and owned.", which no gate can pass or fail. Reprinted as O55's BBB- rows.

Data — eleven sheets

Entities, attributes, components, mastering, metadata, migration, mapping. Two belong to Phase F, not Phase C.

Sheet, exactly as the tab reads	Rows	Columns, in sheet order
O20 Data Entity	14	Ref, Data entity, Class, Definition, Classification, Owner, Master system, Lineage
O21 Logical Data Component	12	Ref, Logical data component, Entity, Description
O22 Physical Data Component	12	Ref, Physical data component, Application, Type
O36 Data Entity Relationship	11	Ref, From entity, Relationship, To entity, Cardinality
O37 Data Attribute	60	Ref, Entity, Attribute, Type, Key, Classification
O38 Data Set	6	Ref, Data set, Description
O39 Data Migration	6	Ref, Entity, From system, To system, Status
O40 Source To Target Mapping	12	Ref, Entity, Source field, Target field, Rule
O42 Metadata Element	14	Ref, Metadata element, Category, What it records, Coverage
O43 Master Data Rule	10	Ref, Entity, Rule type, Description, Golden source
O44 Data Building Block	10	Ref, Data building block, Entity, What it must do, Acceptance criteria

Sheet	What it is for	Item	What is wrong with it
O20	Fourteen data entities with owner and master.	5	Master system is wrong on eight of fourteen, by rotation. E03 Subject is Internal and three rows are Public.
O21	Twelve logical data components.	5	Covers E01 – E12 only. Every description reads "Logical grouping of <entity> data."
O22	Twelve physical components, one per application.	5	Type is rotated: the safety database is Time series, the ERP is Document, the laboratory system is File.
O36	Eleven entity-to-entity relationships.	5	A positional chain E01 → E02 →…→ E12 with the verb cycling through three values. REL-07 composes a marketing authorisation of adverse events.
O37	Sixty data attributes across ten entities.	5	Six generated attributes each for E01 – E10; E11 – E14 have none. ATR-002 Compound · Name is Public, Key: Yes.
O38	Six data sets.	5	DST-02 Customer analytics set and DST-05 Risk measurement set are imported from other sectors.
O39	Six migrations with a from and a to.	9	Every row is impossible: MIG-01 moves Compound from CTMS to the safety database. The pairs are O20's rotation, one on.
O40	Twelve field mappings with a rule.	9	Both field columns hold the string <system> field, and the target is always one position further round the rotation.
O42	Fourteen metadata elements with coverage.	5	MTD-03 Steward, MTD-07 Consent basis and MTD-08 Lineage read Absent. Two rows use Recorded, not held as a coverage.
O43	Ten master data rules with a golden source.	5	Disagrees with O20.Master system on all ten entities it covers. Two sheets, two answers, no reconciliation.
O44	Ten data building blocks with criteria.	5	Every criterion reads "No <entity> record without a named owner" — ownership standing in for mastering.

Application — three sheets

Twelve applications, twelve services, ten building blocks. The smallest domain, and the one an inspector opens first.

Sheet, exactly as the tab reads	Rows	Columns, in sheet order
O23 Application Component	12	Ref, Application, Type, Description, Disposition, Criticality, Health, In service, Users, Owner
O24 Information System Service	12	Ref, Information system service, Application, Description, Style, Published
O45 Application Building Block	10	Ref, Application building block, Application, What it must do, Acceptance criteria

Sheet	What it is for	Item	What is wrong with it
O23	Twelve applications with disposition and health.	6	Four rows contradict themselves and Owner holds units, five wrong. **No validation state, review date or GxP flag.**
O24	Twelve information system services.	6	Five of twelve are Unpublished, contradicting O15's claim that every service is externally consumable.
O45	Ten application building blocks with criteria.	6	A11 and A12 have no block. "No unmanaged interface to or from <application>" is the only testable criterion here.

Technology — eight sheets

Services, components, platform services, environments, resilience, network. O47 is filed under Governance, which is why nobody reconciles it.

Sheet, exactly as the tab reads	Rows	Columns, in sheet order
`025 Technology Service`	12	`Ref`, `Technology service`, `Description`, `Category`
`026 Logical Technology Componen` [cut]	7	`Ref`, `Logical technology component`, `Rationale`, `Status`
`027 Physical Technology Compone` [cut]	12	`Ref`, `Physical technology component`, `Vendor`, `Class`, `Support end`, `Position`, `Location`
`028 Platform Service`	8	`Ref`, `Platform service`, `Description`
`046 Technology Building Block`	7	`Ref`, `Technology building block`, `Ref` (holds `T01` – `T07`), `What it must do`, `Acceptance criteria`
`048 Environment`	4	`Ref`, `Environment`, `Description`, `Personal data present`
`049 Resilience Tier`	12	`Ref`, `Application`, `Tier`, `Recovery objective`, `Tested`
`050 Network Link`	10	`Ref`, `From`, `To`, `Bandwidth`, `Diverse path`

Sheet	What it is for	Item	What is wrong with it
`025`	Twelve technology services with a category.	7	`Category` is rotated: `TSV-04 Identity` reads `Data`, `TSV-05 Integration` reads `Security`.
`026`	Seven logical technology components.	7	Real sector building blocks. Its status vocabulary is not `047`'s for the same seven, so no pair reconciles.
`027`	Twelve physical components with support end.	7	Four are in the wrong building, including `PTC-01 GMP MES servers` at the `Distribution Hub`. `PTC-07` and `PTC-11` are past support.
`028`	Eight platform services.	7	Generic. `PSV-02 API gateway` is listed as a service while `O65 EP-03` records the product as not yet selected.
`046`	Seven technology building blocks with criteria.	7	**Two columns headed `Ref`.** The second holds `T01` – `T07`, which nothing in the file resolves against. No status column.
`048`	Four environments with a personal-data flag.	7	Inverted: production reads `No`, pre-production and development read `Yes`, and `SEC-07` masking is `Partial`.
`049`	Twelve resilience tiers with a recovery objective.	7	**Nine of twelve are `Never tested`.** `RES-04 Safety Database`, which carries the expedited clock, is the lowest tier.
`050`	Ten network links with a diverse-path flag.	7	Ten rows over eight pairs. `NWL-02` and `NWL-10` are the same pair with two different answers.

Security — three sheets

Controls, policies and enforcement points. Three small sheets whose `Position` columns are the most honest text in the file.

Sheet, exactly as the tab reads	Rows	Columns, in sheet order
`041 Security Control`	8	`Ref`, `Security control`, `What it does`, `Position`, `Enforced at`
`064 Security Policy`	4	`Ref`, `Security policy`, `Scope`, `Position`
`065 Enforcement Point`	5	`Ref`, `Enforcement point`, `Acts on`, `Position`

Sheet	What it is for	Item	What is wrong with it
`041`	Eight security controls with a position.	7	`SEC-01` and `SEC-03` are `Absent`, yet `R16` maps `CO-01` to `SEC-01` and calls it `Evidenced`. `SEC-08` meets no obligation.
`064`	Four security policies with a position.	7	Four defensible statements. `POL-03` reads `Breached`, which `R06`'s 31 unlogged interfaces confirm.
`065`	Five enforcement points with a position.	7	The most honest column in the file: `EP-05` reads "Most refreshes are unmasked". `EP-05` is in no relationship.

Realisation — twelve sheets

Gaps, packages, deliverables, standards, contracts, alternatives, transitions, projects, risks, increments. Two are the same list twice.

Sheet, exactly as the tab reads	Rows	Columns, in sheet order
`O29 Gap`	0	`Ref` , `Gap` , `Domain` , `Building block` , `Severity` , `Status` , `Due`
`O30 Work Package`	12	`Ref` , `Work package` , `Domains` , `Cost` , `Start` , `End` , `Approval` , `Delivery`
`O31 Deliverable`	20	`Ref` , `Deliverable` , `Phase` , `Status`
`O32 Standard`	16	`Ref` , `Standard` , `Category` , `Status` , `Conformance`
`O33 Architecture Contract`	8	`Ref` , `Architecture contract` , `Work package` , `Compliance status`
`O52 Implementation Factor`	8	`Ref` , `Implementation factor` , `Category` , `Deduction`
`O53 Transition Architecture`	3	`Ref` , `Transition architecture` , `Description` , `Work packages` , `Exit criteria`
`O54 Architecture Alternative`	5	`Ref` , `Alternative` , `Description` , `Cost` , `Recommendation`
`O55 Building Block`	37	`Ref` , `Domain` , `Building block` , `What it must do` , `Acceptance criteria`
`O56 Project`	12	`Ref` , `Project` , `Work package` , `Cost` , `Start` , `End` , `Lead`
`O62 Delivery Risk`	8	`Ref` , `Risk` , `Likelihood` , `Impact` , `Owner`
`O66 Architecture Increment`	6	`Ref` , `Increment` , `Lands` , `What is added`

Sheet	What it is for	Item	What is wrong with it
`O29`	The gap register, with a `Due` date.	8	**Empty.** Seven columns, no rows — and the dashboard still reports `Gaps overdue: 0` .
`O30`	Twelve work packages with cost and dates.	8	Costs sum to $84.3m against the $46m alternative marked `Recommended` . `WP-08` is `Not started` and `Delivered` at once.
`O31`	Twenty TOGAF deliverables with a phase.	8	No Preliminary and no Phase H deliverable, no approver and no date. Thirteen read `Approved` , including `DLV-13 Consolidated Gaps` .
`O32`	Sixteen standards with status and conformance.	19	**Not one GxP standard.** `STD-04 Data retention` and `STD-15 Records management` contradict each other.
`O33`	Eight architecture contracts, `WP-01` – `WP-08` .	10	Four packages have no contract. Six read `Not yet assessable` , and none can be assessed while `O59` is empty.
`O52`	Eight implementation factors with a deduction.	8	`IF-03 Peak trading freeze` is a `Constraint` , and there is no trading peak here. The quality review cycle is absent.
`O53`	Three transition states with exit criteria.	9	`Exit criteria` reads `5 exit criteria` , `6 exit criteria` , `7 exit criteria` — a count, not criteria.
`O54`	Five alternatives with a cost and a verdict.	8	Two live answers: `ALT-B` at $46m is `Recommended` and `ALT-E` at $31m is `Partly adopted` . Neither scores revalidation load.
`O55`	The union building-block sheet, with a `Domain` .	8	`R09` and `R13` resolve against **this** sheet, not the four domain sheets. One of the 37 criteria passes today.
`O56`	Twelve projects with a cost and a lead.	9	The same twelve items as `O30` , same names, costs, dates. `PRJ-11 Customer experience` is led by the Internal Auditor.
`O62`	Eight delivery risks with likelihood.	8	Three rows read `Certain` — `RSK-01` , `RSK-07` , `RSK-08` . A certainty is an issue; `RSK-01` duplicates `IF-01` .
`O66`	Six increments with a landing month.	9	Three are the three `O53` states by exact name identity. The other three have no state.

Governance — seven sheets

Reference models, bodies, gates, assessments, decisions, change requests, obligations. Both empty object sheets sit here.

Sheet, exactly as the tab reads	Rows	Columns, in sheet order
`O47 Reference Model Entry`	11	`Ref` , `Reference model entry` , `Description` , `Status`
`O57 Governance Body`	5	`Ref` , `Governance body` , `Cadence` , `Membership` , `Mandate`
`O58 Governance Gate`	4	`Ref` , `Gate` , `When` , `What it tests`
`O59 Compliance Assessment`	0	`Ref` , `Building block` , `Block name` , `Acceptance criterion` , `Gate` , `Verdict` , `Evidence`
`O60 Governance Decision`	7	`Ref` , `Decision` , `Body` , `Status`
`O61 Governance Change Request`	5	`Ref` , `Change request` , `Against` , `Type` , `Status` , `Expires`
`O63 Control Objective`	7	`Ref` , `Obligation` , `Source` , `Control objective` , `Evidence`

Sheet	What it is for	Item	What is wrong with it
`O47`	Eleven reference model entries with a status.	7	Filed under `Governance` , not Technology. Its seven sector entries duplicate `O26` 's names and contradict their statuses.
`O57`	Five governance bodies with a mandate.	2	**None is a quality body.** No change control board, no validation committee, no data integrity forum.
`O58`	Four governance gates and what each tests.	2	The four gates are sound, but `GT-3 G3 Benefit` tests a benefit against a date no sheet carries.
`O59`	The compliance assessment register.	10	**Empty**, while `O63 CO-01` reads `Evidenced` and `DLV-19` reads `Draft` . Item 10 opens it at `CA-001` .
`O60`	Seven governance decisions with a status.	10	Four are `Outstanding` , including `DEC-01` . **No date column**, so nothing can be overdue and nothing has an author.
`O61`	Five change requests with an expiry.	11	`Expires` is filled on one row of five. `DEC-07` grants a dispensation with no expiry; `CR-G01` expires 2029-06.
`O63`	Seven obligations with a control objective.	10	Every objective restates its obligation. **No South African obligation appears**: no SAHPRA duty, single exit price or POPIA.

The industry extension, and the six objects that are not there

One sheet, three columns, six rows, and the single most important fact in the repository.

Sheet, exactly as the tab reads	Rows	Columns, in sheet order
`EXT Industry Extension`	6	`Ref` , `Industry extension type` , `What it holds and why the core metamodel is not enough`

Ref	Industry extension type	What the repository holds of it
`X01`	Compound	Nothing. `O20 E01 Compound / Molecule` is an *entity*, not an instance
`X02`	Study / Trial	Nothing. `O20 E02` is the entity
`X03`	Batch / Lot	Nothing. `O20 E09` is the entity
`X04`	Regulatory Submission	Nothing. `O20 E06` is the entity
`X05`	GxP Computerised System	**Nothing, and there is no entity either**
`X06`	Adverse Event	Nothing. `O20 E08` is the entity

Six object types this sector needs beyond the TOGAF core are declared, and no sheet holds one instance of any. Five have an entity standing in for them in `O20` . The sixth, `X05 GxP Computerised System` , whose description here reads "A validated system with its validation state and owner", has no entity, sheet, column or row anywhere. Item 2 creates the register; item 5 creates the entity [5].

The twenty-five relationship sheets

These are the useful half and the half nobody opens. Only `R06` and `R19` carry a `Ref` of their own; the rest resolve a pair of refs and add a column or two of judgement. **Ten of them are index-order joins** rather than statements about the world: `R04` , `R07` , `R09` , `R13` , `R17` , `R18` , `R20` , `R21` , `R22` and `R23` . Read a positional join as a question, never an answer [6].

Sheet, exactly as the tab reads	Rows	Columns, in sheet order
R01 Capability decomposition	90	Parent, Parent name, Child, Child name, Level
R02 Value stream uses Capabilit [cut]	15	Value stream, Name, Capability, Capability name
R03 Capability realised by Busi [cut]	10	Capability, Capability name, Business service, Service name
R04 Business Service supported [cut, and ends in a space]	10	Business service, Service name, Application, Application name
R05 Application manages Data En [cut]	28	Application, Application name, Data entity, Entity name, Role
R06 Application interfaces with [cut]	36	Ref, Source, Source name, Target, Target name, Type, Frequency, Managed, Logged
R07 Application deployed on Tec [cut]	12	Application, Application name, Platform component, Component name, Environment
R08 Requirement derives from Dr [cut]	32	Requirement, Requirement name, Driver, Driver name
R09 Requirement satisfied by Bu [cut]	32	Requirement, Requirement name, Building block, Block name, Traced
R10 Objective measured by Measu [cut]	14	Objective, Objective name, Measure, Measure name, Unit
R11 Goal realised by Objective	14	Goal, Goal name, Objective, Objective name
R12 Gap closed by Work Package	0	Gap, Gap name, Severity, Work package, Package name
R13 Work Package delivers Build [cut]	37	Work package, Package name, Building block, Domain, Block name

Sheet, exactly as the tab reads	Rows	Columns, in sheet order
R14 Project delivers Work Packa [cut]	12	Project, Project name, Work package, Package name, Cost
R15 Transition contains Work Pa [cut]	7	Transition, Name, Work package, Package name
R16 Control Objective met by Co [cut]	7	Control objective, Obligation, Control, Control name, Position
R17 Policy enforced at Enforcem [cut]	4	Policy, Policy name, Enforcement point, Point name, Position
R18 Standard applies to Applica [cut]	16	Standard, Standard name, Application, Application name, Conformance
R19 Data Entity relates to Data [cut]	11	Ref, From, Relationship, To, Cardinality
R20 Application located at Loca [cut]	12	Application, Application name, Location, Location name
R21 Role performed by Actor	18	Role, Role name, Actor, Actor name, Status
R22 Org Unit owns Capability	10	Org unit, Unit name, Capability, Capability name
R23 Increment delivered by Proj [cut]	6	Increment, Increment name, Lands, Project, Project name
R24 Compliance Assessment tests [cut]	0	Assessment, Building block, Block name, Criterion, Verdict
R25 Application supports Produc [cut]	4	Product, Product name, Application, Application name

Sheet	What it is for	Item	What is wrong with it
R01	Capability parent to child.	4	90 rows, thirty level-2 and sixty level-3. A faithful reflection of a generated tree.
R02	Value stream to the capabilities it uses.	4	Fifteen rows sliding one capability per stream. C08, C09 and C10 are used by no value stream.
R03	Capability to the service realising it.	4	C01 → BSV-01 … C10 → BSV-10. An index join, and the only one the index gets right.
R04	Business service to the supporting application.	6	BSV-01 → A01 … BSV-10 → A10. Ten rows for twelve applications, which is all of RPT13.
R05	Application to the entity it masters or consumes.	5	Fourteen Master and fourteen Consumer. The first column holds names, not refs. The Master rows reproduce O20's errors.
R06	The interface register: type, frequency, managed, logged.	6	The most useful sheet here. 17 Managed, 19 Unmanaged, 5 Logged, and all six direct database reads unlogged.
R07	Application to platform component, with environment.	7	A01 → PTC-01 … A12 → PTC-12, and **every Environment cell reads Production**.
R08	Requirement to the driver it derives from.	12	32 rows that wrap, which is why twenty-five requirements derive from four business drivers.
R09	Requirement to the block satisfying it.	12	Resolved against O55. TBB-03 – TBB-07 satisfy no requirement, and Traced reads No on every fourth row including REQ-001.
R10	Objective to the measure measuring it.	3	Fourteen rows, OBJ-nn → MSR-nn. MSR-15 and MSR-16 measure no objective, and their only source sheet is empty.
R11	Goal to the objectives realising it.	3	Fourteen rows, consistent with O03.Goal. One of the few that is a reading rather than a join.
R12	Gap to the work package closing it.	8	**Empty**, because O29 is empty.
R13	Work package to the block it delivers.	8	Twelve packages cycled round the 37 O55 blocks, so WP-10 Compliance mechanisms delivers no compliance mechanism.

Sheet	What it is for	Item	What is wrong with it
R14	Project to the work package it delivers.	9	Twelve rows, PRJ-nn → WP-nn, costs identical to both parents. It exists because O30 and O56 are one list twice.
R15	Transition state to its work packages.	9	Seven rows for twelve packages. WP-06, WP-08, WP-09, WP-10 and WP-12 — $29.1m — sit in no state.
R16	Control objective to the control meeting it.	10	Position is a verbatim copy of O63.Evidence, never a reading of O41.Position; the two disagree on four of seven.
R17	Security policy to its enforcement point.	7	Four rows for five points; EP-05 appears nowhere. Position copies the point's, so POL-04, which O64 calls Met, does not.
R18	Standard to the application it applies to.	6	Sixteen rows touching the odd-numbered applications only. **No Critical application has any standard applied to it.**
R19	Entity to entity, with a cardinality.	5	A verbatim duplicate of O36 with the headers renamed. It resolves nothing.
R20	Application to the location it runs at.	7	A01 → LOC-01 … A08 → LOC-08, then wrapping. Contradicts R07 plus O27.Location for nine of twelve.
R21	Role to the actor performing it.	4	Eighteen rows, ROL-nn → ACT-nn, every one Filled. This join is *why* O10 and O11 hold the same titles.
R22	Organisation unit to the capability it owns.	4	Ten rows by index, putting Regulatory Affairs under Human Resources and Pharmacovigilance under Technology.
R23	Increment to the project delivering it.	9	Six rows touching the odd-numbered projects only. Six projects deliver no increment.
R24	Compliance assessment to the criterion tested.	10	**Empty**, because O59 is empty.
R25	Product to the application supporting it.	4	Four rows striding by three. Eight applications support no product.

The thirteen reports

Every report recomputes — more often reprints — object sheets you have already read. Item 14 catalogues them properly. What you need here is the tab-level reference and the footer trap.

Nine of the thirteen end with a prose sentence sitting alone in the first column — `RPT01`, `RPT04`, `RPT05`, `RPT06`, `RPT08`, `RPT09`, `RPT10`, `RPT11` and `RPT13`. Only `RPT02`, `RPT03` and `RPT12` end on a record, and `RPT07` is empty, so it cannot. The counts below exclude that footer. The rule to script: **drop the last row of any report if only its first cell is filled** [2].

Report, exactly as the tab reads	Records	Columns, in sheet order
`RPT01 Dashboard`	18 + footer	`Measure`, `Value`, `What it means`
`RPT02 Application Portfolio`	12	`Ref`, `Application`, `Type`, `Disposition`, `Criticality`, `Health`, `In service`, `Users`, `Owner`
`RPT03 Capability Heatmap`	10	`Ref`, `Capability`, `Criticality`, `Current`, `Target`, `Gap`, `Disposition`
`RPT04 Interface Catalogue`	36 + footer	`Ref`, `Source`, `Target`, `Type`, `Frequency`, `Managed`, `Logged`
`RPT05 Data Ownership`	14 + footer	`Ref`, `Entity`, `Class`, `Classification`, `Owner`, `Master system`, `Lineage`
`RPT06 Requirements Traceability` **[cut]**	32 + footer	`Requirement`, `Name`, `Category`, `Priority`, `Driver`, `Reaches a block`
`RPT07 Gap Analysis`	0	`Ref`, `Gap`, `Domain`, `Severity`, `Status`, `Due`, `Work package`

Report, exactly as the tab reads	Records	Columns, in sheet order
`RPT08 Roadmap`	12 + footer	`Ref`, `Work package`, `Cost`, `Start`, `End`, `Approval`, `Delivery`
`RPT09 Regulatory Coverage`	7 + footer	`Ref`, `Obligation`, `Source`, `Control objective`, `Evidence`
`RPT10 Technology Currency`	12 + footer	`Ref`, `Component`, `Vendor`, `Class`, `Support end`, `Position`, `Location`
`RPT11 Security Position`	8 + footer	`Ref`, `Control`, `What it does`, `Position`, `Enforced at`
`RPT12 Standards Conformance`	16	`Ref`, `Standard`, `Category`, `Status`, `Conformance`
`RPT13 Orphan Analysis`	2 + footer	`Ref`, `Object`, `Disposition`, `Finding`

Report	What it is for	Verified finding
`RPT01`	Eighteen headline measures.	Nine are honest, three wrong, and four are not in `ZZ Checks`.
`RPT02`	`O23` without the `Description` column.	Ends on a record. Carries the four self-contradictions forward.
`RPT03`	The ten level-1 capabilities with a `Gap`.	Every gap is 1 and every disposition `Sustain`, because every row was written with target = current + 1.
`RPT04`	`R06` without the source and target refs.	The original footer trap: 37 data rows, the last a sentence.
`RPT05`	`O20` without the `Definition` column.	Publishes the wrong master system for eight of fourteen entities.
`RPT06`	`O06` with a `Reaches a block` column.	That column is `R09.Traced` copied across, so it reads `No` on `REQ-001` while `R09` in fact maps all 32.
`RPT07`	The gap report.	**Empty**, because `O29` and `R12` are empty.

Report	What it is for	Verified finding
`RPT08`	`O30` without the `Domains` column.	Repeats typed `Past end date` strings as though they were computed.
`RPT09`	`O63` reprinted in full.	The report an inspector asks for first, and the one that would embarrass you.
`RPT10`	`O27` reprinted in full.	Correct as far as it goes, with the locations still wrong.
`RPT11`	`O41` reprinted in full.	Footer: "A control that is absent is not a control." The best sentence in the repository.
`RPT12`	`O32` reprinted in full.	Ends on a record. Shows four `Deprecated` and `Conformant` pairs without comment.
`RPT13`	Objects no relationship reaches.	Two records plus a footer. It tests one join only, `R04`, so it misses every other orphan.

The checks sheet

Sheet, exactly as the tab reads	Rows	Columns, in sheet order
`ZZ Checks`	15 + footer	`Check` , `Result` , `Threshold`

Fifteen checks with a threshold, and a footer reading "The thresholds are what a healthy estate would show. The results are what this one does. That distance is the work." It is **not** the same list as `RPT01 Dashboard` , which has eighteen measures and a `What it means` column; fourteen lines are common. `Applications` , `Interfaces` , `Gaps overdue` and `Level 1 capabilities` are on `RPT01` only; `Records validated against a real organisation` is on `ZZ Checks` only [7].

Nine of the fifteen are correct and checkable. `Records` (850) is wrong by 36, `Object types populated` (65) by at least two, and `Work packages past end date` (3) is a string typed into `O30.Delivery` rather than a computation — the earliest `Start` in `O30` is 2027-02, so on a late-2026 reading nothing has started [3][8].

The four sheets you add

Four sheets this repository needs and does not have, numbered `O67` to `O70` , and nothing else may claim those numbers. Item 2 defines `O67` and `O70` , item 5 `O69` , item 11 `O68` . `VAL-` , `PRS-` , `RC-` and `RGR-` appear nowhere in the 105 sheets, which is why they were chosen — `VS-` would have collided with `O14 Value Stream` .

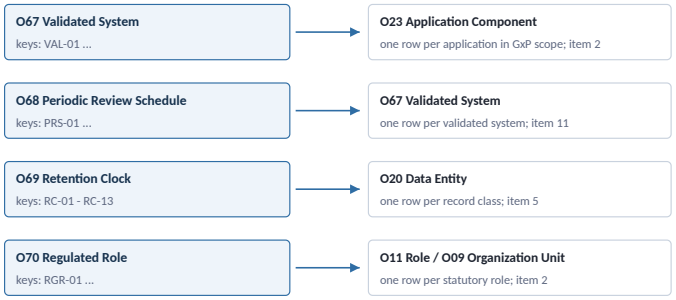


Figure 13.2 The four sheets you add, and the existing sheet each one reconciles against. The numbering is fixed, and all four key prefixes are unused anywhere in the 105 sheets.

Sheet	Rows a first pass needs	Columns, in sheet order
`O67 Validated System`	one per GxP application, so nine to twelve	`Ref` , `Application` , `GxP` , `Validation state` , `Last periodic review` , `Next review due` , `System owner` , `Process owner` , `Validation lead` , `URS reference` , `Audit trail reviewed` , `Decommissioning plan`
`O68 Periodic Review Schedule`	one per row of `O67`	`Ref` , `System` , `Frequency` , `Last review` , `Next due` , `Reviewer` , `Outcome` , `Actions open`
`O69 Retention Clock`	thirteen, `RC-01` to `RC-13`	`Ref` , `Record class` , `Period` , `Clock starts at` , `Instrument` , `System of record` , `Disposal approved by`
`O70 Regulated Role`	six to ten	`Ref` , `Role` , `Statutory basis` , `Named person` , `Deputy` , `Registration number held` , `Signs for`

Three rules. Keep `Ref` in column A, because every other sheet does. Put the header on row 4, or your tooling will read these differently from the other 105. And give each one an owner in `O70` before you populate it, because a register nobody signs is a spreadsheet [5].

Item 10 also opens space inside existing sheets: `CA-001` onward for `O59` , a fifth gate `GT-4` named `G2R Release for use` in `O58` , and `GB-06 Change Control Board` , `GB-07 Validation Committee` and `GB-08 Data Integrity Forum` in `O57` . Additions, not new sheets.

What this repository has no sheet for at all

Worth its own page, because in a regulated estate the absences are the finding. None of the following has a sheet, a column or a row anywhere in the 105 tabs.

- **Validation state.** No application, component or environment carries one. The register is `O67` .
- **A periodic-review due date.** Nothing expires, so nothing is overdue. The schedule is `O68` .
- **A retention clock.** `O32 STD-04` asserts a standard; no sheet holds a period, an instrument or the event the clock runs from. The register is `O69` .

- **A regulated role.** No Responsible Pharmacist, qualified person or QA release officer; `O11` holds eighteen generic corporate titles. The register is `O70`.
- **A named person.** `O23.Owner` holds organisation units and `O11.Performed by` reads `Person` on all eighteen rows. `O20.Owner` holds seven free-text pharmaceutical job titles and is the only sheet in the file that knows this company has a Head of Pharmacovigilance [9].
- **A date on a decision.** `O60` has four columns and none is a date, so nothing is overdue.
- **A benefit date.** `GT-3 G3 Benefit` tests a benefit against a date no sheet carries.
- **An approver on a deliverable.** `O31` has `Phase` and `Status` and nothing else.
- **Any instance of the six industry objects.** `EXT` declares all six; the file holds none.

Nine absences, four new sheets, and two of the nine are columns on sheets that already exist. Add the columns and say in the text that you are adding them. Do not create a fifth sheet [5].

THE CHAPTER IN TEN LINES

- 105 tabs: 00 Index, 64 object sheets O01 to O66 with no O34 and no O51, EXT, 25 relationship sheets, 13 reports and ZZ Checks.
- Headers are on row 4 and the first record on row 5, on every sheet but 00 Index.
- Twenty-three tab names are cut at 31 characters, and R04 Business Service supported also ends in a space. Match on the prefix.
- 814 records across the 64 object sheets plus EXT, against 850 claimed in row 2 of 00 Index. The per-sheet counts in the same sheet sum to 814.
- Two object sheets are empty: O29 Gap and O59 Compliance Assessment. R12, R24 and RPT07 are empty in consequence.
- O46 Technology Building Block has two columns headed Ref. The second holds T01 to T07 and nothing resolves against it.
- O55 Building Block is the union sheet of 37 blocks with a Domain column. R09 and R13 resolve against it, not against O35, O44, O45 or O46.
- Ten relationship sheets are index-order joins: R04, R07, R09, R13, R17, R18, R20, R21, R22 and R23.
- Nine of the thirteen reports end in a typed sentence in the first column. Drop the last row if only its first cell is filled.
- ZZ Checks has 15 checks with a Threshold column; RPT01 Dashboard has 18 measures with a What it means column. Fourteen lines are common.
- EXT declares six industry object types and the file holds no instance of any of them. X05 GxP Computerised System has no entity either.
- The four sheets you add are O67 Validated System, O68 Periodic Review Schedule, O69 Retention Clock and O70 Regulated Role.
- Nine things have no sheet, column or row anywhere: validation state, review due date, retention clock, regulated role, named person, decision date, benefit date, deliverable approver and any instance of the six industry objects.

SOURCES FOR THIS CHAPTER

- **1.** Pharmaceuticals_TOGAF_Metamodel.xlsx — all 105 worksheet names read from the workbook; twenty-three are exactly 31 characters and 'R04 Business Service supported ' ends in a space
- **2.** Pharmaceuticals_TOGAF_Metamodel.xlsx — report sheets RPT01, RPT04, RPT05, RPT06, RPT08, RPT09, RPT10, RPT11 and RPT13, each ending in a prose sentence in the first column; RPT02, RPT03 and RPT12 end on a record and RPT07 is empty
- **3.** Pharmaceuticals_TOGAF_Metamodel.xlsx — sheet 00 Index: row 2 subtitle claiming 850 records, against the 65 per-sheet counts in rows 14 to 78 summing to 814; sheets O29 Gap and O59 Compliance Assessment hold zero rows
- **4.** Pharmaceuticals_TOGAF_Metamodel.xlsx — sheets O35 Business Building Block, O44 Data Building Block, O45 Application Building Block, O46 Technology Building Block and O55 Building Block (37 rows with a Domain column), against R09 Requirement satisfied by Bu and R13 Work Package delivers Build
- **5.** Pharmaceuticals_TOGAF_Metamodel.xlsx — sheet EXT Industry Extension, rows 5 to 10 (X01 to X06, no instances held anywhere), against sheet O20 Data Entity rows 5 to 18, which carries no entity for a validated system
- **6.** Pharmaceuticals_TOGAF_Metamodel.xlsx — relationship sheets R01 to R25, headers row 4; the nine index-order joins verified row by row against both parent object sheets
- **7.** Pharmaceuticals_TOGAF_Metamodel.xlsx — sheets ZZ Checks (15 checks with a Threshold column) and RPT01 Dashboard (18 measures with a What it means column), fourteen check names common to both
- **8.** Pharmaceuticals_TOGAF_Metamodel.xlsx — sheet O30 Work Package rows 5 to 16: the Delivery column holds the typed strings Past end date, In flight and Delivered; the earliest Start is 2027-02 and the earliest End is 2029-02
- **9.** Pharmaceuticals_TOGAF_Metamodel.xlsx — sheet O20 Data Entity, Owner column: seven free-text pharmaceutical job titles that appear in no other sheet, against O09 Organization Unit, O10 Actor and O11 Role

APPENDIX B

The report catalogue

Thirteen reports, none of which computes anything. What each one says, what a healthy number would be, which ones an inspector asks for, the one that would embarrass you, and the four reports this repository does not have and should.

What a report is in this repository

Thirteen tabs prefixed `RPT`. None of them computes anything in Excel: there is no formula anywhere in the file. Each one is a flattened copy of one or two object sheets, written out once by whoever built the workbook, with between zero and one derived column added. That matters in two ways. A report cannot be stale relative to its source, because it is not linked to it — it is **already** stale, from the moment anybody edits the source. And a report cannot be wrong in a way its source is right, because it inherits every defect of the sheet it copies, and in four cases adds one of its own [1].

So read them as published positions, not as calculations. A report is what Calderon Therapeutics would hand an auditor who asked. Which of them you would be willing to hand over is the question this item answers.

Report	Reads	Records
RPT01 Dashboard	counts over every sheet	18
RPT02 Application Portfolio	O23	12
RPT03 Capability Heatmap	O13, level 1 only	10
RPT04 Interface Catalogue	R06	36
RPT05 Data Ownership	O20	14
RPT06 Requirements Traceability	O06 + R09.Traced	32
RPT07 Gap Analysis	O29 + R12 (both empty)	0
RPT08 Roadmap	O30	12
RPT09 Regulatory Coverage	O63	7
RPT10 Technology Currency	O27	12
RPT11 Security Position	O41	8
RPT12 Standards Conformance	O32	16
RPT13 Orphan Analysis	O23 against R04 only	2

Figure 14.1 What each report reads, and how many records it carries. Eight are a straight reprint of one object sheet. Only `RPT01` aggregates. `RPT07` is empty and `RPT13` holds two records.

Eight of the thirteen are a straight reprint of a single object sheet with one or two columns dropped. `RPT03`, `RPT06` and `RPT13` add a derived column. `RPT01` is the only one that aggregates. `RPT07` is empty.

Look it up. Volume 1, chapter 10, section *The reporting layer, and the reconciliations that are always wrong*, is the teaching behind this whole item: a reporting layer that restates its sources without reconciling to them produces numbers that are individually defensible and collectively false.

The thirteen, in two tables

Each row: what the report computes, which sheets it reads, what the number would look like in an estate you would be happy to be inspected in, and what this one shows.

Report	What it computes	From	Healthy looks like	This repository shows
`RPT01 Dashboard`	Eighteen headline measures with a `What it means` gloss	Counts over the whole workbook	Every defect measure at zero against its `ZZ Checks` threshold	19 unmanaged interfaces, 31 unlogged, 6 direct reads, 2 components past support, 2 controls absent, 3 obligations unevidenced, 4 entities without lineage [2]
`RPT02 Application Portfolio`	The system inventory with disposition, criticality, health and owner	`O23` , minus `Description`	A named person owning each system, a disposition consistent with criticality and health, a validation state	Twelve rows owned by organisation units, four self-contradicting, and no validation column to inspect [3]
`RPT03 Capability Heatmap`	Level-1 capability maturity with a computed `Gap`	`O13` , level 1 only	Gaps spread from 0 to 3, with the largest against the most critical capabilities	Every gap exactly 1, every disposition `Sustain` . A heatmap with no heat, because every row was written as target = current + 1 [4]
`RPT04 Interface Catalogue`	Every interface with type, frequency, managed and logged	`R06`	Every interface managed and logged, and no direct database read	19 of 36 unmanaged, 31 unlogged, 6 direct reads, and only 2 both managed and logged [5]
`RPT05 Data Ownership`	Entity ownership, classification, master system and lineage	`O20` , minus `Definition`	One named individual per entity, one master system agreeing with `O43` , lineage `Full`	Eight master systems wrong, `O43` disagreeing on all ten it covers, four `Absent` lineage, and the trial subject classified `Internal` [6]
`RPT06 Requirements Traceability`	Requirements against whether each reaches a building block	`O06` plus `R09.Traced`	Every `Must` requirement reaching a block, with the block named	Eight of thirty-two read `No` , including `REQ-001` , the GxP evidence requirement — while `R09` in fact maps all thirty-two [7]
`RPT07 Gap Analysis`	Gaps with severity, status, due date and the package that closes them	`O29` and `R12` , both empty	One row per agreed gap, each with a severity and a date in the future	Nothing. Seven columns, zero rows — and `RPT01` reports `Gaps overdue: 0` [8]

Report	What it computes	From	Healthy looks like	This repository shows
`RPT08 Roadmap`	Work packages with cost, dates, approval and delivery	`O30` , minus `Domains`	Cost summing to the chosen alternative, approval preceding delivery, nothing past its end date	$84.3m against a $46m `Recommended` alternative; `WP-08` both `Not started` and `Delivered` ; three rows typed `Past end date` before anything has begun [9]
`RPT09 Regulatory Coverage`	Obligations, their source, the control objective and the evidence	`O63`	Every obligation in your jurisdiction, a real control objective, evidence pointing at a dated assessment	Seven EU and US obligations, none South African; every objective a restatement; `CO-01` `Evidenced` through a control `O41` calls `Absent` [10]
`RPT10 Technology Currency`	Components with vendor, class, support end and position	`O27`	No component past support, and a dated decision for anything expiring inside a year	Two past support today, the next expiring 2027-12, and four components recorded in the wrong building [11]
`RPT11 Security Position`	Controls with what each does, its position and where it is enforced	`O41`	No control `Absent` , and nothing `Partial` that an obligation depends on	Two `Absent` — identity lifecycle and privileged access — and four `Partial` , one of which is attributable access [12]
`RPT12 Standards Conformance`	Standards with category, lifecycle status and conformance	`O32`	No deprecated standard carrying a live verdict, and a standard for every regulated record	Seven `Deprecated` , three of them `Conformant` , four `Not assessed` , five in `Breach` , and not one GxP standard in the list [13]
`RPT13 Orphan Analysis`	Objects that no relationship reaches	`O23` against `R04` only	A short list, each row carrying the question it raises	**Two records**, `A11 Statistical Computing` and `A12 Medical Information` , plus a footer. It tests one join, so it misses every other orphan in the file [14]

The three derived columns, and two of them are wrong

Ten of the thirteen reports add nothing to their source. Three add one computed column, and those three columns are the only arithmetic in the workbook. Check all three by hand, because two do not survive it.

`RPT03.Gap` is `Target` minus `Current` , taken from the ten level-1 rows of `O13` . The subtraction is right. The input is not: every level-1 row in `O13` was written with `Target` one step above `Current` — 2 to 3 six times, 3 to 4 three times, 4 to 5 once. So the gap is 1 on all ten rows, and because the disposition rule keys off the gap, every disposition reads `Sustain` . The column is arithmetically correct and tells you nothing. A heatmap in which the most critical capability and the least critical carry the same gap is not a prioritisation instrument; it is a picture of how the sheet was generated [4].

`RPT06.Reaches a block` should be a lookup: does this requirement appear on the left-hand side of any row in `R09` ? Run it and the answer is yes, thirty-two times out of thirty-two. The column in fact reproduces `R09.Traced` , which is a typed value reading `No` on every fourth

row. So the report says `REQ-001 Evidence GMP / GCP / GLP / GVP` reaches no building block, while `R09` row 5 maps it to `BBB-01`. One of the two is wrong and the report is the one you would show somebody [7].

`RPT13.Finding` is the one that works. "No business service maps to it" is true of `A11` and `A12`, because `R04` has ten rows for twelve applications. The defect is scope, not arithmetic: it asks the question of one join, so `EP-05`, `SEC-08`, `TBB-03` to `TBB-07`, `MSR-15`, `MSR-16` and six of the twelve projects are orphans the report never mentions [14].

The trap: nine of thirteen end in a sentence

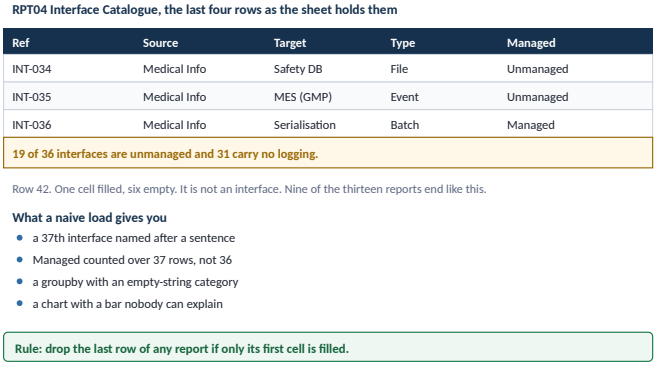


Figure 14.2 The footer trap, drawn. The last populated row of nine of the thirteen reports is a sentence in the first column with every other cell empty. Test the shape of the row, never its index.

The last populated row of nine reports is not a record. It is a prose sentence sitting alone in the first column, with every other cell empty. `RPT04`'s is the one everybody quotes, but it is not alone [1].

Report	The footer sentence, verbatim
`RPT01`	Every red figure is computed from this workbook, not asserted. They are the findings a first pass over this estate would produce.
`RPT04`	19 of 36 interfaces are unmanaged and 31 carry no logging.
`RPT05`	4 of 14 entities have no lineage from report back to source.
`RPT06`	A requirement that reaches no building block is a requirement nothing will deliver.
`RPT08`	3 work packages are past their end date.
`RPT09`	3 obligations have no mechanism or a mechanism in breach.
`RPT10`	2 components are running past vendor support today.
`RPT11`	2 controls are absent. A control that is absent is not a control.
`RPT13`	An orphan is not always wrong. It is always worth a question.

Only `RPT02`, `RPT03` and `RPT12` end on a record. `RPT07` is empty, so it cannot end on anything. `ZZ Checks` has the same footer habit, in row 20.

The rule, and it is the rule for every report you ever republish: drop the last row if only its first cell is filled. Not "drop row 42 of `RPT04`" — the row index moves the moment somebody adds an interface. Test the shape of the row [1].

Four consequences worth knowing before you load anything. A naive read of `RPT04` gives you thirty-seven interfaces, one of them named after a sentence. A count of `Unmanaged` over that frame divides by thirty-seven. A group-by on `Managed` produces an empty-string category. And the chart you paste into a steering pack has a bar nobody can explain, which is the version of this mistake that reaches an executive.

Two of the nine footers are also wrong in content, independently of the shape problem. `RPT08`'s says three packages are past their end date; the earliest `End` in `O30` is 2029-02, so on a late-2026 reading nothing is past anything — the three strings are typed into `O30.Delivery`, not computed [9]. And `RPT01`'s says every red figure is computed from the workbook rather than asserted, which is the claim this whole item exists to test.

The dashboard and the checks sheet are not the same list

They look like the same list and they are not, which is how a reader comes to quote a number from one and a threshold from the other.

`ZZ Checks` has **fifteen checks** and a `Threshold` column saying what a healthy estate would show. `RPT01 Dashboard` has **eighteen measures** and a `What it means` column saying what the figure describes. **Fourteen lines are common to both.** Four are on the dashboard only: applications, interfaces, gaps overdue and level 1 capabilities. One is on the checks sheet only — records validated against a real organisation, result zero, threshold "0, illustrative data throughout" [2][15].

Line	On	Value	Verified
Object types populated	both	65	**Wrong.** At most 63; `O29` and `O59` are empty
Records	both	850	**Wrong.** 814 across 64 object sheets plus `EXT`
Relationship types	both	25	Right as a sheet count; `R12` and `R24` carry no instances
Relationship instances	both	437	**Right.** The sum over `R01` – `R25` is exactly 437
Reports	both	13	Right
Applications	`RPT01`	12	Right
Interfaces	`RPT01`	36	Right
Unmanaged interfaces	both	19	Right against `R06.Managed`
Interfaces with no logging	both	31	Right. Five of thirty-six are logged
Direct database reads	both	6	Right against `R06.Type`, and all six are unlogged
Platform components past support	both	2	Right: `PTC-07` and `PTC-11`
Security controls absent	both	2	Right: `SEC-01` and `SEC-03`
Control objectives with no evidence	both	3	Right: `CO-03`, `CO-04`, `CO-05`, all `In breach`
Data entities with no lineage	both	4	Right: `E04`, `E06`, `E12`, `E14`
Vacant architecture roles	both	0	Right, because `R21` marks all eighteen `Filled`
Gaps overdue	`RPT01`	0	**Meaningless.** `O29` is empty, so nothing can be overdue
Work packages past end date	both	3	**Typed, not computed.** Nothing has started
Level 1 capabilities	`RPT01`	10	Right; `O13` decomposes consistently to level 3
Records validated against a real organisation	`ZZ`	0	Right, and the most honest line in the file

Nine of the eighteen are correct and checkable. Three are wrong. One is meaningless. Hand this table to a new analyst on their first day and they will have learnt more about reading a dashboard than a week of training would teach them [2][15].

Which of them an inspector asks to see

An inspector does not ask for an architecture report. They ask a question, and someone goes looking for the artefact that answers it. Six of these thirteen are the artefact somebody will reach for, and five of the six do not survive the reach.

Report	Asked for	Why	Survives it
`RPT09 Regulatory Coverage`	**First**	It is the coverage statement. The obligation list and the evidence against each	**No.** See below
`RPT02 Application Portfolio`	Early	Stands in for the system inventory, which is the first list in any GxP inspection	**No.** It cannot say which systems are in GxP scope or what state they are validated to [3]
`RPT04 Interface Catalogue`	Yes	An interface into a validated system is inside its validation scope	**No.** 19 unmanaged, 31 unlogged, and the serious-adverse-event path is an unmanaged hourly file [5]
`RPT11 Security Position`	Yes	Attributable access and privileged access are Part 11 and Annex 11 territory	**No.** Identity lifecycle and privileged access management both `Absent` [12]
`RPT05 Data Ownership`	Yes	Data integrity: who owns the record, what class it is, where it comes from	**No.** The trial subject is `Internal` and eight master systems are wrong [6]
`RPT10 Technology Currency`	Sometimes	Infrastructure qualification, and the plan for what is running unsupported	**Partly.** Two past support, honestly reported; the locations are wrong [11]

The other seven — `RPT01`, `RPT03`, `RPT06`, `RPT07`, `RPT08`, `RPT12`, `RPT13` — are internal architecture artefacts. Nobody from a regulator will ask for them, and you should not volunteer `RPT01`: handing over a dashboard that is wrong about three of eighteen figures is worse than handing over nothing, because it puts your own numbers into the record.

The one that would embarrass you

`RPT09 Regulatory Coverage` is the report an inspector reaches for first and the one you would least like them to read. Seven rows. Here is what is wrong with each layer of it [10].

The obligations are the wrong jurisdiction. `CO-03 Falsified Medicines / DSCSA` names the European and United States serialisation regimes. `CO-02 21 CFR Part 11 and Annex 11` names a United States rule and a European annex. There is **no South African obligation anywhere in the sheet**: no registration or variation duty, no single exit price declaration, no scheduling duty, no pharmacy licensing duty, and POPIA is never named — `CO-06 Data protection and health data` is the nearest thing and cites no instrument.

The control objectives are not control objectives. Every cell in that column reads "Control that evidences <the obligation, restated>." That is the obligation with four words in front of it. A control objective says what the control must achieve, in terms you can test. None of these can be tested, which is why `O59 Compliance Assessment` is empty: there is nothing to assess.

The evidence column is an assertion. `CO-01 GMP / GCP / GLP / GVP` reads `Evidenced`. Follow it through `R16`: `CO-01` is met by `SEC-01 Identity lifecycle`, which `O41` records as **`Absent`**. `R16.Position` is a verbatim copy of `O63.Evidence`, never a reading of `O41.Position`, and the two disagree on four of seven rows. The sheet that would hold the actual evidence, `O59`, has no rows at all [10][12].

And the mapping is one control per obligation. Seven obligations, seven `SEC-` controls, one each, which is how an entire good-practice regime gets reduced to an access control. `SEC-08 Data-at-rest encryption` is mapped to no obligation at all.

> If you are asked for `RPT09` in week one, do not print it. Say that the coverage statement is being rebuilt, show the obligation list you are rebuilding it from, and give a date. Item 10 rebuilds `O63` and refills `O59` from `CA-001`; this report republishes itself once that work is done. Printing the version in the file puts a false `Evidenced` into an inspection record, and that is a different category of problem from an incomplete one.

Three questions no report here can answer

Before the missing reports, the missing answers. Three questions get asked in the first fortnight of any pharmaceutical engagement, and not one of the thirteen reports — or the sixty-five object sheets behind them — can answer any of them.

"Which of these systems is validated, and to what?" There is no validation-state column anywhere in the repository. `RPT02` gives you twelve applications with health and criticality, which is an IT portfolio view of a regulated estate. Whether `A07 MES (GMP)` is validated, when its last qualification ran, and what the validation scope covers are not recorded [3].

"When was each system last reviewed, and when is the next review due?" No date column exists for it. Periodic review is the mechanism that keeps a validated state true, and the repository has no place to record that it happened [16].

"Who signs the release?" `O57` has five governance bodies and none of them is a quality body. `O11` holds eighteen generic corporate titles and `Performed by` reads `Person` on every row. The person with the legal duty to certify a batch does not appear in the file [16].

Look it up. Volume 1, chapter 16, section *Who inspects, and what they ask for*, has the twelve-row table of regulator against artefact. Read down its right-hand column and mark which artefacts this repository could produce. The answer is two of twelve, and that is the finding you open your first steering report with.

Four reports this repository does not have, and should

Every report above computes over the sheets that exist. The four most important reports in a regulated estate compute over sheets that do not — `O67` to `O70`, the four this volume tells you to create. Build these four and you have the set an inspection actually runs on.

Report	From	The question it answers	Healthy looks like
Validated system register	`O67`, joined to `O23` on `Application`	Which systems are in GxP scope, what state is each validated to, and who owns it	One row per GxP application; every row `Validated`; every row with a named system owner, a process owner and a URS `reference`
Periodic review due	`O68`, joined to `O67` on `System`	Which validated systems are overdue for periodic review, and by how long	Zero overdue; every system with a `Next due` inside its risk-justified `Frequency`; `Actions open` falling, not rising
Retention clock	`O69`, joined to `O20` and `O22`	For each record class: how long, from what event, under which instrument, in which system, and who approves disposal	Thirteen record classes, each with a period, an `Instrument` and a named `Disposal approved by`; nothing disposed of without approval
Regulated role	`O70`, joined to `O11` and `O57`	Who holds each statutory role today, who deputises, and what they sign for	Every statutory role filled with a named person and a named deputy; `Registration number held` populated where the role requires registration

Four things to notice about that table. Each report reads one new sheet and joins it to one that already exists, so none of them needs anything the repository cannot give you once the four sheets are populated. Each answers a question somebody will ask out loud, which none of the thirteen existing reports does. Each has a healthy number that is zero or complete, not a trend. And every one of them would **end on a record**, because you are writing them, and you now know not to type a sentence into the first column [16].

Number them `RPT14` to `RPT17` and nothing else. Give them the same five-row opening as every other sheet — title in row 1, subtitle in row 2, blank row 3, headers row 4, first record row 5 — so your own tooling reads them with the same constant it uses for the other 105 [16].

Before you publish any report

Six checks. They take about twenty minutes over the whole set and they are the difference between a pack you can defend and one you cannot.

1. **Drop the footer row.** If only the first cell is filled, it is not a record. Nine of thirteen need this today.
2. **Count the source, not the report.** `RPT04` has 37 rows and `R06` has 36. Where they differ, the source wins and the difference is your bug.
3. **Test one derived column by hand.** `RPT03`'s `Gap`, `RPT06`'s `Reaches a block` and `RPT13`'s `Finding` are the only three in the file, and two of the three are wrong.
4. **Say which sheet and which row range the figure came from**, in the report, in the pack, in the mail. Every number in this item carries one; copy the habit.
5. **Name the date the report was taken.** None of the thirteen carries one, which means none of them can be re-derived, and a figure that cannot be re-derived is not evidence.
6. **Ask whether an inspector would be shown this.** If yes, it is not a report any more. It is a quality record, and it needs a version, an approver and a retention period — which is the decision item 2 made in week one [16].

Look it up. Volume 1, chapter 13, section *Data integrity*, for why the version, the approver and the date are properties of the record rather than decoration on it; and chapter 16, section *The retention clocks*, for how long each class of record has to survive. A report that becomes a quality record inherits a retention clock, and the clock runs from an event, not from the date you printed it.

THE CHAPTER IN TEN LINES

- Thirteen reports, no formulae anywhere in the workbook. Eight are a straight reprint of one object sheet.
- Nine of the thirteen end with a prose sentence alone in the first column: RPT01, RPT04, RPT05, RPT06, RPT08, RPT09, RPT10, RPT11 and RPT13.
- The rule is to drop the last row of any report if only its first cell is filled. Test the shape, not the row index.
- RPT13 Orphan Analysis holds two records, not four, and it tests one join only, so it misses every other orphan.
- RPT07 Gap Analysis is empty because O29 and R12 are empty, while RPT01 reports gaps overdue as zero.
- ZZ Checks has 15 checks with a Threshold column; RPT01 Dashboard has 18 measures with a What it means column. Fourteen lines are common.
- Nine of RPT01's eighteen measures are correct, three are wrong and one is meaningless.
- Relationship instances at 437 is correct. Records at 850 is wrong by 36, and object types populated at 65 is wrong by at least two.
- Six of the thirteen are what somebody reaches for in an inspection, and five of the six do not survive the reach.
- RPT09 Regulatory Coverage is the one that would embarrass you: wrong jurisdiction, tautological objectives, asserted evidence and one control per obligation.
- Do not print RPT09 in week one. Say the coverage statement is being rebuilt and give a date.
- The four reports the repository needs are a validated-system register, a periodic-review-due list, a retention-clock report and a regulated-role report.
- All four derive from O67 to O70 joined to a sheet that already exists. Number them RPT14 to RPT17.

SOURCES FOR THIS CHAPTER

- **1.** Pharmaceuticals_TOGAF_Metamodel.xlsx - the thirteen RPT sheets, headers row 4; no cell in the workbook contains a formula. Nine end in a prose sentence in the first column with the remaining cells empty: RPT01, RPT04, RPT05, RPT06, RPT08, RPT09, RPT10, RPT11, RPT13
- **2.** Pharmaceuticals_TOGAF_Metamodel.xlsx - sheet RPT01 Dashboard, 18 measures in rows 5-22 with a What it means column, plus a footer sentence in row 23
- **3.** Pharmaceuticals_TOGAF_Metamodel.xlsx - sheets RPT02 Application Portfolio and O23 Application Component, rows 5-16: twelve rows, Owner holding organisation units, and no validation-state, periodic-review or GxP column
- **4.** Pharmaceuticals_TOGAF_Metamodel.xlsx - sheet RPT03 Capability Heatmap, rows 5-14, against O13 Business Capability level-1 rows: Target is Current plus one on all ten, so Gap is 1 and Disposition is Sustain throughout
- **5.** Pharmaceuticals_TOGAF_Metamodel.xlsx - sheets RPT04 Interface Catalogue (37 data rows) and R06 Application interfaces with (36 rows): Managed holds Managed 17 and Unmanaged 19; Logged holds Yes 5 and No 31; Type holds Direct DB read 6, all unlogged; INT-009 EDC to Safety Database is an unmanaged hourly file
- **6.** Pharmaceuticals_TOGAF_Metamodel.xlsx - sheets RPT05 Data Ownership and O20 Data Entity, rows 5-18, against O43 Master Data Rule rows 5-14: Master system wrong on eight of fourteen, E03 Subject classified Internal, four Absent lineage
- **7.** Pharmaceuticals_TOGAF_Metamodel.xlsx - sheet RPT06 Requirements Traceability, rows 5-36, against R09 Requirement satisfied by Bu rows 5-36: Reaches a block reproduces R09.Traced, which reads No on eight of thirty-two rows including REQ-001, while R09 maps all thirty-two
- **8.** Pharmaceuticals_TOGAF_Metamodel.xlsx - sheets RPT07 Gap Analysis, O29 Gap and R12 Gap closed by Work Package: all three hold zero rows
- **9.** Pharmaceuticals_TOGAF_Metamodel.xlsx - sheets RPT08 Roadmap and O30 Work Package, rows 5-16: the twelve costs sum to $84.3m; Delivery holds the typed strings Past end date, In flight and Delivered; earliest Start 2027-02, earliest End 2029-02
- **10.** Pharmaceuticals_TOGAF_Metamodel.xlsx - sheets RPT09 Regulatory Coverage and O63 Control Objective, rows 5-11, with R16 Control Objective met by Co rows 5-11 and O59 Compliance Assessment (zero rows)
- **11.** Pharmaceuticals_TOGAF_Metamodel.xlsx - sheets RPT10 Technology Currency and O27 Physical Technology Compone, rows 5-16: PTC-07 support end 2026-08 and PTC-11 support end 2026-01 both Past support; next expiry PTC-01 at 2027-12
- **12.** Pharmaceuticals_TOGAF_Metamodel.xlsx - sheets RPT11 Security Position and O41 Security Control, rows 5-12: SEC-01 and SEC-03 Absent, SEC-04, SEC-05, SEC-07 and SEC-08 Partial, SEC-08 mapped to no obligation in R16
- **13.** Pharmaceuticals_TOGAF_Metamodel.xlsx - sheets RPT12 Standards Conformance and O32 Standard, rows 5-20: seven Deprecated of which three are Conformant, four Not assessed, five in Breach, and no computerised-system validation, electronic records, data-integrity or audit-trail-review standard in the list
- **14.** Pharmaceuticals_TOGAF_Metamodel.xlsx - sheet RPT13 Orphan Analysis: two data rows, A11 Statistical Computing and A12 Medical Information, both 'No business service maps to it', plus a footer sentence; derived from R04 Business Service supported, which holds ten rows
- **15.** Pharmaceuticals_TOGAF_Metamodel.xlsx - sheet ZZ Checks, 15 checks in rows 5-19 with a Threshold column and a footer sentence in row 20; fourteen check names common with RPT01 Dashboard
- **16.** Volume 1, Pharmaceuticals - a starter pack for enterprise architects, chapters 13 and 16 and appendix 20: periodic review as the mechanism that keeps the validated state true, the retention clocks by record class, the regulated roles and their statutory basis, and what the inspector asks for

APPENDIX C

Workshop packs, checklists and the ninety-day calendar

Eight sessions you can run from these pages, the ninety days they fit into, and three checklists to print — written for a repository that is already full and mostly wrong.

How this pack works

Eight sessions. Seven are workshops you can run from these pages; the eighth is the executive read-out, which runs four times. Each pack states who is in the room, how long, what you bring, the agenda in timed blocks, the questions in the words to use, the sheet the output lands in, and what a bad outcome looks like — because in this estate most workshops fail in a way that feels like success.

Two rules run through all eight.

Name the jobs as the industry names them. Responsible Pharmacist, QA release officer, pharmacovigilance officer, regulatory affairs manager, validation lead, clinical operations lead, head of quality. The repository holds the same eighteen generic corporate titles in both `O10 Actor` and `O11 Role`, with every `Performed by` cell reading `Person` [1]. Not one of them signs anything a regulator would ask about. Invite the eighteen and you get a workshop about information technology.

Put the validation lead in every session that touches a system, from the first one. They are the only person who can say in the room that the thing on the board is a six-month revalidation, which saves the four weeks the group would otherwise spend designing it [2].

Calderon Therapeutics is the invented company in the repository you were handed; it describes no actual company [3].

> **You are not starting from an empty desk.** Volume 1's ninety days assume nothing exists. Here, 814 records across 64 object sheets already exist and most are wrong in ways you can demonstrate [3]. So you are not eliciting, you are disproving: print the sheet, put it on the wall, ask the room whether it is true. Faster, with the opposite risk — the room defers to the printed page because it looks authoritative.

WORKSHOP	SHEETS IT FILLS
1 GxP scoping	→ O67
2 Capability and organisation	→ O09 · O10 · O11 · O70 · R22
3 Data ownership	→ O20 · O37 · O43 · O69
4 Portfolio and interface	→ O23 · O24 · R06 · O67
5 Validated estate	→ O67 · O68 · O48 · O49
6 Obligation and evidence	→ O41 · O59 · O63 · R16
7 Gap and sequencing	→ O29 · O30 · O52 · O53 · O66 · R12
8 Executive read-out	→ O60, dated

Figure 15.1 Eight sessions, and the sheets each one fills. A workshop whose right-hand column is not written up the same day did not happen. Workshops 1 and 4 both touch `O67 Validated System`, which is why workshop 4 runs after it and not before.

Workshop 1 — GxP scoping

In the room. The Head of Quality, who chairs the scope decision and is not you. The validation lead. One QA release officer. The Responsible Pharmacist, for the last forty minutes only, for the systems that touch batch certification. One named owner per candidate system. Six to nine people. **Not the Chief Information Officer**: the moment validation scope looks like an IT opinion it stops binding anybody.

Length. Three hours, week one or two. Do not split it: the argument in the second hour depends on the rule set in the first.

You bring. The twelve rows of `O23 Application Component` printed one per A4 sheet and stuck to the wall. `EXT Industry Extension` row 9 on its own: *X05 GxP Computerised System — a validated system with its validation state and owner* [4]. Forms 2-A and 2-B from item 2. The validation master plan, which names the systems the company *believes* are in scope.

Agenda.

Time	Block
0:00–0:15	One slide: `X05` is declared and has no instances anywhere in 105 sheets
0:15–0:35	The rule: **the record decides scope, not the software**
0:35–1:40	The twelve wall sheets, one at a time, five minutes each
1:40–1:50	Break. Leave the wall up
1:50–2:25	The question that adds rows: what else holds a GxP record?
2:25–2:45	The three contestable systems, argued out loud
2:45–3:00	Count the `Never` rows; agree who signs `O67`

ASK THESE, IN THESE WORDS

Which records in this system are GxP records, and under which regime — GMP, GCP, GLP, GVP, GDP or good documentation practice?

If this system stopped tomorrow, which regulated record could you not produce?

What else in this department holds a record a regulator would ask about? A spreadsheet, an instrument, a vendor portal, a monitor.

Who is the named person who owns this system, and can I ring them this afternoon?

Where the output lands.

> **`O67 Validated System`, defined in item 2, rows 5 onward**
>
> Fill `Ref` (`VAL-01` onward), `Application` copied character for character from `O23.Application`, `GxP` as a comma-separated list of regimes or `None`, `Validation state` from the six permitted values, `System owner` as a person, and `URS reference` as a document number. The review dates and `Audit trail reviewed` come from documents afterwards, not from the room. Rows with no counterpart in `O23` take the system's own name.

Look it up. Volume 1, chapter 13, section *Which GxP, and for which record*; and chapter 19, section *Week one: find out what is validated*.

A bad outcome looks like. The room answers *is it validated?* and you leave with a yes/no column, which cannot carry *which GxP, and for which record*. Or every row maps one-to-one onto `O23`'s twelve, which means nobody answered the third question — the spreadsheet that calculates an assay result is in scope and is not an application. Or the Head of Quality sends a delegate, in which case the scope is a draft and you say so in writing.

Workshop 2 — Capability and organisation

In the room. The Head of Quality. The regulatory affairs manager. The pharmacovigilance officer — in South Africa, the responsible person for pharmacovigilance, who should reside in the country [5]. The clinical operations lead. The site head or head of manufacturing. The Head of Supply Chain. Human resources, for the headcount reconciliation only, in the last thirty minutes. Seven to nine people.

Length. Three and a half hours, run as two plus ninety minutes with a real break, the same day. Week two or three.

You bring. Three printed pages that make the argument for you. First, `O09 Organization Unit`'s twelve units, which include Customer Service and Legal and no Quality, Regulatory Affairs, Pharmacovigilance, Manufacturing or Clinical Development [6]. Second, `O17 Business Function`'s ten rows, which name Regulatory Affairs, Pharmacovigilance, Manufacturing & Supply, Quality Assurance & Control, Medical Affairs and Clinical Development [7]. Third, `R22 Org Unit owns Capability` as it actually resolves.

The second page is the whole workshop. **The repository already names the five functions the organisation sheet omits**, in the function sheet and in the capability tree, and nobody read the two side by side. You are not asking the room to invent an organisation. You are asking who runs the functions the repository has already named.

> `R22` assigns the ten level-1 capabilities to the twelve units in index order: `ORG-01` → `C01`, `ORG-02` → `C02`, down to `ORG-10` → `C10` [8]. Read out, that puts **Regulatory Affairs under Human Resources, Pharmacovigilance under Technology, Manufacturing & Supply under Data & Analytics, Quality Assurance & Control under Operations and Medical Affairs under Customer Service.** Read it out. It takes ninety seconds and it ends the argument about whether the repository can be trusted.

Agenda.

Time	Block
0:00–0:20	Read `R22` out loud, pair by pair. Say nothing else
0:20–1:00	Who actually runs each of `O17`'s ten functions? One name each
1:00–1:40	The jobs that sign: build the regulated-role list from what people sign
1:40–2:00	Break
2:00–2:40	Deputies, and what happens when the named person is on leave
2:40–3:10	Re-score the ten level-1 capabilities, with the people who run them
3:10–3:30	Headcount: 1,879 against 1,215, and where the difference sits

ASK THESE, IN THESE WORDS

Who runs Regulatory Affairs, and who do they report to? Not the function — the person.

What do you personally sign, and what is the consequence if you do not sign it?

Who signs when you are on leave, and where is that delegation written down?

Which of these ten capabilities is run twice, differently, at two sites?

Our headcount sheet says Legal has 240 people and Technology 116. Is that right?

Where the output lands.

> **`O09`, `O10`, `O11`, `O70 Regulated Role` and `R22`**
>
> `O09`: add from `ORG-13` — Quality, Regulatory Affairs, Pharmacovigilance, Manufacturing, Clinical Development. `O10 Actor` and `O11 Role`: add from `ACT-19` and `ROL-19`, and stop writing `Person` in `Performed by` where the performer is an organisation or a collaboration. `O70 Regulated Role`, defined in item 2: one `RGR-` row per role that signs something, with `Signs for` filled from what the person told you, not from a job description. `R22`: re-resolve every one of the ten pairs.

Look it up. Volume 1, chapter 5, sections *Quality and manufacturing* and *The interview order*; and chapter 3, section *Qualified Person versus Responsible Pharmacist*, because conflating the two in `O70` will embarrass you in this jurisdiction.

A bad outcome looks like. You leave with a new organisation chart. The output is a list of jobs with what each one signs, because that is what Phase G needs and what no chart carries. The second failure is a headcount debate, which is the least useful forty minutes available to you — hence last, and twenty minutes.

Workshop 3 — Data ownership and classification

In the room. The seven heads the repository already names as data owners in `O20 Data Entity` — Head of Pharmacovigilance, Head of Regulatory Affairs, Head of Clinical Development, Head of Preclinical Development, Head of Discovery Research, Head of Supply Chain, Head of Commercial [9]. Seven of those eight distinct owners exist in no other sheet in the file. Plus the information officer, for the privacy instruments; the data steward; the clinical operations lead; and QA.

Length. Three hours, week three or four. It is the workshop most likely to need a second sitting; book that now and cancel it if you do not need it.

You bring. `RPT05 Data Ownership` with its final row struck through — that row is a prose sentence in the `Ref` column, not a record [10]. `O20` rows 5 to 18 with the `Master system` column covered up. `O43 Master Data Rule`'s ten `Golden source` values on a separate page. The retention clock table from volume 1 chapter 16. And `O37 Data Attribute`'s `ATR-001` and `ATR-002`, a thirty-second demonstration: the compound identifier is `Internal` and the compound **name** is `Public` and marked `Key: Yes` [11].

Agenda.

Time	Block
0:00–0:10	The `ATR-001 / ATR-002` pair. Ask whether a compound name is public
0:10–0:50	Classification, entity by entity, with the instrument named each time
0:50–1:30	Uncover `Master system`. One entity at a time, ask who told you
1:30–1:40	Break
1:40–2:10	`O43` against `O20`: two sheets, two answers, which one dies
2:10–2:45	Retention: which clock, which event starts it, which system holds the event
2:45–3:00	Owners: a person per entity

ASK THESE, IN THESE WORDS

Under which instrument is this entity classified, and who signed the classification?

Subject data is recorded as Internal in this repository. Is a trial participant's pseudonymised record an internal document?

Which system is the master for a compound, and who told you so?

When does the retention clock on a batch record start, and which system holds the date it starts from?

If this number is wrong in a submission, who finds out, and how?

Where the output lands.

> **`O20`, `O43`, `O37` and `O69 Retention Clock`**
>
> `O20`: `Classification`, `Owner` and `Master system`, one row at a time, with the correction and the reason recorded in your findings log — eight of the fourteen `Master system` values are wrong by rotation, which is a defect of the column, not of eight independent cells [9]. `O43`: retire it or reconcile it; two sheets naming a golden source and disagreeing on every entity is worse than one. `O69 Retention Clock`, defined in item 5: `RC-01` onward, one row per record class, `Clock starts at` naming an event and `Instrument` naming the source. Do not restate the periods — cite the `RC-` row.

Look it up. Volume 1, chapter 9, section *Subject — the most sensitive record the company holds*; chapter 16, sections *POPIA and health data* and *The retention clocks*; and appendix 20, section *Retention clock by record class*.

A bad outcome looks like. The room agrees a classification scheme and never names an instrument, which produces a policy nobody can defend when asked why. Or `E03 Subject` stays `Internal` because the clinical side did not come — in which case do not correct it yourself. Record that the entity with the strongest statutory protection in the estate is classified below three entities marked `Public`, and take that to the read-out.

Workshop 4 — Application portfolio and interface

In the room. The named owner of each application — people, not units. The integration lead. The validation lead. One QA representative. And the person who actually runs the nightly file transfers, who is usually two grades below everyone else in the room and knows more than all of them about what crosses.

Length. Four hours, split two and two with lunch between, week four or five. The portfolio and interface halves are different conversations.

You bring. `RPT02 Application Portfolio`'s twelve rows. `RPT04 Interface Catalogue` with the 37th row struck through, because it is the sentence *19 of 36 interfaces are unmanaged and 31 carry no logging* sitting in the `Ref` column [10]. All 36 rows of `R06 Application interfaces with`, sorted by target. `O24 Information System Service`'s `Published` column.

Four rows do the work of the whole first half:

Ref	Application	Criticality	Health	Disposition
`A10`	Serialisation	Critical	Adequate	Retire
`A08`	QMS	Critical	Healthy	Replace
`A12`	Medical Information	Critical	At risk	Consolidate
`A06`	eTMF	High	Poor	Invest

Only the last of the four is coherent [12].

Agenda.

Time	Block
0:00–0:20	The four contradictory rows. Ask the owner of each to explain their row
0:20–1:10	Replace the `Owner` column with named people
1:10–1:50	Disposition: what actually happens to each, with a date
1:50–2:30	Lunch
2:30–3:20	The 36 interfaces: whose validation scope is each one in?
3:20–3:45	The six direct database reads, named and dated for closure
3:45–4:00	`INT-009`, on its own, as the session's closing item

> Save `INT-009` for last and give it fifteen minutes. `EDC` to `Safety Database`, type `File`, frequency `Hourly`, `Managed: Unmanaged` [13]. That is the path a serious adverse event takes from the trial site to the system carrying the expedited reporting clock, recorded as an unmanaged hourly file. The room will not argue about integration governance again.

ASK THESE, IN THESE WORDS

Your row says this system is critical, healthy and being replaced. Which of the three is wrong?

Who is the named person who owns this application? The sheet says an organisation unit owns it.

Is this interface inside anybody's validation scope, and whose?

Where is the interface specification — the one naming every field, type, unit and transformation?

Which of these interfaces would you not know had stopped, and for how long?

Where the output lands.

> **`O23`, `R06`, `O24` and `O67`**
>
> `O23.Owner`: a person per row. `O23.Disposition`: unchanged unless a dated decision changes it, and a changed row goes to `O60 Governance Decision` with a date. `R06.Managed` holds the words `Managed` and `Unmanaged`, not yes and no, and `Logged` holds `Yes` and `No` — keep the vocabulary [13]. Add a validation-scope column to `R06` and say in the text that you are adding it. `O24.Published`: five of twelve read `Unpublished` while every description reads *Service exposed by* … [14].

Look it up. Volume 1, chapter 12, sections *What is actually installed* and *Interfaces are routinely in nobody's validation scope*; and chapter 13, section *Computerised system validation, as it is actually practised*.

A bad outcome looks like. The group reviews the interface list for completeness, finds it complete, and goes home. Completeness was never the question — the questions are whose scope each one is in and what would be noticed if it stopped. The other failure is an owner column filling with the same four names, which records the four people who came, not the twelve who own the systems.

Workshop 5 — The validated estate and periodic review

In the room. The validation lead chairs, not you. The Head of Quality. One QA release officer. The infrastructure or qualification lead, because qualification and validation are different activities with different owners and different evidence. The owners of whichever systems came out of workshop 1 reading `Never`.

Length. Three hours, week five or six. It runs after workshop 1 and cannot run before it.

You bring. `O67` as workshop 1 left it. `O27 Physical Technology Compone`'s twelve rows, with `PTC-07 Laboratory instruments network` (support ended 2026-08) and `PTC-11 Enterprise storage array` (2026-01) highlighted [15]. `O48 Environment`. `O49 Resilience Tier`, nine of whose twelve rows read `Never tested` [16]. The audit-trail review plan, if one exists, and the last three periodic review reports.

Agenda.

Time	Block
0:00–0:25	Qualified or validated? Sort `O27`'s twelve components into the two piles
0:25–1:00	Periodic review: a date per system, and who holds the report
1:00–1:30	Audit-trail review: who does it, how many hours a month, and the last exception
1:30–1:40	Break
1:40–2:10	`O48`: production says no personal data, development says yes
2:10–2:40	Resilience, re-tiered against the regulatory clock rather than the label
2:40–3:00	The two past-support components, to a date

> `ENV-01 Production` records `Personal data present: No`. `ENV-02 Pre-production` and `ENV-04 Development` record `Yes` [17]. Read literally, the company holds trial participants' data in development and none in production. Either the column is reversed or non-production is full of unmasked subject data — and `O41 Security Control` records `SEC-07 Non-production masking` as `Partial` while `O65` records `EP-05 Non-production provisioning` as *Most refreshes are unmasked*, which points hard at the second reading [18]. Ask the question in the room and write down which answer you got.

ASK THESE, IN THESE WORDS

Is this component qualified, is the system on top of it validated, or neither? Who holds each document?

When is the next periodic review due for this system, and what happens if the date passes?

Who reviews the audit trails, how many hours a month do they have for it, and what did they do with the last exception they found?

The safety database is tier 3 with a four-hour recovery objective and has never been tested. Is that the intention?

Where the output lands.

> **`O68 Periodic Review Schedule`, `O67`, `O48` and `O49`**
>
> `O68`, defined in item 11: `PRS-01` onward, one row per system, `Frequency` risk-justified and `Next due` **a date, never a frequency**. Back-fill `O67.Last periodic review`, `Next review due` and `Audit trail reviewed` from the reports, not from the room. Correct `O48.Personal data present` or record that you could not establish it. Re-tier `O49` and put the regulatory clock in the justification: the safety database carries the expedited reporting clock and sits at `Tier 3` [16].

Look it up. Volume 1, chapter 13, sections *Periodic review* and *Audit-trail review, the most under-resourced obligation in the regime*.

A bad outcome looks like. `O68` fills with the word `Annually` and no dates, which is a policy, not a schedule — nothing can be overdue against a frequency. The subtler failure: the room agrees the environments column is merely reversed, you correct it, and the unmasked-refresh problem disappears from the record. Correct nothing you have not verified.

Workshop 6 — Obligation and evidence

In the room. The regulatory affairs manager. The Responsible Pharmacist. The pharmacovigilance officer. The information officer, for the privacy instruments. The Head of Quality. Internal audit, the only assurance function `O09` carries [6].

Length. Three hours, week seven or eight. It reframes the engagement, so run it before the week-ten read-out, not after.

You bring. `O63 Control Objective`'s seven rows with the `Control objective` column **covered up**, because every cell in it reads *Control that evidences* followed by the obligation restated [19]. `R16 Control Objective met by Co`, which maps each obligation to exactly one `SEC-` control. `O41 Security Control`, with `SEC-01 Identity lifecycle` and `SEC-03 Privileged access management` marked `Absent` [18]. Volume 1 chapter 16's inspection table. The manufacturing licence and one product's certificate of registration.

Agenda.

Time	Block
0:00–0:15	Uncover one `Control objective` cell. Let the room read it
0:15–1:00	Seven obligations: what control would actually evidence each?
1:00–1:40	The jurisdiction: name the South African instruments the sheet does not hold
1:40–1:50	Break
1:50–2:20	`CO-01` reads `Evidenced` through a control recorded `Absent`. Where is the evidence?
2:20–2:45	One obligation, many controls: break the one-to-one mapping
2:45–3:00	What an inspector would ask for, and which report you would not show them

> The obligations in `O63` are European and American: `CO-02 21 CFR Part 11 and Annex 11`, `CO-03 Falsified Medicines / DSCSA`. **There is no South African obligation anywhere in the sheet** — no registration or variation duty, no single exit price declaration, no privacy instrument by name, no scheduling duty, no pharmacy licensing [19]. `CO-06 Data protection and health data` is the nearest thing and names no instrument. That is the finding, and this room is where it gets corrected. The instruments and their dates come from volume 1 chapter 16, never from memory and never from the room.

ASK THESE, IN THESE WORDS

This cell says the control is a control that evidences the obligation. What control, specifically?

Which South African instrument binds us here, and what does it oblige us to produce?

The sheet records this obligation as evidenced through identity lifecycle management, which another sheet records as absent. Which is true?

If an inspector asked for the evidence for this obligation this afternoon, what document would you hand over?

Which of our thirteen reports would you not want an inspector to see, and why?

Where the output lands.

> **`O63`, `R16`, `O59 Compliance Assessment` and `O41`**
>
> `O63`: rewrite `Control objective` as a statement of what a control must achieve, and add rows for the South African obligations, with `Source` naming the instrument and its date. `R16`: many controls per obligation, and stop copying `O63.Evidence` into `Position` — the two columns disagree on four of seven today [20]. `O59`: `CA-001` onward, per item 10, with every verdict derived from a criterion rather than asserted. `O41`: nothing changes here, but `SEC-01` and `SEC-03` get owners and dates.

Look it up. Volume 1, chapter 16, sections *Who inspects, and what they ask for* and *Inspection response*; and appendix 20, section *Regulator to evidence*.

A bad outcome looks like. You leave with seven better-worded tautologies. The test is whether each `Control objective` cell can be read by somebody who has not seen the obligation and still makes sense as an instruction. If it only makes sense next to the obligation, it is the obligation with words in front of it.

Workshop 7 — Gap and sequencing

In the room. You chair. The person who owns quality assurance capacity, which is the binding constraint in this estate and not the budget. The validation lead. The delivery lead. The heads of the five functions from workshop 2. Finance, second half only.

Length. Four hours in two halves, weeks eleven to twelve. Gaps in the morning, sequence after lunch.

You bring. One page with nothing on it but a header row: `O29 Gap`'s seven columns and no records [21]. It is the strongest prop in this pack. The gap set derived in item 8, each row traceable to a cell in another sheet. `O30 Work Package`'s twelve rows summing to **$84.3m** against `ALT-B Foundations first` at $46m marked `Recommended`, and `ALT-E Compliance led` at $31m marked `Partly adopted` [22]. `O53 Transition Architecture`'s three states, whose `Exit criteria` column reads `5 exit criteria`, `6 exit criteria` and `7 exit criteria` — a count, not criteria [23]. And `O52 Implementation Factor` with `IF-03 Peak trading freeze` struck through, because a pharmaceutical company has no trading peak; it has an annual product quality review cycle and an inspection window [24].

Agenda.

Time	Block
0:00–0:10	Hold up the empty gap sheet. Ask whether the estate has no gaps
0:10–1:10	Walk the derived gap set; the room corrects it, you do not defend it
1:10–1:45	Severity: what would a regulator call this, and what is the patient consequence?
1:45–2:15	Break
2:15–2:55	Capacity: how many validations can quality assurance absorb in a year?
2:55–3:30	The calendars that bind: review cycle, inspection, renewals
3:30–4:00	Three transition states, with exit criteria written in the room

ASK THESE, IN THESE WORDS

Is this a gap, or is it how the business has decided to work? Say which.

How many system validations can quality assurance absorb in a year, alongside the annual product quality reviews?

What has to be true before this transition state is finished? Give me the test, not the intention.

Our roadmap costs $84.3m and the alternative we recommended costs $46m. Which number is the plan?

Where the output lands.

> **`O29`, `R12`, `O30`, `O53`, `O52` and `O66`**
>
> `O29`: `GAP-001` onward, `Gap` naming the difference and not the remedy, `Building block` resolving against `O55 Building Block` — the 37-row union sheet — and `Due` a date that can actually expire. `R12 Gap closed by Work Package`, empty today, gets a row per closure. `O53.Exit criteria`: the eighteen criteria the column currently counts, written out. `O52`: `IF-03` replaced; the review cycle and the inspection window added. `O66`'s six increments mapped onto the states, three of which match by exact name identity already [25].

Look it up. Volume 1, chapter 7, sections *The Annual Product Quality Review* and *The regulatory commitment calendar*; and chapter 19, section *What ninety days will not achieve*.

A bad outcome looks like. The room produces twenty-six gaps. Nobody acts on twenty-six; six to eight with a named remedy owner each is the output. The other failure is capacity discussed in money. The constraint here is the people who would run the validations, and they are the same people assembling the annual product quality reviews.

Workshop 8 — The executive read-out, four times

In the room. The executive committee. The Head of Quality must be present at all four, because two of the four carry quality findings and a decision taken without them is taken again later.

Length. Sixty minutes, at the ends of weeks 2, 6, 10 and 13.

You bring. One page to the first; the counted baseline and three facts to the second; six to eight findings and the reframed brief to the third; three transition states, five decisions and the honest year to the fourth.

Two of the four carry no recommendations at all, deliberately. Week 2 says what you are doing, who you have spoken to and whether a validated-system register exists. Week 6 says what the estate contains, in numbers nobody disputes. An architect who recommends in week two has produced a preference, and a preference does not survive a quality review.

ASK THE COMMITTEE THESE, AND WRITE THE ANSWERS DOWN

Is an architecture standard a controlled document in this company? Under what document number?

Who signs the release of a validated system for use, and is that person in this room?

How many system validations will you fund next year, and who told you that number?

Which of these decisions can be taken today, and which needs a body that does not exist yet?

Where the output lands.

> **`O60 Governance Decision`**
>
> `O60` has four columns — `Ref`, `Decision`, `Body`, `Status` — and **no date column**, which is why four of its seven rows can read `Outstanding` indefinitely [26]. Add `Decided on` and `Review due`, per item 2, then write one row per decision taken in the room, the same day. A decision that leaves the room without a date did not happen.

A bad outcome looks like. You present conclusions in week two and the room recognises a guess. Or the fourth session turns into a funding conversation before the three transition states have exit criteria, in which case you are being asked to price something nobody has defined.

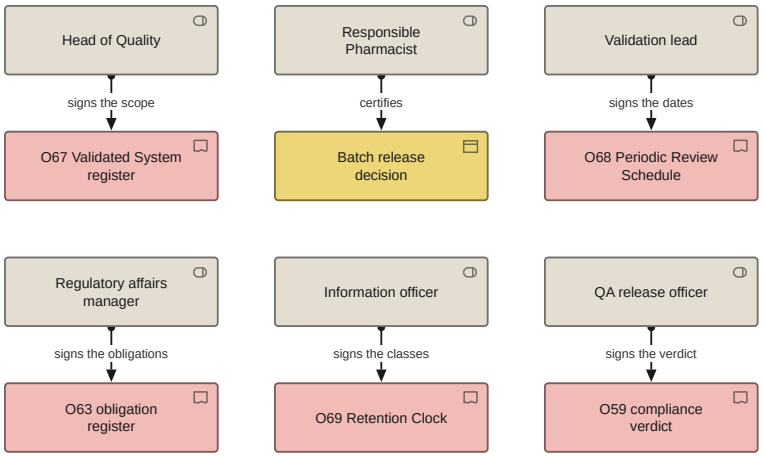


Figure 15.2 Who signs what, and none of them is you. Six regulated or quality roles against the six artefacts they sign. The element is `Role`; there is no layer-prefixed variant in ArchiMate 4. Not one of these six exists in `O10 Actor` or `O11 Role` as the repository stands.

The ninety-day calendar

Volume 1 chapter 19 sets the shape: orientation and the register in weeks 1 to 2, the counted baseline in weeks 3 to 6, the findings in weeks 7 to 10, the target and the decisions in weeks 11 to 13, four executive sessions, two of them carrying no recommendations [2]. That shape holds. What changes is that you are working a populated repository, so the count is a verification rather than a discovery, the findings are mostly already in the file, and the first argument arrives in week six.



Figure 15.3 The ninety days against the repository. Four phases, nine workshop sittings, four new sheets opened, and four read-outs. The **signed** row is the one an executive asks about, and weeks 9 to 11 are deliberately empty in it.

Week	What is done	What is signed	What is presented
1	Read the four documents. Count the repository: 64 object sheets, 814 records. Interview the validation lead	—	—
2	Workshop 1. `O67` opened and populated. Decide whether an architecture decision is a quality record	`DEC-08`, the quality-record decision, by the Executive Committee	One page: what you are doing, who you spoke to, whether a register exists
3	Workshop 2. `O09`, `O10`, `O11` and `O70` rebuilt. `R22` re-resolved	`O70` 's content, by the Head of Quality and the Responsible Pharmacist	—
4	Workshop 3, first half. `O20` reclassified with the instrument named per row	—	—
5	Workshop 3 second half and workshop 4 first half. `O69` opened. `O23.Owner` replaced with people	`O69` 's classes, by the information officer and Head of Quality	—
6	Workshop 4 second half. 36 interfaces assigned a validation scope. Reconcile `O67` against the quality function's list	`O67` as the baseline register, by the Head of Quality	**The counted baseline and three facts.** No recommendations
7	Workshop 5. `O68` opened with dates. `O48` and `O49` corrected or marked unestablished	`O68`, by the validation lead	—

Week	What is done	What is signed	What is presented
8	Workshop 6. `063` rewritten; the South African obligations added with their instruments	`063` 's new rows, by the regulatory affairs manager	—
9	Findings written: six to eight, each traceable to a cell and shown privately to the person whose area it concerns	—	—
10	Corrections absorbed. The cost framed in the committee's own terms	—	**The findings and the reframed brief**
11	Workshop 7, first half. `029` populated from differences already recorded. `R12` filled	—	—
12	Workshop 7, second half. Capacity established. Three transition states with eighteen written exit criteria	`053.Exit criteria` , by the Architecture Board and validation lead	—
13	The year's arithmetic. Five decisions, each with the cost of deferring it. Register handed to a named internal owner	`060` rows, dated, in the room	**The transition states, the decisions and the honest year**

A dash means nothing signed and nothing presented. Three notes, each of which has cost somebody a quarter.

Week 6 is the meeting that buys the next one. The numbers that land are the ones contradicting the repository's own front page: `00 Index` row 2 and `RPT01 Dashboard` both claim 850 records, and the per-sheet counts in rows 15 to 79 of `00 Index` sum to 814, which is what the sheets hold [3]. The detail is right and the headline is wrong. Show that first; everything afterwards is believed.

Nothing is signed in weeks 9, 10 and 11. Those weeks are for correction, and a signature closes an argument you want open.

Thirteen weeks does not hold a validation. No system is revalidated inside this calendar, no safety database re-coded, no interface moved onto managed integration. Ninety days delivers a known estate, four registers that did not exist, a gap set and a sequence. Say so in week two and again in week thirteen.

Checklist 15-A — the first-day checks

Print this and work it before you open Excel. It feeds your findings log, not a sheet. Ten checks, about three hours. Nobody signs it.

Check	Where you look	Done
Headers on row 4, first record on row 5	Any object sheet	
`O34` and `O51` do not exist, so there are 64 object sheets, not 66	Tab bar	
Twenty-three tab names are cut at 31 characters, and one ends in a space	`R04 Business Service supported`	
`O46` has two columns headed `Ref`; the second holds `T01` to `T07`	`O46`, row 4	
`O55` is the 37-row union sheet, and `R09` and `R13` resolve against it	`O55`, `Domain` column	
Four sheets are empty: `O29`, `O59`, `R12`, `R24`	Each sheet's row 5	
Nine of the thirteen reports end in a prose sentence in the first column	`RPT04`, last row	
The six industry object types in `EXT` have no instances anywhere	`EXT` rows 5 to 10	
No sheet anywhere carries a validation state, a GxP flag or a review date	`O23`, all ten columns	
Your own record count against the 850 claimed	`00 Index` rows 15 to 79	

Checklist 15-B — the thirteen afternoon checks, against the repository

Volume 1 chapter 19 sets these thirteen questions for a company. This version asks them of the file. Each is answerable in an afternoon. It feeds your findings log and the week-6 read-out. Signed by nobody; verified by someone internal, per check.

Check	Which sheet would answer it	What this repository says	Done
1. Is there a register of validated systems?	None exists; `O67` is the sheet you create	Nothing. This is the engagement's first finding	
2. Which systems hold GxP records, and which have had a periodic review?	`O23`, which has no GxP and no review column	Unanswerable from the file. Route the scope to quality assurance	
3. Is the safety dictionary version current, and when was history re-coded?	No sheet holds a dictionary version	Unrecorded. `O42` has no dictionary-version element	
4. Can the company produce its commitment list with due dates?	`O63`, five columns, no date column	No. Seven obligations, no dates, no owners	
5. Is the trial master file complete and filed contemporaneously?	`O23 A06 eTMF`: High, Poor, Invest	The only coherent row in the portfolio	
6. Is there an audit-trail review plan, and is it resourced?	No sheet; `O67.Audit trail reviewed`	Nothing. Ask who does it and for how many hours	
7. Are interfaces in anybody's validation scope?	`R06`, nine columns, none of them scope	19 of 36 unmanaged, 31 unlogged, 6 direct database reads	

Check	Which sheet would answer it	What this repository says	Done
8. Does any reporting layer join trial and commercial data?	`O38 Data Set`	`DST-02 Customer analytics set` sits beside `DST-01 Regulatory reporting set`	
9. Is there an adequacy determination per cross-border transfer?	`O12`, `O27.Location`, `O48`	No transfer register. `LOC-08` is a cloud region with 82 people	
10. What is the retention policy, and which clocks does it breach?	`O32`: `STD-04` Current and `STD-15` Deprecated, both `Conformant`	Two retention standards contradicting each other	
11. A measure definition register, or just a dashboard?	`O04 Measure`: no definition, formula or grain	A dashboard. `MSR-01` says 54% where `O63` shows two of seven	
12. For each guideline relied on: which version, checked when?	No sheet holds a version or a date	Nothing. Volume 1 chapter 16 is your source	
13. Who is the Responsible Pharmacist, and when does each licence expire?	`O10`, eighteen titles, none of them regulated	Nobody. `O19 Contract` holds ten vendor agreements and no licence	

Checklist 15-C — before you present

Work this the afternoon before each of the four read-outs. Eleven checks. It stops you being wrong in public. Signed by you, and dated, on the pack.

Check	Done
Every number on every slide was counted by you from the file, not quoted from `RPT01` or `ZZ Checks`	
Every finding has been shown privately to the person whose area it concerns, and at least one was corrected	
No count of GxP systems appears anywhere. There is no GxP column; any count is a judgement, and not yours	
Every regulatory claim carries the instrument, its date and where you read it	
No pharmaceutical job title appears in a signature block unless a named person holds it	
No printed page carries a professional registration number, a subject identifier or an instance of trial data	
Every report you quote has had its last row checked for a prose sentence in the first column	
The two numbers for the same measure are both shown, with the reconciliation	
Anything you could not establish is on a named slide, with what would settle it and who owns the re-check	
No recommendation appears in the week-2 or week-6 pack	
Every decision you are asking for names a body that can take it, and that body exists	

THE CHAPTER IN TEN LINES

- Seven workshops and one read-out that runs four times. Each pack names the room, the hours, the props, the agenda in timed blocks and the questions in the words to use.
- Name the jobs as the industry names them. The eighteen titles in O10 and O11 sign nothing a regulator asks about.
- Put the validation lead in every session that touches a system, from the first one.
- You are disproving, not eliciting: 814 records already exist and most are wrong in ways you can demonstrate from the file.
- Workshop 2 is won by one printed page. O17 Business Function already names the five functions O09 omits.
- Read R22 out loud, pair by pair. It puts Pharmacovigilance under Technology and Regulatory Affairs under Human Resources.
- Save INT-009 for the end of workshop 4. The serious adverse event path is recorded as an unmanaged hourly file.
- Workshop 7 opens with a page that has nothing on it but O29 Gap's seven column headers.
- The ninety-day shape is volume 1 chapter 19's, operating on a repository: the count is verification, and the first argument lands in week six.
- Two of the four read-outs carry no recommendations. Week 2 says what you are doing; week 6 says what the estate contains.
- Nothing is signed in weeks 9, 10 and 11. Those weeks are for correction.
- Thirteen weeks holds no validation. It holds a known estate, four registers, a gap set and a sequence.
- Three checklists to print: the first-day checks, the thirteen afternoon checks against the file, and what to verify before you present.

SOURCES FOR THIS CHAPTER

- **1.** Pharmaceuticals_TOGAF_Metamodel.xlsx, Calderon Therapeutics — sheets O10 Actor (rows 5-22), O11 Role (rows 5-22) and R21 Role performed by Actor (rows 5-22): the same eighteen generic corporate titles in both object sheets, joined ROL-nn to ACT-nn down the index, every Performed by cell reading Person and every Status reading Filled
- **2.** Volume 1, Pharmaceuticals — a starter pack for enterprise architects, chapter 19, Your first ninety days: the validation lead as the first interview, the ninety-day shape, the four executive sessions of which two carry no recommendations, the thirteen afternoon checks and the three workshop habits
- **3.** Pharmaceuticals_TOGAF_Metamodel.xlsx, Calderon Therapeutics — sheet 00 Index rows 1-12 and 15-79, where row 2 claims 850 records and the per-sheet counts sum to 814; and ZZ Checks rows 4-21 and RPT01 Dashboard rows 4-24, which repeat the 850. Calderon Therapeutics is stated on 00 Index to be fictional
- **4.** Pharmaceuticals_TOGAF_Metamodel.xlsx, Calderon Therapeutics — sheet EXT Industry Extension rows 5-10: X01 Compound, X02 Study / Trial, X03 Batch / Lot, X04 Regulatory Submission, X05 GxP Computerised System and X06 Adverse Event, declared as object types with no instances held in any of the other 104 sheets
- **5.** SAHPRA, SAHPGL-CEM-PV-04 v10, Guideline on Post-Marketing Reporting of Adverse Drug Reactions to Human Medicines in South Africa, 22 September 2025; the responsible person for pharmacovigilance and the back-up should reside in South Africa, and expedited clocks run from first awareness by anyone in the company. Cited as volume 1 cited it https://www.sahpra.org.za/wp-content/uploads/2025/09/SAHPGL-CEM-PV-04_v10-Guideline-on-Post-Marketing-Reporting-of-ADRs-to-Human-Medicines-in-South-Africa.pdf
- **6.** Pharmaceuticals_TOGAF_Metamodel.xlsx, Calderon Therapeutics — sheet O09 Organization Unit rows 5-16: twelve units including Customer Service (211 people) and Legal (240) against Technology (116), with no Quality, Regulatory Affairs, Pharmacovigilance, Manufacturing or Clinical Development unit; Internal Audit is the only assurance unit, at 29
- **7.** Pharmaceuticals_TOGAF_Metamodel.xlsx, Calderon Therapeutics — sheet O17 Business Function rows 5-14: BFN-01 to BFN-10, whose names are the ten level-1 capabilities of O13 Business Capability and include Regulatory Affairs, Pharmacovigilance, Manufacturing & Supply, Quality Assurance & Control, Medical Affairs and Clinical Development
- **8.** Pharmaceuticals_TOGAF_Metamodel.xlsx, Calderon Therapeutics — sheets R22 Org Unit owns Capability rows 5-14 and O13 Business Capability rows 5-104: the ten level-1 capabilities assigned to ORG-01 to ORG-10 in index order, and RPT03 Capability Heatmap, which gives all ten a gap of exactly 1 and a disposition of Sustain
- **9.** Pharmaceuticals_TOGAF_Metamodel.xlsx, Calderon Therapeutics — sheet O20 Data Entity rows 5-18: eight of fourteen Master system values inconsistent with the entity by a single rotation; the Owner column holding seven pharmaceutical head titles that appear in no other sheet; E03 Subject classified Internal while E07, E12 and E14 are Public
- **10.** Pharmaceuticals_TOGAF_Metamodel.xlsx, Calderon Therapeutics — the nine report sheets whose last row is a prose sentence in the first column: RPT01, RPT04, RPT05, RPT06, RPT08, RPT09, RPT10, RPT11 and RPT13. RPT04 Interface Catalogue carries 37 data rows, the last being the sentence on unmanaged and unlogged interfaces
- **11.** Pharmaceuticals_TOGAF_Metamodel.xlsx, Calderon Therapeutics — sheet O37 Data Attribute rows 5-64: ATR-001 Compound / Molecule Identifier classified Internal, and ATR-002 Compound / Molecule Name classified Public with Key: Yes; the sheet covers ten of the fourteen entities, E11 to E14 having none
- **12.** Pharmaceuticals_TOGAF_Metamodel.xlsx, Calderon Therapeutics — sheet O23 Application Component rows 5-16: A10 Serialisation Critical, Adequate, Retire; A08 QMS Critical, Healthy, Replace; A12 Medical Information Critical, At risk, Consolidate; A06 eTMF High, Poor, Invest. The Owner column holds organisation units, and no column carries a GxP flag, a validation state or a review date
- **13.** Pharmaceuticals_TOGAF_Metamodel.xlsx, Calderon Therapeutics — sheet R06 Application interfaces with rows 5-40: 36 interfaces, Managed holding the words Managed (17) and Unmanaged (19), Logged holding Yes (5) and No (31), six of type Direct DB read and all six unlogged; INT-009 EDC to Safety Database is File, Hourly, Unmanaged
- **14.** Pharmaceuticals_TOGAF_Metamodel.xlsx, Calderon Therapeutics — sheet O24 Information System Service rows 5-16: twelve services, five Unpublished, every description reading Service exposed by the application
- **15.** Pharmaceuticals_TOGAF_Metamodel.xlsx, Calderon Therapeutics — sheet O27 Physical Technology Compone rows 5-16: PTC-07 Laboratory instruments network, support end 2026-08, Past support; PTC-11 Enterprise storage array, support end 2026-01, Past support; and four components recorded at locations their class contradicts
- **16.** Pharmaceuticals_TOGAF_Metamodel.xlsx, Calderon Therapeutics — sheet O49 Resilience Tier rows 5-16: twelve applications, nine Never tested; RES-04 Safety Database at Tier 3 with a four-hour objective, RES-08 QMS at Tier 2 with fifteen minutes
- **17.** Pharmaceuticals_TOGAF_Metamodel.xlsx, Calderon Therapeutics — sheet O48 Environment rows 5-8: ENV-01 Production, Personal data present No; ENV-02 Pre-production and ENV-04 Development, Yes
- **18.** Pharmaceuticals_TOGAF_Metamodel.xlsx, Calderon Therapeutics — sheets O41 Security Control rows 5-12 (SEC-01 Identity lifecycle and SEC-03 Privileged access management both Absent, SEC-07 Non-production masking Partial) and O65 Enforcement Point rows 5-9 (EP-05 Non-production provisioning reading Most refreshes are unmasked)
- **19.** Pharmaceuticals_TOGAF_Metamodel.xlsx, Calderon Therapeutics — sheet O63 Control Objective rows 5-11: seven obligations, every Control objective cell reading Control that evidences followed by the obligation restated; the obligations are European and American framed and no South African instrument is named anywhere in the sheet
- **20.** Pharmaceuticals_TOGAF_Metamodel.xlsx, Calderon Therapeutics — sheet R16 Control Objective met by Co rows 5-11: each obligation mapped to exactly one SEC-control, with Position a verbatim copy of O63.Evidence rather than a reading of O41.Position; the two disagree on four of seven
- **21.** Pharmaceuticals_TOGAF_Metamodel.xlsx, Calderon Therapeutics — sheets O29 Gap (header row 4, no records), R12 Gap closed by Work Package (no records) and RPT07 Gap Analysis (no records), against RPT01 Dashboard reporting Gaps overdue: 0
- **22.** Pharmaceuticals_TOGAF_Metamodel.xlsx, Calderon Therapeutics — sheets O30 Work Package rows 5-16, whose twelve costs sum to $84.3m, and O54 Architecture Alternative rows 5-9, where ALT-B Foundations first at $46m is Recommended and ALT-E Compliance led at $31m is Partly adopted
- **23.** Pharmaceuticals_TOGAF_Metamodel.xlsx, Calderon Therapeutics — sheet O53 Transition Architecture rows 5-7: TA-1, TA-2 and TA-3, with the Exit criteria column reading 5 exit criteria, 6 exit criteria and 7 exit criteria
- **24.** Pharmaceuticals_TOGAF_Metamodel.xlsx, Calderon Therapeutics — sheet O52 Implementation Factor rows 5-12: IF-03 Peak trading freeze, category Constraint, deduction No change in the peak window; beside IF-01 Vendor selection not made and IF-04 Regulatory deadline
- **25.** Pharmaceuticals_TOGAF_Metamodel.xlsx, Calderon Therapeutics — sheets O66 Architecture Increment rows 5-10 and R15 Transition contains Work Pa rows 5-11: six increments against three transition states, three matching by exact name identity, and only seven of twelve work packages allocated to a state
- **26.** Pharmaceuticals_TOGAF_Metamodel.xlsx, Calderon Therapeutics — sheet O60 Governance Decision rows 5-11: seven decisions in four columns with no date column of any kind, four of them Outstanding

APPENDIX D

The repository in ArchiMate 4

Which element each of the sixty-four object sheets maps onto, where the mapping is an approximation you have to declare, where there is no element at all — and five views drawn from Calderon's own rows.

What this appendix is for

Volume 1 chapter 17 teaches the notation through this industry's problems: the four identities of a product, the protocol version the notation cannot carry, why a batch genealogy is not drawable, and the three objects that have no ArchiMate home at all [1]. Do not re-read it here. This appendix does the other half of the job: it takes the **105 sheets you were actually handed** and says, sheet by sheet, which ArchiMate 4 element each one maps onto, where that mapping is clean, where it is an approximation you must declare, and where there is no element and the rows belong in a register instead.

Then it draws five views from Calderon Therapeutics' own rows, so you can see what the repository looks like when it is rendered rather than tabulated. Calderon Therapeutics is invented and describes no actual company [2].

One warning before any of it. **Drawing a defective sheet draws the defect at scale.** Ten of the twenty-five relationship sheets are index-order joins rather than statements about the world [3]. A view generated from `R22 Org Unit owns Capability` will put Pharmacovigilance under Technology on a wall, in colour, in front of the head of pharmacovigilance. Check the sheet before you render it.

Four rules that decide most of the mapping

The element is `Role`

There is no layer-prefixed variant of it in ArchiMate 4. The element that holds *Responsible Pharmacist*, *QA release officer*, *principal investigator* and *marketing authorisation holder* is called `Role` and nothing else, and the same collapse applies to `Service`, `Process`, `Function`, `Event` and `Collaboration` — one element each, used at whichever layer you are working in [1].

This matters more in a pharmaceutical repository than it sounds, because `O11 Role` and `O10 Actor` hold the **same eighteen titles** and `R21 Role performed by Actor` joins them `ROL-01` to `ACT-01` down the index [4]. A role and an actor are different elements for a reason: the actor is the person or the organisation, the role is the accountability they hold. Here they have been entered twice and joined by position, which is why the repository cannot say that one named pharmacist holds the Responsible Pharmacist role at two sites, or that the role is vacant while the person is still employed.

Look it up. Volume 1, chapter 17, section *The ground rules, stated once*; and chapter 5, section *The interview order*, for the jobs that need `Role` elements.

There is no Constraint element

`O07 Constraint` is a sheet with eight rows and **no element behind it** [5]. Express each one as a `Requirement`: state the obligation the prohibition implies and move on.

Ref	`O07` row, as written	As a `Requirement`
`CON-01`	Flat technology budget	Change is funded from the existing run cost envelope
`CON-02`	Delivery capacity unrecorded	Concurrent delivery capacity is recorded before a plan is approved
`CON-03`	Legacy platform skills scarce	A change to the core is made only by a named qualified person
`CON-04`	Regulatory change in flight	Every design states the guideline version it was built against
`CON-05`	Data residency	Classified data remains in the jurisdiction, per record class
`CON-06`	Change freeze windows	No change is released inside a declared freeze window

`CON-07 Vendor roadmap dependency` becomes *no capability depends on an unreleased vendor feature*, and `CON-08 Union and works council consultation` becomes *operational change is consulted before release*. Six of the eight are stated in the sheet as nouns; all eight restate as obligations without losing anything, which is the test that the `Requirement` substitution is honest.

> The one rewrite to get right is `CON-05 Data residency`, because in this jurisdiction it is the constraint with a criminal consequence attached. Write it as *classified data remains in the jurisdiction, per record class, unless an adequacy determination exists for the recipient country*, and cite the instrument [6]. A `Requirement` that names no record class cannot be tested.

There is no metric element, and `Value` is not a number

`O04 Measure` has sixteen rows, six columns — `Ref`, `Measure`, `Unit`, `Domain`, `Baseline`, `Cadence` — and **not one of them is an ArchiMate element** [7]. `Value` is an element about worth, not a container for 54%. A measure is a property of the thing measured, and a baseline is a dated reading of that property.

So `MSR-01 Obligations with evidence` at a baseline of `54` does not go in the model. It goes in a measure definition register, which `O04` becomes once you add the five columns it lacks — `Definition`, `Formula`, `Grain`, `Source system` and `As at` — and that is a column addition to an existing sheet, not a new sheet. Then the contradiction becomes visible rather than arguable: `MSR-01` says 54% while `O63 Control Objective` shows two of seven obligations evidenced, which is 29% [8]. Two sheets, two numbers, one measure, no definition anywhere to adjudicate between them.

Look it up. Volume 1, chapter 17, section *Specification, test, result — and the fact that `Value` is not a number*; and appendix 20, section *Pharmaceutical object to ArchiMate 4 element*.

Three more in `O04` fail the same test. `MSR-07 Components within vendor support` reads 22% against `O27`'s ten of twelve, which is 83%. `MSR-13 Recovery objectives tested` reads `53` in unit `count` against a population of twelve applications. `MSR-15` and `MSR-16` can only be computed from `O59 Compliance Assessment`, which is empty [7].

> A specification limit has the same problem and a worse consequence. *Not less than 80% dissolved in 30 minutes* is a release criterion with legal force and there is nowhere in ArchiMate to put it [9]. Draw the `Business Object` for the specification, draw a `Requirement` that the batch meets the specification in force at manufacture, and keep the limit where it belongs — in the quality system, versioned, approved and retained. An architect who puts a limit on a view has created a second, uncontrolled copy of a controlled number.

Junctions exist, and a view is a selection

Junctions are part of the notation and they are the only honest way to draw *all four of these, and then that* or *either of these*. Use one wherever a sheet aggregates: `R15 Transition contains Work Pa` puts four work packages into `TA-1`, which is an and-junction and not four separate arrows, and `R16 Control Objective met by Co` ought to put several controls behind one obligation, which is the whole correction that item 10 makes to it.

And a view is a selection over the repository, not the model itself. The 105 sheets are the model. Each of the five views below is a query you could re-run, which is why each one names the sheets and rows it was built from. Regenerate; never edit the exported picture [1].

The mapping, sheet by sheet

Sixty-four object sheets, `O01` to `O66` with no `O34` and no `O51`, plus `EXT Industry Extension` [2]. `Fit` reads `Clean` where the element carries the sheet's meaning, `Approx` where it carries some of it and you must say which part is lost, and `None` where there is no element and the rows belong in a register. The tally is **twenty-four clean, twenty-three approximate, seventeen with no element** — and the seventeen are not the unimportant ones.

Motivation and governance intent.

Sheet	Element	Fit	What the element cannot carry
`O01 Driver`	`Driver`	Clean	The instrument, its version and its date. `Pressure` is a property
`O02 Goal`	`Goal`	Clean	Nothing material. This is the cleanest sheet in the file
`O03 Objective`	`Outcome`	Approx	`Horizon`. ArchiMate has no Objective; using `Goal` for both flattens the chain `R11` records
`O04 Measure`	—	**None**	Everything. No metric element; `Value` is not a number
`O05 Principle`	`Principle`	Clean	`Status`. A draft principle and an approved one are one element with a property
`O06 Requirement`	`Requirement`	Clean	`Priority`, and the evidence burden a regulated requirement carries
`O07 Constraint`	`Requirement`	**None**	There is no Constraint element. Eight rows restate as obligations
`O08 Assumption`	`Assessment`	Approx	An assumption is a view note and a dated register row, not an element
`O63 Control Objective`	`Driver` plus `Requirement`	Approx	One sheet, two elements: the obligation is a `Driver`, the objective a `Requirement`
`O64 Security Policy`	`Requirement`	Approx	The policy document itself, which is a `Business Object` under document control
`O41 Security Control`	`Requirement` plus the mechanism	Approx	The sheet conflates what must be true with what makes it true
`O65 Enforcement Point`	`Process`, or `Node`	Approx	One sheet, two elements: `EP-01` is a design gate, `EP-03` an unselected product
`O62 Delivery Risk`	`Assessment`	Approx	Likelihood and impact. Three rows read `Certain`, which makes them issues

Strategy and business.

Sheet	Element	Fit	What the element cannot carry
`O09 Organization Unit`	`Business Actor`	Clean	`Headcount`, and that five of the industry's functions are missing
`O10 Actor`	`Business Actor`	Clean	`Status: Filled`, which is a property of the assignment, not of the actor
`O11 Role`	`Role`	Clean	Statutory basis, professional registration, one-per-licence
`O12 Location`	`Location`, or `Facility`	Approx	The licence scope. `LOC-08 Validated Cloud Region` is neither, and has 82 people
`O13 Business Capability`	`Capability`	Clean	`Current` and `Target`, which are properties and do not roll up
`O14 Value Stream`	`Value Stream`	Clean	`Stages` as a count, and the decade between spend and revenue
`O15 Business Service`	`Service`	Clean	`Exposure`. There is no layer-prefixed variant of `Service`
`O16 Business Process`	`Process`	Clean	Whether the step is GxP, which is the property that decides everything
`O17 Business Function`	`Function`	Clean	Nothing material
`O18 Product`	`Product`	Approx	The four identities and their cardinality chain. Volume 1 chapter 17 has the pattern
`O19 Contract`	`Contract`, a contract-type `Business Object`	Approx	Conditions as separately dated obligations. All ten rows are vendor agreements
`O35`, `O44`, `O45`, `O46`, `O55`	`Grouping`, or the element it stands for	**None**	A building block is a TOGAF concept. There is no building-block element

Data.

Sheet	Element	Fit	What the element cannot carry
`O20 Data Entity`	`Business Object`	Clean	Classification, owner, master system and lineage — all four are properties
`O21 Logical Data Component`	`Grouping` of `Data Object`	Approx	The grouping has no semantics and enforces nothing
`O22 Physical Data Component`	`Artifact`	Clean	`Type`, which here reads `File` for four relational stores
`O36 Data Entity Relationship`	`Composition`, `Association`	Approx	Cardinality. `N:M` has nowhere to live in a view
`O37 Data Attribute`	—	**None**	Sixty rows, zero elements. Attributes are properties
`O38 Data Set`	`Data Object`, or `Grouping`	Approx	Nothing material. `DST-02 Customer analytics set` is a vocabulary problem, not a notation one
`O39 Data Migration`	`Work Package`	Approx	A migration is work, not data. Verification of value and meaning is a `Requirement`
`O40 Source To Target Mapping`	`Artifact`	**None**	A field mapping is a specification document and a validation deliverable
`O42 Metadata Element`	—	**None**	Fourteen rows of properties. `Recorded, not held` is a coverage value
`O43 Master Data Rule`	`Requirement`	Approx	`Rule type`. And it disagrees with `O20` on every entity it names
`O69 Retention Clock`	`Requirement` per record class	**None**	The period, because three of the longest clocks are computations over operational data

Application.

Sheet	Element	Fit	What the element cannot carry
`O23 Application Component`	`Application Component`	Clean	Validation state, GxP class, periodic review date, electronic-records applicability
`O24 Information System Service`	`Service` plus `Application Interface`	Approx	`Style` and `Published`, both properties of the interface
`O48 Environment`	`Node`, or `Grouping`	**None**	An environment is a deployment property set. `Personal data present` is its most important column
`O49 Resilience Tier`	—	**None**	Tier, recovery objective and `Tested` are properties of an `Application Component`
`O67 Validated System`	—	**None**	Every column. This is exactly why validation state has no element
`R06` 's interfaces	`Application Interface`, `Flow`	Clean	`Managed`, `Logged`, and whose validation scope it is in

Technology.

Sheet	Element	Fit	What the element cannot carry
`O25 Technology Service`	`Service`	Clean	`Category`, which is wrong on eight of twelve rows
`O26 Logical Technology Componen`	`Node`	Approx	Three of the seven rows are standards and one is a data model, and a standard is not a `Node`
`O27 Physical Technology Compone`	`Node`, `Device`, `Equipment`, `Communication Network`	Approx	One sheet, four elements. `PTC-07` is a network; `PTC-11` is storage
`O28 Platform Service`	`Service`, or `System Software`	Clean	Nothing material
`O47 Reference Model Entry`	`Grouping`	**None**	A catalogue row is not an element. Three sheets give these seven blocks three different statuses
`O50 Network Link`	`Path`, or `Communication Network`	Clean	`Bandwidth` and `Diverse path`. Two pairs are duplicated with different answers

Implementation, migration and governance.

Sheet	Element	Fit	What the element cannot carry
`O29 Gap`	—	**None**	A gap is the difference between two `Plateau` elements. It is a register row
`O30 Work Package`	`Work Package`	Clean	Cost, dates, approval and delivery state — four properties
`O31 Deliverable`	`Deliverable`	Clean	An approver and a date, which the sheet also lacks
`O32 Standard`	`Requirement` plus `Artifact`	Approx	`Status` against `Conformance`. Four deprecated standards read `Conformant`
`O33 Architecture Contract`	`Contract`-type `Business Object`	Approx	The compliance verdict, which belongs on an `Assessment`
`O52 Implementation Factor`	`Assessment`, or `Requirement`	Approx	`IF-03`'s category is literally `Constraint`, and there is no such element
`O53 Transition Architecture`	`Plateau`	Clean	Exit criteria, which become one `Requirement` each
`O54 Architecture Alternative`	`Course of Action`	Approx	Cost, and that two of the five are live answers at once
`O56 Project`	`Work Package`	Clean	Nothing — which is the finding: it is `O30` entered twice
`O57 Governance Body`	`Collaboration`	Clean	`Cadence` and `Mandate`. None of the five is a quality body
`O58 Governance Gate`	`Process`, or `Event`	**None**	A gate is a decision point in a process. `GT-3` cannot be passed by anything
`O59 Compliance Assessment`	`Assessment`	Clean	`Verdict` and `Evidence`, both properties. The sheet is empty
`O60 Governance Decision`	`Deliverable`	**None**	A decision is a dated record. The sheet has no date column
`O61 Governance Change Request`	`Business Object`	**None**	A change request is a quality-system record with a lifecycle
`O66 Architecture Increment`	`Deliverable`	Approx	`Lands`, a month offset with no anchor date
`O68`, `O70`	— and `Role`	Mixed	`O70` is `Role` and clean. `O68` is dates and outcomes: no element

And the industry extension. `EXT` declares six object types and holds no instances of any of them [10]. Five of the six have a home; the sixth is the reason this appendix exists.

`EXT` type	Element	What is lost
`X01 Compound`	`Business Object`	The normalisation rule version that decided identity
`X02 Study / Trial`	`Business Object` plus `Work Package`	The protocol version, and the amendment that invalidates downstream structures
`X03 Batch / Lot`	`Business Object`, type only	The genealogy. A graph of instances is not drawable in a type model
`X04 Regulatory Submission`	`Business Object` plus `Artifact`	The dossier state as a fold over an append-only sequence log
`X05 GxP Computerised System`	**none**	All of it. It is an `Application Component` with four governed properties
`X06 Adverse Event`	`Business Object`	The coded term's dictionary version, and the bi-temporality of a case

Where there is no element at all

Collected in one place, because a client will ask and the answer should be immediate.

The thing	Why there is no element	Where it lives instead
A measure, a baseline, a specification limit	No metric element, and `Value` is not a number	`O04` extended into a measure definition register; the quality system for a limit
Validation state, GxP class, periodic review date	Properties of an `Application Component`	`O67 Validated System` and `O68 Periodic Review Schedule`
A retention period	Three of the longest clocks are computations over operational data	`O69 Retention Clock`, with `Clock starts at` naming an event
A batch genealogy	A directed graph of instances with quantities on the edges	The executed batch records and their export format
A gap	The difference between two `Plateau` elements	`O29 Gap`, with a `Due` date that can expire
A governance decision, a change request	A dated controlled record, not a structural element	`O60` and `O61`, both of which need the date columns they lack
A building block	A TOGAF concept with no ArchiMate counterpart	`O55 Building Block`, the 37-row union sheet
An assumption, a risk	Not elements. `Assessment` is the nearest and it loses the dating	`O08` and `O62`, as dated registers with owners
A statutory basis	A `Role` cannot carry the instrument that creates its duty	`O70 Regulated Role`, `Statutory basis` column

> Say all of this early in an engagement. An architect who says *validation state is a property, not an element* on day three is credible. One who is still drawing round it in month three is not [1]. **Look it up.** Volume 1, chapter 17, section *What has no ArchiMate home at all*.

The relationship sheets

Twenty-three of the twenty-five carry instances; `R12` and `R24` are empty [11]. Most map onto a relationship cleanly: `R01 Capability decomposition` is `Composition`, `R03 Capability realised by Busi` and `R09 Requirement satisfied by Bu` are `Realization`, `R04 Business Service supported` and `R07 Application deployed on Tec` are `Serving` and `Assignment`, `R06` is `Flow`, `R08 Requirement derives from Dr` is `Influence`, `R21 Role performed by Actor` is `Assignment`.

Three do not. `R05 Application manages Data En` carries a `Role` column holding `Master` and `Consumer`, which is not a relationship property in ArchiMate — split it into `Access` with a read or write direction. Its first fourteen rows also put an application **name** in the `Application` reference column instead of an `A` reference, and they reproduce `O20`'s rotated `Master system` values exactly; the fourteen `Consumer` rows beneath them use proper references [12]. `R19 Data Entity relates to Data` carries a `Cardinality` column with nowhere to go. And `R16 Control Objective met by Co` is a many-to-many entered one-to-one, so rendering it draws a compliance regime reduced to a single access control.

> Before you render any relationship sheet, check whether it is a join or a statement. **Ten** are index-order joins: `R04`, `R07`, `R09`, `R13`, `R17`, `R18`, `R20`, `R21`, `R22` and `R23` [3]. `R18 Standard applies to Applica` hits only the odd-numbered applications, so no `Critical` application in `O23` has a standard against it. `R20 Application located at Loca` contradicts `R07` plus `O27.Location` for nine of twelve applications, so two views drawn from the same repository disagree about where nine systems run.

Which viewpoints this repository can feed today

Volume 1 chapter 17 lists the viewpoints that earn their place in a pharmaceutical engagement, and the three you have to invent because no standard viewpoint covers them [1]. This table asks a narrower question: with the file as handed over, which of them can you actually draw, and from which sheets?

Viewpoint	Sheets that feed it	Can you draw it today?
Motivation	`O01`, `O02`, `O03`, `O06`, `O07`, `R08`, `R11`	Yes, and it is the best view in the file. The drivers are external, dated and citable
Organisation	`O09`, `O10`, `O11`, `O12`, `R21`	No. Eighteen generic titles joined by position, and five functions missing
Capability map	`O13`, `R01`, `R22`, `RPT03`	The tree, yes. The heat, no: every gap is 1 and every disposition is `Sustain`
Value stream	`O14`, `O16`, `R02`	The streams, yes — they are genuinely pharmaceutical. The stages, no
Information structure	`O20`, `O21`, `O36`, `R19`	Yes, but the `Master system` property is wrong on eight of fourteen
Application co-operation	`O23`, `O24`, `R06`	Yes, and it is the second most useful view. Validation scope is absent
Technology	`O26`, `O27`, `O50`, `R07`, `R20`	No. `R07` and `R20` disagree for nine of twelve applications
Migration	`O53`, `O66`, `O30`, `R15`	Partly. Three plateaus, no exit criteria, and $29.1m unallocated
Requirements realisation	`O06`, `O55`, `R09`, `RPT06`	Draw it and label it generated. `R09` is an index-order join
Validated boundary (invented)	`O67`, `O23`, `O48`, `R06`	Only after item 2 and item 7. No sheet in the original file supports it
Clinical and commercial firewall (invented)	`O38`, `O41`, `O48`, `R05`	No. `DST-02 Customer analytics set` sits beside `DST-01 Regulatory reporting set` with nothing between them
Submission and commitment (invented)	`O06`, `O31`, `O63`	No. Nothing in the file carries a dated commitment or a dossier state

Four of the twelve are drawable as the file stands, three more become drawable after the two registers in item 2 exist, and the remaining five need data nobody has collected. That ratio is a better summary of the repository's maturity than any count of records, and it is the honest answer to *can you show me the architecture?* in week one.

Five views, drawn from Calderon's own rows

Each one names the sheets it was built from, so you can re-run it. Each is under fifteen elements, which is the upper limit for a view in a regulated estate — every element on a view invites a question with a regulatory consequence [1].

One: the motivation chain, and the measure that is not in it

The cleanest thing in the repository is the chain from an external obligation to a requirement. `O01 Driver` has seven regulatory drivers, all `External`, all owned by the Chief Compliance Officer. `R08 Requirement derives from Dr` traces each to a requirement, and `REQ-001 Evidence GMP / GCP / GLP / GVP` is the right first requirement in a pharmaceutical repository [13].

Two things are in the sheets and cannot be in the view. `CON-05 Data residency` is in `O07 Constraint`, which has no element, so it appears as a `Requirement`. And `MSR-01` is in `O04 Measure`, which has no element at all, so it appears as a labelled `Grouping` pointing at the register — because leaving it out entirely would hide the contradiction, and putting a number in a box would create a second copy of it.

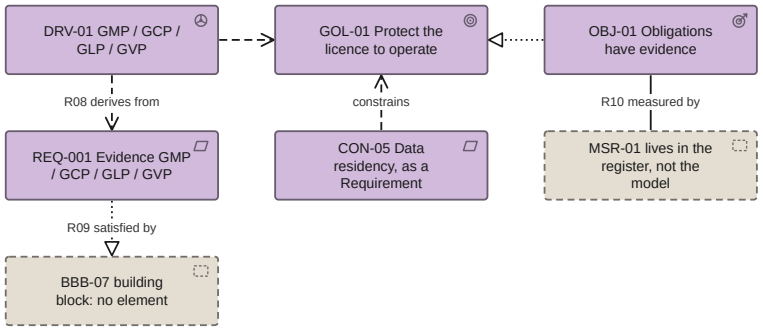


Figure 16.1 The motivation chain, from `O01`, `O02`, `O03`, `O06`, `O07`, `R08` and `R11`. Two of the seven boxes are there to show what has no element: `CON-05` appears as a `Requirement` because there is no Constraint element, and `MSR-01` appears as a labelled `Grouping` because there is no metric element and `Value` is not a number.

> `RPT06 Requirements Traceability` reports `REQ-001` as reaching no building block, and it is right in form and arbitrary in substance: `R09` is an index-order join, and `Traced` reads `No` on every fourth row [3]. Eight of thirty-two requirements fail traceability by position. Draw the chain, then say in the view's notes that the traceability column is generated and the real figure is unknown.

Two: the application landscape, with the interface that matters

Twelve applications, 36 interfaces, `Managed` holding the words `Managed` and `Unmanaged`, and `Logged` holding `Yes` and `No` [14]. A landscape view with all twelve components and all 36 flows on it is a report, not a view. Draw the one path that decides a regulatory outcome.

`INT-009` runs from `A03 EDC` to `A04 Safety Database`, type `File`, frequency `Hourly`, `Unmanaged`. That is the route a serious adverse event takes from a trial site to the system that carries the expedited reporting clock, and the clock runs from first awareness by anyone in the company [15]. The `Application Interface` element is the right one and it carries none of what matters: `Managed`, `Logged` and whose validation scope it sits in are three properties, and the third has no column in the repository at all.

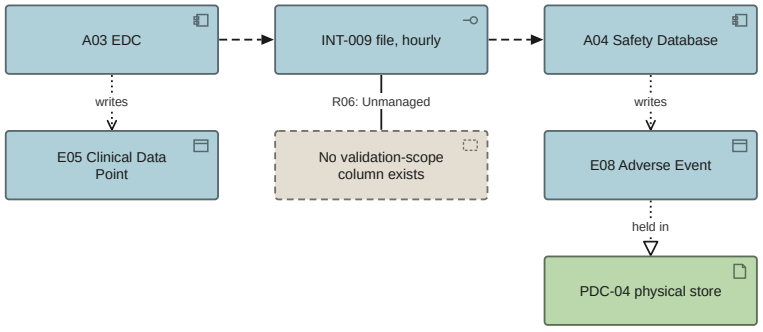


Figure 16.2 One interface, from `O23`, `O20`, `O22` and `R06`. `INT-009` is the route a serious adverse event takes from the trial site to the system carrying the expedited reporting clock, and `R06` records it as an unmanaged hourly file. `Application Interface` is the right element and carries none of `Managed`, `Logged` or validation scope.

Three: the validated boundary

`Grouping` for the boundary, per volume 1 chapter 17, and the warning that goes with it: a `Grouping` has no semantics and enforces nothing. It is a box with a dashed border. Its job is to make the boundary visible so a reviewer notices when a proposed interface crosses it [1].

Declaring `A09 ERP` outside the validated estate is a legitimate architectural act, provided the declaration is written, dated and enforced. What is not legitimate is the state the repository is in: no sheet says which systems are inside, so the boundary cannot be drawn from the file at all. This view is drawn from `O67 Validated System` as item 2 and item 7 populate it, against `O23 Application Component` and `O48 Environment` [16].

Look it up. Volume 1, chapter 17, section *The validated boundary and the clinical/commercial firewall*; and chapter 11, section *The capability no other industry has: the register of validated systems*.

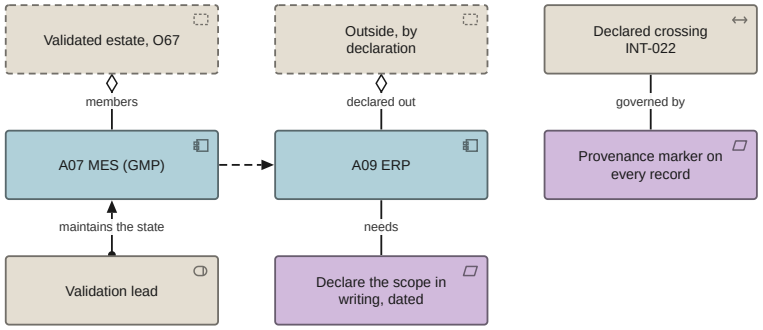


Figure 16.3 The validated boundary, from `O67`, `O23` and `O48`. Both boundaries are `Grouping` elements, which have no semantics and enforce nothing — their job is to make a crossing visible to a reviewer. The declaration that `A09 ERP` sits outside is legitimate; it has to be written, dated and enforced.

> The direction of travel is the trap. A reporting layer that was outside the boundary while it produced management information comes **inside** it the day a trend from it is used to justify a specification. Two tests settle it: does anyone make a GxP decision from this output, and is the output cited in a regulated record? Either yes, and the pipeline needs verification, change control and traceability to source [1].

Four: the obligation-to-evidence chain, with a junction

Seven obligations in `O63`, one `SEC-` control each in `R16`, and an empty `O59 Compliance Assessment` [8]. The view that matters is the one that shows the break: `CO-01` is recorded as `Evidenced` through `SEC-01 Identity lifecycle`, which `O41 Security Control` records as `Absent` [17]. An obligation cannot be evidenced by a control that does not exist.

The junction is the correction. One obligation needs several controls, and an and-junction says so without pretending that any single control discharges it. The `Assessment` element is `O59`'s home and the evidence itself is a `Deliverable` — a document somebody hands over — signed by a named person, which is the part the repository has no row for.

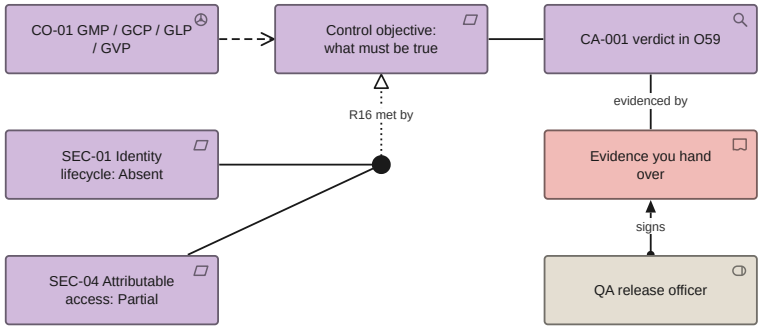


Figure 16.4 Obligation to evidence, from `O63`, `R16`, `O41` and `O59`. The and-junction is the correction `R16` needs: one obligation, several controls. Today `CO-01` reads `Evidenced` through `SEC-01`, which `O41` records as `Absent`, and the `Assessment` element has no rows to sit on.

Five: the transition states

Three `Plateau` elements in `O53`, six increments in `O66`, and seven of twelve work packages allocated to a state in `R15` [18]. Three of the six increments match a transition state by exact name identity — `INC-1` with `TA-1`, `INC-2` with `TA-2`, `INC-4` with `TA-3` — which is a

coincidence of naming you should not promote to a mapping without asking.

The five unallocated packages are the view's point. `WP-06`, `WP-08`, `WP-09`, `WP-10` and `WP-12` total **$29.1m** and sit in no transition state, so they appear on the view as a `Grouping` connected to nothing. Leave them unconnected. A floating box on a migration view is the most efficient way to ask a steering committee which plateau it is paying for.

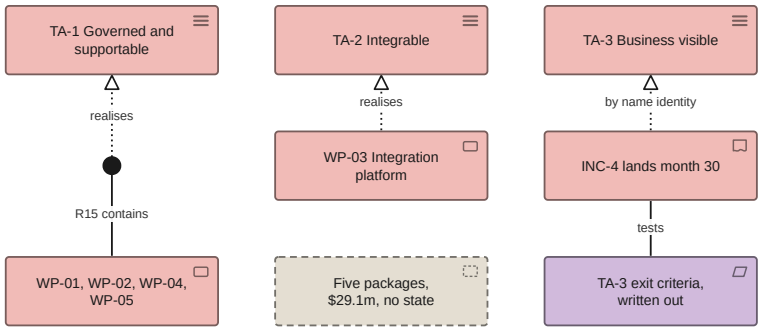


Figure 16.5 Transition states, from `053`, `066`, `030` and `R15`. The junction aggregates the four packages `R15` puts in `TA-1`. The floating `Grouping` is deliberate: five work packages worth $29.1m belong to no plateau, and a box connected to nothing is the fastest way to ask a committee which state it is funding.

A view is a selection, not the model

Four habits, and the fourth has no analogue outside this industry.

Look it up. Volume 1, chapter 17, sections *The discipline that keeps a view useful* and *The viewpoints that actually get used*.

Regenerate; never edit the export. A view is a query over the repository. The failure mode is always the same: a view is exported to a slide, the slide is edited, the edited slide becomes the version people quote, and six months later the repository and the circulating picture disagree. Date every exported view in its own footer and name the sheets it came from, as the five above do.

Fifteen elements, not forty. If a view needs forty elements it is two views or it is a report. The repository has thirteen reports; use them. `RPT04 Interface Catalogue` is the interface view — drop its last row, which is a sentence in the `Ref` column [14].

Say what the view cannot show, on the view. The validated-boundary view cannot show whether the periodic reviews are current; that is a report run from `O68`. The application co-operation view cannot show whether an interface is in anyone's validation scope. The migration view cannot show revalidation effort. Write the sentence in the notes rather than waiting to be asked.

Decide whether the view is a controlled document, and mark it. A view attached to a validation package, a Site Master File, a submission module or a standard operating procedure is inside the quality system, and changing it is a change-control event with an assessment, an approval and a document version. That is correct and it is slow, which is why **most architecture views should deliberately not be controlled documents.** Mark them *not a controlled document*. Get it wrong in the other direction and you have put an uncontrolled drawing inside a regulated record, where an inspector will find it and ask which version was operative [1].

THE CHAPTER IN TEN LINES

- Volume 1 chapter 17 teaches the notation. This appendix maps the 105 sheets you were handed onto it, sheet by sheet.
- The element is Role. There is no layer-prefixed variant, and the same collapse applies to Service, Process, Function, Event and Collaboration.
- There is no Constraint element. O07's eight rows become Requirement elements, restated as the obligations they imply.
- There is no metric element and Value is not a number. O04 Measure's sixteen rows belong in a measure definition register, which O04 becomes once you add Definition, Formula, Grain, Source system and As at.
- Junctions exist, and they are the honest way to draw an aggregation like R15 or the many-to-many R16 should have been.
- Twenty-four of the sixty-four object sheets map cleanly, twenty-three are approximations you must declare, and seventeen have no element at all.
- Nine kinds of thing have no element: a measure, a validation state, a retention period, a genealogy, a gap, a decision, a building block, an assumption and a statutory basis.
- Ten of the twenty-five relationship sheets are index-order joins. Rendering one draws the defect at wall scale and in colour.
- Four of twelve useful viewpoints can be drawn from the file as handed over; three more need the registers item 2 creates; five need data nobody has collected.
- The five worked views each name the sheets and rows they came from, so each one can be re-run.
- A view is a selection over the repository, not the model. Regenerate it; never edit the exported picture.
- Fifteen elements is the upper limit in a regulated estate, because every element invites a question with a regulatory consequence.
- Decide for every view whether it is a controlled document, and mark it. The default should be outside the quality system.

SOURCES FOR THIS CHAPTER

- **1.** Volume 1, Pharmaceuticals — a starter pack for enterprise architects, chapter 17, Modelling pharmaceuticals in ArchiMate: the ground rules, the four identities of a product, the protocol version the notation cannot carry, the undrawable genealogy, what has no ArchiMate home at all, the viewpoints that actually get used, and the discipline that keeps a view useful
- **2.** Pharmaceuticals_TOGAF_Metamodel.xlsx, Calderon Therapeutics — 105 sheets: 00 Index, 64 object sheets O01 to O66 with no O34 and no O51, EXT Industry Extension, 25 relationship sheets, 13 reports and ZZ Checks; headers on row 4, first record on row 5, first column Ref. 00 Index states that Calderon Therapeutics is fictional
- **3.** Pharmaceuticals_TOGAF_Metamodel.xlsx, Calderon Therapeutics — the ten index-order relationship sheets: R04 (BSV-nn to A-nn), R07 (A-nn to PTC-nn), R09 (REQ-nnn to building block by position, with Traced reading No on every fourth row), R13, R17, R18 (odd-numbered applications only), R20 (A01 to LOC-01 then wrapping), R21, R22 and R23 (odd-numbered projects only)
- **4.** Pharmaceuticals_TOGAF_Metamodel.xlsx, Calderon Therapeutics — sheets O10 Actor rows 5-22, O11 Role rows 5-22 and R21 Role performed by Actor rows 5-22: the same eighteen generic corporate titles in both sheets, joined ROL-nn to ACT-nn down the index, every Performed by cell reading Person
- **5.** Pharmaceuticals_TOGAF_Metamodel.xlsx, Calderon Therapeutics — sheet O07 Constraint rows 5-12: CON-01 Flat technology budget, CON-02 Delivery capacity unrecorded, CON-03 Legacy platform skills scarce, CON-04 Regulatory change in flight, CON-05 Data residency, CON-06 Change freeze windows, CON-07 Vendor roadmap dependency, CON-08 Union and works council consultation, with the Type column
- **6.** Protection of Personal Information Act 4 of 2013 — s 26 and s 27 on special personal information, s 57 on prior authorisation and s 72 on transborder flows, with the Information Regulator's Guidance Note on Applications for Prior Authorisation. Cited as volume 1 chapter 16 cited it; the Act assigns no body the task of determining adequacy https://inforegulator.org.za/wp-content/uploads/2020/07/Revised-Guidance-Note-on-applications-for-Prior-Authorisation.pdf
- **7.** Pharmaceuticals_TOGAF_Metamodel.xlsx, Calderon Therapeutics — sheet O04 Measure rows 5-20: sixteen measures in six columns with no definition, formula, grain or source; MSR-01 Obligations with evidence at baseline 54, MSR-07 Components within vendor support at 22 against O27's ten of twelve, MSR-13 Recovery objectives tested at 53 in unit count against a population of twelve, MSR-15 and MSR-16 computable only from the empty O59
- **8.** Pharmaceuticals_TOGAF_Metamodel.xlsx, Calderon Therapeutics — sheets O63 Control Objective rows 5-11 (two of seven Evidenced), R16 Control Objective met by Co rows 5-11 (one SEC- control per obligation, Position copied from O63.Evidence) and O59 Compliance Assessment (header row 4, no records)
- **9.** Volume 1, appendix 20, The cross-maps — pharmaceutical object to ArchiMate 4 element, including the objects for which there is no element: batch genealogy, specification limit and acceptance criterion, yield and compliance rate, validation state and periodic review date, and retention period
- **10.** Pharmaceuticals_TOGAF_Metamodel.xlsx, Calderon Therapeutics — sheet EXT Industry Extension rows 5-10: X01 Compound, X02 Study / Trial, X03 Batch / Lot, X04 Regulatory Submission, X05 GxP Computerised System and X06 Adverse Event, declared with no instances held in any of the other 104 sheets
- **11.** Pharmaceuticals_TOGAF_Metamodel.xlsx, Calderon Therapeutics — the 25 relationship sheets R01 to R25 with their header rows; R12 Gap closed by Work Package and R24 Compliance Assessment tests carry a header row and no records, so 23 carry instances, 437 in total
- **12.** Pharmaceuticals_TOGAF_Metamodel.xlsx, Calderon Therapeutics — sheet R05 Application manages Data En rows 5-32: 28 rows, of which the first fourteen carry an application name rather than an A- reference in the Application column and a Role of Master reproducing O20's Master system values, and the second fourteen carry proper references and a Role of Consumer
- **13.** Pharmaceuticals_TOGAF_Metamodel.xlsx, Calderon Therapeutics — sheets O01 Driver rows 5-17 (seven regulatory drivers, all External, all owned by the Chief Compliance Officer), O06 Requirement rows 5-36 (REQ-001 Evidence GMP / GCP / GLP / GVP first) and R08 Requirement derives from Dr rows 5-36
- **14.** Pharmaceuticals_TOGAF_Metamodel.xlsx, Calderon Therapeutics — sheet R06 Application interfaces with rows 5-40 (36 interfaces; Managed holds Managed and Unmanaged, Logged holds Yes and No; INT-009 EDC to Safety Database is File, Hourly, Unmanaged) and RPT04 Interface Catalogue, whose 37th data row is a prose sentence in the Ref column
- **15.** SAHPRA, SAHPGL-CEM-PV-04 v10, Guideline on Post-Marketing Reporting of Adverse Drug Reactions to Human Medicines in South Africa, 22 September 2025; expedited reporting clocks run from first awareness by anyone in the company. Cited as volume 1 cited it https://www.sahpra.org.za/wp-content/uploads/2025/09/SAHPGL-CEM-PV-04_v10-Guideline-on-Post-Marketing-Reporting-of-ADRs-to-Human-Medicines-in-South-Africa.pdf
- **16.** Pharmaceuticals_TOGAF_Metamodel.xlsx, Calderon Therapeutics — sheets O23 Application Component rows 5-16, which carry no GxP flag, validation state or review date, and O48 Environment rows 5-8, where ENV-01 Production records Personal data present No while ENV-02 and ENV-04 record Yes; against O67 Validated System as defined in item 2 of this volume
- **17.** Pharmaceuticals_TOGAF_Metamodel.xlsx, Calderon Therapeutics — sheet O41 Security Control rows 5-12: SEC-01 Identity lifecycle and SEC-03 Privileged access management both Absent, SEC-04 Attributable access Partial, SEC-08 Data-at-rest encryption mapped to no obligation
- **18.** Pharmaceuticals_TOGAF_Metamodel.xlsx, Calderon Therapeutics — sheets O53 Transition Architecture rows 5-7, O66 Architecture Increment rows 5-10, O30 Work Package rows 5-16 and R15 Transition contains Work Pa rows 5-11: three plateaus, six increments, seven of twelve packages allocated, and WP-06, WP-08, WP-09, WP-10 and WP-12 totalling $29.1m in no state

APPENDIX E

Where to look in volume 1

Both directions. For each of the twenty items here, the volume 1 chapters and the exact sections behind it; for each of volume 1's twenty-three items, where in this workbook you act on it. Every heading printed has been checked against the source.

How to use this index

Volume 1 teaches the industry. This volume works the repository. Where a step here has teaching behind it, the step carries a **Look it up.** line naming the volume 1 chapter and the exact section heading — and this item collects every one of them, both ways round.

Read it in whichever direction you are travelling. **Forward** if you are about to work an item of this workbook and want the twenty minutes of reading that makes the steps land. **Backward** if you have read a chapter of volume 1 and want to know where you put it to work. There is a five-section reading list at the end for the case where you have an hour before a workshop and no more.

Three notes on the mechanics. Every heading printed below exists in volume 1 exactly as printed — it was checked against the heading index, twice, because a dangling cross-reference is the defect a reader notices first and the one that makes them stop trusting the rest. Section headings are italicised; chapter numbers are volume 1's own numbering, in which items 20 to 23 are appendices A to D. And a section cited by three or more items is doing real work: there are nine of those, listed near the end [1].

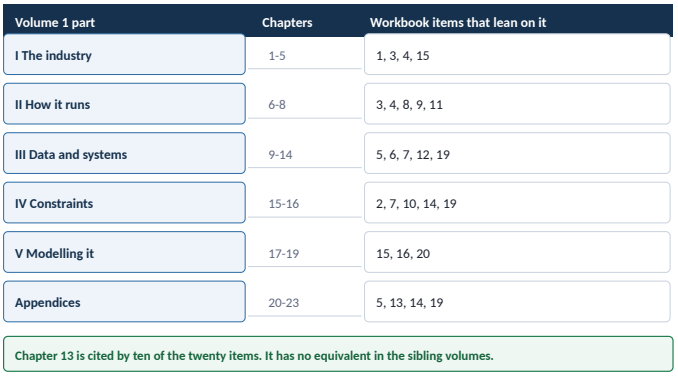


Figure 17.1 Volume 1's five parts and its appendices, against the workbook items that lean on them. Chapter 13 is cited by half the volume; chapter 17 by one item, deliberately.

Forward: what to read before each item

Item	Volume 1	The exact sections
1 How to use this workbook	19, 9, 13, 1	ch 19 *The checks that take an afternoon* · ch 9 *The object most architects leave out: the validated system itself* · ch 13 *The system itself is regulated* · ch 1 *What the rest of this book does*
2 Standing the practice up in a regulated estate	13, 3, 5, 19	ch 13 *What the architect is actually expected to produce*, *The validation master plan and the deliverable set* · ch 3 *Qualified Person versus Responsible Pharmacist* · ch 5 *Quality and manufacturing* · ch 19 *The ninety-day shape*
3 Architecture Vision	19, 16, 2, 7	ch 19 *Week one: find out what is validated*, *What to say when you do not know* · ch 16 *The spine: Act 101 of 1965 and SAHPRA* · ch 2 *How to tell which archetype you are in, in the first week* · ch 7 *The Annual Product Quality Review*
4 Business Architecture	5, 8, 7	ch 5 *Read this before you book the workshop*, *The interview order*, *A role-to-system map* · ch 8 *The four levels, and who reads each*, *Running a process-modelling workshop in a GxP environment* · ch 7 *The cadence, which is the spine of this chapter*
5 Data Architecture	9, 10, 20	ch 9 *The object that breaks most models: "a product" is four things*, *Subject — the most sensitive record the company holds*, *The dictionary-version problem* · ch 10 *Master data governance, and why it is a compliance matter*, *Retention as a data-model constraint* · ch 20 *System class to what it masters*, *Retention clock by record class*

Item	Volume 1	The exact sections
6 Application Architecture	11, 12	ch 11 *The capability no other industry has: the register of validated systems*, *Three questions that decide every pharmaceutical application architecture* · ch 12 *What is actually installed*, *Interfaces are routinely in nobody's validation scope*, *The eTMF, at length, because it is the inspection artefact*
7 Technology Architecture and the validated estate	3, 13, 15	ch 3 *Validation, qualification, verification* · ch 13 *Computerised system validation, as it is actually practised* · ch 15 *Cloud regions, and the check this book teaches instead of a table*, *Health-data residency under POPIA*, *What goes stale first, in order*
8 Opportunities and Solutions	18, 19, 7	ch 18 *3. The findings*, *4. What it costs to be wrong*, *5. The target and the transition* · ch 19 *What ninety days will not achieve* · ch 7 *The regulatory commitment calendar*
9 Migration Planning	10, 18, 7, 13	ch 10 *Data migration in a validated estate* · ch 18 *6. The roadmap and what the year allows* · ch 7 *Variation planning, and the correction that inverts a workflow* · ch 13 *The V-model, and how it actually runs*
10 Implementation Governance, validation and release	13, 16, 20	ch 13 *The Responsible Pharmacist*, *Seven pieces of folklore, punctured* · ch 16 *Licensing, and the Responsible Pharmacist*, *Inspection response* · ch 20 *Regulator to evidence*

Item	Volume 1	The exact sections
11 Architecture Change Management and the validated state	13, 3, 8, 7	ch 13 *Periodic review*, *Decommissioning, which nobody budgets* · ch 3 *Deviation, OOS, OOT, CAPA, change control, complaint* · ch 8 *Change control* · ch 7 *The Annual Product Quality Review*
12 Requirements Management	13, 9, 16	ch 13 *The V-model, and how it actually runs*, *The validation master plan and the deliverable set* · ch 9 *Controlled vocabularies as architectural dependencies* · ch 16 *Registration, variation and renewal*
13 The sheet-by-sheet reference	9, 10, 20	ch 9 *The object most architects leave out: the validated system itself* · ch 10 *What changes when you go logical* · ch 20 *Pharmaceutical object to ArchiMate 4 element*
14 The report catalogue	10, 16, 13	ch 10 *The reporting layer, and the reconciliations that are always wrong* · ch 16 *Who inspects, and what they ask for*, *The retention clocks* · ch 13 *Data integrity*
15 Workshop packs, checklists and the ninety-day calendar	19, 5, 8	ch 19 *Before week one: what to read, and what to put down*, *How to run the workshops*, *The ninety-day checklist* · ch 5 *The interview order* · ch 8 *Running a process-modelling workshop in a GxP environment*

Item	Volume 1	The exact sections
16 The repository in ArchiMate 4	17	ch 17 *The ground rules, stated once*, *The validated system itself*, *The validated boundary and the clinical/commercial firewall*, *What has no ArchiMate home at all*, *The viewpoints that actually get used*, *The BPMN exception, stated once*
17 Where to look in volume 1	20, 23	ch 20 *How to use this appendix* · ch 23 *The sources most worth re-checking before this goes in front of a client*, *The categories most likely to have moved*
18 The forms	5, 19, 13	ch 5 *A role-to-system map* · ch 19 *The ninety-day checklist* · ch 13 *What the architect is actually expected to produce*
19 The principles and standards starter set	13, 14, 22, 20	ch 13 *Data integrity*, *GAMP 5, with the version facts right*, *Audit-trail review, the most under-resourced obligation in the regime* · ch 14 *What is genuinely mandatory* · ch 22 *How to check a standard's version* · ch 20 *Regulator-mandated standard to what it constrains*
20 Running the maturity assessment	19, 18, 6	ch 19 *The checks that take an afternoon* · ch 18 *1. What the architect walks into*, *7. What the architect hands over* · ch 6 *What the architect takes from this chapter*

Two patterns in that table are worth saying out loud. **Chapter 13 appears against ten of the twenty items**, which is the single clearest statement of how this volume differs from its siblings: in no other industry does one chapter of the teaching volume touch half the workbook. And **chapter 17 appears against exactly one item**, because item 16 is the modelling item and the notation rules belong in one place rather than scattered through the phases [1].

Backward: where each chapter of volume 1 gets used

Volume 1	What it teaches	Acted on in items
1 The industry in one chapter	The shape of the sector, single exit price, approval is not launch	1, 3
2 The archetypes	Which of eight businesses you are actually in, and the three failure modes	3
3 The vocabulary	Validation against qualification; Qualified Person against Responsible Pharmacist	2, 7, 11
4 How the business model works	Four customers at once; what the price regime deletes from the architecture	3, 8
5 Who does what	The roles the repository has no row for, and the order to interview them in	2, 4, 15
6 How pharmaceutical companies make money	The portfolio as the unit; why the architect cares about attrition at all	3, 8, 20
7 The operating model and the portfolio cadence	The eleven cadences that bind a plan, and the registered site as a regulated object	4, 8, 9, 11
8 The process model	Nine processes drawn, and how to run a modelling workshop under GxP	4, 11
9 The conceptual data model	Every entity the industry needs, including the one the repository omits	5, 13, 16
10 The information model	Mastering, data integrity as a constraint, migration, retention, reporting	5, 9, 14
11 The conceptual application model	Seven domains and the register of validated systems	6, 7
12 The application landscape	What is really installed, and interfaces in nobody's validation scope	6, 7

Volume 1	What it teaches	Acted on in items
13 GxP, validation and data integrity	The whole of it. The teaching half of items 2, 7, 10, 11 and 12	1, 2, 5, 6, 7, 9, 10, 11, 12, 19
14 The reference architecture	What is genuinely mandatory, and where every published model stops	7, 19
15 Technology providers	What goes stale first; cloud regions; health-data residency	7, 8
16 Regulation, risk and what the inspector asks for	The obligation set, the twelve-row inspector table, the retention clocks	2, 3, 10, 14, 19
17 Modelling pharmaceuticals in ArchiMate	Every modelling rule this volume applies, and what has no element	16
18 Veldtkruin Group, end to end	A worked engagement: findings, cost of being wrong, target, roadmap, handover	8, 9, 20
19 Your first ninety days	The checks, the shape, the workshops, and what ninety days will not achieve	1, 15, 20
20 Appendix A, The cross-maps	Ten cross-maps, including retention clock by record class and regulator to evidence	5, 10, 13, 19
21 Appendix B, Glossary	184 terms. Print the A to Z and take it to every workshop	4, 15
22 Appendix C, Standards, bodies and where to find them	Which body owns which standard, and how to check a version yourself	19
23 Appendix D, References	Every source with a retrieval date, and which ones to re-check first	14, 19

Nothing in volume 1 is unused, which is not a coincidence: the volume was written first and this one was written against it. The thinnest links are chapters 1 and 2, which set the framing rather than the method, and chapter 6, which an architect reads once and then stops needing. The heaviest are chapters 13, 16 and 19.

The nine sections cited by three or more items

If you were to print nine pages of volume 1 and staple them inside the back cover of this one, these are the nine. Each is cited by at least three items of this workbook.

Volume 1	Section	Why it keeps coming back
ch 13	*The system itself is regulated*	Every phase changes when the system is itself regulated
ch 13	*Which GxP, and for which record*	It decides the evidence burden on every system under a process
ch 13	*Periodic review*	The mechanism that keeps a validated state true, and the repository has no column for it
ch 13	*What the architect is actually expected to produce*	The deliverable set you are expected to produce, and the one an inspector asks for first
ch 5	*Read this before you book the workshop*	Stops you booking the wrong eighteen people, which is what `O10` would have you do
ch 16	*Who inspects, and what they ask for*	The twelve-row table of regulator against artefact
ch 16	*The retention clocks*	Periods, and the events they run from, for `O69`
ch 9	*The object most architects leave out: the validated system itself*	The object the repository has no row for anywhere
ch 19	*The checks that take an afternoon*	A day's work that tells you more than a fortnight of reading

If you only read five sections before your first workshop

An hour, in this order. They are chosen so that each one changes what you will say in the room, not what you know in the abstract.

1. **ch 13, *The system itself is regulated.*** Six pages. It is why "just patch it" does not exist here, and it is the sentence you will repeat most often in your first month. Read it before you promise anybody a date.
2. **ch 13, *Which GxP, and for which record.*** Which regime applies to which record, and therefore what evidence each system under it has to produce. You cannot mark a process step GxP or not until you have read this, and item 4 asks you to mark twenty of them.
3. **ch 5, *Read this before you book the workshop.*** The roles the industry actually has, and the order to interview them in. `O10 Actor` and `O11 Role` hold eighteen generic corporate titles and not one pharmaceutical job; this section is the replacement list.
4. **ch 16, *Who inspects, and what they ask for.*** The twelve-row table of who inspects and what they ask for. Read down its artefact column and you have your first finding: this repository can produce two of twelve.
5. **ch 9, *The object most architects leave out: the validated system itself.*** The entity `O20` has no row for. It is the first thing you create in Phase C and the reason item 2 builds `O67 Validated System` in week one.

That is about fifty pages. If you have a second hour, add ch 7, *The cadence, which is the spine of this chapter*, for the eleven calendars that will decide your roadmap whatever you put in `O53` [1].

If a reference does not resolve

It should. Every heading in this item was machine-checked against volume 1's own heading index before this page was typeset, and so was every **Look it up.** line in items 1 to 20. But books get revised, and a section can be renamed between editions while the workbook keeps the old name. Three things to try, in order.

Check whether it is a sub-section. Volume 1 nests two levels. *Periodic review* is a sub-section of *Computerised system validation, as it is actually practised* in chapter 13, and a reader scanning only the top-level contents will not see it. The chapter contents page lists both levels.

Check the appendices. Volume 1 numbers its items 1 to 23, and items 20 to 23 are appendices A to D. This index uses the item numbers, so "ch 20" means appendix A, the cross-maps. A reference to *Retention clock by record class* is in appendix A and will not be found by looking for a chapter 20.

Check whether the edition moved. If the heading is genuinely gone, work from the chapter rather than the section and say in your own notes which reference failed. A dangling cross-reference in a printed volume is a defect, and it is the kind a reader fixes for you if you ask them to.

One standing rule while you are reading both volumes together. **Never state a standard's version from memory.** Volume 1 states the version it read, with the date it retrieved it. Where volume 1 and a research note disagree, volume 1 is what the reader has in their hand: cite volume 1, and say in the same sentence that the version may have moved [2].

Where volume 1 has no answer, and says so

Four places where looking it up will not help, because volume 1 deliberately leaves the question open. Know them before somebody in a workshop asks you to settle one.

- **ch 14, *The finding: where every one of them stops.*** No standards body publishes an enterprise reference architecture for this industry. Item 19's principle and standard set is therefore written here rather than cited from anywhere, and you should say so when you present it.

- **ch 16, *Section 57, and the question nobody has answered.*** Whether a sponsor needs prior authorisation for its own trial data is unresolved. Do not resolve it in a design review; route it to legal and record the open question.
- **ch 16, *The research code of conduct, which is the highest-value open question.*** A code of conduct would change the lawful basis for research processing, and none is in force. Item 5's classification work has to stand without it.
- **ch 22, *A closing statement about what is not known.*** Standard versions move. Volume 1 states the version it read with a retrieval date; this volume never states one from memory. Where the two disagree, volume 1 is what the reader has in their hand [2].

And one gap that is this volume's, not volume 1's: **volume 1 says nothing about the repository.** It was written before anybody handed you `Pharmaceuticals_TOGAF_Metamodel.xlsx`, so there is no chapter to look up for a sheet name, a column or a report. That is what items 13 and 14 are for.

THE CHAPTER IN TEN LINES

- Volume 1 teaches the industry; this volume works the repository. Every step with teaching behind it carries a Look it up line.
- Read this index forward before working an item, and backward after reading a chapter of volume 1.
- Every section heading printed here exists in volume 1 exactly as printed, checked twice against the heading index.
- Volume 1 chapter 13, on GxP, validation and data integrity, is cited by ten of the twenty items.
- Volume 1 chapter 17, on ArchiMate, is cited by one item, because the notation rules belong in one place.
- The heaviest chapters are 13, 16 and 19. The thinnest links are chapters 1, 2 and 6, which set framing rather than method.
- Nine sections are cited by three or more items. Print those nine and staple them inside the back cover.
- The five to read before a first workshop: the system itself is regulated, which GxP for which record, who to book for the workshop, who inspects and what they ask for, and the validated system as an object.
- That reading is about fifty pages and an hour. Add the cadence section if you have a second hour.
- Four questions volume 1 leaves open: no published enterprise reference architecture, prior authorisation for a sponsor, the research code of conduct, and standard versions that move.
- Never state a standard version from memory. Volume 1 states the version it read with a retrieval date.
- Volume 1 says nothing about the repository, because it was written before anybody handed you the file. Items 13 and 14 cover it.
- Nothing in volume 1 goes unused by this workbook, which is a consequence of writing the teaching volume first.

SOURCES FOR THIS CHAPTER

- **1.** Pharmaceuticals - a starter pack for enterprise architects (volume 1), Thotnr Consulting, October 2026: 246 pages, 23 items; every section heading cited in this item verified against the chapter sources, including chapters 1 to 19 and appendices 20 to 23
- **2.** Volume 1, appendix 22, Standards, bodies and where to find them - the practice of stating a standard's version with the date it was retrieved, and the closing statement on what is not known

APPENDIX F

The forms

Forty-seven printable worksheets, each one keyed to a real sheet and a real set of columns, and six of them for sheets you have to create first.

Every form this volume asks you to fill in, in one place, so that this appendix is the part you photocopy. Each names the sheet it feeds, the columns it covers in that sheet's own order and spelling, how many rows a first pass needs, and who signs it; each carries one example row marked `EXAMPLE` — delete it — and then blank ruled rows. Where a sheet has more than six columns the form splits into an identify half and an assess half, keyed by `Ref`, and the text says which columns are typed straight into Excel. Six of these forms feed the four sheets **the repository does not have**: they say so on their face and give the columns of the sheet you are to create. Thirty-three further forms are printed inside the phase items, and the second index table says where. A column appears on a form because a step transcribes it [1].

The index

Forms in this appendix, part one.

Form	Feeds	Signed by
18-01	`O67 Validated System` — identify	Head of Quality Assurance
18-02	`O67 Validated System` — assess	Head of QA and the system owner
18-03	`O68 Periodic Review Schedule` — schedule	Validation lead and system owner
18-04	`O68 Periodic Review Schedule` — outcome	Reviewer, countersigned by the system owner
18-05	`O69 Retention Clock`	Legal, with the information officer
18-06	`O70 Regulated Role`	Head of QA and the Responsible Pharmacist
18-07	`O01 Driver`	The executive named in `Owner`
18-08	`O02 Goal`	Chief Executive
18-09	`O03 Objective`	The goal owner
18-10	`O04 Measure`	The measure's domain owner
18-11	`O05 Principle`	`GB-01 Architecture Board`
18-12	`O06 Requirement`	The requirement's category owner
18-13	`O07 Constraint`	Chief Financial Officer, or the constraint's owner

Part two.

Form	Feeds	Signed by
18-14	`O08 Assumption`	Chief Enterprise Architect
18-15	`O10 Actor`	Chief Operating Officer
18-16	`O12 Location`	Head of Operations, plus the Responsible Pharmacist per site
18-17	`O19 Contract`	Legal, with Head of QA on quality agreements
18-18	`O13 Business Capability` — identify	The capability's executive owner
18-19	`O13 Business Capability` — score	`GB-05 Architecture Function`
18-20	`O14 Value Stream`	The value stream owner
18-21	`O15 Business Service`	The service owner
18-22	`O18 Product`	Head of Regulatory Affairs
18-23	`O37 Data Attribute`	The entity owner from `O20`
18-24	`O43 Master Data Rule`	`GB-03 Data Governance Council`
18-25	`O42 Metadata Element`	Chief Data Officer

Part three.

Form	Feeds	Signed by
18-26	`O39 Data Migration`	Validation lead and receiving system owner
18-27	`O23 Application Component` — identify	Chief Information Officer
18-28	`O23 Application Component` — ownership	The named system owner
18-29	`R06 Application interfaces with` — identify	The integration lead
18-30	`R06 Application interfaces with` — assess	The validation lead
18-31	`O55 Building Block`	The domain architect, with `GB-01`
18-32	`O48 Environment`	The information officer
18-33	`O26 Logical Technology Componen`	Domain Architect, Technology
18-34	`O46 Technology Building Block`	Domain Architect, Technology
18-35	`O30 Work Package` — scope	`GB-02 Investment Committee`
18-36	`O30 Work Package` — status	The project lead from `O56`

Part four.

Form	Feeds	Signed by
18-37	`O66 Architecture Increment`	Chief Enterprise Architect
18-38	`R12 Gap closed by Work Package`	`GB-01 Architecture Board`
18-39	`O62 Delivery Risk`	The risk owner
18-40	`O33 Architecture Contract`	The work package's delivery lead
18-41	`O41 Security Control`	Security Architect
18-42	`O32 Standard`	`GB-01` , with `GB-07` on validation rows
18-43	`O60 Governance Decision`	The body named in `Body`
18-44	`O61 Governance Change Request`	`GB-01 Architecture Board`
18-45	`O57 Governance Body`	`GB-04 Executive Committee`
18-46	`O58 Governance Gate`	`GB-01 Architecture Board`
18-47	`O31 Deliverable`	The phase owner



Figure 18.1 The four sheets this volume creates, and what cites them. The six forms on the left are the ones to fill first, because every other form in this appendix carries a column that points at one of the four boxes in the middle. The right-hand column is where each new sheet is read from.

Forms printed inside the phase items rather than here. Use those, not a second version of them.

Form	Item	Feeds
1-A Repository count sheet	1	Nothing — your own count against `00 Index`
2-A, 2-B Validated system register	2	`O67 Validated System` , Preliminary scope
2-C Regulated role register	2	`O70 Regulated Role`
A-1 GxP classification of the portfolio	3	`O67` , Phase A subset
A-2 Driver register, South African rows	3	`O01 Driver`
A-3 Regulated role register, Phase A subset	3	`O70 Regulated Role`
B-1 Organisation unit register	4	`O09 Organization Unit`
B-2 Role register	4	`O11 Role`
B-3 Process register with the GxP marking	4	`O16 Business Process` , plus two added columns
5-A Entity classification and ownership	5	`O20 Data Entity`
5-B Master system and lineage	5	`O20 Data Entity`
5-C Retention clock	5	`O69 Retention Clock` , the `Period` column

Second half of the same table.

Form	Item	Feeds
6-A Validated system register, identify	6	`O67` , application view
6-B Periodic review position	6	`O68` , application view
7-A Physical component qualification register	7	`O27 Physical Technology Compone`
7-B Resilience re-tier worksheet	7	`O49 Resilience Tier`
7-C Validated system register, technology half	7	`O67` , per environment
8-A Gap register, first pass	8	`O29 Gap`
8-B Implementation factors, corrected	8	`O52 Implementation Factor`
8-C Alternative scoring	8	`O54 Architecture Alternative` , plus three added columns
F-1 Transition state exit criteria	9	`O53 Transition Architecture`
F-2 Field mapping specification	9	`O40 Source To Target Mapping`
G-1, G-2 Compliance assessment	10	`O59 Compliance Assessment`
G-3 Control objective	10	`O63 Control Objective`
G-4 Release-relevant validation state	10	`O67` , release view

Third part, same columns, from the items after Phase G.

Form	Item	Feeds
11-A Periodic review schedule, schedule it	11	`O68 Periodic Review Schedule`
11-B Change triage record	11	`O61 Governance Change Request` , plus four added columns
11-C Periodic review outcome	11	`O68` , and it updates two `O67` columns
12-A Requirement as a user requirement specification	12	`O06 Requirement` , after item 12 widens it
12-B Traceability line	12	`R08` , `R09` , `O06` and `O59` together
12-C Acceptance criterion rewrite	12	`O55 Building Block` , column `Acceptance criteria`

Item 15 also carries three checklists, 15-A to 15-C, and item 20 carries the maturity scoring sheet, form 20-A. Neither set feeds a sheet, so neither is indexed above. Some sheets get no form, deliberately. `O56 Project` is `O30 Work Package` twice over, with the same twelve names, the same costs and the same dates [10] — fill `O30` and generate `O56` , or delete one of them. `O13` 's level-2 and level-3 rows are generated names rather than findings, so form 18-18 is for level 1 and a re-cut level 2 only. `O21` , `O22` , `O24` , `O25` , `O28` , `O35` , `O36` , `O38` , `O44` , `O45` , `O47` , `O50` , `O64` and `O65` are derived from, or small enough to type directly against, a sheet that does have a form; work them in Excel beside the parent sheet. And `O59 Compliance Assessment` is worked in item 10, where the gate that consumes it lives.

> **TWO PREFIXES THAT ARE NOT INTERCHANGEABLE**
>
> The validated-system register is keyed `VAL-01` onward, fixed in item 2. `VS-` is not available: `O14 Value Stream` already holds `VS01` to `VS05` without a hyphen, and the two collide in any sort, filter or lookup [6]. The other three new sheets are keyed `PRS-` , `RC-` and `RGR-` , and none of the four prefixes is used anywhere in the 105 sheets you were handed.

The four sheets the repository does not have

These four come first because without them the rest of the set describes an estate that could be in any industry. Nothing in the 105 sheets carries a validation state, a periodic-review date, a retention clock or a statutory role [2].

Form 18-01 — Validated system register, identify

Feeds `O67 Validated System`, **a sheet the repository does not have**; item 2 creates it with twelve columns. This form covers the first six in sheet order: `Ref`, `Application`, `GxP`, `Validation state`, `System owner` and `URS reference`. `Process owner` is the one column on no form — type it straight into Excel, because it changes more often than anything else on the sheet. A first pass needs twelve to eighteen rows: the twelve applications in `O23` are the floor, and standalone instruments, spreadsheets and laboratory data systems are computerised systems too [8]. `GxP` takes the regimes, not a yes: `GMP`, `GCP`, `GLP`, `GVP`, `GDP`, `GDocP`. Signed by the Head of Quality Assurance; the architecture function drafts it and does not own it.

Ref	Application	GxP	Validation state	System owner	URS reference
VAL-01	EXAMPLE — delete this row. MES (GMP)	GMP, GDocP	Validated, review overdue		

Form 18-02 — Validated system register, assess

Feeds `O67 Validated System`, the remaining five columns keyed by `Ref` to form 18-01: `Last periodic review`, `Next review due`, `Validation lead`, `Audit trail reviewed` and `Decommissioning plan`. Fill one row for every row of 18-01, including the rows whose answer is `Never` — an empty cell reads as unknown and `Never` reads as a finding. `Audit trail reviewed` takes a date and a scope. `Decommissioning plan` is a document reference or `None`; where it reads `None` for an application whose `O23` disposition is `Retire`, you have found the gap that `A10 Serialisation` carries today [8]. Signed by the Head of QA and the system owner jointly.

Ref	Last periodic review	Next review due	Validation lead	Audit trail reviewed	Decommissioning plan
VAL-01	EXAMPLE — delete this row. Never			No	None

Form 18-03 — Periodic review schedule

Feeds `O68 Periodic Review Schedule`, **a sheet the repository does not have**; item 11 creates it with eight columns. This form covers six in sheet order: `Ref`, `System`, `Frequency`, `Last review`, `Next due` and `Reviewer`. `Frequency` is a period the quality system has approved, and it is risk-based rather than uniform. `System` cites a `VAL-` reference from `O67`, not an application name, so that a renamed application does not orphan its schedule. A first pass is one row per `O67` row, so twelve to twenty. Signed by the validation lead and the system owner jointly, with `GB-07 Validation Committee` — the body item 2 adds to `O57 Governance Body` — owning the schedule [12]. Item 11 prints the same form as 11-A, in the step that creates the sheet; this is the copy you photocopy.

Ref	System	Frequency	Last review	Next due	Reviewer
PRS-01	EXAMPLE — delete this row. VAL-01	2 years	None recorded	Overdue	

Form 18-04 — Periodic review outcome

Feeds `O68 Periodic Review Schedule`, the last two columns plus the two that identify the row: `Ref`, `System`, `Outcome` and `Actions open`. One form per review cycle, not per system. `Outcome` takes `Confirmed`, `Confirmed with actions` or `Not confirmed`, matching item 11's form 11-C — the third is a quality event and leaves this form immediately. `Actions open` is a count and a reference to where the actions are tracked, which is the quality system, not the architecture repository. Signed by the reviewer and countersigned by the system owner.

Ref	System	Outcome	Actions open
PRS-01	EXAMPLE — delete this row. VAL-01	Confirmed with actions	4, in QMS CAPA-__

Form 18-05 — Retention clock

Feeds `O69 Retention Clock`, **a sheet the repository does not have**; item 5 creates it with seven columns and populates thirteen rows, `RC-01` to `RC-13`, from the periods volume 1 verified [13]. This form covers `Ref`, `Record class`, `Clock starts at`, `Instrument`, `System of record` and `Disposal approved by`. `Period` is deliberately not here: it is a computation, not a value — *one year after batch expiry, or five years after certification, whichever is longer* does not fit a ruled cell — so take it from form 5-C in item 5 and cite `RC-nn` rather than retyping it. `Clock starts at` names an event, and the system in `System of record` must supply that event's date; if it cannot, the period is unimplementable and that is the finding. Thirteen rows plus whatever the company's own legal register adds. Signed by legal for the instrument column and the information officer for rows carrying personal information. **Look it up.** Volume 1 chapter 16, section *The retention clocks*.

Ref	Record class	Clock starts at	Instrument	System of record	Disposal approved by
RC-01	EXAMPLE — delete. Batch documentation	Batch expiry or certification	Mfg guide Part I §4.11	MES (GMP)	

Form 18-06 — Regulated role register

Feeds `O70 Regulated Role`, **a sheet the repository does not have**; item 2 creates it with seven columns. This form covers six: `Ref`, `Role`, `Statutory basis`, `Named person`, `Deputy` and `Signs for`. `Registration number held` is deliberately absent — type it into Excel and restrict the sheet, because a printed form leaves the room. A role with no deputy is a single point of failure with a legal duty attached. Eight to twelve rows is a first pass: the Responsible Pharmacist, the qualified person for pharmacovigilance, the quality assurance release officer, the study monitor, the validation lead, the information officer [14]. None of them appears in `O10 Actor` or `O11 Role`, which hold the same eighteen generic corporate titles twice over [4]. Signed by the Head of Quality Assurance and the Responsible Pharmacist. **Look it up.** Volume 1 chapter 16, section *Licensing, and the Responsible Pharmacist*.

Ref	Role	Statutory basis	Named person	Deputy	Signs for
RGR-01	EXAMPLE — delete this row. Responsible Pharmacist	Medicines Act 101 of 1965, s 22C			Batch release

Motivation: drivers, goals, objectives, measures, principles, requirements

Seven sheets, 93 rows, and the regulatory half of them is sound while the business half was written for a company that sells to consumers [3].

Form 18-07 — Driver register

Feeds `O01 Driver`, six of its seven columns in sheet order: `Ref`, `Driver`, `Type`, `Origin`, `Owner`, `Pressure`. `Description` is typed straight into Excel. Thirteen rows today: seven `Regulatory`, all `External`, all owned by the Chief Compliance Officer and all at `High`, which is right; and six `Business` rows at `Medium` owned by `Group Executive`, of which `DRV-09 Customer expectation` and `DRV-13 Sustainability reporting` are not this industry's pressures [3]. A first pass adds six to ten rows and reassigns the business six. Form A-2 in item 3 covers the South African rows; this form is the whole register. Signed by the executive named in `Owner`, not the architect who wrote the row.

Ref	Driver	Type	Origin	Owner	Pressure
DRV-15	EXAMPLE — delete this row. SAHPRA registration and variation duty	Regulatory	External		High

Form 18-08 — Goal register

Feeds `O02 Goal`, all five columns. Six rows today, every one `Active`, each owned by a different chief officer, and `GOL-01 Protect the licence to operate` — *meet every obligation with evidence, not assertion* — is the right first goal for this industry [3]. `GOL-03 Grow revenue from existing customers` is the one that belongs to another sector's workbook. Six to nine rows; more than nine goals and you have objectives. Signed by the Chief Executive.

Ref	Goal	Description	Owner	Status
GOL-07	EXAMPLE — delete this row. Supply without interruption	Meet the registered supply obligation on every authorised product		Active

Form 18-09 — Objective register

Feeds `O03 Objective`, all five columns. Fourteen rows today across the six goals, with `Horizon` in months or years and `Status` one of `Not started` or `In progress` [3]. `OBJ-05 Single view of customer across channels` and `OBJ-06 Cross-sell rate improved from baseline` are the two to replace. Twelve to eighteen rows. Every row must be measurable by a row of `O04 Measure` through `R10 Objective measured by Measu`, or it is a wish. Signed by the goal owner.

Ref	Objective	Goal	Horizon	Status
OBJ-15	EXAMPLE — delete this row. Every GxP system has a current periodic review	GOL-01	18 months	Not started

Form 18-10 — Measure register

Feeds `O04 Measure`, all six columns. Sixteen rows today, and four of them contradict the sheets they measure: `MSR-01 Obligations with evidence` at a baseline of 54% against `O63 Control Objective`'s two of seven evidenced; `MSR-07 Components within vendor support` at 22% against `O27`'s ten of twelve supported; `MSR-13 Recovery objectives tested` at 53 in unit `count` against a population of twelve; and `MSR-15` and `MSR-16` whose only possible source, `O59 Compliance Assessment`, is empty [3][11]. Before you add a row, write down which sheet and which column the number is computed from; a baseline with no source is the defect this sheet teaches. Twelve to sixteen rows. Signed by the measure's domain owner.

Ref	Measure	Unit	Domain	Baseline	Cadence
MSR-17	EXAMPLE — delete this row. GxP systems with a current periodic review	%	Compliance		Monthly

Form 18-11 — Principle register

Feeds `O05 Principle`, all six columns. Twenty-four rows today, every one owned by `Architecture Board`, every `Category` one of `Business`, `Application`, `Data`, `Technology` or `Security`, and not one of them mentions validated state, data integrity, patient safety, retention or the regulator [3]. Item 2 adds the `Regulatory` category and three rows, `PRN-25` to `PRN-27`. **Do not draft new principles on this form.** Item 19 carries the full starter set, written in these six columns with its rationale and its conformance test; this form transcribes that set and the rows your own company adds beyond it. Nine to twelve new rows, from `PRN-28`. Signed by `GB-01 Architecture Board`, whose approval is recorded in `Status`.

Ref	Principle	Category	Statement	Owner	Status
PRN-28	EXAMPLE — delete. Retention Runs From an Event	Regulatory	Computed from an event, not a number of years	Architecture Board	Draft

Form 18-12 — Requirement register

Feeds `O06 Requirement`, six of its seven columns: `Ref`, `Requirement`, `Category`, `Priority`, `Driver`, `Status`. `Statement` is typed straight into Excel — in this industry that sentence is the first document in the validation life cycle. Thirty-two rows today, opening on `REQ-001 Evidence GMP / GCP / GLP / GVP`, which is the right first requirement; the trap is downstream, where `R09 Requirement satisfied by Bu` pairs `REQ-001` to `BBB-01` by index and marks `Traced: No` on every fourth row, `REQ-001` among them [3][10]. `Priority` takes `Must`, `Should` or `Could`. Next free reference `REQ-033`. Signed by the requirement's category owner. **Look it up.** Volume 1 chapter 13, section *What the architect is actually expected to produce*.

Ref	Requirement	Category	Priority	Driver	Status
REQ-033	EXAMPLE — delete this row. Validated state visible per system	Regulatory	Must	DRV-01	Proposed

Form 18-13 — Constraint register

Feeds `O07 Constraint`, all four columns. Eight rows today with `Type` one of `Financial`, `Resource`, `External`, `Regulatory`, `Operational` or `Organisational` [3]. `CON-06 Change freeze windows` reads *peak periods are closed to change*, which is the wrong freeze for this industry — the windows that close change here are the inspection window and the annual product quality review cycle. And **ArchiMate 4 has no Constraint element**, so when item 16 draws the repository these rows become `Requirement` elements; write them so that they survive the translation. Eight to twelve rows. Signed by the Chief Financial Officer for financial rows and by the constraint's owner otherwise.

Ref	Constraint	Description	Type
CON-09	EXAMPLE — delete this row. Validation capacity	Quality can validate a bounded number of systems a quarter	Resource

Form 18-14 — Assumption register

Feeds `O08 Assumption`, all three columns. Six rows today, four `Open` and two `At risk`, and they are the shortest rows in the repository because the sheet has nowhere to record who would be hurt if the assumption failed [3]. Add two columns in Excel, not on this form: the date the assumption is next tested, and the row of `O62 Delivery Risk` it becomes if it fails. Six to ten rows. Signed by the Chief Enterprise Architect, who is the only person who can be held to an architecture assumption.

Ref	Assumption	Status
ASM-07	EXAMPLE — delete this row. Quality will accept the architecture decision as a controlled record	Open

Organisation, roles, locations and contracts

`O09 Organization Unit` and `O11 Role` have their forms in item 4. These four cover the rest, and every one of them is a sheet where the repository describes a generic corporate [4].

Form 18-15 — Actor register

Feeds `O10 Actor`, all four columns. Eighteen rows today, and they are the same eighteen titles as `O11 Role` because `R21 Role performed by Actor` joins `ROL-nn` to `ACT-nn` by index — the duplication is a property of how the file was generated, not a statement about the company [10]. Every `Status` reads `Filled`, which is why the dashboard can show *vacant architecture roles: 0*. Use `Status` for `Vacant`, `Interim` or `Shared`: an interim Responsible Pharmacist is a different risk from a permanent one. Twenty to thirty rows, so continue on a second copy. Next free reference `ACT-20`. Signed by the Chief Operating Officer. **Look it up.** Volume 1 chapter 5, section *A role-to-system map*.

Ref	Actor	Type	Status
ACT-20	EXAMPLE — delete this row. Qualified person for pharmacovigilance	Regulated	Filled

Form 18-16 — Location register

Feeds `O12 Location`, all four columns. Eight rows today with `Type` one of `Corporate`, `R&D`, `GMP`, `Logistics` or `Technology`, and `Scale` in people — which is why `LOC-08 Validated Cloud Region` reads *82 people* [4]. A region is not staffed. Record cloud regions with a `Scale` of `n/a` and a note of the jurisdiction, because health-data residency is a POPIA question before it is a hosting one. The eight sites sum to 1,215 people against `O09`'s headcount of 1,879, so 664 people have no location, or 746 once the region's 82 come out. `Type` needs a `Licensed site` value, because a registered manufacturing site is a dated regulatory object. Eight to twelve rows. Signed by the Head of Operations, with the Responsible Pharmacist signing each licensed site. **Look it up.** Volume 1 chapter 7, section *A registered manufacturing site is a regulated object*.

Ref	Location	Type	Scale
LOC-09	EXAMPLE — delete this row. Validated Cloud Region, eu-central	Technology	n/a — no people

Form 18-17 — Contract register

Feeds `O19 Contract`, all five columns. Ten rows today, every one a vendor agreement, and `Counterparty` holds **application names rather than legal entities** — `CTR-01` names its counterparty as `ELN / LIMS` [4]. What is missing is every contract this industry governs by: no quality agreement, no contract manufacturing agreement, no clinical trial agreement, no supply agreement, no pharmacovigilance agreement. Put the legal entity in `Counterparty` and the system in the contract name. Twelve to twenty rows. Signed by legal, with the Head of Quality Assurance signing every quality agreement.

Ref	Contract	Counterparty	Type	Expiry
CTR-11	EXAMPLE — delete this row. Quality agreement, fill and finish		Quality agreement	

Capabilities, value streams, services and products

`O13 Business Capability` is the best-built sheet in the file and the least informative: 100 rows decomposed to level 3, structurally consistent, and scored in a way that does not roll up [5].

Form 18-18 — Capability register, identify

Feeds `O13 Business Capability`, five of its nine columns: `Ref`, `Capability`, `Level`, `Parent`, `Criticality`. `Description` is typed straight into Excel. Use this form for level 1 and a re-cut level 2 only — the existing thirty level-2 names are a level-1 name plus one word from a pool of ten, and the sixty level-3 names add — *Define* or — *Operate*, so they are generated, not gathered [5]. The ten level-1 names are genuinely pharmaceutical and worth keeping. Ten rows at level 1 and twenty-five to thirty-five at level 2, which is two copies of this form. Signed by the capability's executive owner, which `R22 Org Unit owns Capability` currently gets wrong by index — it puts Regulatory Affairs under Human Resources and Pharmacovigilance under Technology [10].

Ref	Capability	Level	Parent	Criticality
C04.01	EXAMPLE — delete this row. Submission management	2	C04	High

Form 18-19 — Capability register, score

Feeds `O13 Business Capability`, the four scoring columns plus the name: `Ref`, `Capability`, `Current`, `Target`, `Disposition`. Scores run 1 to 5 and `Disposition` takes `Sustain`, `Build`, `Buy` or `Retire`. **Scores must roll up**: today `C01 Discovery Research` is `Criticality: Low` while its own descendant `C01.01.02` is `Critical`, and a parent cannot be less critical than its child [5]. And a gap is a number you can defend: `RPT03 Capability Heatmap` gives every one of the ten level-1 capabilities a gap of exactly 1 and a disposition of `Sustain`, which is a heatmap with no heat [11]. Score level 1 first and let level 2 argue with it. Ten rows per pass. Signed by `GB-05 Architecture Function`, which publishes the criteria.

Ref	Capability	Current	Target	Disposition
C04	EXAMPLE — delete this row. Regulatory Affairs	2	4	Build

Form 18-20 — Value stream register

Feeds `O14 Value Stream`, all five columns. Five rows today — Target to Candidate, Candidate to Approval, Plan to Release, Signal to Action, Launch to Patient — and they are the strongest pharmaceutical content in the repository [5]. Keep them. Two defects to fix on this form. `Stages` holds a count written as text, *4 stages*, and the five counts total 24 against `O16 Business Process`'s 20 rows. And `Capabilities` disagrees with `R02 Value stream uses Capabilit`, leaving `C08`, `C09` and `C10` used by no value stream — in a repository whose fifth stream is called Launch to Patient [10]. Write the stage names, not a count. Five to seven rows. Signed by the value stream owner.

Ref	Value stream	Scope	Stages	Capabilities
VS03	EXAMPLE — delete this row. Plan to Release	Production plan through manufacture, testing and batch release	Plan, make, test, review, release, ship	C06, C07, C10

Form 18-21 — Business service register

Feeds `O15 Business Service`, all six columns. Ten rows today, one per level-1 capability, and three problems on every one of them. Every `Description` reads *Externally consumable service realising …* while `Exposure` is `Internal` on seven of ten. `Criticality` disagrees with the capability the service realises on eight of ten. And `A11 Statistical Computing` and `A12 Medical Information` support no service at all, because `R04 Business Service supported` pairs `BSV-01` to `A01` down to `BSV-10` to `A10` and stops — which is the whole content of `RPT13 Orphan Analysis` [10][11]. Write `Exposure` from who consumes the service. Ten to fourteen rows. Signed by the service owner.

Ref	Business service	Capability	Description	Exposure	Criticality
BSV-11	EXAMPLE — delete. Batch release service	C07	Disposition and release certification for a batch	Internal	Critical

Form 18-22 — Product register

Feeds `O18 Product`, all four columns. Four rows today, `PRD-01` to `PRD-04`, and the sheet cannot carry what this industry means by a product, because a product here is four things with four different cardinalities: the molecule, the formulation, the registration and the pack. Use `Type` to say which identity the row is, and expect four rows per marketed medicine rather than one. Twelve to twenty rows for a small portfolio, and `R25 Application supports Produc` grows with it. Signed by the Head of Regulatory Affairs, because the registration identity is a regulatory object. **Look it up.** Volume 1 chapter 9, section *The object that breaks most models: "a product" is four things*.

Ref	Product	Type	Description
PRD-05	EXAMPLE — delete this row. Registration, 10 mg tablet	Registration	One marketing authorisation with its conditions and its renewal date

Data

`O20 Data Entity` has its two forms in item 5 and `O40 Source To Target Mapping` has form F-2 in item 9. These five cover the rest of the data domain.

Form 18-23 — Data attribute register

Feeds `O37 Data Attribute`, all six columns. Sixty rows today, six per entity — `Identifier`, `Name`, `Status`, `Effective date`, `Owner`, `Classification` — for ten of the fourteen entities; `E11` to `E14` have no attributes at all [7]. The classifications were assigned by pattern: `ATR-002 Compound / Molecule · Name` is `Public` and `Key: Yes` while `ATR-001 Identifier` is `Internal`, and the name of a compound in development is usually the most commercially sensitive string in the company. `Classification` takes `Public`, `Internal`, `Confidential` or `Restricted`; add the class item 5 introduces for subject data. Six to twelve attributes per entity, one copy of this form each. Next free reference `ATR-062`. Signed by the entity owner named in `O20`.

Ref	Entity	Attribute	Type	Key	Classification
ATR-062	EXAMPLE — delete this row. Adverse Event	Dictionary version	String	No	Internal

Form 18-24 — Master data rule register

Feeds `O43 Master Data Rule`, all five columns. Ten rows today with `Rule type` one of `Golden source`, `Match rule`, `Survivorship` or `Validation`, and the sheet **disagrees with `O20 Data Entity`'s `Master system` column on every one of the ten entities they both cover** — `MDR-01` puts Compound in `ELN / LIMS` while `O20` puts it in `Safety Database`; `MDR-08` puts Adverse Event in `MES (GMP)` while `O20` puts it in `ERP` [7]. Two sheets, two answers, no reconciliation. Fill this form and form 5-B in the same workshop, with the same people, and do not leave the room until `Golden source` and `Master system` agree. Fourteen to twenty rows. Signed by `GB-03 Data Governance Council`.

Ref	Entity	Rule type	Description	Golden source
MDR-11	EXAMPLE — delete. Adverse Event	Golden source	The safety database holds the case; every other copy is a read	Safety Database

Form 18-25 — Metadata element register

Feeds `O42 Metadata Element`, all five columns. Fourteen rows today across `Business`, `Governance`, `Technical` and `Operational` categories, with `Coverage` taking `Full`, `Partial`, `Absent` or — on two rows — `Recorded, not held`, which is a status masquerading as a coverage value [7]. `MTD-03 Steward`, `MTD-07 Consent basis` and `MTD-08 Lineage` all read `Absent`, and the third is why four entities in `O20` have no lineage. `MTD-06 Retention rule` reads `Partial` and can only reach `Full` once `O69 Retention Clock` exists. Add the two elements the sheet lacks: the audit-trail position and the controlled vocabulary version. Sixteen to twenty rows. Signed by the Chief Data Officer.

Ref	Metadata element	Category	What it records	Coverage
MTD-15	EXAMPLE — delete this row. Dictionary version	Governance	Which version of the coding dictionary produced the coded value	Absent

Form 18-26 — Data migration register

Feeds `O39 Data Migration`, all five columns. Six rows today, and not one of the pairings is possible: `MIG-01` moves Compound from `CTMS` to `Safety Database`, `MIG-03` moves Subject from `Safety Database` to `eTMF` [7]. The rows were generated by rotating the application list, the same way `O40`'s target field sits one position further round the rotation than its source. A migration here is a validation activity, so type two things into Excel that the sheet does not carry: the verification approach, and who signs that value **and meaning** were both preserved. Six to twelve rows. Signed by the validation lead and the receiving system owner. **Look it up.** Volume 1 chapter 10, section *Data migration in a validated estate*.

Ref	Entity	From system	To system	Status
MIG-07	EXAMPLE — delete this row. Adverse Event	Legacy safety database	Safety Database	Planned

Applications, interfaces and building blocks

`O23 Application Component` is ten columns wide and carries no validation state, no GxP flag and no review date, and its `Owner` column holds organisation units rather than people [8].

Form 18-27 — Application register, identify

Feeds `O23 Application Component`, six of its ten columns in sheet order: `Ref`, `Application`, `Type`, `Disposition`, `Criticality`, `Health`. Twelve rows today, and four of them contradict themselves: `A10 Serialisation` is `Critical`, `Adequate` and `Retire` with no successor named anywhere; `A08 QMS` is `Critical`, `Healthy` and `Replace` with no reason recorded; `A12 Medical Information` is `Critical`, `At risk` and `Consolidate`; `A06 eTMF` is `High`, `Poor` and `Invest`, the only honest row of the four [8]. A disposition that contradicts the criticality and the health needs a sentence of reasoning, and the sheet has nowhere to put it, so put it in the governance decision. Twelve to twenty rows. Signed by the Chief Information Officer, with `Criticality` countersigned by the business owner of the process the application serves.

Ref	Application	Type	Disposition	Criticality	Health
A13	EXAMPLE — delete this row. Validation lifecycle tool	Package	Invest	High	Adequate

Form 18-28 — Application register, ownership

Feeds `O23 Application Component`, the remaining four columns plus the name, with `System owner` added by item 6: `Ref`, `Application`, `In service`, `Users`, `Owner`, `System owner`. `Description` is typed straight into Excel. `Owner` holds an organisation unit today and holds the wrong one: `A04 Safety Database` sits under Legal, `A07 MES (GMP)` under Internal Audit, `A10 Serialisation` under Human Resources, `A01 ELN / LIMS` under Executive and `A03 EDC` under Supply & Logistics [8]. `System owner` takes **one named person with a deputy**: an owner who cannot be telephoned is not an owner. Twelve to twenty rows. Signed by the named system owner — the signature is the point of the form. **Look it up.** Volume 1 chapter 12, section *What is actually installed*.

Ref	Application	In service	Users	Owner	System owner
A04	EXAMPLE — delete this row. Safety Database	2017	301	Pharmacovigilance	

Form 18-29 — Interface register, identify

Feeds `R06 Application interfaces with`, six of its nine columns in sheet order: `Ref`, `Source`, `Source name`, `Target`, `Target name`, `Type`. Thirty-six rows today, `INT-001` to `INT-036`, with `Source` and `Target` holding `A`-references and the two name columns holding the application names [9]. `Type` takes `API`, `Event`, `Batch`, `File` or `Direct DB read`, and six rows read `Direct DB read` — a direct read into another application's database is outside that application's interface contract and inside its validation scope. Thirty-six rows, so three copies of this form. Signed by the integration lead. **Look it up.** Volume 1 chapter 12, section *Interfaces are routinely in nobody's validation scope*.

Ref	Source	Source name	Target	Target name	Type
INT-037	EXAMPLE — delete this row. A03	EDC	A04	Safety Database	API

Form 18-30 — Interface register, assess

Feeds `R06 Application interfaces with`, its last three columns plus the two item 6 adds: `Ref`, `Frequency`, `Managed`, `Logged`, `Validation scope`, `Interface specification`. `Managed` takes the words `Managed` and `Unmanaged`, not a yes — nineteen of thirty-six read `Unmanaged`; `Logged` takes `Yes` and `No`, and thirty-one read `No`. Sixteen are both unmanaged and unlogged, only two of thirty-six are both managed and logged, and all six direct database reads are unlogged while three are marked `Managed` [9]. Look first at `INT-009 EDC → Safety Database`, the serious adverse event path, which is an unmanaged hourly file. `Validation scope` names the system whose validation the interface sits inside. Thirty-six rows. Signed by the validation lead.

Ref	Frequency	Managed	Logged	Validation scope	Interface specification
INT-009	EXAMPLE — delete this row. Hourly	Unmanaged	No	VAL-__	None

Form 18-31 — Building block register

Feeds `O55 Building Block`, all five columns, and note that `O55` is the **union sheet** — thirty-seven rows carrying `BBB-01` to `BBB-10`, `DBB-01` to `DBB-10`, `ABB-01` to `ABB-10` and `TBB-01` to `TBB-07`, with a `Domain` column — and that `R09` and `R13` resolve their building-block column against `O55`, not against the four domain sheets [9]. The column that matters is `Acceptance criteria`, and **exactly one of the thirty-seven criteria passes today**: `ABB-08 QMS target`, because QMS is the only application with no unmanaged interface — and it passes while `O23` has `A08 QMS` at `Disposition: Replace`, so the criterion does not test what the block must do. Of the other thirty-six, nine are non-conformant, twenty are not testable as written and seven are not assessable. Write criteria that name a sheet, a column and a value. Thirty-seven rows. Signed by the domain architect, accepted by `GB-01 Architecture Board`.

Ref	Domain	Building block	What it must do	Acceptance criteria
ABB-08	EXAMPLE — delete	QMS target	Change control, deviation and CAPA	Every R06 row touching A08 reads Managed

Technology, environments and the network

`O27 Physical Technology Compone` has form 7-A and `O49 Resilience Tier` has form 7-B, both of them in item 7 of this workbook. These four cover the rest, and three are sheets that two other sheets disagree with [9].

Form 18-32 — Environment register

Feeds `O48 Environment`, all four columns. Four rows today, and the `Personal data present` column is **inverted**: `ENV-01 Production` reads `No` while `ENV-02 Pre-production` and `ENV-04 Development` read `Yes` [9]. Read literally, the company holds trial participants' data in development and none in production. Either the column is reversed or non-production is full of unmasked subject data, and `O41 Security Control` points at the second reading — `SEC-07 Non-production masking` is `Partial` and `O65 EP-05 Non-production provisioning` reads *most refreshes are unmasked* [11]. Settle it with the information officer in the room, and add two columns in Excel: the masking method and the date of the last refresh. Four to eight rows. Signed by the information officer. **Look it up.** Volume 1 chapter 16, section *POPIA and health data*.

Ref	Environment	Description	Personal data present
ENV-02	EXAMPLE — delete this row. Pre-production	Release validation for a GxP system	Yes — unmasked, a POPIA finding

Form 18-33 — Logical technology component register

Feeds `O26 Logical Technology Componen`, all four columns, with `Status` taking `Current` or `Emerging`. Seven rows today, naming real sector building blocks — GxP-validated cloud, the electronic records and signatures control set, the clinical data standards, the product identification data model, cold chain telemetry, the scientific data lake, the secure enclave — which is the strongest technology content in the file [9]. The defect is that `O26`, `O46 Technology Building Block` and `O47 Reference Model Entry` **give a different status for all seven**: `O26` calls GxP-validated cloud `Current`, `O47` calls it `Candidate`. One status per component, in one sheet. Seven to twelve rows. Signed by the Domain Architect, Technology.

Ref	Logical technology component	Rationale	Status
LTC-01	EXAMPLE — delete this row. GxP-validated cloud	Qualified environments with documented validation	Current

Form 18-34 — Technology building block register

Feeds `O46 Technology Building Block`, all five columns — and read the header row before you start, because **`O46` has two columns headed `Ref`**. The first holds `TBB-01` to `TBB-07`; the second holds `T01` to `T07`, a key space nothing else in the 105 sheets resolves [9]. Call the second `Ref (T)` where you must distinguish them, and decide here whether to populate it or delete it. The `Acceptance criteria` column reads … *in production for at least* … on all seven rows, which is a duration, not a criterion. Seven to twelve rows. Signed by the Domain Architect, Technology.

Ref	Technology building block	Ref (T)	What it must do	Acceptance criteria
TBB-01	EXAMPLE — delete. GxP-validated cloud	T01	Qualified environments, documented validation	Qualification report signed for every region in use

Gaps, work packages, increments and risk

`O29 Gap` has form 8-A, `O52` form 8-B, `O54` form 8-C and `O53` form F-1. These six cover the rest of the realisation domain, which is where the repository is most completely placeholder [10].

Form 18-35 — Work package register, scope

Feeds `O30 Work Package`, six of its eight columns in sheet order: `Ref`, `Work package`, `Domains`, `Cost`, `Start`, `End`. Twelve rows today, and they cost an alternative nobody chose: the twelve sum to **$84.3m** against `ALT-B Foundations first` at $46m marked `Recommended`, and close to the rejected `ALT-D Big bang replacement` at $88m [10]. `R15 Transition contains Work Pa` allocates only seven of the twelve, leaving `WP-06`, `WP-08`, `WP-09`, `WP-10` and `WP-12` — $29.1m — in no transition state. Costs here are in dollars because the sheet is; convert once, state the rate and the date, and do not mix the two. Twelve to eighteen rows. Signed by `GB-02 Investment Committee`.

Ref	Work package	Domains	Cost	Start	End
WP-13	EXAMPLE — delete this row. Validated system register and periodic review	Data			

Form 18-36 — Work package register, status

Feeds `O30 Work Package`, its last two columns plus the name, with `Benefit date` added: `Ref`, `Work package`, `Approval`, `Delivery`, `Benefit date`. `Approval` takes `Approved`, `Not started` or `In flight`; `Delivery` takes `Not started`, `In flight`, `Delivered` or `Past end date`. Both are typed strings, not computations, which is how `WP-08 Process automation` comes to be `Approval: Not started` and `Delivery: Delivered`, and how the dashboard line *work packages past end date: 3* survives a sheet whose earliest start is 2027-02 [10]. `Benefit date` is the column `GT-3 G3 Benefit` needs and no sheet carries; `CR-G04` in `O61` already asks for it. Twelve to eighteen rows. Signed by the project lead named in `O56 Project`.

Ref	Work package	Approval	Delivery	Benefit date
WP-08	EXAMPLE — delete this row. Process automation	Not started	Delivered	

Form 18-37 — Architecture increment register

Feeds `O66 Architecture Increment`, all four columns, with `Transition state` added: `Ref`, `Increment`, `Lands`, `What is added`, `Transition state`. Six rows today against three transition states in `O53`, with no mapping between them in any relationship sheet — but **three of the six map to the three states by exact name identity**: `INC-1` to `TA-1`, `INC-2` to `TA-2`, `INC-4` to `TA-3` [10]. Decide whether three further states exist; do not invent a mapping. `Lands` holds a month number from programme start, so record the start date or the column is meaningless. Six to nine rows. Signed by the Chief Enterprise Architect.

Ref	Increment	Lands	What is added	Transition state
INC-3	EXAMPLE — delete this row. Compliant by mechanism	Month 22	Retention, masking, consent, glossary	__ or new TA-4

Form 18-38 — Gap closed by work package

Feeds `R12 Gap closed by Work Package`, all five columns — and note that its first column is `Gap`, not `Ref`, because relationship sheets name both ends rather than carrying their own key. The sheet is **empty**, and so is `O29 Gap`, and so is `RPT07 Gap Analysis` in consequence; the dashboard line *gaps overdue: 0* is therefore meaningless rather than good [10][11]. Fill form 8-A in item 8 first, then this form, one row per gap per package. A gap that no package closes is the finding this sheet exists to produce: leave `Work package` blank and raise it. Twenty to forty rows. Signed by `GB-01 Architecture Board`.

Gap	Gap name	Severity	Work package	Package name
GAP-002	EXAMPLE — delete this row. No validation state recorded for any application	Patient		

Form 18-39 — Delivery risk register

Feeds `O62 Delivery Risk`, all five columns, with `Likelihood` taking `Unlikely`, `Possible`, `Likely` or `Certain` and `Impact` taking `Medium`, `High` or `Critical`. Eight rows today, and **three of them read `Certain`** — `RSK-01 Integration product not selected`, `RSK-07 Benefit not measurable at G3` and `RSK-08 Unmanaged interfaces proliferate` [10]. A certainty is not a risk. It is an issue or a constraint, and `IF-01 Vendor selection not made` in `O52 Implementation Factor` already records the first of the three. Move every `Certain` row off this sheet first. Eight to fourteen rows, and `Owner` is a named person. Signed by the risk owner.

Ref	Risk	Likelihood	Impact	Owner
RSK-09	EXAMPLE — delete this row. Periodic review backlog grows faster than quality capacity	Likely	High	

Form 18-40 — Architecture contract register

Feeds `O33 Architecture Contract`, all four columns. Eight rows today, covering `WP-01` to `WP-08` only, so the last four work packages are governed by nothing; six of the eight read `Not yet assessable` and two read `Non-compliant` [10]. `Not yet assessable` is honest only while `O59 Compliance Assessment` is empty — once item 10 fills it, every row here has a verdict available and the value becomes a refusal to look. One contract per work package, so twelve to eighteen rows, and each one names the validation deliverables the package must produce. Signed by the work package's delivery lead, countersigned by `GB-07 Validation Committee` where the package touches a validated system.

Ref	Architecture contract	Work package	Compliance status
ACO-09	EXAMPLE — delete this row. Resilience uplift architecture contract	WP-09	Not yet assessable

Obligations, controls and standards

`O63 Control Objective` has form G-3 and `O59 Compliance Assessment` forms G-1 and G-2, all three of them in item 10. These cover the controls and standards the obligations are supposed to rest on [11].

Form 18-41 — Security control register

Feeds `O41 Security Control`, all five columns, with `Position` taking `In place`, `Partial` or `Absent`. Eight rows today: `SEC-01 Identity lifecycle` and `SEC-03 Privileged access management` read `Absent`, four read `Partial`, and `SEC-08 Data-at-rest encryption` is mapped to no obligation at all [11]. The defect to carry into this form sits in the relationship sheet: `R16 Control Objective met by Co` records a `Position` per obligation that is a **verbatim copy of `O63`'s `Evidence` column**, and the two disagree on four of seven — `CO-01` is `Evidenced` through `SEC-01`, which this sheet records as `Absent`. Fill `Position` here, then re-derive `R16` from it. Ten to sixteen rows. Signed by the Security Architect.

Ref	Security control	What it does	Position	Enforced at
SEC-09	EXAMPLE — delete this row. Audit trail review	Risk-targeted review of changes to GxP records	Absent	Application

Form 18-42 — Standard register

Feeds `O32 Standard`, all five columns, with `Conformance test` added: `Ref`, `Standard`, `Category`, `Status`, `Conformance`, `Conformance test`. Sixteen rows today and **not one of them is a GxP standard** [11]. Three rows contradict themselves or each other: `STD-02 Event schema standard` is `Deprecated` and `Conformant`, `STD-07 Logging and monitoring standard` is `Deprecated` and in `Breach` — a deprecated standard can be neither, because it is not in force — and `STD-04 Data retention standard` is `Current` and `Conformant` while `STD-15 Records management standard` is `Deprecated` and `Conformant`, which cannot both be true of one estate. **Do not draft the new standards on this form.** Item 19 carries the nine validation-category standards with their conformance tests, `STD-17` to `STD-25`. `Conformance test` is the column that makes the sheet mean something: a sentence naming a sheet, a column and a value. Signed by `GB-01 Architecture Board`, with `GB-07 Validation Committee` owning the validation rows. **Look it up.** Volume 1 appendix 22, section *How to check a standard's version*.

Ref	Standard	Category	Status	Conformance	Conformance test
STD-20	EXAMPLE — delete. Audit trail review	Validation	Emerging	Not assessed	Every VAL- row has a dated review within its frequency

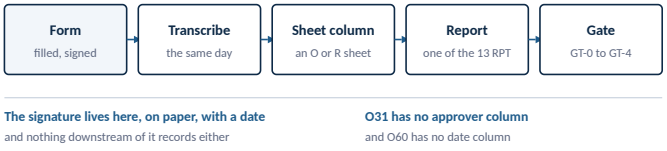


Figure 18.2 Where a signature actually lives. The form carries the signature and the date; the repository carries neither. `O31 Deliverable` has no approver column and `O60 Governance Decision` has no date column, so the paper is the record until forms 18-43 and 18-47 add those columns to the sheets.

Governance, decisions and deliverables

Five sheets, and the common defect is that none of them carries a date, so nothing in the governance domain can be overdue [12].

Form 18-43 — Governance decision register

Feeds `O60 Governance Decision`, all four columns, with `Decided on` and `Expires` added: `Ref`, `Decision`, `Body`, `Status`, `Decided on`, `Expires`. Seven rows today, four of them `Outstanding` including `DEC-01 Adopt the recommended architecture alternative` — so the alternative is recommended in `O54` and unadopted here, while `ALT-E Compliance led` is marked `Partly adopted`, giving two live answers [10][12]. The sheet has **no date column**, which is why nothing can be overdue; until the two added columns exist, item 6's interim rule applies and the date goes in the decision text. `DEC-07` grants the point-to-point dispensation with no expiry while `CR-G01` in `O61` records the same dispensation expiring 2029-06: a dispensation without an expiry is a standard. Next free reference `DEC-11`. Twelve to twenty rows. Signed by the body named in `Body`.

Ref	Decision	Body	Status	Decided on	Expires
DEC-11	EXAMPLE — delete. An architecture decision is a quality record	GB-04	Outstanding		n/a

Form 18-44 — Governance change request register

Feeds `O61 Governance Change Request`, all six columns, with `Type` taking `Dispensation`, `Scope`, `Correction` or `Compliance` and `Status` taking `Granted`, `Approved`, `Raised` or `Board decision required`. Five rows today, and `Expires` is populated on exactly one of them — `CR-G01`, the point-to-point dispensation, at 2029-06 [12]. Every `Dispensation` row needs an expiry and a named condition for its removal; every `Correction` row needs the sheet and column it corrects. `CR-G03 Raise unfunded building blocks as gaps` and `CR-G04 Record benefit realisation dates per package` are both corrections this workbook asks for, and both are still open. Eight to sixteen rows. Signed by `GB-01 Architecture Board`.

Ref	Change request	Against	Type	Status	Expires
CR-G06	EXAMPLE — delete this row. Add validation state and review date to O23	Repository	Correction	Raised	n/a

Form 18-45 — Governance body register

Feeds `O57 Governance Body`, all five columns. Five rows today and **none of them is a quality body**: no change control board, no validation committee, no data integrity forum [12]. Item 2 adds `GB-06 Change Control Board`, `GB-07 Validation Committee` and `GB-08 Data Integrity Forum`, taking the sheet to eight. `Cadence` comes from the meeting invitation, not the terms of reference. `Membership` names roles from `O11 Role` and `O70 Regulated Role` by reference, never a list of people. `Mandate` says which gate the body holds, if any. Eight to ten rows. Signed by `GB-04 Executive Committee`, because a new standing body is an executive act.

Ref	Governance body	Cadence	Membership	Mandate
GB-07	EXAMPLE — delete. Validation Committee	Monthly	RGR-01, RGR-02, ROL-08	Owns O67; co-signs validation evidence at G2

Form 18-46 — Governance gate register

Feeds `O58 Governance Gate`, all four columns. Four rows today, `GT-0` to `GT-3`, and they are sound: charter, design, build complete, benefit [12]. Item 10 adds a fifth, `GT-4`, gate name `G2R Release for use`, which is the gate that matters in this industry because its signatory is a named person with a legal duty. Test every row for whether the gate can actually be passed: `GT-3 G3 Benefit` tests a stated benefit confirmed against a date, and no sheet carries a benefit date, so today it cannot [10]. Write `What it tests` so that it names the artefact the gate must see, the validation artefact included. Five to six rows. Signed by `GB-01 Architecture Board`. **Look it up.** Volume 1 chapter 13, section *The V-model, and how it actually runs*.

Ref	Gate	When	What it tests
GT-4	EXAMPLE — delete. G2R Release for use	Before first GxP use	Validation summary report signed, audit trail reviewed, signatory named

Form 18-47 — Deliverable register

Feeds `O31 Deliverable`, all four columns, with `Approver` and `Approved on` added: `Ref`, `Deliverable`, `Phase`, `Status`, `Approver`, `Approved on`. Twenty rows today, correctly phased `A` to `G` plus `RM`, and it is one of the best-built sheets in the file — but it has **no approver and no date column**, no Preliminary deliverable and no Phase H deliverable, and thirteen of the twenty already read `Approved`, including `DLV-13 Consolidated Gaps` while `O29 Gap` is empty [12]. Item 2 adds `DLV-21 Validation Plan`, `DLV-22 User Requirement Specification` and `DLV-23 Validation Summary Report`, so the next free reference is `DLV-24` and the sheet still needs a Preliminary and a Phase H row. `Approved` with no approver and no date is the cheapest false assurance in the repository. Twenty-four to thirty rows. Signed by the phase owner.

Ref	Deliverable	Phase	Status	Approver	Approved on
DLV-24	EXAMPLE — delete this row. Periodic Review Schedule	H	Draft		

Before you photocopy anything

Four rules that save a reprint.

1. **Fill the first six forms before the other forty-one.** Every later form has a column pointing at `O67`, `O68`, `O69` or `O70`, and a form pointing at a sheet that does not exist yet produces answers nobody can transcribe.
2. **Transcribe the same day.** A form typed up a week after the workshop loses the hedging, and the hedging is the finding.
3. **Delete the example row before you copy the form**, or somebody transcribes `VAL-01 MES (GMP)` into the repository and it survives two review cycles.
4. **Where a form adds a column, add it to the sheet and say in the text that you added it.** This volume adds `System owner` to `O23`, `GxP` and `Regime` to `O16`, `Validation scope` and `Interface specification` to `R06`, `Benefit date` to `O30`, `Transition state` to `O66`, `Conformance test` to `O32`, `Decided on` and `Expires` to `O60`, and `Approver` and `Approved on` to `O31`. There is no fifth new sheet and there must not be one.

Calderon Therapeutics, whose data every example row above is drawn from, is an invention that appears only inside this repository; Veldtkruin Group, which volume 1 uses, is a different invention, and their data never mix.

THE CHAPTER IN TEN LINES

- Forty-seven forms here, twenty-seven more inside the phase items, and the second index table says which item holds which.
- Fill forms 18-01 to 18-06 first: they create O67 Validated System, O68 Periodic Review Schedule, O69 Retention Clock and O70 Regulated Role, and every other form points at one of them.
- The validated-system register is keyed VAL-. VS- is not available, because O14 Value Stream already holds VS01 to VS05.
- A form's columns are the real columns of the real sheet, in sheet order. Where a sheet has more than six, the form splits into identify and assess and the text says what is typed straight into Excel.
- Every form names the rows a first pass needs and the person who signs it. A form with no signatory is a questionnaire.
- Delete the example row before you photocopy, and transcribe the filled form the same day.
- Nine forms carry a column the repository does not have. Add the column to the sheet and say in the text that you added it. There is no fifth new sheet.
- O56 Project is O30 Work Package twice over, so it gets no form: fill O30 and generate O56.
- Exactly one of the thirty-seven acceptance criteria in O55 Building Block passes today, and it passes for the wrong reason.
- Nineteen of thirty-six interfaces are unmanaged and thirty-one carry no logging, so forms 18-29 and 18-30 are the two with the most work behind them.
- O48 Environment records personal data as absent from production and present in development. Settle that with the information officer before anything else in the technology set.
- Calderon Therapeutics is an invention that exists only inside this repository, and Veldtkruin Group is volume 1's separate invention.

SOURCES FOR THIS CHAPTER

- **1.** Pharmaceuticals_TOGAF_Metamodel.xlsx, sheet 00 Index rows 1 to 79 and sheet ZZ Checks rows 4 to 21 — 105 sheets, header row 4 and first record row 5 on every object sheet, the per-sheet record counts summing to 814 against the 850 stated in row 2
- **2.** Pharmaceuticals_TOGAF_Metamodel.xlsx, header row 4 of all 64 object sheets, EXT Industry Extension, the 25 relationship sheets and the 13 reports — no column anywhere records a validation state, a periodic-review date, a retention clock or a statutory role; EXT rows 5 to 10 declare X01 to X06 and hold no instances
- **3.** Pharmaceuticals_TOGAF_Metamodel.xlsx, sheets O01 Driver rows 5 to 17, O02 Goal rows 5 to 10, O03 Objective rows 5 to 18, O04 Measure rows 5 to 20, O05 Principle rows 5 to 28, O06 Requirement rows 5 to 36, O07 Constraint rows 5 to 12 and O08 Assumption rows 5 to 10 — 13, 6, 14, 16, 24, 32, 8 and 6 records; O05 owned entirely by Architecture Board; MSR-01 at 54, MSR-07 at 22, MSR-13 at 53 in unit count
- **4.** Pharmaceuticals_TOGAF_Metamodel.xlsx, sheets O09 Organization Unit rows 5 to 16, O10 Actor rows 5 to 22, O11 Role rows 5 to 22, O12 Location rows 5 to 12 and O19 Contract rows 5 to 14 — the same eighteen titles in O10 and O11, every Status reading Filled, Scale in people including LOC-08 Validated Cloud Region at 82, and Counterparty holding application names
- **5.** Pharmaceuticals_TOGAF_Metamodel.xlsx, sheets O13 Business Capability rows 5 to 104, O14 Value Stream rows 5 to 9, O15 Business Service rows 5 to 14, O16 Business Process rows 5 to 24, O17 Business Function rows 5 to 14 and O18 Product rows 5 to 8 — C01 at Criticality Low against C01.01.02 at Critical; Stages held as text totalling 24 against O16's 20 rows; every O15 description reading Externally consumable
- **6.** Pharmaceuticals_TOGAF_Metamodel.xlsx, sheet O14 Value Stream column Ref rows 5 to 9 — VS01 to VS05, unhyphenated, which is why the validated-system register is keyed VAL- and not VS-
- **7.** Pharmaceuticals_TOGAF_Metamodel.xlsx, sheets O20 Data Entity rows 5 to 18, O36 Data Entity Relationship rows 5 to 15, O37 Data Attribute rows 5 to 64, O39 Data Migration rows 5 to 10, O40 Source To Target Mapping rows 5 to 16, O42 Metadata Element rows 5 to 18 and O43 Master Data Rule rows 5 to 14 — O43's Golden source disagreeing with O20's Master system on all ten shared entities; ATR-002 Public and Key Yes; MTD-03, MTD-07 and MTD-08 Absent
- **8.** Pharmaceuticals_TOGAF_Metamodel.xlsx, sheets O23 Application Component rows 5 to 16 and O24 Information System Service rows 5 to 16 — ten columns with no validation state, GxP flag or review date; Owner holding organisation units; A10 Critical, Adequate and Retire, A08 Critical, Healthy and Replace, A12 Critical, At risk and Consolidate, A06 High, Poor and Invest; four ISS rows Unpublished
- **9.** Pharmaceuticals_TOGAF_Metamodel.xlsx, sheets R06 Application interfaces with rows 5 to 40, O26 Logical Technology Componen rows 5 to 11, O46 Technology Building Block rows 5 to 11, O47 Reference Model Entry rows 5 to 15, O48 Environment rows 5 to 8, O50 Network Link rows 5 to 14 and O55 Building Block rows 5 to 41 — R06 Managed holding Managed (17) and Unmanaged (19), Logged holding Yes (5) and No (31), six Direct DB read rows; O46 carrying two columns headed Ref; ENV-01 Production recording Personal data present No
- **10.** Pharmaceuticals_TOGAF_Metamodel.xlsx, sheets O29 Gap (empty), O30 Work Package rows 5 to 16, O33 Architecture Contract rows 5 to 12, O52 Implementation Factor rows 5 to 12, O53 Transition Architecture rows 5 to 7, O54 Architecture Alternative rows 5 to 9, O56 Project rows 5 to 16, O62 Delivery Risk rows 5 to 12, O66 Architecture Increment rows 5 to 10, R04, R09, R12 (empty), R13, R15, R21 and R22 — O30 summing to $84.3m against ALT-B at $46m; R15 allocating seven of twelve packages; the index-order joins in R04, R09, R13, R21 and R22
- **11.** Pharmaceuticals_TOGAF_Metamodel.xlsx, sheets O32 Standard rows 5 to 20, O41 Security Control rows 5 to 12, O59 Compliance Assessment (empty), O63 Control Objective rows 5 to 11, O64 Security Policy rows 5 to 8, O65 Enforcement Point rows 5 to 9, R16 Control Objective met by Co rows 5 to 11, RPT03 Capability Heatmap rows 5 to 14 and RPT13 Orphan Analysis rows 5 to 8 — STD-02 Deprecated and Conformant, STD-07 Deprecated and in Breach, STD-04 against STD-15; SEC-01 and SEC-03 Absent; R16.Position a copy of O63.Evidence; every RPT03 gap exactly 1
- **12.** Pharmaceuticals_TOGAF_Metamodel.xlsx, sheets O31 Deliverable rows 5 to 24, O57 Governance Body rows 5 to 9, O58 Governance Gate rows 5 to 8, O60 Governance Decision rows 5 to 11 and O61 Governance Change Request rows 5 to 9 — no approver or date column on O31 with thirteen of twenty reading Approved, no quality body in O57, no date column in O60, and Expires populated on one of five rows in O61
- **13.** Pharmaceuticals — a starter pack for enterprise architects (volume 1), chapter 16, section The retention clocks, and appendix 20, section Retention clock by record class — the verified periods, their triggers and their instruments, from which item 5 of this workbook builds RC-01 to RC-13
- **14.** Pharmaceuticals — a starter pack for enterprise architects (volume 1), chapter 16, section Licensing, and the Responsible Pharmacist, and chapter 13, section The Responsible Pharmacist — the statutory roles and the signatures they carry, none of which appears in O10 Actor or O11 Role

APPENDIX G

The principles and standards starter set

Twenty-four generic principles and sixteen generic standards, judged row by row, plus nine pharmaceutical principles and nine validation standards written in the repository's own columns.

`O05 Principle` holds twenty-four principles and `O32 Standard` holds sixteen standards, and between them they are the two sheets a reader is most tempted to accept as they stand. Do not. The twenty-four are the generic TOGAF set, every one owned by `Architecture Board`, and not one of them mentions validated state, data integrity, patient safety, retention or the regulator [1]. The sixteen are generic IT standards, and **not one of them is a GxP standard** [2]. This item is the starter set you adapt: which of the twenty-four to keep, which to delete and why, nine pharmaceutical principles written in `O05`'s own six columns, and nine standards written in `O32`'s own five with a conformance test each. Transcribe, argue, amend. A principle you cannot lose an argument with is not a principle, and a standard with no conformance test is a filename. The printable forms for both sheets are in item 18 — form 18-11 feeds `O05` and form 18-42 feeds `O32` — so this item supplies the content and that one supplies the paper.

What `O05` holds today, row by row

Six columns — `Ref`, `Principle`, `Category`, `Statement`, `Owner`, `Status` — twenty-four rows, `PRN-01` to `PRN-24`, nineteen `Approved` and five `Draft`, every `Owner` cell reading `Architecture Board`, every `Category` one of `Business`, `Application`, `Data`, `Technology` or `Security` [1]. The verdicts below are the starting position; the first column of each table is the row you are judging.

Principle	Verdict	Why
`PRN-01` Primacy of Principles	**Delete**	A statement about the document, not a design rule. In a regulated company the quality system has primacy and says so in its own manual
`PRN-02` Maximise Benefit to the Enterprise	**Delete**	It loses the argument it most needs to win: a GxP control is a local cost with no enterprise benefit, and this principle votes against it
`PRN-03` Compliance is Designed, Not Added	**Keep**	The best row in the sheet. `PRN-25` to `PRN-33` are what it means in practice
`PRN-04` Business Continuity	**Amend**	Add the recovery objective and the clock it serves. `O49 Resilience Tier` has nine of twelve applications `Never tested` [3]
`PRN-05` Common Use Applications	**Amend**	Add that consolidating two validated systems is a revalidation of both, not a saving
`PRN-06` Technology Independence	**Delete**	False here. The qualified platform version sits inside the validation scope of the system above it
`PRN-07` Ease of Use	**Keep**	A system that is worked around produces a data-integrity finding, which makes this a compliance principle
`PRN-08` Service Orientation	**Amend**	The contract must name the interface's validation scope, not only its schema
`PRN-09` Data is an Asset	**Merge**	Merge into `PRN-13`. "An owner, a definition and a lifecycle" is `PRN-13` with two nouns added
`PRN-10` Data is Shared	**Amend**	Add that a shared copy inherits the source's classification and its retention clock
`PRN-11` Data is Accessible	**Delete**	"Those entitled can reach it without a project" cannot hold across the clinical and commercial boundary, and subject data is the case it fails on
`PRN-12` Common Vocabulary	**Amend**	A controlled vocabulary here is versioned and licensed. See `PRN-30`

Second half, same columns.

Principle	Verdict	Why
PRN-13 Data Ownership is Individual	**Keep**	"An owner is a person, not a committee." The repository breaks it in every Owner column it has [4]
PRN-14 Classification Travels with Data	**Keep**	True, and already broken: O48 Environment is the proof [3]
PRN-15 Requirements-Based Change	**Amend**	Name the artefact: the stated requirement is the user requirement specification
PRN-16 Responsive Change Management	**Delete**	"The estate changes at the pace the business needs" is not true of a validated system, and writing it down licenses the wrong argument
PRN-17 Control Technical Diversity	**Keep**	One approved technology per capability, and here each extra one is an extra qualification to maintain
PRN-18 Interoperability	**Amend**	Standards come from the authority's own data standards catalogue before anyone's preference
PRN-19 Security is Built In	**Keep**	O65 EP-01 Design review gate is the enforcement point, and it reads Newly established [2]
PRN-20 Least Privilege	**Keep**	Draft today. Approve it: SEC-03 Privileged access management reads Absent [2]
PRN-21 Attributable Access	**Keep**	Nearly PRN-26, and keep both — this one governs access, that one governs the record
PRN-22 Defence in Depth	**Keep**	Uncontroversial, cheap to keep, and R16 reducing each obligation to one control is what it forbids [2]
PRN-23 Models are Governed	**Amend**	Add that a model used in a GxP decision is a computerised system and is validated as one
PRN-24 Sustainable by Design	**Delete**	Draft, owned by nobody who can act on it, and no gate tests it. It belongs in a corporate policy

That leaves seventeen of the twenty-four, nine of them amended, and takes the sheet to twenty-six rows once the nine below are added.

The six to delete, and the argument for each

Deleting a principle is a governance act, so take each one to GB-01 Architecture Board with the reason, not with a tidier list.

- **PRN-01 Primacy of Principles** says the principles apply to every part of the enterprise without exception. In a company with a pharmaceutical quality system that is false and harmful: the quality manual, the validation master plan and the licence conditions outrank the architecture principle set, and a principle that claims otherwise invites the first argument you will lose.
- **PRN-02 Maximise Benefit to the Enterprise** says enterprise benefit outranks local optimisation. Every GxP control is a local optimisation with no enterprise benefit — a second review, a signature, an audit trail nobody reads until an inspection. Keep the principle and somebody will cite it to remove one.
- **PRN-06 Technology Independence** says applications are independent of the platform beneath them. A validated application is not. Its qualification covers named operating system, database and middleware versions, and a platform upgrade is a change-control record before it is a patch. Replace the row with the honest statement, which is that platform change is inside validation scope.
- **PRN-11 Data is Accessible** says those entitled to data can reach it without a project. The estate's largest data class is trial participants' records, and reaching those requires a lawful basis, a protocol, an ethics position and often a project. Delete the row and let PRN-27 carry the access question.
- **PRN-16 Responsive Change Management** says the estate changes at the pace the business needs. The pace is set by change control, validation and release, and the gap between the two statements is where every unrealistic programme date in this industry comes from [5].
- **PRN-24 Sustainable by Design** is Draft, has no owner who can act on it, is tested by no gate in O58 Governance Gate and duplicates a corporate reporting obligation that DRV-13 already carries [1]. A principle no gate tests is decoration.

PRN-09 Data is an Asset is not a deletion but a merge: fold its definition-and-lifecycle clause into PRN-13 Data Ownership is Individual, which is the row that does the work, and free the reference for reuse. Record all seven changes as one decision in O60 Governance Decision rather than seven, because the board will approve the set or none of it.

The pharmaceutical principle set

Nine rows, written in O05's own six columns so that they transcribe straight in. Add the sixth Category value, Regulatory, before you paste them, or they land under Business and read as aspiration. PRN-25 to PRN-27 are written by item 2 of this workbook, in Preliminary, because the phase cannot proceed without them; they are repeated here so that the set transcribes as one block. Owner is Architecture Board on all nine, because that is what every other row in the sheet says, and Status is Draft until the board approves them — so those two columns are stated here once rather than repeated in the tables. Note that Owner means the approving body, which is a different thing from the named individual PRN-31 demands. The six columns are split across two tables below because Statement is a sentence and the rest are labels.

Ref	Principle	Category
PRN-25	Validated State is Maintained	Regulatory
PRN-26	Integrity is a Property of the Record	Regulatory
PRN-27	The Patient's Data is the Most Protected Class	Regulatory
PRN-28	Retention Runs From an Event	Regulatory
PRN-29	An Interface Is Inside a Validation Scope	Regulatory
PRN-30	The Dictionary Version Is Part of the Datum	Regulatory
PRN-31	An Owner Is a Named Person	Regulatory
PRN-32	A Control Objective States a Control	Regulatory
PRN-33	Non-Production Holds No Patient Data	Regulatory

The `Statement` column of the same nine rows, keyed by `Ref`. First five.

Ref	Statement
PRN-25	A validated system stays validated only while its reviews and change records are current
PRN-26	The record carries its own attribution, timestamp and history; a report cannot restore them
PRN-27	Subject data outranks every other classification in the estate
PRN-28	A retention period is computed from a recorded event, not set as a number of years
PRN-29	Every interface into or out of a validated system belongs to one named validation scope

Last four, same column.

Ref	Statement
PRN-30	A coded value is incomplete without the version of the vocabulary that produced it
PRN-31	Every owner cell holds one named individual with a deputy, never a unit and never a committee
PRN-32	A control objective names what the control does; the obligation restated is not a control objective
PRN-33	Personal data leaves production only masked, and the masking is evidenced for every refresh

Principle	What a gate can test, and where
PRN-25 Validated state is maintained	O67 Next review due — none in the past
PRN-26 Integrity is a property of the record	RPT05 Lineage — no entity reads Absent
PRN-27 Patient data is the most protected class	O20 E03 Subject in the highest class
PRN-28 Retention runs from an event	O69 Clock starts at — an event
PRN-29 Interfaces sit inside a validation scope	R06 Validation scope — no empty cell
PRN-30 Dictionary version is part of the datum	O37 a version attribute per coded value
PRN-31 An owner is a named person	O23 System owner — a person, not a unit
PRN-32 A control objective states a control	O63 no cell restating the obligation
PRN-33 Non-production holds no patient data	O48 and SEC-07 — masking In place

Figure 19.1 Nine principles and the nine tests that make them governable. A principle a gate cannot test is decoration, so each row's right-hand column names a sheet, a column and the value the gate reads. Transcribe the left column into `O05 Principle` and the right into `O32 Standard`'s added `Conformance test` column.

Each of the nine now gets the treatment a principle needs to survive its first argument: the statement, why it holds, what follows from it, and how you would know it was being followed. The last of those four is the one most principle sets leave out, and it is the only one a gate can use. **Look it up.** Volume 1 chapter 13, section *The system itself is regulated*, is the teaching half of all nine.

PRN-25 Validated State is Maintained

Statement. A validated system stays validated only while its reviews and change records are current.

Rationale. Validation is widely treated as a project deliverable: a summary report, signed, filed, done. It is a state, and the state decays. Volume 1 is explicit that periodic review is the mechanism that keeps it true, and that decommissioning is part of the same life cycle rather than a tidy-up afterwards [5]. The repository you were handed has no column for a validation state anywhere, which means it cannot represent the decay even in principle [4].

Implications.

- `O67 Validated System` exists and carries `Validation state`, `Last periodic review` and `Next review due` for every GxP system.
- A change to a validated system raises a change-control record before it reaches a backlog.
- An overdue periodic review is a finding with a name against it, not a scheduling problem.

How you would know. Every row of `O67` has a `Next review due` in the future, and `O68 Periodic Review Schedule` has no row whose `Next due` is in the past. The measure is the percentage of `O67` rows with a current review, reported monthly in `O04 Measure`.

PRN-26 Integrity is a Property of the Record

Statement. The record carries its own attribution, timestamp and history; a report cannot restore them.

Rationale. Data integrity is bought at the point of capture or it is not bought at all. The attributes volume 1 sets out — attributable, legible, contemporaneous, original, accurate, and the additions after them — are properties of the record, and a reporting layer that aggregates records cannot add any of them back [6]. The repository's own `O42 Metadata Element` records lineage as `Absent`, which is the same failure stated as metadata coverage [4].

Implications.

- An audit trail is a design input for every GxP system, not a configuration option.
- A report is evidence only where it names the records behind it and the version of the query.
- A spreadsheet that recalculates a GxP result is a computerised system and is treated as one.

How you would know. For any figure in a regulatory report, you can reach the source record, its author, its timestamp and its change history in three steps, and you can do it in front of an inspector. `RPT05 Data Ownership` shows no entity at `Lineage: Absent`.

PRN-27 The Patient's Data is the Most Protected Class

Statement. Subject data outranks every other classification in the estate.

Rationale. A trial participant's pseudonymised record is special personal information under the national data protection statute and the subject of the source-data rules in the clinical practice guidelines [7]. The repository classifies `E03 Subject` as `Internal` — the same class as a protocol — while `E08 Adverse Event` is the only `Confidential` row in the sheet [4]. That single cell is the clearest test of whether a classification scheme was written by someone who knew the industry.

Implications.

- Subject data gets the estate's highest class, and the class travels into every copy, extract and analysis dataset.
- Access is by protocol and lawful basis, not by seniority.
- The clinical and commercial boundary is an architectural control, not an agreement between teams.

How you would know. `O20 Data Entity` records `E03 Subject` in the highest class, every `O37 Data Attribute` row for that entity matches or exceeds it, and no analysis environment holds subject data outside the approved basis. **Look it up.** Volume 1 chapter 9, section *Subject — the most sensitive record the company holds*.

PRN-28 Retention Runs From an Event

Statement. A retention period is computed from a recorded event, not set as a number of years.

Rationale. Volume 1's verified clocks run from three years to forty and three of the longest are triggered by events — batch expiry, trade-item expiry, final trial closure or discontinuation of development [8]. You cannot evaluate "one year after batch expiry" without the batch expiry date, so retention is a join across operational data rather than a policy setting. A single corporate period is either unlawfully short for the longest clock or needlessly expensive for everything else, and in both cases unimplementable.

Implications.

- `O69 Retention Clock` carries `Clock starts at` as an event and `System of record` as the system that can supply that event's date.
- Where an instrument states no period, the cell reads `No period stated` and a working period is assigned with its basis recorded.
- An archive format must outlive the application: four or five times over, on the longest clocks.

How you would know. Every entity in `O20` maps to exactly one `O69` row, every `O69` row names an instrument or an explicit absence, and for one sampled record you can compute its disposal date from live data rather than from a policy document.

PRN-29 An Interface Is Inside a Validation Scope

Statement. Every interface into or out of a validated system belongs to one named validation scope.

Rationale. Volume 1 states the failure plainly: interfaces are routinely in nobody's validation scope [9]. The repository demonstrates it. Nineteen of thirty-six interfaces are `Unmanaged`, thirty-one carry no logging, and six are direct database reads into another application — including `INT-009 EDC → Safety Database`, the serious adverse event path, which is an unmanaged hourly file [10]. A direct read is outside the target's interface contract and inside its validation scope, which is the worst of both positions.

Implications.

- `R06 Application interfaces with` carries `Validation scope` and `Interface specification` on every row.
- A new interface to a GxP system is a change to that system's validated state.
- A dispensation for a point-to-point transfer carries an expiry and a named condition for removal.

How you would know. No row of `R06` has an empty `Validation scope`, no GxP interface reads `Unmanaged`, and the interface count on `RPT04 Interface Catalogue` equals the count of interfaces inside a scope.

PRN-30 The Dictionary Version Is Part of the Datum

Statement. A coded value is incomplete without the version of the vocabulary that produced it.

Rationale. The safety and clinical dictionaries are licensed products on fixed release cycles — one is released twice yearly and the other on the first of March and the first of September [11] — and a term's position, and sometimes its existence, changes between releases. A coded event stored without its dictionary version cannot be re-derived, re-aggregated or defended, and a signal detected under one version is not the same signal under the next. Volume 1 treats this as a modelling problem, not an operational one [12].

Implications.

- Every coded attribute is stored with the version that coded it, and `O37 Data Attribute` carries that attribute explicitly.
- Recoding is a dated, recorded activity with its own evidence, not a background job.
- Licence scope is an architectural constraint: a contract research organisation needs its own licence, and a non-production environment may not be permitted to hold the dictionary at all.

How you would know. For any coded value in the safety database you can state the dictionary version that produced it, and an aggregate report names the version it was run under. Never print the term list itself; the licence does not allow it. **Look it up.** Volume 1 chapter 9, section *The dictionary-version problem*.

PRN-31 An Owner Is a Named Person

Statement. Every owner cell holds one named individual with a deputy, never a unit and never a committee.

Rationale. The repository holds four kinds of owner and none of them is a person. `O23 Application Component`'s `Owner` holds organisation units, and the wrong ones — the safety database under Legal, the manufacturing execution system under Internal Audit, serialisation under Human Resources. `O11 Role`'s `Performed by` reads `Person` on all eighteen rows, which is a type, not a name. `O05`'s own `PRN-13` already says an owner is a person, not a committee, and the sheet that states the principle is the sheet that breaks it [1][4]. In an industry where a release decision is signed by a named individual with a legal duty, this is not a cosmetic gap.

Implications.

- Every `Owner`, `System owner`, `Head` and `Reviewer` cell holds a name and a deputy.
- `O70 Regulated Role` holds the statutory roles with the person, the deputy and what they sign for.
- A vacancy is recorded as a vacancy, not as the unit that would fill it.

How you would know. No owner column in the repository contains an organisation unit or the string `Person`, and for any sampled row somebody can telephone the owner inside a working day.

PRN-32 A Control Objective States a Control

Statement. A control objective names what the control does; the obligation restated is not a control objective.

Rationale. All seven rows of `O63 Control Objective` read `Control that evidences <the obligation restated>` — the obligation with four words in front of it [2]. That is a tautology table, and it passes every automated check because it is internally consistent. Worse, `R16 Control Objective met by Co` maps each obligation to exactly one `SEC-` control, which reduces a compliance regime to an access control, and its `Position` column is a copy of `O63`'s `Evidence` rather than a reading of `O41 Security Control` — the two disagree on four of seven [2].

Implications.

- A control objective states a testable control: what is prevented or detected, where, and by what.

- An obligation may need several controls, and one control may serve several obligations.
- `Evidence` cites an assessment in `O59 Compliance Assessment`, not an opinion.

How you would know. No cell in `O63` begins with the phrase *Control that evidences*, every row cites at least one assessment, and `R16.Position` is derived from `O41.Position` rather than typed.

PRN-33 Non-Production Holds No Patient Data

Statement. Personal data leaves production only masked, and the masking is evidenced for every refresh.

Rationale. `O48 Environment` records `ENV-01 Production` as holding no personal data and `ENV-02 Pre-production` and `ENV-04 Development` as holding it [3]. Read literally, the company keeps trial participants' records in development and none in production. Two sheets say which reading is live: `SEC-07 Non-production masking` is `Partial`, and `O65 EP-05 Non-production provisioning` reads *most refreshes are unmasked* [2]. That is a data protection exposure with a named enforcement point and no policy attached to it, because `R17` leaves `EP-05` in no relationship at all.

Implications.

- Masking is enforced at provisioning, and an unmasked refresh is a recordable event.
- The subject-data class travels into every lower environment or the data does not go there.
- A validated system's test environment is inside its validation scope, so the masking method is itself verified.

How you would know. `O48` records personal data as present only where a lawful basis exists, `SEC-07` reads `In place`, and every environment refresh in the last quarter has a masking record against it.

What `O32 Standard` holds today, row by row

Five columns — `Ref`, `Standard`, `Category`, `Status`, `Conformance` — sixteen rows, categories `Integration`, `Data`, `Security`, `Platform`, `Delivery` and `Governance`, `Status` one of `Current`, `Emerging` or `Deprecated`, and `Conformance` one of `Conformant`, `Partial`, `Breach` or `Not assessed` [2]. `RPT12 Standards Conformance` is this sheet copied, which makes it one of the three reports in the file that end on a record rather than a sentence. Not one of the sixteen is a GxP standard: there is no computerised-system validation standard, no electronic records standard, no data-integrity standard and no audit-trail-review standard.

Standard	Status · Conformance	Verdict
`STD-01` API design standard	Current · Breach	Keep. A breach against a current standard is an honest row
`STD-02` Event schema standard	Deprecated · Conformant	**Reinstate or clear.** A deprecated standard cannot be conformed to, because it is not in force
`STD-03` Data classification standard	Current · Not assessed	Assess it first. `O20` and `O37` are the test, and both fail it today
`STD-04` Data retention standard	Current · Conformant	**Reconcile with `STD-15`.** Retention here runs from events and differs by record class, so one conformant retention standard is improbable
`STD-05` Identity and access standard	Emerging · Breach	Keep. `SEC-01` and `SEC-03` read `Absent`, so the breach is right
`STD-06` Encryption standard	Emerging · Not assessed	Assess. `SEC-08 Data-at-rest encryption` is `Partial` and mapped to no obligation
`STD-07` Logging and monitoring standard	Deprecated · Breach	**Reinstate.** Thirty-one of thirty-six interfaces carry no logging; this is the one standard that should be current and in breach
`STD-08` Cloud landing zone standard	Deprecated · Partial	Amend. A qualified landing zone is validation scope, so this is a validation standard in disguise

Second half, same columns.

Standard	Status · Conformance	Verdict
`STD-09` Container image standard	Current · Breach	Keep
`STD-10` Database standard	Deprecated · Not assessed	Retire or reinstate. `O22 Physical Data Component` types four GxP stores as `File` [3]
`STD-11` Network segmentation standard	Current · Not assessed	Assess. Six of ten links in `O50 Network Link` read `Diverse path: No` [3]
`STD-12` Coding and review standard	Emerging · Partial	Amend. In a validated estate, code review evidence is validation evidence
`STD-13` Environment standard	Deprecated · Conformant	**Reinstate.** `O48 Environment` 's personal-data column is the reason, and `PRN-33` is the principle
`STD-14` Naming convention	Current · Partial	Keep
`STD-15` Records management standard	Deprecated · Conformant	**Reconcile with `STD-04` .** One of the two is the retention standard; the other is withdrawn
`STD-16` Supplier security standard	Deprecated · Breach	**Reinstate and widen** into `STD-24` below, which covers quality as well as security

Three defects to settle before you add a row. **`STD-02` is `Deprecated` and `Conformant`** and **`STD-07` is `Deprecated` and in `Breach`** : a withdrawn standard can be neither, because conformance is measured against something in force, and a verdict against a withdrawn standard teaches every reader that the column means nothing. **`STD-04 Data retention standard` is `Current` and `Conformant` while `STD-15 Records management standard` is `Deprecated` and `Conformant`** , and the two cover the same obligation from different sides. And `R18 Standard applies to Applica` ties each standard to the odd-numbered applications only, so **no application marked `Critical` in `O23` has a single standard applied to it** [2].

The nine standards a pharmaceutical estate needs

Add the `Validation` category, then these nine rows. Items 2 and 19 of this workbook share the reference space: item 2 creates `STD-17` to `STD-21` in the Preliminary phase, because the gate regime needs them on day one, and the four from `STD-22` are added here. `Status` is `Emerging` and `Conformance` is `Not assessed` on all nine, because both values are already in use and both are honest on day one. Then add the column `O32` does not have and cannot do without: a `Conformance test` , which is a sentence naming a sheet, a column and a value.

Ref	Standard	Category	Status	Conformance
STD-17	Computerised system validation standard	Validation	Emerging	Not assessed
STD-18	Electronic records and signatures standard	Validation	Emerging	Not assessed
STD-19	Data integrity standard	Validation	Emerging	Not assessed
STD-20	Audit trail review standard	Validation	Emerging	Not assessed
STD-21	Periodic review standard	Validation	Emerging	Not assessed

Second half, same columns.

Ref	Standard	Category	Status	Conformance
STD-22	Data migration verification standard	Validation	Emerging	Not assessed
STD-23	Dictionary version management standard	Validation	Emerging	Not assessed
STD-24	Supplier and service-provider assessment standard	Validation	Emerging	Not assessed
STD-25	Decommissioning and archive standard	Validation	Emerging	Not assessed

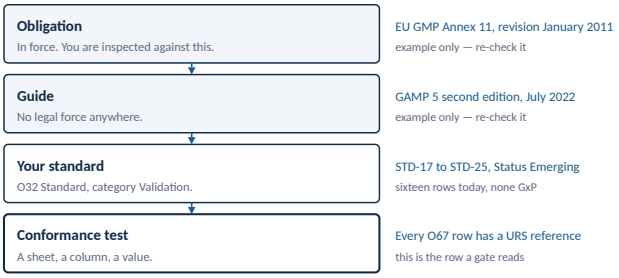


Figure 19.2 Four layers, and only the bottom one can be failed at a gate. The obligation is what you are inspected against; the guide has no legal force; your own standard is the row in `O32 Standard`; the conformance test is the only one of the four that names a sheet, a column and a value. The two versions shown are examples carried from volume 1 — re-check both before printing them.

> **WRITE NO VERSION NUMBER FROM MEMORY**
>
> Every version fact below is the one volume 1 cited, with volume 1's own source and date, and where volume 1 says a version could not be sourced this item says so and prints nothing. That is not caution for its own sake: volume 1 found a national guideline superseded inside its own stated validity window, and found a standards body's publications listing that does not mark superseded entries at all [19][16]. Follow volume 1's own procedure before you put any of these into a client document [20].

STD-17 Computerised system validation standard

What it governs. How a GxP computerised system is specified, risk-assessed, verified and released for use, and what evidence each step leaves behind. It is the standard the validation master plan and every validation plan cite.

What volume 1 cites. The guide everyone names is **GAMP 5, second edition, published July 2022, 404 pages, by ISPE through its GAMP community of practice** — ISPE's own publication page is volume 1's source [13]. Three corrections volume 1 insists on. **The guide has no legal force anywhere**: the obligation is to the regulation, and an architect who says "GAMP 5 requires..." is wrong in a way an inspector notices. **There is no GAMP Category 2** — it covered firmware and was withdrawn — and volume 1 states that the current category scheme in the second edition is **unverified**, so print no category numbers [5]. And the second edition does not mandate the V-model; it states that specification and verification can be iterative. For qualification and validation recommendations volume 1 cites **PIC/S PI 006-4, dated 1 October 2026** [16]. The United States computer software assurance guidance **went final in September 2025, docket FDA-2022-D-0795**, and is a device guidance under the device quality regulation, not drug good manufacturing practice — adopt its thinking by analogy and say that you are doing so [18].

Conformance test. Every row of `O67 Validated System` has a `URS reference` and a `Validation state` that is not blank, and every GxP work package in `O30` names a validation plan in its architecture contract in `O33`.

Where it is tested. `GT-1 G1 Design` for the specification and risk assessment, `GT-2 G2 Build complete` for the verification evidence, and the release gate item 10 adds as `GT-4`.

STD-18 Electronic records and signatures standard

What it governs. When an electronic record may stand in place of paper, what an electronic signature must carry, and how printouts and hybrid records are controlled.

What volume 1 cites. Two instruments, and they are not interchangeable. **The United States rule on electronic records and signatures has been in force since 1997**, its scope was narrowed by guidance in 2003, and the final guidance on electronic systems, records and signatures in clinical investigations issued in **October 2024** under docket FDA-2017-D-1105 [15]. Volume 1 states that it did **not re-confirm the current text of the rule itself**, so no clause numbers appear here. **The European computerised-systems annex is in force in its revision of January 2011**, on the European Commission's own index, which carries no adoption or entry-into-operation entry for any revision [14]. A draft was consulted jointly from **7 July to 7 October 2025**, alongside a revised documentation chapter and a **new annex on artificial intelligence**, and **no final text and no date have been published** [17]. One sentence to carry into every vendor conversation: **no product can be "Part 11 compliant"**, because the rule applies to records and signatures, not to software [5].

Conformance test. Every `O67` row for a system that holds a GxP record names the signature mechanism it uses, and no record class in `O69 Retention Clock` is held only as a printout.

Where it is tested. `GT-1` for the design, and `O59 Compliance Assessment` for the evidence per building block.

STD-19 Data integrity standard

What it governs. The properties a GxP record must have at the point of capture, and the controls that keep them: attribution, contemporaneity, originality, accuracy, legibility and the rest.

What volume 1 cites. The guidance stack, in volume 1's order of usefulness: **PIC/S PI 041-1, "Good Practices for Data Management and Integrity in Regulated GMP/GDP Environments", dated 1 July 2021, 63 pages**, which is the final document and not its 2016 or 2018 drafts [16]; and the United Kingdom regulator's GXP data integrity guidance, **Revision 1, March 2018** [6]. Two corrections. **The manufacturing guide's Part I never uses the term "ALCOA"** in the edition volume 1 read — **PE 009-17, dated 25 August 2023** — so a slide citing "Chapter 4 ALCOA" is citing something that is not there in those words [16]. And the verified South African statement of the mnemonic is **ALCOA-CCEA, in the national good clinical practice guidelines, third edition**, which makes it a clinical requirement rather than a manufacturing one [6][7].

Conformance test. `O42 Metadata Element` records no element at `Coverage: Absent` for lineage, steward or consent basis, and `RPT05 Data Ownership` shows no entity at `Lineage: Absent`.

Where it is tested. `GT-1` for the design input, `GB-08 Data Integrity Forum` for the standing review, and `RPT05` monthly. **Look it up.** Volume 1 chapter 13, section *Audit-trail review, the most under-resourced obligation in the regime*.

STD-20 Audit trail review standard

What it governs. Which audit trails are reviewed, by whom, on what risk-based frequency, against what, and what the review leaves behind.

What volume 1 cites. Volume 1 calls audit-trail review **the most under-resourced obligation in the regime** [6]. The position is moving: the 2025 draft of the European computerised-systems annex is reported, by the consultancy source volume 1 read and marked as such, to make audit trails mandatory wherever users can change data and to expect the review to be **normally completed before batch release** [17]. That is a draft. **It is not in force, it has no published date, and volume 1 says so explicitly** — so design for the stricter reading and do not cite it as law. The reviewable surface in this estate is already larger than the review capacity: six direct database reads and thirty-one unlogged interfaces [10].

Conformance test. Every `O67` row has `Audit trail reviewed` holding a date and a scope, not a yes, and no GxP system's last review is older than the frequency its risk assessment set.

Where it is tested. `GT-4` before release, and `O68 Periodic Review Schedule` thereafter.

STD-21 Periodic review standard

What it governs. The cycle that keeps a validated state true: frequency per system, what the review examines, who performs it, and what happens when the outcome is not "maintained".

What volume 1 cites. Periodic review is a lifecycle activity rather than a project closedown task, and volume 1 treats it as the mechanism by which validation stays true [5]. The 2025 draft annex is reported to carry a twelve-item periodic review; again, a draft [17]. The adopted manufacturing guide is where the related annual cycle lives: volume 1 cites the **product quality review as normally annual, at §1.10 of PE 009-17** [16], and notes that **PE 009-18 was adopted 22 September 2026 and came into force 24 September 2026**, which volume 1 did not read — so re-check the clause before you cite it [16]. The national guideline is not the place to look: volume 1 records that **SAHPGL-INSP-02 version 9 of 17 November 2025 and version 10 of 13 April 2026 contain no computerised-systems content, no reference to the computerised-systems annex and no data-integrity section** [19].

Conformance test. `O68` has a row for every `O67` row, no `Next due` is in the past, and every `Outcome` of `Not maintained` has a quality record against it.

Where it is tested. `GB-07 Validation Committee` monthly, and `O04 Measure` as the percentage of systems with a current review.

STD-22 Data migration verification standard

What it governs. How data is moved between GxP systems, and what evidence shows that **value and meaning** both survived the move.

What volume 1 cites. Volume 1 is direct about the failure mode: migrating a result without its specification and method version destroys its meaning, and this is the most expensive mistake in pharmaceutical system replacement [5]. The European annex's project-phase clause is the obligation behind it [14]. What volume 1 asks you to record is the set of facts a migration report must carry: which records moved, when, under what script version, with what verification outcome, and what happened to the ones that failed. The repository shows what happens without a standard: `O39 Data Migration`'s six rows move a compound from the trial management system to the safety database and a subject from the safety database to the trial master file, and neither pairing is possible [4].

Conformance test. Every row of `O39` names a verification approach and a signatory, and every field in `O40 Source To Target Mapping` has a rule that is not the word `Direct` by default.

Where it is tested. `GT-2` for the migration evidence, and `O59` per building block.

STD-23 Dictionary version management standard

What it governs. How licensed controlled vocabularies are versioned, upgraded, recorded against coded data, and licensed across the company and its partners.

What volume 1 cites. The safety dictionary is maintained under the international harmonisation council and **released twice yearly**; volume 1 cites the maintenance organisation's own "what's new" material for **version 29.0, released March 2026**, with training material for the following release dated **21 August 2026**, and notes it is a **licensed subscription whose fee bands could not be sourced** [18]. The drug dictionary is maintained by the Uppsala Monitoring Centre, **released on 1 March and 1 September**, covering products from nearly 150 countries, delivered as character-separated files under subscription, and **the United States authority mandates it for studies starting after 15 March 2019** — which volume 1 calls the clearest example available of a regulator making a paid subscription product compulsory [11]. Volume 1 treats the versioning consequence as a data-model problem [12]. **Never reproduce a hierarchy, a term list or a category table from either**; the licence does not permit it and this volume prints none.

Conformance test. Every coded attribute in `O37 Data Attribute` has a companion version attribute, and `O19 Contract` holds a dictionary licence row for the company and for every partner that codes on its behalf.

Where it is tested. `GT-1` for the design, and `GB-03 Data Governance Council` at each release.

STD-24 Supplier and service-provider assessment standard

What it governs. How a vendor, a hosting provider, a contract manufacturer or a contract research organisation is assessed before it is relied on, and what the agreement must carry when the service sits inside a validated scope.

What volume 1 cites. The second edition of the GAMP guide shifted emphasis onto **service providers and supplier documentation**, which is one of the reasons volume 1 cites the ISPE page for the edition's own description [13]. Volume 1's comparison of the two computerised-systems instruments is the practical point: a system designed for the United States rule usually needs **supplier agreements, periodic evaluation, business continuity and the printout clause** added for the European annex [5][14]. And the licensed-dictionary consequence arrives during contract negotiation and is always late: a partner that codes on your behalf needs its own licence [11]. The repository's `O19 Contract` holds ten vendor agreements and **no quality agreement, no contract manufacturing agreement and no clinical trial agreement** [4].

Conformance test. Every `O67` row whose system is hosted or operated by a third party names the agreement in `O19 Contract`, and every such agreement has a dated assessment behind it.

Where it is tested. `GT-0 G0 Charter` for the contract existence, and `GB-07` for the validation implications.

STD-25 Decommissioning and archive standard

What it governs. How a GxP system is retired: what is exported, in what format, how long it stays readable, who approves disposal, and what evidence the retirement itself leaves.

What volume 1 cites. Volume 1 gives decommissioning its own section and the title says the finding: nobody budgets it [5]. The reason it cannot be deferred is the clock arithmetic. Volume 1's verified retention periods run from three years to forty, and the longest sit on the least glamorous data, so **the obligation outlives the application four or five times over** — which makes the requirement an **export format carrying the audit trail and the dictionary version, readable without the source application** [8]. The repository shows the gap: `A10 Serialisation` is `Critical` / `Adequate` and marked `Retire` with no successor and no plan named anywhere [4], while the traceability clock runs to five years after the expiry of a trade item [8].

Conformance test. Every `O67` row has `Decommissioning plan` holding a document reference or an explicit `None`, and no `O23` row marked `Retire` has `None` against it.

Where it is tested. `GB-07 Validation Committee` at approval, `GB-08 Data Integrity Forum` for the archive readability test, and `O69 Retention Clock` for the disposal authority. **Look it up.** Volume 1 chapter 13, section *Decommissioning, which nobody budgets*.

Before this goes in front of a client

Two jobs, and both are recurring rather than one-off. **Re-check every version.** Volume 1's own procedure is seven steps and the first is the one most often skipped: go to the publishing body's own page for that standard, not to a search result, and record the URL and the retrieval date in the same cell as the fact [20]. The national guideline index is a standing job rather than a task: volume 1 found two generations of the same manufacturing guideline inside one five-year validity window, and its two research passes disagreed about which was operative [19].

And re-check the conformance tests, not the standards. A standard with no test is the defect this item exists to fix, and a test that names no sheet and no column is the same defect wearing a sentence. Nine standards, nine tests, each one a query somebody can run against the repository on a Friday afternoon — that is the whole ambition. Calderon Therapeutics, whose sheets every test above names, is an invention that exists only inside this repository; volume 1's Veldtkruin Group is a separate invention, and their data never mix.

THE CHAPTER IN TEN LINES

- O05 Principle holds the twenty-four generic TOGAF principles, all owned by Architecture Board, and none of them mentions validated state, data integrity, patient safety, retention or the regulator.
- Delete six: Primacy of Principles, Maximise Benefit to the Enterprise, Technology Independence, Data is Accessible, Responsive Change Management and Sustainable by Design. Each loses an argument you need to win.
- Merge Data is an Asset into Data Ownership is Individual, and amend eight more. Seventeen of the twenty-four survive.
- Add the Regulatory category to O05 before you paste anything, or the new rows land under Business and read as aspiration.
- Nine pharmaceutical principles, PRN-25 to PRN-33, written in O05's own six columns. Items 2 and 19 share the reference space.
- Every principle carries a statement, a rationale, its implications and a test. The test is the part most principle sets leave out and the only part a gate can use.
- O32 Standard holds sixteen generic IT standards and not one GxP standard.
- STD-02 is Deprecated and Conformant, STD-07 is Deprecated and in Breach, and STD-04 contradicts STD-15 on status. A withdrawn standard can carry no verdict.
- Nine validation standards, STD-17 to STD-25, each with a conformance test that names a sheet, a column and a value.
- Add the Conformance test column to O32. A standard with no test is a filename.
- Never state a version from memory. Every version here is the one volume 1 cited, with volume 1's source and date, and where volume 1 could not source one, nothing is printed.
- GAMP 5 is a guide with no legal force, no product can be compliant with an electronic-records rule, and the 2025 draft of the computerised-systems annex is a draft with no published date.

SOURCES FOR THIS CHAPTER

- **1.** Pharmaceuticals_TOGAF_Metamodel.xlsx, sheet O05 Principle, header row 4 and rows 5 to 28 — twenty-four records, columns Ref, Principle, Category, Statement, Owner and Status; every Owner cell reading Architecture Board; nineteen Approved and five Draft (PRN-05, PRN-10, PRN-15, PRN-20, PRN-24); categories Business, Application, Data, Technology and Security
- **2.** Pharmaceuticals_TOGAF_Metamodel.xlsx, sheets O32 Standard rows 5 to 20, O41 Security Control rows 5 to 12, O63 Control Objective rows 5 to 11, O65 Enforcement Point rows 5 to 9, R16 Control Objective met by Co rows 5 to 11, R17 rows 5 to 8, R18 Standard applies to Applica rows 5 to 20 and RPT12 Standards Conformance rows 5 to 20 — STD-02 Deprecated and Conformant, STD-07 Deprecated and Breach, STD-04 against STD-15; every O63 control objective reading Control that evidences the obligation restated; R16.Position a verbatim copy of O63.Evidence; R18 pairing odd-numbered applications only
- **3.** Pharmaceuticals_TOGAF_Metamodel.xlsx, sheets O22 Physical Data Component rows 5 to 16, O48 Environment rows 5 to 8, O49 Resilience Tier rows 5 to 16 and O50 Network Link rows 5 to 14 — ENV-01 Production recording Personal data present No against ENV-02 and ENV-04 recording Yes; nine of twelve applications Never tested; six of ten links Diverse path No
- **4.** Pharmaceuticals_TOGAF_Metamodel.xlsx, sheets O11 Role rows 5 to 22, O19 Contract rows 5 to 14, O20 Data Entity rows 5 to 18, O23 Application Component rows 5 to 16, O37 Data Attribute rows 5 to 64, O39 Data Migration rows 5 to 10 and O42 Metadata Element rows 5 to 18 — E03 Subject classified Internal; Owner holding organisation units; Performed by reading Person on all eighteen rows; MTD-03, MTD-07 and MTD-08 at Coverage Absent; MIG-01 and MIG-03 pairing systems that cannot hold the entity
- **5.** Pharmaceuticals — a starter pack for enterprise architects (volume 1), chapter 13: Computerised system validation, as it is actually practised; Periodic review; Decommissioning, which nobody budgets; GAMP 5, with the version facts right; 21 CFR Part 11 and EU GMP Annex 11 — including the no-legal-force position, the withdrawal of GAMP Category 2 with the current scheme marked unverified, and the comparison of the two instruments
- **6.** Pharmaceuticals — a starter pack for enterprise architects (volume 1), chapter 13, sections Data integrity and Audit-trail review, the most under-resourced obligation in the regime — citing PIC/S PI 041-1 of 1 July 2021 (63 pages) and the MHRA GXP Data Integrity Guidance and Definitions, Revision 1, March 2018, and recording ALCOA-CCEA as the verified South African statement, in the South African Good Clinical Practice guidelines, third edition, §5.10.1.2 https://assets.publishing.service.gov.uk/government/uploads/system/uploads/attachment_data/file/687246/MHRA_GxP_data_integrity_guide_March_edited_Final.pdf
- **7.** Pharmaceuticals — a starter pack for enterprise architects (volume 1), chapter 16, section POPIA and health data, and the 2026 health-data regulations — health information as special personal information; the regulations signed 27 February 2026, Gazette 54268, Notice 7198, in force 6 March 2026 with no transitional period, and volume 1's warning that whether a manufacturer, authorisation holder or sponsor falls within the eight covered categories is unverified
- **8.** Pharmaceuticals — a starter pack for enterprise architects (volume 1), chapter 16, section The retention clocks, and appendix 20, section Retention clock by record class — periods from three years to forty; batch documentation one year after expiry or five years after certification, whichever is longer; trial essential documents at least ten years after final closure; traceability records up to five years after the expiry of a trade item
- **9.** Pharmaceuticals — a starter pack for enterprise architects (volume 1), chapter 12, section Interfaces are routinely in nobody's validation scope
- **10.** Pharmaceuticals_TOGAF_Metamodel.xlsx, sheet R06 Application interfaces with, rows 5 to 40 — 36 rows; Managed holding Managed (17) and Unmanaged (19); Logged holding Yes (5) and No (31); six rows of Type Direct DB read, all unlogged; INT-009 EDC to Safety Database an unmanaged hourly file
- **11.** Pharmaceuticals — a starter pack for enterprise architects (volume 1), appendix 22, section The licensed dictionaries — the safety dictionary released twice yearly under the international harmonisation council's maintenance organisation, and WHODrug Global from the Uppsala Monitoring Centre released 1 March and 1 September, covering products from nearly 150 countries, delivered as UTF-8 character-separated files under subscription, with the United States authority mandating it for studies starting after 15 March 2019 https://who-umc.org/whodrug/whodrug-global/what-is-whodrug-global/
- **12.** Pharmaceuticals — a starter pack for enterprise architects (volume 1), chapter 9, sections The dictionary-version problem and Controlled vocabularies as architectural dependencies
- **13.** ISPE, GAMP 5 Guide 2nd Edition publication page — published July 2022, 404 pages, produced by the ISPE GAMP Community of Practice, with its emphasis on service providers and supplier documentation and its statement that specification and verification need not be linear; the page does not describe the software category scheme. Cited by volume 1 chapter 13 and appendix 22 https://ispe.org/publications/guidance-documents/gamp-5-guide-2nd-edition
- **14.** European Commission, EudraLex Volume 4 — EU GMP guidelines index, listing Annex 11 Computerised Systems, revision January 2011, with no draft, consultation, adoption or entry-into-operation entry for a revision. Cited by volume 1 chapter 13 and appendix 22 https://health.ec.europa.eu/medicinal-products/eudralex/eudralex-volume-4_en
- **15.** US Federal Register, Vol. 89 No. 191, 2 October 2024 — notice of availability of the final guidance Electronic Systems, Electronic Records, and Electronic Signatures in Clinical Investigations: Questions and Answers, docket FDA-2017-D-1105. Cited by volume 1 chapter 13 https://www.govinfo.gov/content/pkg/FR-2024-10-02/html/2024-22562.htm
- **16.** PIC/S, publications listing and revision notices — PE 009-17 (Part I) of 25 August 2023, the edition volume 1 read, including the annual product quality review at §1.10, the batch documentation retention clause at §4.11 and the absence of the term ALCOA from Part I; PE 009-18 adopted 22 September 2026 and in force 24 September 2026, not read by volume 1; PI 006-4 of 1 October 2026; PI 041-1 of 1 July 2021; PI 054-1 of 15 July 2021; the listing does not mark superseded entries https://picscheme.org/en/publications
- **17.** PIC/S, Joint stakeholders consultation on the revision of Chapter 4 and Annex 11 of the PIC/S GMP Guide and on the new Annex 22 — consultation 7 July to 7 October 2025, run jointly with the European Commission, with no outcome, adoption or entry into force reported. The reading of the draft's audit-trail and periodic-review content is volume 1's consultancy source, marked as such https://picscheme.org/en/news/joint-stakeholders-consultation-on-the-revision-of-chapter-4
- **18.** MedDRA MSSO, What's New with MedDRA Version 29.0, dated 3 March 2026, with training material for Version 29.1 dated 21 August 2026; released twice yearly as a licensed subscription whose fee bands volume 1 could not source. Cited by volume 1 appendix 22 https://files.meddra.org/www/Training%20Materials/2026/Materials/001363_Whats_New_with_MedDRA_V29.0.pdf
- **19.** SAHPRA, SAHPGL-INSP-02 Guidelines to Good Manufacturing Practice for Medicines — version 9 of 17 November 2025, superseded by version 10 of 13 April 2026 inside its own five-year validity window; both adopt the PIC/S guide at §2.1 and both carry no Annex 11 reference, no computerised-systems content, no data-integrity section and no record-retention periods. Cited by volume 1 chapter 13 https://www.sahpra.org.za/wp-content/uploads/2025/11/SAHPGL-INSP-02_v9-Guidelines-to-Good-Manufacturing-Practice-for-Medicines.pdf
- **20.** Pharmaceuticals — a starter pack for enterprise architects (volume 1), appendix 22, section How to check a standard's version — the seven-step procedure, and the standing warning that South African guideline versions move faster than their stated validity

APPENDIX H

Running the maturity assessment

Eleven domains, thirty-five questions, and an instrument that scores the sheets the repository does not have — because in a regulated estate the absences are the finding.

What this instrument is for

You run it twice: in your first month, and then once a year. The first run tells you what you have walked into. Every run after it tells you whether anything moved, and that is the only reason the first one is worth the effort.

It produces three things worth having. A score per domain, each attached to a named piece of evidence you have seen. A candidate gap row in `O29 Gap` for every domain scoring below 2. And a weighted mean written into `O04 Measure` with a target and a date, so next year has something to compare against. If no gap severity changed and no work package moved as a result, you scored yourself for nothing.

It scores eleven domains: the ten the ADM covers, and the one this industry adds. The eleventh is the validated estate, and it carries the joint heaviest weight, because in a pharmaceutical company the validated state of the systems is the thing every other domain depends on and the thing no generic repository has a column for.

And it scores the sheets that are not there. That is the design decision that matters most here. Seven things are absent from the repository you were handed, and each absence is a finding rather than a blank: there is no validated-system register, no periodic-review schedule, no retention clock, no regulated-role register, no South African obligation in `O63 Control Objective`, no quality body in `O57 Governance Body`, and `O29 Gap` and `O59 Compliance Assessment` are both empty [1]. An instrument that skipped those would score this estate at about 2.5 and tell you nothing.

Calderon Therapeutics is the invented company in the repository; its scores below are worked from its own rows and describe no actual company [2].

WHAT IS ABSENT	WHERE IT WOULD SIT	SCORE
No validated-system register	O67, to be created V1	● ○ ○ ○ ○
No periodic-review schedule	O68, to be created V2	● ○ ○ ○ ○
No retention clock	O69, to be created C4	● ○ ○ ○ ○
No regulated-role register	O70, to be created P3	● ○ ○ ○ ○
No South African obligation	O63, seven rows, none local RM2	● ○ ○ ○ ○
No quality body	O57, five bodies, none quality P2	● ○ ○ ○ ○
No gap and no assessment	O29 and O59, both empty E1, G1	● ○ ○ ○ ○

Figure 20.1 The seven absences, scored. Each one scores 1 and gets a sentence, never a blank. Four of the seven are the same absence seen from four directions, which is why one sheet created in week two moves four questions.

The scoring rule

- **Five levels, integers only.** No halves. A 2.5 is a disagreement you have hidden; record both scores and who gave each.
- **Levels 1, 3 and 5 are written out.** Levels 2 and 4 are interpolations and are deliberately not defined, because defining them produces an argument about the rubric instead of about the estate.
- **A score needs evidence you have seen.** Not evidence that exists, not evidence somebody says exists. A sheet and a row, a document with a date, or a person who showed you something and the date they showed you.
- **Score 1 where there is nothing.** The temptation is to score 2 out of politeness. A 1 with a reason is actionable; a 2 out of politeness is noise.
- **An absence scores 1 and gets a sentence.** Never leave the row blank, because a blank hides the finding and a reader assumes the question was not asked.
- **Weights sum to 100.** The domain score is the mean of its question scores; the weighted mean is the sum of weight times domain score, divided by 100, and reads on the same 1-to-5 scale.

Level	What you observe
1 Absent	Nothing exists, or what exists is one person's private knowledge. No artefact you could be shown
2 Ad hoc	It exists where somebody needed it. Undated, unmaintained, inconsistent between the projects that made it
3 Defined	Written down, owned by a named person, and used in the work it was written for
4 Managed	Used routinely, measured, and the measure is reported to somebody who acts on it
5 Measured	The measure drives change, and the mechanism itself is reviewed on a cycle

> **Score the repository as it was handed to you.** Items 2, 5, 10 and 11 of this volume create `O67 Validated System`, `O69 Retention Clock`, `O70 Regulated Role` and `O68 Periodic Review Schedule`, add three quality bodies to `O57` and open `O59` at `CA-001`. If you score after doing that work you are scoring your own output, which is not an assessment. Take the day-one baseline first, date it, and keep it. The re-run a year later is where those sheets earn their scores.

The eleven domains

#	Domain	Weight	What it asks	Evidence
1	Preliminary and standing	10	Does the practice bind anything in the quality system?	`O57`, `O60`, `O10`, `O11`
2	Vision and stakeholders	7	Is the promise jurisdictionally and arithmetically possible?	`O01`, `O02`, `O03`, `O30`
3	Business architecture	9	Does the model describe a pharmaceutical company?	`O09`, `O13`, `O16`, `R22`, `RPT03`
4	Data architecture	11	Who owns each entity, which system masters it, how long is it kept?	`RPT05`, `O20`, `O37`, `O43`
5	Application architecture	10	Is the portfolio true, and are the interfaces governed?	`RPT02`, `O23`, `R06`, `RPT04`
6	Technology architecture	8	Qualified, supported, and are the environments honest?	`O27`, `O48`, `RPT10`
7	Opportunities and solutions	8	Is there a gap set, and has an option been chosen?	`O29`, `O54`, `RPT07`
8	Migration planning	7	Do the plateaus have exit criteria and does the cost match the choice?	`O53`, `R15`, `O30`, `O39`
9	Implementation governance	10	Has a gate ever refused anything?	`O58`, `O59`, `O55`, `O33`
10	Requirements management	9	Does an obligation reach a thing that gets built?	`RPT06`, `O63`, `O04`
11	**The validated estate**	11	Can you answer the first question an inspector asks?	No sheet exists. The absence is the evidence

Three weights are worth defending out loud, because a board will try to move them.

The validated estate and data architecture carry 11 each. They are the two domains every other domain depends on. A system whose validation state is unknown cannot be changed, migrated, retired or released; an entity with no owner has nobody to ask when the number in a submission is wrong.

Implementation governance carries 10 despite nothing to score. It is the only domain that measures whether the practice has ever changed an outcome. `O55 Building Block` carries thirty-seven testable acceptance criteria and `O59 Compliance Assessment` carries none, and that ratio is the whole argument [3].

The validated estate's weight is not reduced because there is no sheet behind it. Weighting a domain at zero because the repository has no column for it is the same mistake the repository made. A domain with no evidence source is still a domain, and giving it a weight is how it gets a sheet.

The thirty-five questions

Domain 1 — Preliminary and standing, weight 10

Ref	Question	1	3	5
P1	Is an architecture output a controlled quality record?	Nobody has asked	A dated decision says which outputs are controlled	Change records cite your standards by document number
P2	Does a quality body hold any architecture mandate?	No quality body exists	A change control board or validation committee holds a gate	The architecture function is named in the quality manual
P3	Is there a register of regulated roles with named people?	None	O70 exists with Signs for filled	Every validated system has a named signatory and a deputy

Calderon today: 1.0. P1 scores 1: O60 Governance Decision has four columns and **no date column**, so nothing in it can be recorded as decided when, let alone as late; four of its seven rows read Outstanding [4]. P2 scores 1: O57 holds five bodies — Architecture Board, Investment Committee, Data Governance Council, Executive Committee, Architecture Function — and not one of them is a quality body [5]. P3 scores 1: O10 Actor and O11 Role hold the same eighteen generic corporate titles, every Performed by cell reads Person, and ZZ Checks reports Vacant architecture roles: 0, which is true and uninformative [6].

Domain 2 — Vision and stakeholders, weight 7

Ref	Question	1	3	5
A1	Do the drivers name the jurisdiction you work in?	Generic or foreign only	Local instruments named and dated	Each driver has an owner who re-checks its date
A2	Can the promised date survive a validation cycle?	No arithmetic was done	Dates tested against revalidation effort	Validation capacity is a planning input with a number
A3	Does the stakeholder map include the regulated roles?	Generic corporate titles	The pharmacist, the safety officer and quality are on it	Each has a named person and a recorded interest

Calderon today: 1.33. A1 scores 2: O01 Driver carries seven regulatory drivers, correctly typed External, all owned, all Pressure: High — which is real credit — and **not one names a South African instrument** [7]. A2 scores 1: O03 Objective carries horizons of 12 months to three years with no validation arithmetic anywhere, while O30 Work Package's earliest start is 2027-02 and its earliest end 2029-02, so on a late-2026 reading nothing has begun [8]. A3 scores 1, for the reason P3 does.

Domain 3 — Business architecture, weight 9

Ref	Question	1	3	5
B1	Does the organisation model name the industry's functions?	Generic corporate units	Quality, regulatory, safety, manufacturing and clinical all exist with heads	Headcount reconciles and capability ownership is correct
B2	Is there a process model, or only stage names?	<stream> — stage N	Level-1 processes per stream, each marked GxP or not	Level 2 in lanes that resolve to roles
B3	Does the capability map carry a credible score?	No scores, or scores nobody set	Scores set by the people who run the capability	Scores roll up and drive the roadmap

Calderon today: 1.33. B1 scores 1: twelve units including Customer Service and Legal, no Quality, Regulatory Affairs, Pharmacovigilance, Manufacturing or Clinical Development, and R22 Org Unit owns Capability assigning the ten level-1 capabilities in index order [9]. B2 scores 1: twenty rows named Target to Candidate — stage 1 and the like, descriptions reading Stage N of <value stream>, and O14.Stages totalling 24 against 20 rows [10]. B3 scores 2: the 100-row tree is correctly and consistently decomposed to level 3, which is genuine credit, and RPT03 Capability Heatmap then gives **all ten level-1 capabilities a gap of exactly 1 and a disposition of Sustain** — a heatmap with no heat — while the scores do not roll up: C01 is Low and its own descendant C01.01.02 is Critical [11].

Domain 4 — Data architecture, weight 11

Ref	Question	1	3	5
C1	Does every entity have a named individual owner?	None, or committees	A person per entity	Owners act on quality results
C2	Is the master system right, and agreed by one sheet?	No master recorded	One master per entity, agreed by its owner	Enforced in the interface design
C3	Is classification defensible against an instrument?	No classification, or wrong by class	Per entity, with the instrument named	Enforced at read, and tested
C4	Is there a retention clock per record class?	One corporate period, or none	O69 populated with event triggers and instruments	Computed from operational data

Calderon today: 1.25. C1 scores 2: RPT05 Data Ownership names an owner for all fourteen entities and the titles are plausibly pharmaceutical, but they are **titles, not people**, and seven of the eight distinct owners exist in no other sheet in the file [12]. C2 scores 1: eight of fourteen Master system values are wrong by a single rotation — the safety database recorded as mastering compounds and batches, the manufacturing execution system as mastering clinical data points — and O43 Master Data Rule names a Golden source for ten of the same entities and disagrees with O20 on every one of them [13]. C3 scores 1: E03 Subject is classified Internal while E07, E12 and E14 are Public, and in O37 Data Attribute the compound **name** is Public with Key: Yes while its identifier is Internal [14]. C4 scores 1: no retention sheet exists, O32 carries two retention standards that contradict each other — STD-04 Data retention Current and Conformant against STD-15 Records management standard Deprecated and Conformant — and O42 MTD-06 Retention rule reads Partial [15].

Look it up. Volume 1, chapter 16, section *The retention clocks*; and appendix 20, section *Retention clock by record class*.

Domain 5 — Application architecture, weight 10

Ref	Question	1	3	5
AP1	Is the portfolio internally consistent?	Rows contradict themselves	Criticality, health and disposition agree, or a dated decision explains why not	Disposition drives funding
AP2	Is an application owner a person?	An organisation unit	A named person who can be rung up	The owner signs the validation state
AP3	Are interfaces managed, logged and in a validation scope?	Unmanaged majority, no scope concept	Every interface managed, with a specification	Scope recorded and reviewed on cadence

Calderon today: 1.0. AP1 scores 1: four of twelve rows contradict themselves — A10 Serialisation Critical, Adequate and Retire with no successor named anywhere; A08 QMS Critical, Healthy and Replace with no reason recorded; A12 Medical Information Critical, At risk and Consolidate; only A06 eTMF is coherent [16]. AP2 scores 1: the Owner column holds organisation units, and the units are wrong — the safety database owned by Legal, the manufacturing execution system by Internal Audit, serialisation by Human Resources [16]. AP3 scores 1: of 36 interfaces, 19 are Unmanaged, 31 carry no logging, six are direct database reads and all six are unlogged, **two of the 36 are both managed and logged**, and no column anywhere records a validation scope [17].

Domain 6 — Technology architecture, weight 8

Ref	Question	1	3	5
D1	Is qualification distinguished from validation?	Neither is recorded	Components carry a qualification reference, systems a validation state	Both reviewed on cadence by named owners
D2	Is the estate within vendor support?	Unknown	A dated support end per component, with a plan per exception	Currency is a measure with a reader who acts
D3	Are non-production environments inside the privacy boundary?	Unknown, or the column is inverted	Masked, and the masking tested	Provisioning enforces it

Calderon today: 1.67. `D1` scores 1: no sheet distinguishes the two, and `O27` carries no qualification reference. `D2` scores **3** — the best score in this assessment: `O27 Physical Technology Compone` gives a dated `Support end` for all twelve components and flags the two that are past it, `PTC-07 Laboratory instruments network` at `2026-08` and `PTC-11 Enterprise storage array` at `2026-01` [18]. It stops at 3 because the measure contradicts the sheet: `MSR-07 Components within vendor support` reads 22% against the sheet's ten of twelve, which is 83% [19]. `D3` scores 1: `ENV-01 Production` records `Personal data present: No` while `ENV-02` and `ENV-04` record `Yes`, and `SEC-07 Non-production masking` is `Partial` while `O65 EP-05` reads *Most refreshes are unmasked* [20].

Domain 7 — Opportunities and solutions, weight 8

Ref	Question	1	3	5
E1	Is there a gap register with dates that can expire?	Empty, or no `Due` column	One row per difference, each with a date and an owner	Gaps close and the closure is recorded
E2	Has an option been chosen and the others closed?	Two live answers	One recommended, one dated decision, the rest rejected	The decision is reviewed at a stated date

Calderon today: 1.0. `E1` scores 1: `O29 Gap`, `R12 Gap closed by Work Package` and `RPT07 Gap Analysis` are all empty, while `RPT01 Dashboard` reports `Gaps overdue: 0` — which is not a reassuring number but a meaningless one [21]. `E2` scores 1: `ALT-B Foundations first` at $46m is `Recommended` and `ALT-E Compliance led` at $31m is **`Partly adopted`**, and `DEC-01 Adopt the recommended architecture alternative` is `Outstanding` on a sheet with no date column [22].

Domain 8 — Migration planning, weight 7

Ref	Question	1	3	5
F1	Do the transition states have written exit criteria?	A count, or nothing	Each criterion written as a testable statement	Criteria tested at a gate before the state is declared
F2	Is every work package in a plateau, at the chosen cost?	Most unallocated, cost unreconciled	Every package in a state, cost matching the chosen option	Re-planned when a package moves
F3	Is a migration treated as a validation activity?	Pairings that are not possible	Verification of value and meaning planned per entity	Protocol written before the first extract

Calderon today: 1.0. `F1` scores 1: `O53.Exit criteria` reads `5 exit criteria`, `6 exit criteria` and `7 exit criteria` — a count, not criteria [23]. `F2` scores 1: `R15 Transition contains Work Pa` allocates seven of twelve packages, leaving `WP-06`, `WP-08`, `WP-09`, `WP-10` and `WP-12` — **$29.1m** — in no state, while `O30`'s twelve rows sum to $84.3m against the $46m of the alternative marked `Recommended` [22]. `F3` scores 1: `O39 Data Migration` moves Compound from CTMS to the Safety Database and Subject from the Safety Database to the eTMF, neither of which is a possible pairing, and `O40` maps `CTMS field` to `EDC field` with rule `Direct` [24].

Domain 9 — Implementation governance, weight 10

Ref	Question	1	3	5
G1	Has a gate ever refused anything?	No assessment records at all	Verdicts recorded per criterion, with evidence	A refusal is on record, with what changed
G2	Is an acceptance criterion testable as written?	Restatements of the block's name	Each criterion has a test and a threshold	Tested at the gate and re-tested after change
G3	Is there a named release signatory?	No release gate	A gate for release with a named signatory	The signature is traceable to a validation state

Calderon today: 1.0. `G1` scores 1: four sound gates in `O58`, and `O59 Compliance Assessment` and `R24 Compliance Assessment tests` both empty; `O33 Architecture Contract` covers only `WP-01` to `WP-08`, six reading `Not yet assessable` and two `Non-compliant` [25]. `G2` scores 1: of the thirty-seven acceptance criteria in `O55 Building Block`, **exactly one passes today** — `ABB-08 QMS target`, because QMS is the only application with no unmanaged interface — and it passes while `O23` has `A08 QMS` at `Disposition: Replace`, because the criterion does not test what the block must do [3]. `G3` scores 1: the four gates end at `G3 Benefit`, there is no release gate, and `GT-3` tests a benefit confirmed against a date that no sheet carries [26].

Look it up. Volume 1, chapter 16, section *Who inspects, and what they ask for*; and chapter 13, section *What the architect is actually expected to produce*.

Domain 10 — Requirements management, weight 9

Ref	Question	1	3	5
`RM1`	Does a requirement reach a block that gets built?	Traceability generated by position	Each requirement traced to a block and a package by hand	Untraced requirements are refused at a gate
`RM2`	Is a control objective more than the obligation restated?	*Control that evidences …*	A statement of what a control must achieve	Tested, with the evidence named
`RM3`	Is there a measure definition register?	A dashboard over undefined measures	Definition, formula, grain, source and owner per measure	Measures reconcile with the sheets they measure

Calderon today: 1.0. `RM1` scores 1: `RPT06 Requirements Traceability` reports eight of thirty-two requirements reaching no building block, which is computed correctly from `R09 Requirement satisfied by Bu` — an index-order join whose `Traced` column reads `No` on every fourth row, including `REQ-001` [27]. `RM2` scores 1: every `Control objective` cell in `O63` reads *Control that evidences* followed by the obligation restated [28]. `RM3` scores 1: `O04 Measure` has six columns with no definition, formula, grain or source, and four of its sixteen measures contradict the sheets they measure [19].

Domain 11 — The validated estate, weight 11

Ref	Question	1	3	5
`V1`	Is there a register of validated systems?	None	`O67` populated, with GxP regimes and a state per system	Reconciled against the quality function's own list
`V2`	Is periodic review scheduled, with dates, and current?	No schedule	`O68` with a date per system and a named reviewer	No review overdue, and overdue triggers an action
`V3`	Is audit-trail review planned and resourced?	Nobody does it	A plan, a reviewer and recorded hours	Exceptions found, investigated and closed
`V4`	Is there a GxP classification per system and record type?	No column anywhere	Regimes recorded per system, decided by quality	Re-decided when the records change
`V5`	Is resilience tested against the regulatory clock?	Untested, and the tier set by label	Tested, with the objective justified by an obligation	Tested on cadence, with the clock named

Calderon today: 1.0. `V1` to `V4` each score 1 for the same reason, which is the single most important fact about this repository: `EXT Industry Extension` declares `X05 GxP Computerised System` — *a validated system with its validation state and owner* — and **no sheet anywhere holds one instance of it** [29]. `O23`'s ten columns carry no GxP flag, no validation state and no review date, so none of the four questions is answerable from the file. `V5` scores 1: `O49 Resilience Tier` records **nine of twelve applications as `Never tested`**, and the tier does not track the objective — `RES-08 QMS` is Tier 2 with a fifteen-minute objective and tested, while `RES-04 Safety Database`, which carries the expedited reporting clock that runs from any employee's first knowledge, is **Tier 3 with four hours and never tested** [30].

Look it up. Volume 1, chapter 13, sections *Periodic review* and *Audit-trail review, the most under-resourced obligation in the regime*; and chapter 19, section *Week one: find out what is validated*.

Form 20-A — maturity scoring sheet

One row per question. Thirty-five questions need three copies of this form. `Score` is an integer 1 to 5, `Evidence seen` is a sheet and row, a document with a date, or a person and the date they showed you. **A row with an empty `Evidence seen` is an opinion with a number attached.** The weighted mean goes into `004 Measure` as a new row with a target and a date. Signed by you, and challenged by somebody outside the architecture function before it goes anywhere.

Ref	Question	Weight	Score	Evidence seen	Date
P1	EXAMPLE — delete. Architecture output a quality record?	10	1	`060` , no date column	

Calderon's result

#	Domain	Weight	Score	Weighted
1	Preliminary and standing	10	1.0	10.0
2	Vision and stakeholders	7	1.33	9.3
3	Business architecture	9	1.33	12.0
4	Data architecture	11	1.25	13.8
5	Application architecture	10	1.0	10.0
6	Technology architecture	8	1.67	13.3
7	Opportunities and solutions	8	1.0	8.0
8	Migration planning	7	1.0	7.0
9	Implementation governance	10	1.0	10.0
10	Requirements management	9	1.0	9.0
11	The validated estate	11	1.0	11.0
	Weighted mean	**100**		**1.13**

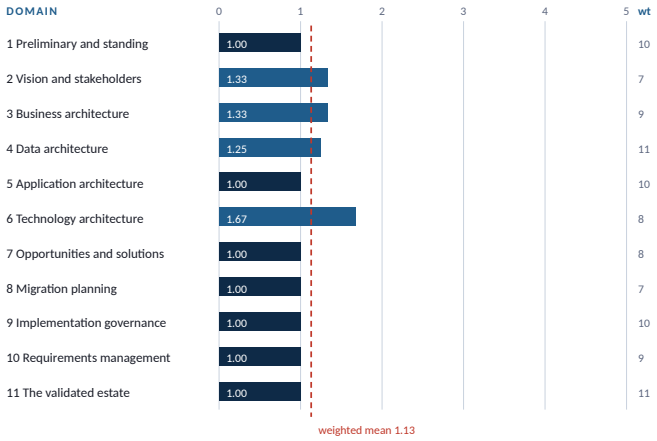


Figure 20.2 Calderon Therapeutics, eleven domains, weighted mean 1.13 of 5. Four domains clear the floor, each lifted by a single question, and only technology currency reaches 3. The scale runs 0 to 5; weights are on the right.

Band	Range	What it means, and what to do first
1 Absent	below 1.5	It does not exist in a form anyone could inspect. Do not write a target architecture. Build the record: one validated-system register, one owner per entity, one gap set with dates
2 Initial	1.5 to 2.49	It exists on paper and has not been exercised. Most inherited estates sit here. Move the two highest-weighted domains scoring 1 up one level each, and nothing else
3 Defined	2.5 to 3.49	Defined, owned and running, and not yet measured. Add the measure before adding the next capability
4 Managed	3.5 to 4.49	Enforced, with consequences somebody can point at. Check that the domain carrying the others is not one person
5 Measured	4.5 and above	Measured, reported and improving, with the movement attributable to something done. Re-weight: the judgement now lives in the weights

1.13 puts Calderon in band 1, and three readings follow from the shape rather than the number.

The estate scores lowest where the industry's risk is highest. The validated estate, application architecture, implementation governance, requirements management, opportunities and migration all sit at exactly 1.0 — thirty-one of the thirty-five questions score 1. The four questions that score above the floor are the driver list, the capability tree, the ownership column and technology currency, which are the four a repository generator can populate without knowing what industry it is in.

The only 3 in the assessment contradicts its own measure. `D2` scores 3 because `O27` gives a dated support end for every component and flags the two past it. `MSR-07` then reports 22% within support where the sheet shows 83% [19]. A dimension that scores 3 on the sheet and 1 on the dashboard is the clearest argument available for doing the measure definition register before the dashboard.

Four of the seven absences are the same absence. `V1` to `V4` all score 1 because there is no validated-system register, and everything downstream of it — the review schedule, the audit-trail plan, the GxP classification — cannot exist before it does. One sheet, created in week two, moves four questions. That is where the first cycle goes, and it is the whole argument for the sequence items 2 to 12 of this volume take.

What a maturity score is and is not worth

It is worth three things and no more. It tells you **where to look**, by ranking eleven domains that otherwise compete for attention on the basis of who complained most recently. It gives you a **dated baseline**, so that next year's run is a comparison rather than another opinion. And it converts into **gap rows**: every domain scoring below 2 becomes a candidate row in `O29 Gap`, which is the only output of this exercise that anybody downstream acts on.

It is not worth anything as a benchmark. There is no defensible South African pharmaceutical maturity benchmark for most of these domains, and for several there is no published figure at all. A score of 1.13 against an invented industry average of 2.8 is a number that will be quoted at you for two years, by people who were not in the room when either number was made.

It is not a finding, either. **A domain scoring 1 is an instruction to go and investigate, not a conclusion.** The finding is what you discover when you look: not *data architecture scores 1.3* but *the safety database is recorded as the master for compounds and for batches, and the two people who would know have never been asked*. The first sentence funds nothing. The second one starts a workshop.

Look it up. Volume 1, chapter 19, sections *The checks that take an afternoon* and *What to say when you do not know*.

And it is not an audit. Nothing in this instrument is evidence of compliance, nothing in it discharges an obligation, and a score of 5 on the validated estate would not survive five minutes of an inspection if the underlying reviews were overdue. The regulator does not ask what

your maturity is. It asks which of your systems hold GxP records, what state each one is in, who owns it, and when it was last reviewed [31]. This instrument scores whether you can answer that. It is not the answer.

> Run it, date it, keep it, and have every score challenged by somebody outside the architecture function before it leaves your laptop. A self-assessed maturity score with no challenge is worth nothing to a board, and the first person who disagrees with an unchallenged number will be right about at least one of them.

THE CHAPTER IN TEN LINES

- Eleven domains: the ten the ADM covers, plus the validated estate, which this industry adds and which carries the joint heaviest weight.
- Thirty-five questions. Levels 1, 3 and 5 are written out; 2 and 4 are interpolations left undefined on purpose.
- Integers only, no halves, and no score without evidence you have seen: a sheet and a row, a dated document, or a person and the date.
- An absence scores 1 and gets a sentence. A blank hides the finding.
- Score the repository as it was handed to you, before items 2, 5, 10 and 11 create the four missing sheets. Otherwise you are scoring your own output.
- Calderon scores 1.13 of 5, which is band 1: do not write a target architecture, build the record.
- Thirty-one of the thirty-five questions score 1, and the lowest scores sit where the industry's risk is highest.
- The only 3 in the assessment is technology currency, and its own measure contradicts it, 22 per cent against 83.
- The only two scores above the floor are the two a generic generator can produce without industry knowledge.
- Every domain below 2 becomes a candidate row in O29 Gap. That conversion is the only output anybody downstream acts on.
- The weighted mean goes into O04 Measure with a target and a date, so next year is a comparison rather than another opinion.
- It is not a benchmark, it is not a finding and it is not an audit. The regulator asks which systems hold GxP records, what state each is in, who owns it and when it was last reviewed.
- Have every score challenged by somebody outside the architecture function before it leaves your laptop.

SOURCES FOR THIS CHAPTER

- **1.** Pharmaceuticals_TOGAF_Metamodel.xlsx, Calderon Therapeutics — the seven absences, verified against all 105 sheets: no validated-system register and no GxP, validation-state or review-date column on any sheet; no periodic-review schedule; no retention sheet; no regulated-role sheet; no South African instrument named anywhere in O63 Control Objective; no quality body among O57's five; and O29 Gap and O59 Compliance Assessment both carrying a header row on row 4 and no records
- **2.** Pharmaceuticals_TOGAF_Metamodel.xlsx, Calderon Therapeutics — sheet 00 Index rows 1-12, which states that Calderon Therapeutics is fictional and that the figures are internally consistent and describe no real organisation; 105 sheets with headers on row 4 and the first record on row 5
- **3.** Pharmaceuticals_TOGAF_Metamodel.xlsx, Calderon Therapeutics — sheet O55 Building Block rows 5-41: thirty-seven acceptance criteria across ten business, ten data, ten application and seven technology blocks; ABB-08 QMS target is the only one satisfied today, against O59 Compliance Assessment and R24 Compliance Assessment tests, both empty
- **4.** Pharmaceuticals_TOGAF_Metamodel.xlsx, Calderon Therapeutics — sheet O60 Governance Decision rows 5-11: seven decisions in four columns with no date column, four reading Outstanding including DEC-01 Adopt the recommended architecture alternative
- **5.** Pharmaceuticals_TOGAF_Metamodel.xlsx, Calderon Therapeutics — sheet O57 Governance Body rows 5-9: GB-01 Architecture Board, GB-02 Investment Committee, GB-03 Data Governance Council, GB-04 Executive Committee and GB-05 Architecture Function, with their cadences and mandates; no change control board, validation committee or data integrity forum
- **6.** Pharmaceuticals_TOGAF_Metamodel.xlsx, Calderon Therapeutics — sheets O10 Actor rows 5-22, O11 Role rows 5-22, R21 Role performed by Actor rows 5-22 and ZZ Checks: the same eighteen generic corporate titles in both object sheets, every Performed by cell reading Person, every Status reading Filled, and Vacant architecture roles reported as 0 against a threshold of 0
- **7.** Pharmaceuticals_TOGAF_Metamodel.xlsx, Calderon Therapeutics — sheet O01 Driver rows 5-17: thirteen drivers of which seven are Regulatory, all Origin External, all owned by the Chief Compliance Officer, all Pressure High; the seven are GMP / GCP / GLP / GVP, 21 CFR Part 11 and Annex 11, Falsified Medicines / DSCSA, pharmacovigilance reporting, clinical trial transparency, data protection and health data, and anti-bribery and transfer of value
- **8.** Pharmaceuticals_TOGAF_Metamodel.xlsx, Calderon Therapeutics — sheets O03 Objective rows 5-18 (horizons of 12 months to three years, eleven of fourteen Not started) and O30 Work Package rows 5-16 (earliest Start 2027-02, earliest End 2029-02, three rows typed Past end date in the Delivery column)
- **9.** Pharmaceuticals_TOGAF_Metamodel.xlsx, Calderon Therapeutics — sheets O09 Organization Unit rows 5-16 and R22 Org Unit owns Capability rows 5-14: twelve units with no Quality, Regulatory Affairs, Pharmacovigilance, Manufacturing or Clinical Development, and the ten level-1 capabilities assigned ORG-01 to C01 down the index
- **10.** Pharmaceuticals_TOGAF_Metamodel.xlsx, Calderon Therapeutics — sheets O16 Business Process rows 5-24 (twenty rows named value stream plus stage number, descriptions reading Stage N of the value stream) and O14 Value Stream rows 5-9, whose Stages column totals 24 against O16's twenty rows
- **11.** Pharmaceuticals_TOGAF_Metamodel.xlsx, Calderon Therapeutics — sheets O13 Business Capability rows 5-104 (ten level-1 capabilities decomposed consistently to level 3; C01 Low while its descendant C01.01.02 is Critical) and RPT03 Capability Heatmap, which gives all ten a Gap of exactly 1 and a Disposition of Sustain
- **12.** Pharmaceuticals_TOGAF_Metamodel.xlsx, Calderon Therapeutics — sheets RPT05 Data Ownership and O20 Data Entity rows 5-18: an Owner for all fourteen entities, held as a job title rather than a person, and seven of the eight distinct owner values appearing in no other sheet in the file
- **13.** Pharmaceuticals_TOGAF_Metamodel.xlsx, Calderon Therapeutics — sheets O20 Data Entity rows 5-18 and O43 Master Data Rule rows 5-14: eight of fourteen Master system values inconsistent with the entity by a single rotation, and O43's Golden source disagreeing with O20 on every one of the ten entities it names
- **14.** Pharmaceuticals_TOGAF_Metamodel.xlsx, Calderon Therapeutics — sheets O20 Data Entity rows 5-18 (E03 Subject Internal; E07, E12 and E14 Public; E08 the only Confidential row) and O37 Data Attribute rows 5-64 (ATR-001 identifier Internal, ATR-002 name Public with Key: Yes)
- **15.** Pharmaceuticals_TOGAF_Metamodel.xlsx, Calderon Therapeutics — sheets O32 Standard rows 5-20 (STD-04 Data retention standard Current and Conformant against STD-15 Records management standard Deprecated and Conformant) and O42 Metadata Element rows 5-18 (MTD-06 Retention rule Partial, MTD-07 Consent basis and MTD-08 Lineage both Absent)
- **16.** Pharmaceuticals_TOGAF_Metamodel.xlsx, Calderon Therapeutics — sheet O23 Application Component rows 5-16: A10 Serialisation Critical, Adequate, Retire; A08 QMS Critical, Healthy, Replace; A12 Medical Information Critical, At risk, Consolidate; A06 eTMF High, Poor, Invest; and the Owner column holding organisation units, with A04 to Legal, A07 to Internal Audit and A10 to Human Resources
- **17.** Pharmaceuticals_TOGAF_Metamodel.xlsx, Calderon Therapeutics — sheets R06 Application interfaces with rows 5-40 (36 interfaces, 19 Unmanaged, 31 with Logged: No, six of type Direct DB read and all six unlogged, two both Managed and Logged) and RPT04 Interface Catalogue, whose 37th data row is a prose sentence in the Ref column
- **18.** Pharmaceuticals_TOGAF_Metamodel.xlsx, Calderon Therapeutics — sheets O27 Physical Technology Compone rows 5-16 and RPT10 Technology Currency: a dated Support end for all twelve components, PTC-07 Laboratory instruments network at 2026-08 and PTC-11 Enterprise storage array at 2026-01, both Past support
- **19.** Pharmaceuticals_TOGAF_Metamodel.xlsx, Calderon Therapeutics — sheet O04 Measure rows 5-20 against the sheets measured: MSR-01 Obligations with evidence at 54 against O63's two of seven; MSR-07 Components within vendor support at 22 against O27's ten of twelve; MSR-13 Recovery objectives tested at 53 in unit count against a population of twelve; MSR-15 and MSR-16 computable only from the empty O59. The sheet has no definition, formula, grain or source column
- **20.** Pharmaceuticals_TOGAF_Metamodel.xlsx, Calderon Therapeutics — sheets O48 Environment rows 5-8 (ENV-01 Production Personal data present No; ENV-02 and ENV-04 Yes), O41 Security Control rows 5-12 (SEC-07 Non-production masking Partial) and O65 Enforcement Point rows 5-9 (EP-05 Non-production provisioning reading Most refreshes are unmasked)
- **21.** Pharmaceuticals_TOGAF_Metamodel.xlsx, Calderon Therapeutics — sheets O29 Gap, R12 Gap closed by Work Package and RPT07 Gap Analysis, all three carrying a header row and no records, against RPT01 Dashboard reporting Gaps overdue: 0
- **22.** Pharmaceuticals_TOGAF_Metamodel.xlsx, Calderon Therapeutics — sheets O54 Architecture Alternative rows 5-9 (ALT-B Foundations first $46m Recommended, ALT-E Compliance led $31m Partly adopted), O30 Work Package rows 5-16 summing to $84.3m, and R15 Transition contains Work Pa rows 5-11 allocating seven of twelve packages, leaving WP-06, WP-08, WP-09, WP-10 and WP-12 worth $29.1m in no transition state
- **23.** Pharmaceuticals_TOGAF_Metamodel.xlsx, Calderon Therapeutics — sheet O53 Transition Architecture rows 5-7: the Exit criteria column reading 5 exit criteria, 6 exit criteria and 7 exit criteria
- **24.** Pharmaceuticals_TOGAF_Metamodel.xlsx, Calderon Therapeutics — sheets O39 Data Migration rows 5-10 (Compound from CTMS to Safety Database; Subject from Safety Database to eTMF) and O40 Source To Target Mapping rows 5-16 (CTMS field to EDC field, rule Direct, the target one position further round the same rotation as the source)
- **25.** Pharmaceuticals_TOGAF_Metamodel.xlsx, Calderon Therapeutics — sheets O58 Governance Gate rows 5-8, O59 Compliance Assessment and R24 Compliance Assessment tests (both empty) and O33 Architecture Contract rows 5-12, covering only WP-01 to WP-08 with six Not yet assessable and two Non-compliant
- **26.** Pharmaceuticals_TOGAF_Metamodel.xlsx, Calderon Therapeutics — sheet O58 Governance Gate row 8: GT-3 G3 Benefit, When After operation, What it tests Stated benefit confirmed against a date; no sheet in the file carries a benefit realisation date, and O61 CR-G04 records the correction as Approved
- **27.** Pharmaceuticals_TOGAF_Metamodel.xlsx, Calderon Therapeutics — sheets RPT06 Requirements Traceability (32 requirement rows plus a footer sentence; eight read Reaches a block: No) and R09 Requirement satisfied by Bu rows 5-36, an index-order join whose Traced column reads No on every fourth row, including REQ-001

- **28.** Pharmaceuticals_TOGAF_Metamodel.xlsx, Calderon Therapeutics — sheet O63 Control Objective rows 5-11: every Control objective cell reading Control that evidences followed by the obligation restated, with CO-01 recorded as Evidenced while O59, the sheet that would hold the evidence, is empty
- **29.** Pharmaceuticals_TOGAF_Metamodel.xlsx, Calderon Therapeutics — sheet EXT Industry Extension row 9: X05 GxP Computerised System, a validated system with its validation state and owner, declared as an object type with no instance held in any of the other 104 sheets; against sheet O23 Application Component's ten columns, which carry no GxP flag, validation state or review date
- **30.** Pharmaceuticals_TOGAF_Metamodel.xlsx, Calderon Therapeutics — sheet O49 Resilience Tier rows 5-16: nine of twelve applications Never tested; RES-08 QMS Tier 2 with a fifteen-minute objective and tested; RES-04 Safety Database Tier 3 with four hours and never tested. The expedited reporting clock that runs from first awareness by anyone in the company is from SAHPRA, SAHPGL-CEM-PV-04 v10, 22 September 2025, cited as volume 1 cited it https://www.sahpra.org.za/wp-content/uploads/2025/09/SAHPGL-CEM-PV-04_v10-Guideline-on-Post-Marketing-Reporting-of-ADRs-to-Human-Medicines-in-South-Africa.pdf
- **31.** European Commission, EudraLex Volume 4 — Annex 11 Computerised Systems, revision January 2011 (in force), which expects an up-to-date listing of all relevant systems and their GMP functionality; with volume 1 chapters 13 and 19 on what an inspector asks for and on validation as a maintained state. A draft revision was consulted jointly with PIC/S from 7 July to 7 October 2025 and no final text had been published as at the date volume 1 was written https://health.ec.europa.eu/medicinal-products/eudralex/eudralex-volume-4_en